Eisenstein sheaves, Hall categories and Canonical bases

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0 Introduction

0.1 Background and Motivation

The present work is an attempt to study the stacks of coherent sheaves on smooth projective curves, the automorphic sheaves over these stacks and their relationship with quantum algebras. Let X be a smooth projective curve over a field $\mathbf{k}$ and $\mathbf{Bun}_G$ be the stack of G-bundles over X, then the space of functions on $\mathbf{Bun}_G$ is referred as (unramified) G-automorphic forms over the function field $\mathbf{k}(X)$, which contributes to one side of the Langlands correspondence. Correspondingly, the category of constructible sheaves on $\mathbf{Bun}_G$ will fabricate into one side of geometric Langlands correspondence. Especially, when $G = \mathrm{GL}_n$, the stack $\mathbf{Bun}_G$ classifies rank n vector bundle on the curve X, which can be enlarged by considering $\mathbf{Coh}_n$ the stack of rank n coherent sheaves instead. It was Laumon's insight [Lau87] that one can build automorphic sheaves of $\mathbf{Bun}_{\mathrm{GL}_n}$ out of $\mathbf{Coh}_n$ and that $\mathbf{Coh}_0$ plays a role of Hecke operators. We can devide the automorphic sheaves into two kinds: for an irreducible localy system σ on X, it was realized in [FGV01] the construction of a cuspidal eigensheaf with eigenvalue σ via a series of Fourier transform on $\mathbf{Coh}$'s. On the complementary part, there are automorphic sheaves built from certain parabolic induction procedure, which are referred as Eisenstein sheaves in [Lau90]. More precisely, they are constructed via pull-push along the following diagram:

$$\mathbf{Coh}_r \times \mathbf{Coh}_s \xleftarrow{\mathsf{q}^{\mathrm{Coh}}} \widetilde{\mathbf{Coh}}_{r,s} \xrightarrow{\mathsf{p}^{\mathrm{Coh}}} \mathbf{Coh}_{r+s} \tag{0.1.1}$$

where the stack $\widetilde{\mathbf{Coh}}_{r,s}$ in the middle keeps track of extensions of coherent sheaves of rank r by those of rank s. Furthermore, the above diagram hinges implicitly an algebraic structure of automorphic forms: let $\mathbf{H}_X$ be the collection of spaces of function over $\mathbf{Coh}_n$, ranging over all rank $n \geqslant 0$, then we can get a graded multiplication in $\mathbf{H}_X$. The algebra $\mathbf{H}_X$ is called the **Hall algebra** associated with X, in the seminal work of Karpranov [Kap96], he shows that (a completion of) $\mathbf{H}_X$ is in fact a Hopf algebra and reveals a deep connection between $\mathbf{H}_X$ and quantum affine algebras. In particular, when X is taken to be $\mathbb{P}^1$ over $\mathbb{F}_q$, there is an algebraic isomorphism between the Hall algebra of $\mathbb{P}^1$ and positive half of (q-deformed) quantum affine algebra of $\mathfrak{sl}_2$. When X is taken to be an elliptic curve, then the resulting Hall algebra of X is called an elliptic Hall algebra, which is known to have deep relation with Cherednik double affine Hecke algebras and with the equivariant K-theory of Hilbert schemes of points on $\mathbb{A}^2$, see [BS12], [Sch12], [SV13] etc.

On the other hand, the discovery of the **canonical bases** by Kashiwara and Lusztig is a milestone in the theory of quantum groups. As an analog of Kazhdan-Lusztig bases for quantum groups, they carry favorable properties for example the positivity and integrality, they also project to bases on each of integrable highest weight module. In a series of work [Lus90a], [Lus90b], [Lus91] (see also [Lus10]), Lusztig constructed and studied the canonical bases $\mathbf{B}$ of quantized enveloping algebra $\mathbf{U}_v^+(\mathfrak{g})$ of Kac-

Moody algebra $\mathfrak{g}$, based on an observation that the quantized enveloping algebra can be realized as the Hall algebra associated with the underlying quiver $\overrightarrow{Q}$ of the Dynkin diagram of $\mathfrak{g}$. More concretely, let $\mathfrak{M}_{\overrightarrow{Q},\alpha}$ be the stack of representations of $\overrightarrow{Q}$ with weight vector α, then there is an indcution diagram

$$\mathfrak{M}_{\overrightarrow{Q},\alpha} \times \mathfrak{M}_{\overrightarrow{Q},\beta} \xleftarrow{\mathsf{q}_{\alpha,\beta}} \widetilde{\mathfrak{M}}_{\overrightarrow{Q},\alpha,\beta} \xrightarrow{\mathsf{p}_{\alpha,\beta}} \mathfrak{M}_{\overrightarrow{Q},\alpha+\beta} \tag{0.1.2}$$

then the canonical bases correspond to perverse sheaves which are simple objects arise in the induction of constant sheaves along the above diagram. Due to the similitude between category of quiver representations and of coherent sheaves on curves (notably the example of the Kronecker quiver and of $\mathbb{P}^1$) triggers the construction of canonical bases of quantum affine algebras related to curves. In [Sch06] and [Sch12], Schiffmann constructed canonical bases for the spherical parts of Hall algebras of curves with genus less or equal than one via the Eisenstein sheaves.

From another aspect, for the case of quiver $\overrightarrow{Q}$, the set of canonical bases can be arranged by geometric objects: we can consider the so-called Lusztig nilpotent stacks in the cotangent space $\Lambda_\alpha \subseteq T^*\mathfrak{M}_{\overrightarrow{Q},\alpha}$. The cotangent space $T^*\mathfrak{M}_{\overrightarrow{Q},\alpha}$ is symplectic and admits a moment map $\mu : T^*\mathfrak{M}_{\overrightarrow{Q},\alpha} \to \mathfrak{g}_\alpha^*$, where $\mathfrak{g}_\alpha$ is the Lie algebra of the automorphism group acting on $\mathfrak{M}_{\overrightarrow{Q},\alpha}$. Then Λ_α is defined to be the locus of nilpotent elements in $\mu^{-1}(0)$ and is turned out to be an Lagrangian in $T^*\mathfrak{M}_{\overrightarrow{Q},\alpha}$. We denote by Λ the collection of Λ_α by varying the weight vectors and $\mathrm{Irr}(\Lambda)$ the set of irreducible components of Λ. In [KS97], Kashiwara-Saito constructed Kashiwara operators acting on $\mathrm{Irr}(\Lambda)$, which endows $\mathrm{Irr}(\Lambda)$ a crystal structure and thus it is isomorphic to the canonical/crystal bases. The set $\mathbf{B}$ of canonical bases given by perverse sheaves and the set $\mathrm{Irr}(\Lambda)$ are tied via the *singular support* map. However, this is not a bijection of sets, the singular support of a simple object in $\mathbf{B}$ might not be irreducible.
On the contrast, in the setup of curve X, the stacks $\mathfrak{M}_{\overrightarrow{Q},\alpha}$ are replaced by $\mathbf{Coh}_\alpha$, where α now represents the rank and degree of coherent sheaves. The cotangent stack $T^*\mathbf{Coh}_\alpha$ is known to be the stack of Higgs sheaves on X, which classifies data of coherent sheaves of class α on X and Higgs fields, the twisted endomorphisms of those sheaves. Notably, the replacement of Lusztig's nilpotent stacks here is Laumon's global nilpotent cone $\mathbf{\Lambda}_\alpha \subseteq T^*\mathbf{Coh}_\alpha$ (see [Lau88]), obtained by imposing the nilpotent condition on the Higgs fields. It is also known [Gin01] that the global nilpotent cone $\mathbf{\Lambda}_\alpha$ is Lagrangian in $T^*\mathbf{Coh}_\alpha$. It was conjectured by Laumon that the singular support of Hecke eigensheaves should fall into $\mathbf{\Lambda}$ and becomes one of the major concerns (in fact, solved) of geometric Langlands program.

0.2 Outline of results

Now, the aim of this paper is threefold. In the first part, we study the geometric properties of the stacks $\mathbf{Coh}_\alpha$ and categorical properties of sheaves on $\mathbf{Coh}_\alpha$, in the fashion of modern geometric Langlands setting, developped in the past decade. We will prove the stacks $\mathbf{Coh}_\alpha$ and associated sheaf categories share many common properties as $\mathbf{Bun}_G$, which are crucial in the geometric Langlands program. Second, we consider a categorification of spherical Hall algebras given by Eisenstein sheaves.

In particular, for a curve X_0 over $\mathbb{F}_q$, we will prove that the set of simple Eisenstein sheaves forms the set of canonical bases associted with the spherical Hall algebra of X_0. This generalizes Schiffmann's result to curves of arbitrary genus. For the third part, we study the micro-local behavior of the Eisenstein sheaves, in analogous to quiver cases. We prove an isomorphism between the Grothendieck group of Eisenstein sheaves and the top Borel-Moore homology of global nilpotent cone. In the sequel of this introduction, we are going to outline the main constructions/results and give some ideas of proof.

0.2.1 *On the geometric and categorical properties of* **Coh**. Let X be a curve over $\mathbf{k} = \overline{\mathbb{F}}_q$ or $\mathbb{C}$, we recall for a fixed class α of rank and degree, the stack $\mathbf{Coh}_\alpha$ classifies the coherent sheaves of class α on X and it is an irreducible component in $\mathbf{Coh}_{\mathrm{rk}(\alpha)}$. We denote by $\mathcal{D}\text{-mod}(\mathbf{Coh}_\alpha)$ or $\mathsf{Shv}(\mathbf{Coh}_\alpha)$ the category of $\mathcal{D}$-modules (only when X is over $\mathbb{C}$) or the category of constructible sheaves respectively.
For arbitrary non-quasicompact stack $\mathcal{X}$, it is in general not true that $\mathcal{D}\text{-mod}(\mathcal{X})$ is compactly generated. However, it is a surprising fact [DG15] that $\mathcal{D}\text{-mod}(\mathbf{Bun}_G)$ is compactly generated and thus dualizable. The duality between the dual category $\mathcal{D}\text{-mod}(\mathbf{Bun}_G)^\vee$ and $\mathcal{D}\text{-mod}(\mathbf{Bun}_G)$ reflects important properties of automorphic forms, see [Che23]. More precisely, for stack $\mathcal{X}$ such that $\mathcal{D}\text{-mod}(\mathcal{X})$ is dualizable, we say $\mathcal{X}$ is *miraculous* if the functor $\text{Ps-Id}_{\mathcal{X},!} : \mathcal{D}\text{-mod}(\mathcal{X})^\vee \to \mathcal{D}\text{-mod}(\mathcal{X})$ induced by kernel $\Delta_!(\mathbb{C}_\mathcal{X})$ is an equivalence. It was proved in [Gai17] that $\mathbf{Bun}_G$ is miraculous and that the equivalence is related with a 'strange' bilinear form over automorphic functions [DW16] and functional equation of Eisenstein series. In our case of $\mathbf{Coh}_\alpha$, we prove:

Theorem A. (1) (=Proposition A.5). The category $\mathcal{D}\text{-mod}(\mathbf{Coh}_\alpha)$ is compactly generated.
(2) (=Theorem 1.11 + Corollary 1.12). The stack $\mathbf{Coh}_\alpha$ is miraculous.

The proof of the first part of Theorem A follows Drinfeld-Gaitsgory: the stack $\mathbf{Coh}_\alpha$ can be stratified by Harder-Narasimhan filtration such that any quasicompact open in $\mathbf{Coh}_\alpha$ is contained in union of Harder-Narasimhan strata which are *truncative*. For the second part, the equivalence of functor $\text{Ps-Id}_{\mathbf{Coh}_\alpha,!}$ can be reduced to the existence of continuous right adjoint. To wit, we need

Theorem B (=Theorem A.20)**.** For reductive group G with Lie algebra $\mathfrak{g}$, the stack $[\mathfrak{g}/G]$ is miraculous.

The Theorem B describes the Deligne-Lusztig duality for Lie algebras and the proof of Theorem B involves wonderful compactification of $\overline{G}$, follows the idea in [Yom19]. When we take α to be $(0,d)$, the stack $\mathbf{Coh}_{(0,d)}$ classifies torsion sheaves of degree d on X, if we take X to be affine line $\mathbb{A}^1$, then $\mathbf{Coh}_{(0,d)}(\mathbb{A}^1) = [\mathfrak{gl}_d/\mathrm{GL}_d]$, which is miraculous by Theorem B. Eventually, we show that the miraculousity of $\mathbf{Coh}_{(0,d)}(X)$ can be glued from $\mathbb{A}^1$'s. For general class α, the stack $\mathbf{Coh}_\alpha$ admits an open exhausion given by torsion degree. For each part with fixed torsion degree, its miraculousity is provided by both $\mathbf{Bun}_{\mathrm{GL}_n}$ and $\mathbf{Coh}_{(0,d)}$ and this leads to the miraculousity of the whole stack $\mathbf{Coh}_\alpha$.

0.2.2 *On the Hall categories and canonical bases of Hall algebras.* Along the diagram 0.1.1, we can define two functors $\mathsf{Eis}^{\mathrm{Coh}} := (\mathsf{p}^{\mathrm{Coh}})_!\,(\mathsf{q}^{\mathrm{Coh}})^*$ and $\mathsf{CT}^{\mathrm{Coh}} := (\mathsf{q}^{\mathrm{Coh}})_!\,(\mathsf{p}^{\mathrm{Coh}})^*$, which we refer as coherent version of geometric Eisenstein series/constant term functor. Let us denote by $\mathrm{D}(\mathbf{Coh}_\alpha)$ one of $\mathcal{D}\text{-}\mathsf{mod}(\mathbf{Coh}_\alpha)$ or $\mathsf{Shv}(\mathbf{Coh}_\alpha)$, depending on the choice of the setup, then we can prove that both functors $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$ are well-defined for both choice of sheaf theory. The idea to show this is parallel to [DG16], we need to prove $\mathsf{CT}^{\mathrm{Coh}}$ is isomorphic to $\mathsf{CT}^{\mathrm{Coh},-}_* := \left(\mathsf{q}^{\mathrm{Coh},-}\right)_* \left(\mathsf{p}^{\mathrm{Coh},-}\right)^!$, where $\mathbf{Coh}_r \times \mathbf{Coh}_s \xleftarrow{\mathsf{q}^{\mathrm{Coh},-}} \widetilde{\mathbf{Coh}}^-_{r,s} \xrightarrow{\mathsf{p}^{\mathrm{Coh},-}} \mathbf{Coh}_{r+s}$ and the stack $\widetilde{\mathbf{Coh}}^-_{r,s}$ in the middle classifies the extensions in a reverse way. The generic part of $\widetilde{\mathbf{Coh}}_{r,s}$ is given by $\mathbf{Bun}_P$ and correspondingly the generic part of $\widetilde{\mathbf{Coh}}^-_{r,s}$ is given by $\mathbf{Bun}_{P^-}$, where P^- is the opposite parabolic of P. The isomorphism of functors reflects the *second adjunction* in coherent sheaf case. To warp up, we can consider $\mathrm{D}(\mathbf{Coh}) = \prod_\alpha \mathrm{D}(\mathbf{Coh}_\alpha)$ the collection of all $\mathbf{Coh}_\alpha$, then $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$ endows $\mathrm{D}(\mathbf{Coh})$ a monoidal and comonoidal structure respectively. We call $\mathrm{D}(\mathbf{Coh})$, it is an algebra (as well as a co-algebra) object in DGCat (Proposition 2.10). And $\mathrm{D}(\mathbf{Coh})$ can thus be viewed as categorified Hall algebra in the fashion of [PS22]. As we have mentioned above, the classical Hall algebra of curve is a bialgebra, similarly, over $\mathrm{D}(\mathbf{Coh})$, the functors $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$ talk with each other, the compatibility condition of being a bialgebra is now of the following form that we call *coherent Langlands formula*:

Theorem C (=Theorem 2.13)**.** The composition of functor $\mathsf{CT}^{\mathrm{Coh}}_{r,s} \circ \mathsf{Eis}^{\mathrm{Coh}}_{p,q}$ admits a filtration with graded pieces are classified by certain permutations from $[r,s]$ to $[p,q]$. Each graded piece is of the form

$$\left(\mathsf{Eis}^{\mathrm{Coh}}_{a,c} \times \mathsf{Eis}^{\mathrm{Coh}}_{b,d}\right) \circ \mathcal{R}^c_b \circ \left(\mathsf{CT}^{\mathrm{Coh}}_{a,b} \times \mathsf{CT}^{\mathrm{Coh}}_{c,d}\right)$$

which is classified by matrix $\mathbb{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ that sums horizontally to $[p,q]$ and vertically to $[r,s]$, the functor $\mathcal{R}^c_b$ is equivalence induced by swapping isomorphism $\mathbf{Coh}_{b,c} \xrightarrow{\cong} \mathbf{Coh}_{c,b} : (\mathcal{F}_1, \mathcal{F}_2) \mapsto (\mathcal{F}_2, \mathcal{F}_1)$.

We consider the classical Hall algebra $\mathbf{H}_{X_0}$ of curve X_0 defined over $\mathbb{F}_q$, inside of which there is a subalgebra $\mathbf{H}^{\mathrm{sph}}_{X_0}$ called the spherical Hall algebra that generated by constant function on $\mathbf{Coh}_\alpha$ with $\mathrm{rk}(\alpha) \leqslant 1$. The algebra $\mathbf{H}^{\mathrm{sph}}_{X_0}$ has been studied extensively in [SV10], where they proved an isomorphism between vector bundle part of $\mathbf{H}^{\mathrm{sph}}_{X_0}$ (which is known to be a quantum loop algebra) and the algebra of functions over commuting varieties. Such isomorphism reveals a decategorified version of Betti geometric Langlands, at the neighbourhood of trivial local systems. Inspired by this isomorphism and Lusztig's construction of canonical bases, we can give a categorification of $\mathbf{H}^{\mathrm{sph}}_{X_0}$:
Consider $\mathsf{P}^{\mathrm{Coh}}$ the collection of simple objects appears in Laumon's Eisenstein sheaf $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right)$ with $\underline{\alpha} = (\alpha_1, \ldots, \alpha_s)$ is a tuple with each α_i having rank $\leqslant 1$. The elements in $\mathsf{P}^{\mathrm{Coh}}$ are called Lusztig's perverse sheaves. We construct a category $\mathsf{Q}^{\mathrm{Coh}}$ by taking the Karoubi completion of the category that additively (i.e. under direct sum and shift) generated by $\mathsf{P}^{\mathrm{Coh}}$. We prove:

Theorem D (=Theorem 3.31)**.** The trace of Frobenius induces an isomorphism between $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right)$ and the topological completion $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$ of $\mathbf{H}^{\mathrm{sph}}_{X_0}$. For each $\mathbf{P} \in \mathsf{P}^{\mathrm{Coh}}$, denote by $\mathbf{b}_{\mathbf{P}}$ the corresponding element in $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$, then $\{\mathbf{b}_{\mathsf{P}}\}_{\mathsf{P}\in\mathsf{P}^{\mathrm{Coh}}}$ forms the *canonical bases* of spherical Hall algebra. They satisfy the positivity and integrality and the Verdier dual induces an auto-equivalence on the set of canonical bases.

To prove the Theorem D, we use induction method. The constant term operators induced by functor $\mathsf{CT}^{\mathrm{Coh}}$ and the Langlands formula at function level help us reduce to considering functions that support only on finite union of certain Harder-Narasimhan strata that we call *almost semi-stable*. The spaces of functions we want to compare are finite dimensional after restriction, and the problem can be further reduced to show that the functors $\mathsf{CT}^{\mathrm{Coh}}$ are conservative over $\mathsf{Eis}^{\mathrm{Coh}}$-generated part. Such conservativity is known for geometric constant term functors over Eisenstein generated part in geometric Langlands program. To link with our case, note that the restriction of $\mathsf{Eis}^{\mathrm{Coh}}$ to vector bundle part is identical to Braverman-Gaitsgory's *compactified Eisenstein series* in [BG02]. Thus $\mathsf{P}^{\mathrm{Coh}}$ can be decomposed as $\mathsf{P}^{\mathrm{Bun}} \times \mathsf{P}^{\mathrm{Coh}}$, where elements in $\mathsf{P}^{\mathrm{Bun}}$ are Lusztig's perverse sheaves obtained from compactified Eisenstein and $\mathsf{P}^{\mathrm{Tor}}$ contains those supported on $\mathbf{Coh}_0$. We can conclude with an comparison between compactified and geometric Eisenstein seris as in [BG08].

0.2.3 *On the characteristic cycles and global nilpotent cones.* Kashiwara-Saito studied the singular supports of Lusztig's perverse sheaves in quiver case, in [Hen24], Hennecart considered their characteristic cycles, which carry more information than singular support. In particular, for quiver case, he proved that the characteristic cycle map induces an algebra isomorphism between K-theory of Lusztig perverse sheaves and top degree Borel-Moore homology of nilpotent substack in the stack of representation of preprojective algebra, whose algebraic structure also comes from parabolic induction.
In the curve case, the stacks involved are not quasi-compact, so we construct an inductive version of characteristic cycle for Eisenstein sheaves, as locally $\mathbf{Coh}_\alpha$ can be written as quasi-compact quotient stacks. More importantly, there is compatibility between characteristic cycles and $\mathsf{Eis}^{\mathrm{Coh}}$, as a result, we get an algebra morphism $\mathrm{CC}^{\mathsf{ind}} : K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right) \longrightarrow H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$, the right hand side is the top degree Borel-Moore homology of global nilpotent cone. We prove:

Theorem E (=Theorem 4.15)**.** There is a bialgebraic structure on $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$ so that the morphism $\mathrm{CC}^{\mathsf{ind}}$: $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right) \longrightarrow H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$ is an isomorphism of bialgebras.

From an explicit description of irreducible components in $\mathbf{\Lambda}$, we can see easily the surjectivity of $\mathrm{CC}^{\mathsf{ind}}$. The injectivity is more subtle, this relies on the co-algebraic structure of $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$. To construct, we note that the (derived) shifted cotangent space $\mathbf{T}^*[-1]\mathbf{Higgs}$ is the same as $\mathfrak{M}_{T^*X\times\mathbb{A}^1}$ derived stack of coherent sheaves with proper support on 3-Calabi-Yau variety $T^*X \times \mathbb{A}^1$. Thus $\mathbf{T}^*[-1]\mathbf{Higgs}_\alpha$ admits a (-1)-shifted symplectic structure and there is a so-called *DT-sheaves* or *Joyce sheaves* $\boldsymbol{\varphi}$ over it, see for example [Ben+15], [BBJ18]. Moreover, it is proved by Kinjo in [Kin22] that there is a *dimensional reduction* phenomena: let $\pi : \mathbf{T}^*[-1]\mathbf{Higgs} \to \mathbf{Higgs}$ be the projection, then $(\pi)_*\left(\boldsymbol{\varphi}_\alpha\right) = \mathbb{DC}_{\mathbf{Higgs}}[\mathbf{shift}]$, thus the Borel-Moore homology $H^{\mathrm{BM}}_*\left(\mathbf{Higgs}\right)$ can be recovered by cohomology of the complex $R\Gamma\left(\mathbf{T}^*[-1]\mathbf{Higgs}, \boldsymbol{\varphi}\right)$.

We then pass to $\mathfrak{M}^{\mathrm{an}}_{T^*X\times\mathbb{A}^1}$, the analytification of the stack $\mathfrak{M}_{T^*X\times\mathbb{A}^1}$. For U_1, U_2 two disconnected convex open set in $\mathbb{C}$, there is an open substack $\mathfrak{M}^{\mathrm{an}}_{T^*X\times U_1}\times\mathfrak{M}^{\mathrm{an}}_{T^*X\times U_2}$ in $\mathfrak{M}^{\mathrm{an}}_{T^*X\times\mathbb{A}^1}$. Moreover, the DT-sheaf restricts to two copies $\boldsymbol{\varphi}_{U_1}\boxtimes\boldsymbol{\varphi}_{U_2}$ such that over each $\mathfrak{M}^{\mathrm{an}}_{T^*X\times U_i}$, there is an equivalence of complexes $R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{T^*X\times U_i}, \boldsymbol{\varphi}_{U_i}\right)\cong R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{T^*X\times\mathbb{A}^1}, \boldsymbol{\varphi}\right)$. Thus, there is an induced co-factorization structure over $H^{\mathrm{BM}}_*\left(\mathbf{Higgs}\right)\cong H_*\left(\mathbf{Higgs}^{\mathrm{an}}\right)$. that induces the co-algebra structure on $H^{\mathrm{BM}}_*\left(\mathbf{Higgs}\right)$, which also descends to $H^{\mathrm{BM}}_{\mathrm{top}}\left(\boldsymbol{\Lambda}\right)$. On the another side, the co-algebraic structure on $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right)$ is given by $\mathsf{CT}^{\mathrm{Coh}}$, and we show there is a compatibility between coproducts on both sides (via characteristic cycle map), this further implies the injectivity of $\mathrm{CC}^{\mathsf{ind}}$, in combining with the conservativity of $\mathsf{CT}^{\mathrm{Coh}}$ on $\mathsf{Q}^{\mathrm{Coh}}$.

0.3 Structure of the paper

The paper is organized as follows: In Section 1 we recall defintion of stacks of coherent sheaves on curve $\mathbf{Coh}_\alpha$ and basic properties, we also introduce categorical properties for sheaves on $\mathbf{Coh}_\alpha$.

In Section 2 we introduce the (variation of) Hall algebras on a curve and the Hall category. We also prove Theorem C.

In Section 3 we recall the construction of Lusztig perverse sheaves and give the definition of Laumon-Lusztig category, in the end of this Section, we prove Theorem D.

In Section 4 we recall the stack of Higgs sheaves, global nilpotent cone and DT-sheaves. We give the construction of inductive characteristic cylces and prove Theorem E.

In the Appendix A, we recall the notion of second adjunction and Deligne-Lusztig involution, and we put the proof of Theorem A and Theorem B there. In the Appendix B, we recall some basic notions and properties of analytification that we need in Section 4.

0.4 Conventions

1 Stacks and categories of coherent sheaves on curves

1.1 Coherent sheaves and vector bundles on curves

The purpose of this section is to fix notations concerning the stacks of coherent sheaves on a smooth projective curve. Throughout, we let $\mathbf{k} = \overline{\mathbb{F}}_q$ or $\mathbb{C}$ be an algebraically closed field.

1.1.1 *The curve.* Let X be a smooth projective connected curve over $\mathbf{k}$ with genus g_X. We denote by $\mathsf{Coh}(X)$ the abelian category of coherent sheaves on X over $\mathbf{k}$ and write

$$(\mathrm{rk}, \deg) : K_0(\mathsf{Coh}(X)) \to \mathbb{Z}^2$$

the rank and degree functions. We put

$$\mathbf{Z}^+ := \left\{(r,d) \in \mathbb{Z}^2 \,\middle|\, r \geqslant 1 \text{ or } r = 0, d \geqslant 0\right\} \subseteq \mathbb{Z}^2, \qquad \mathbf{Z}^{++} := \mathbb{Z}_{\geqslant 1} \times \mathbb{Z} \subseteq \mathbf{Z}^+$$

Then $\mathbf{Z}^+$ is the image of rank/degree functions. For an object $\mathcal{F} \in \mathsf{Coh}(X)$, we can associate it with $(\mathrm{rk}(\mathcal{F}), \deg(\mathcal{F})) \in \mathbf{Z}^+$ the *numerical class* of $\mathcal{F}$ and write $[\mathcal{F}]$ its class in $K_0(\mathsf{Coh}(X))$. Also, we denote by $\mathsf{Fib}(X), \mathsf{Tor}(X)$ the subcategories of $\mathsf{Coh}(X)$ that stand for vector bundles, i.e. locally free coherent sheaves and torsion coherent sheaves on X respectively. The Euler form

$$K_0(\mathsf{Coh}(X))^{\otimes 2} \to \mathbb{Z}, [\mathcal{F}] \otimes [\mathcal{G}] \mapsto \langle [\mathcal{F}], [\mathcal{G}] \rangle := \sum_{i=0}^{1} (-1)^i \dim \mathrm{Ext}^i (\mathcal{F}, \mathcal{F})$$

is given explicitly by

$$\langle [\mathcal{F}], [\mathcal{G}] \rangle = (1 - g_X)\,\mathrm{rk}(\mathcal{F})\,\mathrm{rk}(\mathcal{G}) + \begin{vmatrix} \mathrm{rk}(\mathcal{F}) & \mathrm{rk}(\mathcal{G}) \\ \deg(\mathcal{F}) & \deg(\mathcal{G}) \end{vmatrix}$$

For coherent sheaves on a smooth curve, the condition of being locally free is the same as being torsion free and any $\mathcal{F} \in \mathsf{Coh}(X)$ fits into an short exact sequence

$$0 \longrightarrow \mathcal{F}^{\mathrm{Tor}} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}^{\mathrm{Bun}} \longrightarrow 0$$

where $\mathcal{F}^{\mathrm{Tor}} \in \mathsf{Tor}(X)$ and $\mathcal{F}^{\mathrm{Bun}} \in \mathsf{Fib}(X)$. Moreover, $\mathcal{F}$ can be written non-canonically as direct sum $\mathcal{F} = \mathcal{F}^{\mathrm{Tor}} \oplus \mathcal{F}^{\mathrm{Bun}}$.

1.1.2 *Harder-Narasimhan filtration.* Next, we introduce the Harder-Narasimhan filtration of coherent sheaves on X. Recall that the *slope* of $\mathcal{F} \in \mathsf{Coh}(X)$ is defined as

$$\mu(\mathcal{F}) = \frac{\deg(\mathcal{F})}{\mathrm{rk}(\mathcal{F})} \in \mathbb{Q} \cup \{\infty\}$$

The slope depends only on the numerical class, we also write $\mu(\alpha)$ for $\alpha \in \mathbf{Z}^+$. In particular for any $\mathcal{T} \in \mathsf{Tor}(X)$, its slope $\mu(\mathcal{T})$ is set to be ∞.

For $\mathcal{F} \in \mathsf{Coh}(X)$, it is said to *semi-stable* if the slope of any subsheaf $\mathcal{F}' \subseteq \mathcal{F}$ satisfies $\mu(\mathcal{F}') \leqslant \mu(\mathcal{F})$. For any $\mu \in \mathbb{Q} \cup \{\infty\}$, let $\mathsf{Coh}^{\mu-\mathsf{ss}}$ be the category of semi-stable coherent sheaves with slope μ, then it is an abelian subcategory of $\mathsf{Coh}(X)$. We have the following Lemma:

Lemma 1.1. *Let $\mathcal{F}_1 \in \mathsf{Coh}^{\mu_1-\mathsf{ss}}$ and $\mathcal{F}_2 \in \mathsf{Coh}^{\mu_2-\mathsf{ss}}$ be semi-stable sheaves with slope μ_1, μ_2 respectively,*
(1) If $\mu_2 > \mu_1$, then $\mathrm{Hom}(\mathcal{F}_2, \mathcal{F}_1) = 0$;
(2) If $\mu_1 + 2g_X - 2 < \mu_2$, then $\mathrm{Ext}^1(\mathcal{F}_1, \mathcal{F}_2) = 0$.

Any coherent sheaf $\mathcal{F} \in \mathsf{Coh}(X)$ admits a unique strict filtration by sub-coherent sheaves

$$0 = \mathcal{F}_0 \subsetneq \mathcal{F}_1 \subsetneq \mathcal{F}_2 \subsetneq \ldots \subsetneq \mathcal{F}_s = \mathcal{F}$$

such that $\mu\left(\mathcal{F}_i/\mathcal{F}_{i-1}\right) \geqslant \mu\left(\mathcal{F}_{i+1}/\mathcal{F}_i\right)$ holds for any $i = 1, \ldots, k-1$ and each quotient $\mathcal{F}_i/\mathcal{F}_{i-1}$ is semi-stable. Such filtration is called the *Harder-Narasimhan filtration* (HN-filtration in short) of $\mathcal{F}$. It is known that the HN filtration exixts for coherent sheaves of curves defined over $\mathbb{F}_q$ and is stable under field extension. For any $\alpha \in \mathbf{Z}^+$, we put

$$\mathrm{HN}(\alpha) = \left\{ (\alpha_1, \ldots, \alpha_s) \,|\, \alpha_i \in \mathbf{Z}^+, \sum_i \alpha_i = \alpha, \mu(\alpha_i) \geqslant \mu(\alpha_{i+1}) \right\}$$

The subset of all partitions of α in $\mathbf{Z}^+$. We call elements in $\mathrm{HN}(\alpha)$ partitions of α of Harder-Narasimhan type. Let $\mathrm{HN} : \mathsf{Coh}_\alpha(X) \to \mathrm{HN}(\alpha)$ be the function associates a coherent sheaf $\mathcal{F} \in \mathsf{Coh}_\alpha(X)$ with $(\alpha_1, \ldots, \alpha_s)$, where α_i is numerical class of $\mathcal{F}_i/\mathcal{F}_{i-1}$ that comes from the HN-filtration of $\mathcal{F}$. In graphic point of view, we can associate any partition $\underline{\alpha}$ with a polygon $\mathrm{Pol}(\underline{\alpha})$ in $\mathbb{Q}^2$ by connecting the points $0, \alpha_1, \alpha_1 + \alpha_2, \ldots, \sum_{k=1}^i \alpha_k$. Then $\underline{\alpha} = (\alpha_1, \ldots, \alpha_s)$ is of HN-type if and only if the corresponding polygon is convex, in which case we call it an HN polygon.
There is a partial order on $\mathrm{HN}(\alpha)$: for $\underline{\alpha_1}, \underline{\alpha_2} \in \mathrm{HN}(\alpha)$, we say $\underline{\alpha_1} \preceq \underline{\alpha_2}$ if the convex hull of the HN polygon $\mathrm{Pol}(\underline{\alpha_1})$ contains $\mathrm{Pol}(\underline{\alpha_2})$.

1.1.3 *Stacks of coherent sheaves and G-torsors on curve.* For $\alpha = (r, d) \in \mathbf{Z}^+$, we define the stack of coherent sheaves on X of class α to be the functor:

$$\mathbf{Coh}_\alpha : S \in \mathsf{Sch}_{\mathbf{k}} \mapsto \left\langle \mathcal{F} \in \mathsf{Coh}(X \times S) \;\middle|\; \begin{array}{l} \mathcal{F} \text{ flat over } S, \\ [\mathcal{F}|_{X_s}] = \alpha, \forall s : \operatorname{Spec} \mathbf{k} \to S \end{array} \right\rangle \in \mathsf{Gpd}$$

When $\alpha \in \mathbf{Z}^{++}$, we can also consider $\mathbf{Bun}_\alpha$ the stack classifies the vector bundles of class α on X:

$$\mathbf{Bun}_\alpha : S \in \mathsf{Sch}_{\mathbf{k}} \mapsto \left\langle \mathcal{E} \in \mathsf{Fib}(X \times S) \;\middle|\; \begin{array}{l} \mathcal{E} \in \mathsf{Fib}_n(X \times S) \\ \deg(E|_{X_s}) = d \end{array} \right\rangle \in \mathsf{Gpd}$$

In particular, $\mathbf{Bun}_\alpha$ is an irreducible open substack of $\mathbf{Coh}_\alpha$, we denote by $\jmath_\alpha : \mathbf{Bun}_\alpha \hookrightarrow \mathbf{Coh}_\alpha$ the open inclusion. In replacing α by rank r, we can define stack of coherent sheaves/vector bundles of rank r on X, denoted by $\mathbf{Coh}_r$ and $\mathbf{Bun}_r$ respectively. They decompose into irreducible components

$$\mathbf{Coh}_r = \bigsqcup_{\alpha \in \mathbf{Z}^+, \mathrm{rk}(\alpha) = r} \mathbf{Coh}_\alpha; \quad \mathbf{Bun}_r = \bigsqcup_{\alpha \in \mathbf{Z}^{++}, \mathrm{rk}(\alpha) = r} \mathbf{Bun}_\alpha.$$

Both stacks $\mathbf{Coh}_\alpha$ and $\mathbf{Bun}_\alpha$ are smooth algebraic stacks of dimension $(g_X - 1)(\mathrm{rk}(\alpha))^2$, which does not depend on the degree of α, this gives the equi-dimensionality of the stacks $\mathbf{Coh}_r$ and $\mathbf{Bun}_r$. In fact, let $C_\alpha \in \mathsf{Coh}(\mathbf{Coh}_\alpha \times X)$ be the universal sheaf, then the tangent complex of $\mathbf{Coh}_\alpha$ is given by $\mathbb{T}_{\mathbf{Coh}_\alpha} = (p_1)_* \mathcal{H}\mathrm{om}(C_\alpha, C_\alpha)[1]$, where $p_1 : \mathbf{Coh}_\alpha \times X \to \mathbf{Coh}_\alpha$ is the projection. The fiber of $\mathbb{T}_{\mathbf{Coh}_\alpha}$ at a

point $\mathcal{F} \in \mathbf{Coh}_\alpha$ can be computed as $\mathbb{T}_{\mathbf{Coh}_\alpha}|_{\mathcal{F}} = \left[\mathrm{End}(\mathcal{F}) \to \mathrm{Ext}^1(\mathcal{F},\mathcal{F})\right]$. The dimension of $\mathbf{Coh}_\alpha$ as well as $\mathbf{Bun}_\alpha$ can thus be calculated using characteristic formula.
Also, it is convenient to write $\tau = (0,1) \in \mathbf{Z}^+$ and denote $\mathbf{Coh}_{\tau d}$ the stack parametrize rank 0, degree d torsion sheaves on X. The stack $\mathbf{Coh}_0 = \bigsqcup_{d \geqslant 0} \mathbf{Coh}_{\tau d}$ is of dimension zero.

Remark 1.2. Later, we will need derived stacks of coherent sheaves/vector bundles on general schemes. The stacks $\mathbf{Coh}_\alpha$ and $\mathbf{Bun}_\alpha$ match with derived ones, due to the fact that X is smooth projective of dimension 1 and we have seen thus the cotangent complexes concentrate in degree $[-1,0]$.

A vector bundle of rank r on X is the same as a GL_r-torsor on X. For an affine group scheme H, we denote by $\mathbf{Bun}_H$ the mapping stack $\mathbf{Maps}(X, \mathrm{pt}/H) = \mathbf{Maps}(X, \mathbf{B}H)$, it is known to be an algebraic stack that parametrize H-torsors on X. For $r \geqslant 1$, the stack $\mathbf{Bun}_{\mathrm{GL}_r}$ coincides with $\mathbf{Bun}_r = \bigsqcup_{d \in \mathbb{Z}} \mathbf{Bun}_{(r,d)}$.

1.1.4 *Two open exhaustions of* **Coh**. We consider the complement of $\mathbf{Bun}_\alpha$ in $\mathbf{Coh}_\alpha$. Recall any coherent sheaf $\mathcal{F}$ on X can be decomposed into torsion and torsion-free parts, For $\alpha \in \mathbf{Z}^+, d \geqslant 0$, we denote by $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}=d}$ the locally closed substack of $\mathbf{Coh}_\alpha$ who classifies coherent sheaves $\mathcal{F}$ of class α and $\deg\left(\mathcal{F}^{\mathrm{tor}}\right) = d$. We get:

$$\mathbf{Coh}_\alpha = \bigsqcup_{d \geqslant 0} \mathbf{Coh}_\alpha^{\mathrm{tor.deg}=d}$$

and we set $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant d} = \bigsqcup_{0 \leqslant d' \leqslant d} \mathbf{Coh}_\alpha^{\mathrm{tor.deg}=d'}$ that parametrizes $\mathcal{F}$ such that $\deg\left(\mathcal{F}^{\mathrm{tor}}\right) \leqslant d$.

Proposition 1.3. *For any torsion degree $d \geqslant 0$, we have*
(1) The substack $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant d}$ is open in $\mathbf{Coh}_\alpha$ and there is a series of open exhaustion:

$$\mathbf{Bun}_\alpha = \mathbf{Coh}_\alpha^{\mathrm{tor.deg}=0} \hookrightarrow \mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant 1} \hookrightarrow \ldots \hookrightarrow \mathbf{Coh}_\alpha$$

(2) The canonical map $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}=d} \mathbf{Bun}_{\alpha-\tau d} \times \mathbf{Coh}_{\tau d}$ is a gerbe of unipotent group of rank $-\mathrm{rk}(\alpha)\cdot d$.

The open substack $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant d}$ are in general not quasicompact. There is another open exhaustion of $\mathbf{Coh}_\alpha$ is given by quasicompact global quotient stack of the form $\left[\mathcal{X}/\mathcal{G}\right]$.
For a fixed line bundle $\mathcal{L} \in \mathrm{Pic}_X$, we say a coherent sheaf $\mathcal{F} \in \mathsf{Coh}(X)$ is *strongly generated* by $\mathcal{L}$ if

$$\mathrm{Hom}(\mathcal{L},\mathcal{F}) \otimes \mathcal{L} \to \mathcal{F} \text{ is surjective, and } \mathrm{Ext}^1(\mathcal{L},\mathcal{F}) = \{0\}$$

We consider $\mathbf{Coh}_\alpha^{>\mathcal{L}} \subseteq \mathbf{Coh}_\alpha$ that parametrize coherent sheaves strongly generated by $\mathcal{L}$:

$$\mathbf{Coh}_\alpha^{>\mathcal{L}} : S \mapsto \left\langle \mathcal{F} \in \mathbf{Coh}_\alpha(S) \,\middle|\, \mathcal{F}|_{X_s} \text{ strongly generated by } \mathcal{L}, \forall s : \mathrm{Spec}\,\mathbf{k} \to S \right\rangle$$

We have the following result of $\mathbf{Coh}_\alpha^{>\mathcal{L}}$:

Proposition 1.4. *[Sch16, Lemma 6.1] (1) Let $\mathcal{L}$ be a line bundle on X with Euler class $c(\mathcal{L})$, then $\mathbf{Coh}_\alpha^{>\mathcal{L}}$ is an open quasicompact substack in $\mathbf{Coh}_\alpha$ which isomorphic to the quotient stack $\left[Q_\alpha^{\mathcal{L}}/\mathrm{G}_\alpha^{\mathcal{L}}\right]$, where $Q_\alpha^{\mathcal{L}}$ is a smooth and reduced scheme and $\mathrm{G}_\alpha^{\mathcal{L}} \cong \mathrm{GL}_{\langle c(\mathcal{L}),\alpha\rangle}$.*

(2) Given $\mathcal{L}'$ strongly generated by $\mathcal{L}$, there is an open inclusion $j^{\mathcal{L}',\mathcal{L}} : \mathbf{Coh}_\alpha^{>\mathcal{L}'} \hookrightarrow \mathbf{Coh}_\alpha^{>\mathcal{L}}$ and $\mathbf{Coh}_\alpha$ can be exhibited as $\operatorname{colim}_{\mathcal{L}\in\mathrm{Pic}_X} \mathbf{Coh}_\alpha^{>\mathcal{L}}$ where the partial order on Pic_X is given by strong generation.

Proof. For (i), we consider the Quot scheme $\mathrm{Quot}_\alpha^{\mathcal{L}} := \mathrm{Quot}_X\left(\mathcal{L}\otimes\mathbf{k}^{\langle\mathcal{L},\alpha\rangle},\alpha\right)$ that classifies surjections $\phi : \mathcal{L}\otimes\mathbf{k}^{\langle c(\mathcal{L}),\alpha\rangle} \twoheadrightarrow \mathcal{F}$ with $\mathcal{F}\in\mathbf{Coh}_\alpha$. We can define an open subscheme $Q_\alpha^{\mathcal{L}} \subseteq \mathrm{Quot}_\alpha^{\mathcal{L}}$ so that

$$Q_\alpha^{\mathcal{L}}(\mathbf{k}) = \left\{\left[\phi : \mathcal{L}\otimes\mathbf{k}^{\langle c(\mathcal{L}),\alpha\rangle} \twoheadrightarrow \mathcal{F}\right] \in \mathrm{Quot}_\alpha^{\mathcal{L}} \mid \phi_* : \mathbf{k}^{\langle c(\mathcal{L}),\alpha\rangle} \xrightarrow{\cong} \mathrm{Hom}(\mathcal{L},\mathcal{F})\right\}.$$

There is a natural action of $\mathrm{G}_\alpha^{\mathcal{L}} := \mathrm{GL}_{\langle c(\mathcal{L}),\alpha\rangle}$ on $\mathbf{k}^{\langle c(\mathcal{L}),\alpha\rangle}$ that gives arises to an action of $\mathrm{G}_\alpha^{\mathcal{L}}$ on $Q_\alpha^{\mathcal{L}}$. In particular, it follows from construction that $\mathbf{Coh}_\alpha^{>\mathcal{L}} \cong \left[Q_\alpha^{\mathcal{L}}/\mathrm{G}_\alpha^{\mathcal{L}}\right]$. Moreover, $Q_\alpha^{\mathcal{L}}$ is smooth as for any $\phi\in Q_\alpha^{\mathcal{L}}$, we have the vanishing $\mathrm{Ext}^1(\mathrm{Ker}(\phi),\mathcal{F}) = 0$.
For (ii), it is not hard to see if given $\mathcal{L}' > \mathcal{L}$, then a coherent sheaf $\mathcal{F}$ strongly generated by $\mathcal{L}'$ is strongly generated by $\mathcal{L}$. We can consider subscheme $Q_\alpha^{\mathcal{L},\mathcal{L}'} \subseteq Q_\alpha^{\mathcal{L}}$ contains those $\left[\phi : \mathcal{L}\otimes\mathbf{k}^{\langle c(\mathcal{L}),\alpha\rangle} \twoheadrightarrow \mathcal{F}\right]$ with $\mathcal{F}\in\mathbf{Coh}_\alpha^{>\mathcal{L}'}$, this is open since being strongly generated is an open condition. Then we have $\mathbf{Coh}_\alpha^{>\mathcal{L}'} \cong \left[Q_\alpha^{\mathcal{L},\mathcal{L}'}/\mathrm{G}_\alpha^{\mathcal{L}}\right] \subseteq \left[Q_\alpha^{\mathcal{L}}/\mathrm{G}_\alpha^{\mathcal{L}}\right]$. Moreover, by [SS20, Lemma 2.2], there is an canonical isomorphism $\left[Q_\alpha^{\mathcal{L},\mathcal{L}'}/\mathrm{G}_\alpha^{\mathcal{L}}\right] \xrightarrow{\cong} \left[Q_\alpha^{\mathcal{L}'}/\mathrm{G}_\alpha^{\mathcal{L}'}\right]$ and thus we get an open inclusion $j^{\mathcal{L},\mathcal{L}'} : \mathbf{Coh}_\alpha^{>\mathcal{L}'} \hookrightarrow \mathbf{Coh}_\alpha^{>\mathcal{L}}$. ∎

The above Proposition establish $\mathbf{Coh}_\alpha$ as the following ind-algebraic stack

$$\mathbf{Coh}_\alpha = \varinjlim_{\mathcal{L}\in\mathrm{Pic}_X} \mathbf{Coh}_\alpha^{>\mathcal{L}}$$

1.2 Stacks of filtration

1.2.1 *Harder-Narasimhan stratification.* The Harder-Narasimhan filtration defines a weak stratification on $\mathbf{Coh}_\alpha$, which is labelled by the set $(\mathrm{HN}(\alpha),\preceq)$. We denote by $\mathbf{Coh}_\alpha^{\mathrm{ss}} \subseteq \mathbf{Coh}_\alpha$ the open substack of $\mathbf{Coh}_\alpha$. Notice when $\alpha = \tau d$, $\mathbf{Coh}_{\tau d}^{\mathrm{ss}} = \mathbf{Coh}_{\tau d}$ and for $\alpha\in\mathbf{Z}^{++}$, $\mathbf{Coh}_\alpha^{\mathrm{ss}} \subseteq \mathbf{Bun}_\alpha$. For each $\underline{\alpha}\in\mathrm{HN}(\alpha)$, we denote by $\mathbf{HN}_{\underline{\alpha}} \subseteq \mathbf{Coh}_\alpha$ the locally closed substack classifies coherent sheaves in $\mathbf{Coh}_\alpha$ whose HN-filtration has underlying type $\underline{\alpha}$, namely

$$\mathbf{HN}_{\underline{\alpha}} : S\in\mathsf{Sch}_{\mathbf{k}} \mapsto \left\langle \mathcal{F}\in\mathsf{Coh}(X\times S) \;\middle|\; \begin{array}{l}\mathcal{F} \text{ flat over } S, [\mathcal{F}|_{X_s}] = \alpha \\ \mathrm{HN}\left(\mathcal{F}|_{X_s}\right) = \underline{\alpha}, \forall s : \mathrm{Spec}\,\mathbf{k}\to S\end{array}\right\rangle$$

Then for any $\alpha\in\mathbf{Z}^+$ and $\underline{\alpha}\in\mathrm{HN}(\alpha)$, we have:

Proposition 1.5. *(1) The canonical map $\mathbf{HN}_{\underline{\alpha}} \to \prod_i \mathbf{Coh}_{\alpha_i}^{\mathrm{ss}}$ is a unipotent gerbe of rank $-\sum_i\langle\alpha_{i+1},\alpha_i\rangle$.*
(2) The locally closed substacks $\mathbf{HN}_{\underline{\alpha}}$ give a weak stratification of $\mathbf{Coh}_\alpha$, in the sense that $\mathbf{Coh}_\alpha = \bigsqcup_{\underline{\alpha}\in\mathrm{HN}(\alpha)} \mathbf{HN}_{\underline{\alpha}}$ and $\mathbf{HN}_{\preceq\underline{\alpha}} := \bigsqcup_{\underline{\beta}\preceq\underline{\alpha}\in\mathrm{HN}(\alpha)} \mathbf{HN}_{\underline{\beta}}$ is a closed substack in $\mathbf{Coh}_\alpha$.

1.2.2 *Very unstable and almost semi-stable sheaves.* For later usage, we two special kinds of Harder-Narasimhan types. According to Lemma 1.1, if $\underline{\alpha} = (\alpha_1,\ldots,\alpha_s)\in\mathrm{HN}(\alpha)$ satisfies $\mu(\alpha_i)-\mu(\alpha_{i+1}) > 2(g_X - 1)$ for some i, the Harder-Narasimhan filtration splits at the position i, namely any $\mathcal{F}\in\mathbf{HN}_{\underline{\alpha}}(\mathbf{k})$ can be written as direct sum $\mathcal{F}^{\leqslant i} \oplus \mathcal{F}^{\geqslant i+1}$ such that $\mathcal{F}^{\leqslant i} \in \mathbf{HN}_{\underline{\beta}}(\mathbf{k}), \mathcal{F}^{\geqslant i+1} \in \mathbf{HN}_{\underline{\gamma}}(\mathbf{k})$, where $\underline{\beta} = (\alpha_1, ..., \alpha_i)$ and $\underline{\gamma} = (\alpha_{i+1}, ..., \alpha_s)$. The HN-type $\underline{\alpha} \in \mathrm{HN}(\alpha)$ is said to be *very unstable* if the slopes satisfies

(vu) $\mu(\alpha_i) > \mu(\alpha_{i+1}) + 2(g_X - 1)$ *for all* i.

Also, we call $\underline{\alpha} \in \mathrm{HN}(\alpha)$ *almost semi-stable* if it satisfies

(as) $\mu(\alpha_i) \leqslant \mu(\alpha_{i+1}) + 2(g_X - 1)$ *for all* i.

We denote the locus of very unstable sheaves and almost semi-stable sheaves by $\mathbf{Coh}_\alpha^{\mathsf{vu}}$ and $\mathbf{Coh}_\alpha^{\mathsf{as}}$ respectively, then

$$\mathbf{Coh}_\alpha^{\mathsf{vu}} = \bigsqcup_{\substack{\underline{\alpha}\in\mathrm{HN}(\alpha)\\ \underline{\alpha}\text{ very unstable}}} \mathbf{HN}_{\underline{\alpha}} \qquad \mathbf{Coh}_\alpha^{\mathsf{as}} = \bigsqcup_{\substack{\underline{\alpha}\in\mathrm{HN}(\alpha)\\ \underline{\alpha}\text{ almost semi-stable}}} \mathbf{HN}_{\underline{\alpha}}$$

In particular, $\mathbf{Coh}_\alpha^{\mathsf{ss}} \subseteq \mathbf{Coh}_\alpha^{\mathsf{as}}$ is a quasicompact open substack. Moreover, since there are finitely many almost semi-stable HN-types, the stack $\mathbf{Coh}_\alpha^{\mathsf{as}}$ is also quasicompact. The rest of HN types are mixture of very unstable ones and almost semi-stable ones, we may refer them as *not very unstable*, we put the corresponding locus:

$$\mathbf{Coh}_\alpha^{\mathsf{nvu}} = \mathbf{Coh}_\alpha - \mathbf{Coh}_\alpha^{\mathsf{vu}} - \mathbf{Coh}_\alpha^{\mathsf{as}}$$

1.2.3 *Stacks of filtration.* Let $\underline{\alpha} = (\alpha_1, ..., \alpha_s)$ be an arbitrary partition of α, not necessarily of HN-type. Consider the stack of filtration $\widetilde{\mathbf{Coh}}_{\underline{\alpha}}$ that parametrize flags of coherent sheaves of type $\underline{\alpha}$:

$$\widetilde{\mathbf{Coh}}_{\underline{\alpha}} : S \in \mathsf{Sch}_{\mathbf{k}} \mapsto \left\langle \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq ... \subseteq \mathcal{F}_s \;\middle|\; \begin{array}{l} \mathcal{F}_i \in \mathbf{Coh}_{\sum_{k=1}^i \alpha_k}(S) \\ [\mathcal{F}_i/\mathcal{F}_{i-1}|_{X_s}] = \alpha_i, \forall i, s : \operatorname{Spec}\mathbf{k} \to X_S \end{array} \right\rangle \in \mathsf{Gpd}$$

morphisms in the above groupoid are given by isomorphisms of flags. We have following diagram

$$\mathbf{Coh}_{\alpha_1} \times ... \times \mathbf{Coh}_{\alpha_s} \xleftarrow{\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}} \widetilde{\mathbf{Coh}}_{\underline{\alpha}} \xrightarrow{\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}} \mathbf{Coh}_\alpha$$

where $\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}} : (\mathcal{F}_1 \subseteq ... \subseteq \mathcal{F}_s) \mapsto \mathcal{F}_s$ and $\mathsf{q}_{\alpha}^{\mathrm{Coh}} : (\mathcal{F}_1 \subseteq ... \subseteq \mathcal{F}_s) \mapsto (\mathcal{F}_1, \mathcal{F}_2/\mathcal{F}_1, ..., \mathcal{F}_s/\mathcal{F}_{s-1})$. We have

Proposition 1.6. *(1) The morphism $\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}$ is a vector bundle stack of relative dimension $-\sum_{i=1}^{s-1}\left\langle \sum_{j=1}^i \alpha_j, \alpha_{i+1}\right\rangle$.*
(2) The morphism $\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}$ is schematic and proper.

When $\alpha_1, ..., \alpha_s \in \mathbf{Z}^{++}$, we have similar stack of filtration for vector bundles:

$$\mathbf{Bun}_{P_{\underline{\alpha}}} : S \in \mathsf{Sch}_{\mathbf{k}} \mapsto \left\langle \mathcal{E}_1 \subseteq \mathcal{E}_2 ... \subseteq \mathcal{E}_s \;\middle|\; \begin{array}{l} \mathcal{E}_i \in \mathbf{Bun}_{\sum_{k=1}^i \alpha_k}(S), \\ \mathcal{E}_i/\mathcal{E}_{i-1} \in \mathbf{Bun}_{\alpha_i}(S) \end{array} \right\rangle \in \mathsf{Gpd}$$

Similarly, we have diagram: $\mathbf{Bun}_{\alpha_1} \times ... \times \mathbf{Bun}_{\alpha_s} \xleftarrow{\mathsf{q}_{\underline{\alpha}}^{\mathrm{Bun}}} \mathbf{Bun}_{P_{\underline{\alpha}}} \xrightarrow{\mathsf{p}_{\underline{\alpha}}^{\mathrm{Bun}}} \mathbf{Bun}_{\sum_{k=1}^s \alpha_k}$ where

$$\mathsf{p}_{\underline{\alpha}}^{\mathrm{Bun}} : (\mathcal{E}_1 \subseteq ... \subseteq \mathcal{E}_s) \mapsto \mathcal{E}_s \qquad \mathsf{q}_{\underline{\alpha}}^{\mathrm{Bun}} : (\mathcal{E}_1 \subseteq ... \subseteq \mathcal{E}_s) \mapsto (\mathcal{E}_1, \mathcal{E}_2/\mathcal{E}_1, ..., \mathcal{E}_s/\mathcal{E}_{s-1})$$

In fact, let $r_i = \mathrm{rk}(\alpha_i)$, then $\prod_{i=1}^s \mathbf{Bun}_{\alpha_i}$ is an irreducible component of $\mathbf{Bun}_{M_{\underline{r}}}$ with $\underline{r} = (r_1, ..., r_s)$ and $M_{\underline{r}}$ the standard Levi subgroup of $\mathrm{GL}_{\sum_{k=1}^s r_k}$. Let $P_{\underline{r}}$ be the standard parabolic, the stack of filtration $\mathbf{Bun}_{P_{\underline{\alpha}}}$ is an irreducible component of $\mathbf{Bun}_{P_{\underline{r}}} = \mathbf{Maps}\left(X, \mathbf{B}P_{\underline{r}}\right)$.

Remark 1.7. Similar to the Proposition 1.6, the map $\mathsf{q}^{\mathrm{Bun}}_{\underline{\alpha}}$ is a vector bundle, however, the schematic map $\mathsf{p}^{\mathrm{Bun}}_{\underline{\alpha}}$ is not proper in general since we have imposed the subbundle condition.

Now let $\underline{\alpha} \in \mathrm{HN}(\alpha)$ be a partition of α of HN-type. We denote

$$\jmath^{\mathrm{HN}}_{\underline{\alpha}} : \mathbf{HN}_{\underline{\alpha}} \longrightarrow \widetilde{\mathbf{Coh}}_{\underline{\alpha}}; \qquad \mathcal{F} \mapsto (\mathcal{F}_1 \subseteq \ldots \subseteq \mathcal{F}_s)$$

the map that sends $\mathcal{F} \in \mathbf{HN}_{\underline{\alpha}}$ to the HN-filtration of $\mathcal{F}$. Due to the properties of Harder-Narasimhan filtration, each $\mathcal{F}_i$ is semi-stable, thus the image of the composition $\mathsf{q}^{\mathrm{Coh}}_{\underline{\alpha}} \circ \jmath^{\mathrm{HN}}_{\underline{\alpha}}$ lies in the product of semi-stable locus $\prod_{k=1}^{s} \mathbf{Coh}^{\mathsf{ss}}_{\alpha_k} \subseteq \prod_{k=1}^{s} \mathbf{Coh}_{\alpha_k}$. Conversely, given a filtration $\mathcal{F}_1 \subseteq \ldots \subseteq \mathcal{F}_s$ in $\mathsf{Coh}(X)$, if each $\mathcal{F}_i/\mathcal{F}_{i-1}$ is semi-stable, then this determines the unique HN-filtration of $\mathcal{F}_s$. As a conclusion, we can relate $\widetilde{\mathbf{Coh}}_{\underline{\alpha}}$ and $\mathbf{HN}_{\underline{\alpha}}$ via the following diagram:

$$\begin{tikzcd}
\prod_{k=1}^{s} \mathbf{Coh}_{\alpha_k} & & \widetilde{\mathbf{Coh}}_{\underline{\alpha}} \arrow[ll, "\mathsf{q}^{\mathrm{Coh}}_{\underline{\alpha}}"'] \arrow[r, "\mathsf{p}^{\mathrm{Coh}}_{\underline{\alpha}}"] & \mathbf{Coh}_{\alpha} \\
\prod_{k=1}^{s} \mathbf{Coh}^{\mathsf{ss}}_{\alpha_k} \arrow[u, hook] & \widetilde{\mathbf{Coh}}_{\underline{\alpha}} \underset{\prod_{k=1}^{s} \mathbf{Coh}_{\alpha_k}}{\times} \prod_{k=1}^{s} \mathbf{Coh}^{\mathsf{ss}}_{\alpha_k} \arrow[l] \arrow[ur, phantom, "\lrcorner"] & & \mathbf{HN}_{\underline{\alpha}} \arrow[u, "\mathsf{p}^{\mathrm{Coh}}_{\underline{\alpha}} \circ \jmath^{\mathrm{HN}}_{\underline{\alpha}}"'] \arrow[ll, leftarrow, "\cong"'] \arrow[ul, "\jmath^{\mathrm{HN}}_{\underline{\alpha}}"]
\end{tikzcd} \tag{1.2.1}$$

In particular, $\mathbf{HN}_{\underline{\alpha}}$ is isomorphic to the fiber product $\widetilde{\mathbf{Coh}}_{\underline{\alpha}} \underset{\prod_{k=1}^{s} \mathbf{Coh}_{\alpha_k}}{\times} \prod_{k=1}^{s} \mathbf{Coh}^{\mathsf{ss}}_{\alpha_k}$ and the inclusion $\mathbf{HN}_{\underline{\alpha}} \subseteq \mathbf{Coh}_{\alpha}$ can be viewed as the composition $\mathsf{p}^{\mathrm{Coh}}_{\underline{\alpha}} \circ \jmath^{\mathrm{HN}}_{\underline{\alpha}}$.

1.3 Categories of sheaves and dualities

In this section, we recall some notions of sheaf categories that we are interested in:

1.3.1 *Constructible sheaves and $\mathcal{D}$-modules.* As in [Ari+22b], let Y be a quasi-compact scheme over $\mathbf{k}$, we use the notation $\mathsf{Shv}_{\mathrm{constr}}(Y)$ for one of the following:

- Constructible sheaves on the topological space $Y(\mathbb{C})$ when $\mathbf{k} = \mathbb{C}$.
- Holonomic $\mathcal{D}$-modules on Y when $\mathrm{char}(\mathbf{k}) = 0$.
- Constructible ℓ-adic sheaves on Y, where $\ell \neq \mathrm{char}(\mathbf{k}) > 0$.

There is a perverse t-structure on $\mathsf{Shv}_{\mathrm{constr}}(Y)$, whose heart $\mathsf{Shv}_{\mathrm{constr}}(Y)^{\heartsuit}$ is the category of perverse sheaves on Y, denoted by $\mathsf{Perv}(Y)$. We have theorem of Beilinson saying that $\mathrm{D}^b\,(\mathsf{Perv}(Y)) \cong \mathsf{Perv(Y)}$. We denote by $\mathsf{Shv}(Y)$ the ind-completion $\mathsf{Shv}(Y) := \mathsf{Ind}\,(\mathsf{Shv}_{\mathrm{constr}}(Y))$. The DG-category $\mathsf{Shv}(Y)$ is self-dual, meaning it is dualizable in $\mathsf{DGCat}_{\mathrm{cont}}$ and the dual object is itself. In fact, since Y is supposed to be quasicompact, the Verdier duality establishes an equivalence: $\mathbb{D} : \mathsf{Shv}_{\mathrm{constr}}(Y)^{\mathrm{op}} \xrightarrow{\cong} \mathsf{Shv}_{\mathrm{constr}}(Y)$ and the counit map $\mathsf{Shv}(Y) \otimes \mathsf{Shv}(Y) \to \mathbf{Vect}$ is given by

$$\mathcal{F}_1 \otimes \mathcal{F}_2 \mapsto \mathrm{C}^{\bullet}\left(Y, \mathcal{F}_1 \overset{!}{\otimes} \mathcal{F}_2\right)$$

Also, the perverse t-structure of $\mathsf{Shv}_{\mathrm{constr}}(Y)$ can be extended compatibly with filtered colimits to $\mathsf{Shv}(Y)$. We can see that the heart $\mathsf{Shv}(Y)^{\heartsuit}$ of the extended perverse t-structure on ind-completion is $\mathsf{Ind}\,(\mathsf{Perv}(Y))$, the ind-completion of $\mathsf{Perv}(Y) \subseteq \mathsf{Shv}_{\mathrm{constr}}(Y)$.

Now for any prestack $\mathcal{X}$, we can define $\mathsf{Shv}_{\mathrm{constr}}(\mathcal{X})$ and $\mathsf{Shv}(\mathcal{X})$ respectively as

$$\mathsf{Shv}_{\mathrm{constr}}(\mathcal{X}) = \varprojlim_{S \to \mathcal{X}} \mathsf{Shv}_{\mathrm{constr}}(S); \qquad \mathsf{Shv}(\mathcal{X}) = \varprojlim_{S \to \mathcal{X}} \mathsf{Shv}(S)$$

where the limit is taken over $\mathsf{Sch}_{\mathrm{ft}}{}_{/\mathcal{X}}$ with !-pullback. Also, there exists six functor formalism of Shv, thus $\mathsf{Shv}(\mathcal{X})$ can also be written as a colimit $\varinjlim \mathsf{Shv}(S)$ with transition functors being !-pushforward. The category $\mathsf{Shv}(\mathcal{X})$ is compactly generated and thus dualizable, its dual category is

$$\mathsf{Shv}(\mathcal{X})^{\vee} = \varinjlim_{S \to \mathcal{X}} \mathsf{Shv}(S)$$

where the transition functors in the colimit are $*$-pushforward. When Y is a quasi-compact scheme, the compact subcategory $\mathsf{Shv}(Y)^c$ is exactly $\mathsf{Shv}_{\mathrm{constr}}(Y)$, however, this is false if we replace Y by an arbitrary prestack $\mathcal{X}$ and thus the category $\mathsf{Shv}(\mathcal{X})$ is no longer self-dual in general. Note that the category perverse sheaves are not always defined, due to lack of perverse t-structure in general. When $\mathcal{X}$ is an Artin stack locally of finite type, there exists a perverse t-strucutre on $\mathsf{Shv}(\mathcal{X})$ and we denote $\mathsf{Perv}(\mathcal{X})$ the heart.

When $\mathcal{X}$ is a prestack over $\mathbb{C}$, we can also consider $\mathcal{D}\text{-}\mathsf{mod}(\mathcal{X})$, the category of all $\mathcal{D}$-modules (not necessarily holonomic) on $\mathcal{X}$, which can be exhibited as a limit

$$\mathcal{D}\text{-}\mathsf{mod}(\mathcal{X}) = \varprojlim_{S \to \mathcal{X}} \mathcal{D}\text{-}\mathsf{mod}(S)$$

where the transition functors are taken over $\mathsf{Sch}_{\mathrm{ft}}{}_{/\mathcal{X}}$ with respect to de Rham $*$-pullback. We need to be cautious here the difference between $\mathcal{D}\text{-}\mathsf{mod}$ and Shv: unlike to the case of constructible sheaves, we do not have six functor formalism for $\mathcal{D}\text{-}\mathsf{mod}(\mathcal{X})$, as a consequence it is not compactly generated nor dualizable in general. In [DG15], they introduce the notion of *QCA stacks*, which is abbreviation of *quasicompact with affine stabilizer*. Then it is known that

Lemma 1.8. *The category $\mathcal{D}\text{-}\mathsf{mod}(\mathcal{Y})$ is compactly generated if $\mathcal{Y}$ is a QCA stack.*

We shall assume $\mathcal{X}$ is a locally QCA stack, namely the automorphism groups of its field-valued points are affine, in this situation, we can refine the index of limit in defining $\mathcal{D}\text{-}\mathsf{mod}(\mathcal{X})$: it is not hard to see there is an equivalence

$$\mathcal{D}\text{-}\mathsf{mod}(\mathcal{X}) = \lim_{U \in \text{op-qc}(\mathcal{X})} \mathcal{D}\text{-}\mathsf{mod}(U)$$

where op-qc$(-)$ denotes the collection of quasi-compact opens inside of $\mathcal{X}$ whose mutual intersection is also quasi-compact open. For $j_{1,2} : U_1 \hookrightarrow U_2$ inclusion of quasi-compact opens in $\mathcal{X}$, the limit is taken as $(j_{1,2})^* : \mathcal{D}\text{-}\mathsf{mod}(U_2) \longrightarrow \mathcal{D}\text{-}\mathsf{mod}(U_1)$. It is also shown in [DG15] that $\mathcal{D}\text{-}\mathsf{mod}(\mathcal{X})$ is compactly generated if $\mathcal{X}$ satisfies a geometric property called *truncatable*, see Appendix A.1. For the stacks $\mathbf{Coh}_{\alpha}$

and $\mathbf{Bun}_G$ that we are mostly interested in, we have the following result:

Theorem 1.9. *Both $\mathcal{D}$-mod$(\mathbf{Bun}_G)$ and $\mathcal{D}$-mod$(\mathbf{Coh}_\alpha)$ are compactly generated hence dualizable.*

The compact generation of $\mathcal{D}$-mod($\mathbf{Bun}_G$) has been proved as a main theorem in [DG15]. When $\alpha = \tau d$, we can deduce from Proposition 1.4 that the stack $\mathbf{Coh}_{\tau d}$ is a QCA stack, then the Lemma 1.8 immediately applies. For general $\alpha \in \mathbf{Z}^+$, we can prove that $\mathbf{Coh}_\alpha$ satisfies truncatable condition, thus $\mathcal{D}$-mod$(\mathbf{Coh}_\alpha)$ is compactly generated, the proof relies on HN-strata $\mathbf{HN}_{\underline{\alpha}}$,and we postpone it to the Appendix, see Proposition A.5.

1.3.2 Miraculous dualities. Next we discuss the dualities of the categories introduced above. To simplify the notation, we use $\mathrm{D}(\mathcal{X})$ for either $\mathsf{Shv}(\mathcal{X})$ or $\mathcal{D}$-mod($\mathcal{X}$). In general, assume $\mathsf{C} = \mathsf{Ind}(\mathsf{C}^c)$ is a compactly generated DG category, then it is dualizable and its dual category can be identified with $\mathsf{C}^\vee = \mathsf{Ind}((\mathsf{C}^c)^{\mathrm{op}})$. There is a canonical equivalence $(\mathsf{C}^c)^{\mathrm{op}} \cong (\mathsf{C}^\vee)^c : \mathbf{c} \mapsto \mathbf{c}^\vee$. We denote by $\mathrm{ev}_{\mathsf{C}} : \mathsf{C} \otimes \mathsf{C}^\vee \longrightarrow \mathbf{Vect}$ the counit map, for $\mathbf{c} \in \mathsf{C}^c$ and any $\xi \in \mathsf{C}^\vee$, the counit is given by

$$\mathrm{ev}_{\mathsf{C}}(\mathbf{c} \otimes \xi) = \mathrm{Hom}_{\mathsf{C}^\vee}(\mathbf{c}^\vee, \xi)$$

Thanks to Theorem 1.9, both $\mathrm{D}(\mathbf{Bun}_G)$ resp. $\mathrm{D}(\mathbf{Coh}_\alpha)$ are dualizable, their dual categories in $\mathsf{DGCat}_{\mathrm{cont}}$ can be described as follows:
We denote by $\mathrm{D}(\mathbf{Bun}_G)_{\mathrm{co}}$ and $\mathrm{D}(\mathbf{Coh}_\alpha)_{\mathrm{co}}$ the $*$-generated category of constructible sheaves/$\mathcal{D}$-modules on $\mathbf{Bun}_G$ and $\mathbf{Coh}_r$ respectively. More precisely, they are the colimits taken in cocomplete DG categories and continuous functors as follows

$$\mathrm{D}(\mathbf{Bun}_G)_{\mathrm{co}} := \mathrm{colim}_{U \in \text{op-qc}(\mathbf{Bun}_G)} \mathrm{D}(U); \qquad \mathrm{D}(\mathbf{Coh}_r)_{\mathrm{co}} := \mathrm{colim}_{U \in \text{op-qc}(\mathbf{Coh}_\alpha)} \mathrm{D}(U)$$

For $j_{1,2} : U_1 \hookrightarrow U_2$, we have the corresponding colimit is taken as $(j_{1,2})_* : \mathrm{D}(U_1) \longrightarrow \mathrm{D}(U_2)$. Since for any quasi-compact U, we have self-duality $\mathrm{D}(U) \cong (\mathrm{D}(U))^\vee$ (given by Verdier dual), we can see that

$$\begin{aligned} \mathrm{Maps}_{\mathsf{DGCat}_{\mathrm{cont}}}(\mathrm{D}(\mathbf{Bun}_G)_{\mathrm{co}}, \mathbf{Vect}) &= \mathrm{Maps}_{\mathsf{DGCat}_{\mathrm{cont}}}\left(\mathrm{colim}_{U \in \text{op-qc}(\mathbf{Bun}_G)} \mathrm{D}(U), \mathbf{Vect}\right) \\ = \lim_{U \in \text{op-qc}} \mathrm{Maps}_{\mathsf{DGCat}_{\mathrm{cont}}}(\mathrm{D}(U), \mathbf{Vect}) &\cong \lim_{U \in \text{op-qc}} \mathrm{D}(U)^\vee \cong \mathrm{D}(\mathbf{Bun}_G) \end{aligned}$$

The same argument works for $\mathbf{Coh}_\alpha$, we have $\mathrm{Maps}_{\mathsf{DGCat}_{\mathrm{cont}}}(\mathrm{D}(\mathbf{Coh}_\alpha)_{\mathrm{co}}, \mathbf{Vect}) \cong \mathrm{D}(\mathbf{Coh}_\alpha)$. Hence we can deduce that the dual categories are

$$\mathrm{D}(\mathbf{Bun}_G)_{\mathrm{co}} \cong \mathrm{D}(\mathbf{Bun}_G)^\vee, \qquad \mathrm{D}(\mathbf{Coh}_\alpha)_{\mathrm{co}} \cong \mathrm{D}(\mathbf{Coh}_\alpha)^\vee$$

Also, the above identification gives us canonical pairings, i.e. the counit maps:

$$\langle -,- \rangle_{\mathbf{Bun}_G} : \mathrm{D}(\mathbf{Bun}_G) \times \mathrm{D}(\mathbf{Bun}_G)_{\mathrm{co}} \longrightarrow \mathbf{Vect}; \quad \langle -,- \rangle_{\mathbf{Coh}_\alpha} : \mathrm{D}(\mathbf{Coh}_\alpha) \times \mathrm{D}(\mathbf{Coh}_\alpha)_{\mathrm{co}} \longrightarrow \mathbf{Vect}$$

In general, let $\mathsf{C}_1, \mathsf{C}_2 \in \mathsf{DGCat}$ and assume C_1 is dualizable, then we have equivalence

$$\mathrm{Maps}_{\mathsf{DGCat}_{\mathrm{cont}}}(\mathsf{C}_1, \mathsf{C}_2) \cong \mathsf{C}_1^\vee \otimes \mathsf{C}_2 : \mathsf{F} \mapsto \left(\mathrm{Id}_{\mathsf{C}_1^\vee} \otimes \mathsf{F}\right)(\mathbf{u}_{\mathsf{C}_1})$$

where $\mathbf{u}_{\mathsf{C}_1} \in \mathsf{C}_1^\vee \otimes \mathsf{C}_1$ is the image under unit map $\mathrm{unit}_{\mathsf{C}_1} : \mathbf{Vect} \to \mathsf{C}_1^\vee \otimes \mathsf{C}_1$. Given an object $\mathbf{x} \in \mathsf{C}_1^\vee \otimes \mathsf{C}_2$, the corresponding functor is given by

$$\mathsf{F}_{\mathbf{x}} : \mathsf{C}_1 \xrightarrow{\mathrm{Id}_{\mathsf{C}_1} \otimes \mathbf{x}} \mathsf{C}_1 \otimes \mathsf{C}_1^\vee \otimes \mathsf{C}_2 \xrightarrow{\mathrm{ev}_{\mathsf{C}_1} \otimes \mathrm{Id}_{\mathsf{C}_2}} \mathsf{C}_2$$

the element $\mathbf{x}$ is called the *kernel* of functor $\mathsf{F}_{\mathbf{x}}$. For stack $\mathcal{Y}$ such that $\mathrm{D}(\mathcal{Y})$ is dualizable, we have $\mathrm{Maps}_{\mathsf{DGCat}}(\mathrm{D}(\mathcal{Y})^\vee, \mathrm{D}(\mathcal{Y})) \cong \mathrm{D}(\mathcal{Y}) \otimes \mathrm{D}(\mathcal{Y})$. In the setup that $\mathrm{D}(\mathcal{Y}) = \mathcal{D}\text{-mod}(\mathcal{Y})$, we have Künneth formula

$$\mathcal{D}\text{-mod}(\mathcal{Y}) \otimes \mathcal{D}\text{-mod}(\mathcal{Y}) \cong \mathcal{D}\text{-mod}(\mathcal{Y} \times \mathcal{Y})$$

thus an element in $\mathcal{D}\text{-mod}(\mathcal{Y} \times \mathcal{Y})$ give rises to functors $\mathrm{D}(\mathcal{Y})^\vee \longrightarrow \mathrm{D}(\mathcal{Y})$. The unit $\mathbf{u}_{\mathcal{D}\text{-mod}(\mathcal{Y})^\vee}$ corresponds to $(\Delta_{\mathcal{Y}})_{*,\mathrm{dR}}(\omega_{\mathcal{Y}})$, where $\Delta : \mathcal{Y} \to \mathcal{Y} \times \mathcal{Y}$ is the diagonal. We denote by $\text{Ps-Id}_{\mathcal{Y},*}$ the functor corresponds to the kernel $\mathbf{u}_{\mathcal{D}\text{-mod}(\mathcal{Y}^\vee)}$. If $\mathcal{Y}$ is quasi-compact, then $\text{Ps-Id}_{\mathcal{Y},*}$ is an equivalence. In fact, in this case, $(\Delta_{\mathcal{Y}})_{*,\mathrm{dR}}(\omega_{\mathcal{Y}})$ is a compact object, which implies $\text{Ps-Id}_{\mathcal{Y},*}$ admits a continuous right adjoint. The functor $\text{Ps-Id}_{\mathcal{Y},*}$ fails to be an equivalence when $\mathcal{Y}$ is no longer quasi-compact. Instead, we can consider another functor $\text{Ps-Id}_{\mathcal{Y},!} : \mathcal{D}\text{-mod}(\mathcal{Y})^\vee \longrightarrow \mathcal{D}\text{-mod}(\mathcal{Y})$ that defined by kernel $(\Delta_{\mathcal{Y}})_!(\mathbf{e}_{\mathcal{Y}})$.
In the setup that $\mathrm{D}(\mathcal{Y}) = \mathsf{Shv}(\mathcal{Y})$, we have no longer Künneth formula, but we can still consider the functor $\text{Ps-Id}_{\mathcal{Y},!} : \mathsf{Shv}(\mathcal{Y})^\vee \longrightarrow \mathsf{Shv}(\mathcal{Y})$ given by the kernel $(\Delta_{\mathcal{Y}})_!(\mathbf{e}_{\mathcal{Y}})$.

Definition 1.10. Assume $\mathrm{D}(\mathcal{Y})$ is dualizable, we say $\mathcal{Y}$ is **miraculous** if $\text{Ps-Id}_{\mathcal{Y},!}$ is an equivalence.

Theorem 1.11. *Both* $\mathbf{Bun}_G$ *and* $\mathbf{Coh}_{\tau d}$ *are miraculous.*

For the stack $\mathbf{Bun}_G$, the miraculousity is the main theorem proved in [Gai17], we will give the proof for $\mathbf{Coh}_{\tau d}$ in the Appendix A.3. The combination of Theorem 1.9 with the open exhaustion introduced in Proposition 1.3 yields

Theorem 1.12. *The stack* $\mathbf{Coh}_\alpha$ *is miraculous for any* $\alpha \in \mathbf{Z}^+$.

2 The Hall algebras and Hall categories of curve

2.1 The Hall algebras of curves

In this section, we discuss the Hall algebra associated with the smooth projective curve X_0 defined over $\mathbb{F}_q$, which can be viewed as an algebra of automorphic forms on X_0.

2.1.1 *Hall algebras of abelian categories.* We briefly recall the definition of Hall algebra of an abelian category A, one can refer to [Sch09] for more details. Assume A has finite global dimension and is finitary, namely $\left|\mathrm{Ext}^i(\mathcal{M}, \mathcal{N})\right| < \infty, \forall i \geqslant 0, \forall \mathcal{M}, \mathcal{N} \in \mathsf{A}$. The **Hall algebra** $\mathbf{H}_{\mathsf{A}}$ of A have the underlying vector space being the space of function with compact support on the isomorphism classes of A, as the following:

$$\mathrm{Funct}_c\left(\bigoplus_{\mathcal{M}\in \mathrm{Obj}(\mathsf{A})/\sim}[\mathcal{M}],\mathbb{C}\right)=\bigoplus_{\gamma\in K_0(\mathsf{A})}\mathbf{H}_{\mathsf{A}}[\gamma]$$

The space of $\mathbf{H}_{\mathsf{A}}$ is spanned by characteristic functions $\mathbf{1}_{[\mathcal{M}]}$ of the isomorphism classes, and is endowed with a multiplication structure given by

$$\mathbf{1}_{[\mathcal{M}]}\star\mathbf{1}_{[\mathcal{N}]}=\left(\prod_{i=0}^{\infty}\left(\left|\mathrm{Ext}^i(\mathcal{M},\mathcal{N})\right|^{\frac{(-1)^i}{2}}\right)\right)\sum_{\mathcal{R}\in \mathrm{Ext}^1(\mathcal{M},\mathcal{N})/\sim}\frac{\mathbf{a}^{\mathcal{R}}_{\mathcal{M},\mathcal{N}}}{\mathbf{a}_{\mathcal{M},\mathcal{N}}}\mathbf{1}_{[\mathcal{R}]}$$

where $\mathbf{a}_{\mathcal{M},\mathcal{N}}=|\mathrm{Aut}_{\mathsf{A}}(\mathcal{M})|\cdot|\mathrm{Aut}_{\mathsf{A}}(\mathcal{N})|$ and $\mathbf{a}^{\mathcal{R}}_{\mathcal{M},\mathcal{N}}$ is the number of the class in $\mathrm{Ext}^1(\mathcal{M},\mathcal{N})$ which are given by $\mathcal{R}$. The number $\frac{\mathbf{a}^{\mathcal{R}}_{\mathcal{M},\mathcal{N}}}{\mathbf{a}_{\mathcal{M},\mathcal{N}}}$ is the same as the $|\{\mathcal{L}\subseteq\mathcal{R}\mid \mathcal{L}\cong\mathcal{N},\mathcal{R}/\mathcal{L}\cong\mathcal{M}\}|$. We denote the finite number $\prod_{i=0}^{\infty}\left(\left|\mathrm{Ext}^i(\mathcal{M},\mathcal{N})\right|^{\frac{(-1)^i}{2}}\right)$ by $\langle\mathcal{M},\mathcal{N}\rangle$, it depends only on the class $[\mathcal{M}],[\mathcal{N}]$ in $K_0(\mathsf{A})$.

Define also a coproduct on $\mathbf{H}_{\mathsf{A}}$ by

$$\Delta\left(\mathbf{1}_{[\mathcal{R}]}\right)=\sum_{\mathcal{M},\mathcal{N}}\frac{\langle\mathcal{M},\mathcal{N}\rangle\,\mathbf{a}^{\mathcal{R}}_{\mathcal{M},\mathcal{N}}}{|\mathrm{Aut}_{\mathsf{A}}(\mathcal{R})|}\mathbf{1}_{[\mathcal{M}]}\otimes\mathbf{1}_{[\mathcal{N}]}\in\mathbf{H}_{\mathsf{A}}\hat{\otimes}\mathbf{H}_{\mathsf{A}}$$

where the sum is taken in the topological completion $\mathbf{H}_{\mathsf{A}}\hat{\otimes}\mathbf{H}_{\mathsf{A}}=\prod_{\gamma_1,\gamma_2\in K_0(\mathsf{A})}\mathbf{H}_{\mathsf{A}}[\gamma_1]\otimes\mathbf{H}_{\mathsf{A}}[\gamma_2]$, note that $\mathbf{H}_{\mathsf{A}}\hat{\otimes}\mathbf{H}_{\mathsf{A}}$ should be regarded as functions on $\mathsf{A}\otimes\mathsf{A}$ and we drop the finite support condition, the reason is $\mathcal{R}\in\mathsf{A}$ might not have finitely many subobjects.

To make $\mathbf{H}_{\mathsf{A}}$ a bialgebra, we need consider a refined version of Hall algebra. Let $\widetilde{\mathbf{H}}_{\mathsf{A}}=\mathbf{H}_{\mathsf{A}}\otimes\mathbb{C}\left[K_0(\mathsf{A})\right]$ be an extension of $\mathbf{H}_{\mathsf{A}}$ by $\mathbb{C}\left[K_0(\mathsf{A})\right]$, we write $\boldsymbol{\kappa}_\alpha$ or $\boldsymbol{\kappa}_{[\mathcal{M}]}$ for element in $\mathbb{C}[K_0(\mathsf{A})]$ and let $\deg(\boldsymbol{\kappa}_\alpha)=0$. The extended multiplication in $\widetilde{\mathbf{H}}_{\mathsf{A}}$ is subject to relations $\boldsymbol{\kappa}_{\alpha_1}\boldsymbol{\kappa}_{\alpha_2}=\boldsymbol{\kappa}_{\alpha_1+\alpha_2}$, $\boldsymbol{\kappa}_\alpha\mathbf{1}_{[\mathcal{M}]}\boldsymbol{\kappa}_\alpha^{-1}=\langle\alpha,\mathcal{N}\rangle\mathbf{1}_{[\mathcal{M}]}$ and extended comultiplication with relations

$$\Delta\left(\boldsymbol{\kappa}_\alpha\right)=\boldsymbol{\kappa}_\alpha\otimes\boldsymbol{\kappa}_\alpha;\qquad \Delta\left(\mathbf{1}_{[\mathcal{R}]}\boldsymbol{\kappa}_\alpha\right)=\sum_{\mathcal{M},\mathcal{N}}\frac{\langle\mathcal{M},\mathcal{N}\rangle\,\mathbf{a}^{\mathcal{R}}_{\mathcal{M},\mathcal{N}}}{|\mathrm{Aut}_{\mathsf{A}}(\mathcal{R})|}\mathbf{1}_{[\mathcal{M}]}\boldsymbol{\kappa}_{[\mathcal{N}]+\alpha}\otimes\mathbf{1}_{[\mathcal{N}]}$$

Endow $\widetilde{\mathbf{H}}_{\mathsf{A}}\hat{\otimes}\widetilde{\mathbf{H}}_{\mathsf{A}}$ with canonical multiplication $(x_1\otimes x_2)\star(y_1\otimes y_2)=(x_1\star y_1)\otimes(x_2\star y_2)$, we have

Theorem 2.1 (Green)**.** *Assume* A *is finitary with* $\mathrm{gl.dim}(\mathsf{A})\leqslant 1$, *then:*

(1) $(\mathbf{H}_{\mathsf{A}},\star)$ *is an associative algebra and* $(\mathbf{H}_{\mathsf{A}},\Delta)$ *is a co-associative co-algebra, same for* $\left(\widetilde{\mathbf{H}}_{\mathsf{A}},\star\right)$ *and* $\left(\widetilde{\mathbf{H}}_{\mathsf{A}},\Delta\right)$.

(2) $\left(\widetilde{\mathbf{H}}_{\mathsf{A}},\star,\Delta\right)$ *is a bi-algebra, namely* $\Delta(f\star g)=\Delta(f)\star\Delta(g),\forall f,g\in\widetilde{\mathbf{H}}_{\mathsf{A}}$.

(3) There is a Hopf pairing on $\mathbf{H}_{\mathsf{A}}$ *as well as on* $\widetilde{\mathbf{H}}_{\mathsf{A}}$, *given by integration with respect to orbifold measure:*

$$(\bullet\,|\,\bullet):\mathbf{H}_{\mathsf{A}}\otimes\mathbf{H}_{\mathsf{A}}\longrightarrow\mathbb{C};\,f\otimes g\mapsto\int_{[\mathsf{A}]}f\cdot g\,\mathrm{d}\mu_{\mathsf{A}}:=\sum_{\mathcal{M}\in\mathrm{Obj}(\mathsf{A})/\sim}\frac{f(\mathcal{M})g(\mathcal{M})}{|\mathrm{Aut}_{\mathsf{A}}(\mathcal{M})|}$$

such that $\star$ *and* Δ *are adjoint with respect to the pairing, i.e.* $\left(f\star g\,\middle|\,h\right)=\left(f\otimes g\,\middle|\,\Delta(h)\right)$.

Remark 2.2. In the sense of [DK19], the Hall algebra of A keeps track of the algebraic structure of the 2-Segal space built from Waldhausen's $\mathcal{S}_\bullet$-construction.

2.1.2 Hall algebras of curves. We can take our finitary abelian category A to be either $\mathsf{Coh}(X_0)$ the category of coherent sheaves on X_0, or $\mathsf{Tor}(X_0) \subseteq \mathsf{Coh}(X_0)$ sub-abelian category of torsion sheaves on X_0, both abelian categories are of global dimension $\leqslant 1$ and finitary. Apply Theorem 2.1, we obtain two Hall algebras $\widetilde{\mathbf{H}}_{\mathsf{Coh}(X_0)}$ and $\widetilde{\mathbf{H}}_{\mathsf{Tor}(X_0)}$. We abbreviate $\widetilde{\mathbf{H}}_{\mathsf{Coh}(X_0)}$ as $\mathbf{H}_{X_0}$ and call it the **Hall algebra** of X_0. Similarly, we abbreviate $\widetilde{\mathbf{H}}_{\mathsf{Tor}(X_0)}$ as $\mathbf{H}_{X_0}^{\mathrm{Tor}}$.

By definition, the algebra $\mathbf{H}_{X_0}$ has underlying vector space $\left(\bigoplus_{[\mathcal{F}]\in\mathsf{Coh}(X_0)} \mathbb{C}\left[\mathbf{1}_{[\mathcal{F}]}\right]\right) \otimes \mathbb{C}[\boldsymbol{\kappa}_\alpha]_{\alpha\in\mathbf{Z}^+}$. Let us denote by ${}^0\mathbf{Coh}_\alpha$ the stack of coherent sheaves on X_0, it is the $\mathbb{F}_q$-form of the stack $\mathbf{Coh}_\alpha$ that associated with $X = X_0 \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$. Then the underlying vector space of $\mathbf{H}_{X_0}$ can also be identified with

$$\mathbf{H}_{X_0} = \left(\bigoplus_{\alpha\in\mathbf{Z}^+} \mathrm{Funct}_c\left({}^0\mathbf{Coh}_\alpha(\mathbb{F}_q), \mathbb{C}\right)\right) \otimes \mathbb{C}[\boldsymbol{\kappa}_\beta]_{\beta\in\mathbf{Z}^+}$$

The algebra $\mathbf{H}_{X_0}^{\mathrm{Tor}}$ is a sub-bialgebra of $\mathbf{H}_{X_0}$ with underlying vector space

$$\mathbf{H}_{X_0}^{\mathrm{Tor}} = \left(\bigoplus_{\mathcal{T}\in\mathsf{Tor}(X_0)/\sim} \mathbb{C}\left[\mathbf{1}_{[\mathcal{T}]}\right]\right) \otimes \mathbb{C}[\boldsymbol{\kappa}_{\tau d}] = \left({\bigotimes_{x\in X}}' \mathbf{H}_{\mathsf{Tor}(X_0)_x}\right) \otimes \mathbb{C}[\boldsymbol{\kappa}_{\tau d}]$$

where the second equality is a restricted tensor product over Hall algebras of $\mathsf{Tor}(X_0)_x$, the category of torsion sheaves that supported at $x \in X$. The subcategory $\mathsf{Tor}(X_0)_x$ carries with indecomposable objects $\mathcal{O}_{r\cdot x} := \mathcal{O}_{X_0}/\mathfrak{m}_x^r$ and the algebra $\mathbf{H}_{\mathsf{Tor}(X_0)_x}$ is isomorphic to the classical Hall algebra. We refer $\mathbf{H}_{X_0}^{\mathrm{Tor}}$ as the **algebra of Hecke operators**. In fact, for $n \geqslant 1$, consider $\mathbf{AF}_n := \mathrm{Funct}\left(\mathbf{Bun}_n(\mathbb{F}_q), \mathbb{C}\right)$ the space of automorphic forms of rank n over X_0, then the (left) action of $\mathbf{H}_{X_0}^{\mathrm{Tor}}$ on $\mathbf{AF}_n$ is given by the Hecke action

$$\mathbf{1}_{[\mathcal{T}]} \cdot f = T_{[\mathcal{T}]} f : \left(\mathcal{E} \mapsto \sum_{\substack{\mathcal{E}'\subseteq\mathcal{E} \\ \mathcal{E}/\mathcal{E}\cong\mathcal{T}}} f(\mathcal{E}')\right)$$

It can be regarded as decategorification of geometric Hecke action, note that for fixed $\alpha \in \mathbf{Z}^{++}$ and torsion degree d, we have the Hecke stack $\mathbf{Hk}_\alpha^d$ that classifies $\langle 0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{T} \to 0 \mid \mathcal{E}' \in \mathbf{Bun}_\alpha, \mathcal{E} \in \mathbf{Bun}_{\alpha+\tau d}\rangle$. The degree d Hecke action takes a function on $\mathbf{Bun}_\alpha$ to a function on $\mathbf{Bun}_{\alpha+\tau d}$, given by $(h^{\rightarrow})_* (h^{\leftarrow})^!$, where the maps $h^{\leftarrow}, h^{\rightarrow}$ come from the following diagram:

$$\begin{tikzcd}
\mathbf{Coh}_{\tau d} \times \mathbf{Bun}_\alpha \arrow[d, hook] & \mathbf{Hk}_\alpha^d \arrow[l, "h^{\leftarrow}"'] \arrow[r, "h^{\rightarrow}"] \arrow[d, hook] & \mathbf{Bun}_{\alpha+\tau d} \arrow[d, hook] \\
\mathbf{Coh}_{\tau d} \times \mathbf{Coh}_\alpha & \widetilde{\mathbf{Coh}}_{\alpha,\tau d} \arrow[l, "\mathsf{q}^{\mathrm{Coh}}_{\alpha,\tau d}"'] \arrow[r, "\mathsf{p}^{\mathrm{Coh}}_{\alpha,\tau d}"] & \mathbf{Coh}_{\alpha+\tau d}
\end{tikzcd}$$

the bottom line is the extension diagram we have seen.

Remark 2.3. In section 1.2, we have seen a version of extension diagram for $\mathbf{Bun}_{\mathrm{GL}_n}$. For $n_1, n_2 \geqslant 1$ and $n = n_1 + n_2$, we have map

$$\mathsf{Eis}^{\mathrm{Bun}}_{n_1,n_2} := (\mathsf{p}^{\mathrm{Bun}}_{n_1,n_2})_!(\mathsf{q}^{\mathrm{Bun}}_{n_1,n_2})^* : \mathbf{AF}_{n_1} \times \mathbf{AF}_{n_2} \longrightarrow \mathbf{AF}_n$$

that sends

$$(f_1, f_2) \mapsto f : \left(\mathcal{E} \mapsto \sum_{\substack{\mathcal{E}' \subseteq \mathcal{E} \\ \text{subbundle of rank } n_1}} f_1(\mathcal{E}') f_2(\mathcal{E}/\mathcal{E}') \right)$$

This is known as (pseudo-)Eisenstein series for GL_n and endows $\mathbf{AF}_{X_0} := \left(\bigoplus_{n \geqslant 1} \mathbf{AF}_n, \mathsf{Eis}^{\mathrm{Bun}} \right)$ with graded algebra structure. Moreover, $\mathsf{Eis}^{\mathrm{Bun}}$ sends $\mathbf{AF}_{n_1,c} \times \mathbf{AF}_{n_2,c}$ to $\mathbf{AF}_{n,c}$ and makes $\mathbf{AF}_c = \bigoplus_{n \geqslant 1} \mathbf{AF}_{n,c}$ a subalgebra of $\mathbf{H}_{X_0}$. Also, there is dual operator known as constant term map, given by

$$\mathsf{CT}^{\mathrm{Bun}}_{n_1,n_2} := (\mathsf{q}^{\mathrm{Bun}}_{n_1,n_2})_* (\mathsf{p}^{\mathrm{Bun}}_{n_1,n_2})^! : \mathbf{AF}_n \longrightarrow \mathbf{AF}_{n_1} \times \mathbf{AF}_{n_2}$$

We say an element in $\mathbf{AF}_n$ is a *cuspidal forms* if it lies in intersection of kernel of $\mathsf{CT}^{\mathrm{Bun}}$ for any proper partition of n, and we denote $\mathbf{AF}^{\mathrm{cusp}}_n$ the space of cuspidal forms. The space of automorphic forms $\mathbf{AF}_n$ is then generated by cuspidal forms of lower ranks under $\mathsf{Eis}^{\mathrm{Bun}}$.

***2.1.3** Spherical Hall algebras.* For $\alpha \in \mathbf{Z}^+$, we denote by $\mathbf{H}_{X_0}[\alpha]$ the graded piece of $\mathbf{H}_{X_0}$ that contains functions supported on ${}^0\mathbf{Coh}_\alpha$. For a tuple $\underline{\alpha} = (\alpha_1, \dots, \alpha_s) \in (\mathbf{Z}^+)^s$, we use $\mathbf{H}_{X_0}\left[\underline{\alpha}\right]$ to denote

$$\mathbf{H}_{X_0}[\alpha_1] \widehat{\otimes} \dots \widehat{\otimes} \mathbf{H}_{X_0}[\alpha_s] \subseteq \mathbf{H}_{X_0} \widehat{\otimes} \dots \widehat{\otimes} \mathbf{H}_{X_0}$$

We also put

$$\mathbf{1}_{\tau d} = \sum_{\substack{\mathcal{T} \in \mathsf{Tor}(X_0) \\ \deg(\mathcal{T}) = d}} \mathbf{1}_{[\mathcal{T}]} \in \mathbf{H}^{\mathrm{Tor}}_{X_0} \quad \text{and} \quad \mathbf{1}_{(1,d)} = \sum_{\substack{\mathcal{F} \in \mathsf{Pic}(X_0) \\ \deg(\mathcal{F}) = d}} \mathbf{1}_{[\mathcal{F}]} \in \mathbf{H}_{X_0}$$

Definition 2.4. The **spherical Hall algebra** $\mathbf{H}^{\mathrm{sph}}_{X_0}$ of X_0 is defined to be the sub-bialgebra of $\mathbf{H}_{X_0}$ that generated under the product in $\mathbf{H}_{X_0}$ by

$$\left\{ \mathbf{1}_{\tau d}, \mathbf{1}_{(1,d')} \right\}_{\substack{d \geqslant 0 \\ d' \in \mathbb{Z}}} \quad \text{and} \quad \{ \boldsymbol{\kappa}_\alpha \}_{\alpha \in \mathbf{Z}^+}$$

Denote $\mathbf{H}^{\mathrm{sph,Bun}}_{X_0}$ the subalgebra of $\mathbf{H}^{\mathrm{sph}}_{X_0}$ generated by rank one cuspidal forms $\left\{ \mathbf{1}_{(1,d)} \right\}$, according to Remark 2.3, it can be regarded as the principal series in $\mathbf{AF}_{X_0}$. The underlying vector space of rank one graded piece $\mathbf{H}^{\mathrm{sph,Bun}}_{X_0}[1]$ can be identified with $\mathbb{C}\left[x^{\pm 1}\right]$ by putting $\mathbf{1}_{(1,d)} \mapsto x^d$. We can see

$$\left(\mathbf{H}^{\mathrm{sph,Bun}}_{X_0}[1] \right)^{\widehat{\otimes} r} \cong \mathbb{C}\left[x_1^{\pm 1}, \dots, x_r^{\pm 1}\right] [\![\frac{x_i}{x_{i+1}}]\!]_{i=1,\dots,r-1}$$

Denote also $\mathbf{H}^{\mathrm{sph,Tor}}_{X_0}$ the subalgebra of $\mathbf{H}^{\mathrm{Tor}}_{X_0}$ that generated by $\{ \mathbf{1}_{\tau d} \}$.

It will be useful to consider a formal completion of $\mathbf{H}_{X_0}$: a function $f \in \mathbf{H}_{X_0}[\alpha]$ has h-adic norm n if the restriction $f|_{{}^0\mathbf{Coh}^{\geq \mathcal{L}}_\alpha} \neq 0$ for some $\mathcal{L}$ with degree $\deg(\mathcal{L}) = n$ and $f|_{{}^0\mathbf{Coh}^{\geq \mathcal{L}'}_\alpha} = 0$ for all $\deg(\mathcal{L}') > n$. We write $\widehat{\mathbf{H}}_{X_0}$ the h-adic completion of $\mathbf{H}_{X_0}$. The bialgebra structure on $\mathbf{H}_{X_0}$ (resp. $\mathbf{H}^{\mathrm{sph}}_{X_0}$) naturally extends to $\widehat{\mathbf{H}}_{X_0}$ (resp. $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$) and $\mathbf{AF}_{X_0} \subseteq \widehat{\mathbf{H}}_{X_0}$ is a subalgebra. The same procedure of completion goes for $\widehat{\mathbf{H}}^{\mathrm{sph,Bun}}_{X_0}$. In particular, the constant function on $\mathbf{Bun}_\alpha$ could be counted as element in $\widehat{\mathbf{H}}^{\mathrm{sph,Bun}}_{X_0}$. For

$\alpha \in \mathbf{Z}^{++}$, let us put

$$\mathbf{1}_\alpha^{\mathsf{ss}} = \sum_{\substack{\mathcal{F} \in \mathsf{Coh}_\alpha(X_0) \\ \mathcal{F} \text{ semi-stable}}} \mathbf{1}_{[\mathcal{F}]}; \qquad \mathbf{1}_\alpha^{\mathrm{Bun}} = \sum_{\mathcal{V} \in \mathsf{Fib}_\alpha(X_0)} \mathbf{1}_{\mathcal{V}}$$

It is clear that $\mathbf{1}_\alpha^{\mathsf{ss}} \in \mathbf{H}_{X_0}$, it is also known to be an element in $\mathbf{H}_{X_0}^{\mathrm{sph}}$. Nevertheless, we have

$$\mathbf{1}_\alpha^{\mathrm{Bun}} = \sum_{\underline{\alpha} \in' \mathrm{HN}(\alpha)} q^{\frac{-\sum_{i<j}\langle \alpha_i, \alpha_j \rangle}{2}} \mathbf{1}_{\alpha_1}^{\mathsf{ss}} \star \ldots \star \mathbf{1}_{\alpha_s}^{\mathsf{ss}}$$

where $'\mathrm{HN}(\alpha) \subseteq \mathrm{HN}(\alpha)$ is the (infinite) subset that does not contain α_i with slope ∞, thus $\mathbf{1}_\alpha^{\mathrm{Bun}}$ only lives in $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph,Bun}}$ when $\mathrm{rk}(\alpha) > 1$. In the comletion $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph}}$, there is an action of $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph,Tor}}$ on $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph,Bun}}$. In particular, $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph,Tor}}$ can be regarded as *global spherical Hecke algebra* and the algebra $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph}}$ is generated by $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph,Tor}}$ and $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph,Bun}}$ modulo Rankin-Selberg relation.

2.2 Geometric construction of Hall algebras

In this section, we assume X is a smooth connective projective curve over $\mathbf{k} = \overline{\mathbb{F}}_q$ or $\mathbb{C}$. We will give a geometric formalism of Hall algebra (and more generally Hall category) associated with curve X.

2.2.1 *Coherent Eisenstein series/constant term functors.* Recall for $\alpha, \beta \in \mathbf{Z}^+$, we have the following diagram of extensions:

$$\mathbf{Coh}_{\alpha,\beta} = \mathbf{Coh}_\alpha \times \mathbf{Coh}_\beta \xleftarrow{\mathsf{q}_{\alpha,\beta}^{\mathrm{Coh}}} \widetilde{\mathbf{Coh}}_{\alpha,\beta} \xrightarrow{\mathsf{p}_{\alpha,\beta}^{\mathrm{Coh}}} \mathbf{Coh}_{\alpha+\beta}$$

where $\widetilde{\mathbf{Coh}}_{\alpha,\beta}$ is the stack of filtration introduced in section 1.2. In analogous to the bi-algebraic structure of $\mathbf{H}_{X_0}$, we consider the following two functors between $\mathrm{D}(\mathbf{Coh}_{\alpha,\beta})$ and $\mathrm{D}(\mathbf{Coh}_{\alpha+\beta})$:

$$\mathsf{Eis}_{\alpha,\beta}^{\mathrm{Coh}} = \left(\mathsf{p}_{\alpha,\beta}^{\mathrm{Coh}}\right)_! \left(\mathsf{q}_{\alpha,\beta}^{\mathrm{Coh}}\right)^* [\mathbf{shift}]; \qquad \mathsf{CT}_{\alpha,\beta}^{\mathrm{Coh}} = \left(\mathsf{q}_{\alpha,\beta}^{\mathrm{Coh}}\right)_! \left(\mathsf{p}_{\alpha,\beta}^{\mathrm{Coh}}\right)^* [\mathbf{shift}]$$

where the $[\mathbf{shift}]$ comes from the relative dimension of $\mathsf{q}_{\alpha,\beta}^{\mathrm{Coh}}$ as in Proposition 1.6. In general, for a sequence of classes $\underline{\alpha} = (\alpha_1, \ldots, \alpha_s)$, we can define $\mathsf{Eis}_{\underline{\alpha}}^{\mathrm{Coh}} = \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_! \left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)^* [\mathbf{shift}]$ for

$$\mathbf{Coh}_{\underline{\alpha}} = \mathbf{Coh}_{\alpha_1} \times \ldots \times \mathbf{Coh}_{\alpha_s} \xleftarrow{\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}} \widetilde{\mathbf{Coh}}_{\underline{\alpha}} \xrightarrow{\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}} \mathbf{Coh}_{\sum_{i=1}^s \alpha_i}$$

and similarly $\mathsf{CT}_{\underline{\alpha}}^{\mathrm{Coh}}$ is defined to be $\left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_! \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}\right)^* [\mathbf{shift}]$.

Remark 2.5. In the set-up of $\mathcal{D}$-mod $(\mathbf{Coh}_\alpha)$, since $\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}$ is not proper and $\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}$ is not smooth in general, the functor $\left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_! \circ \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}\right)^*$ is not *a priori* globally defined. However, we can show that the functor $\mathsf{CT}^{\mathrm{Coh}}$ is well-defined for $\mathcal{D}$-mod $(\mathbf{Coh}_\alpha)$ using hyperbolic localization, the prove is in Appendix A.10. As a consequence of Proposition 1.6, $\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}$ is smooth and $\mathsf{p}_{\alpha}^{\mathrm{Coh}}$ is proper, thus we have adjoint pairs:

$$\left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)^! = \left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)^* \dashv \left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_* \qquad \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_* = \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_! \dashv \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}\right)^!$$

and we can deduce that $\mathsf{Eis}^{\mathrm{Coh}}$ is left adjoint to $\mathsf{CT}_{\underline{\alpha},*}^{\mathrm{Coh}} := \left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_* \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}\right)^! [\mathbf{shift}]$.

As an analog to Theorem 2.1, we have

Proposition 2.6. *The functor* $\mathsf{Eis}^{\mathrm{Coh}}$ *is associative, namely for* $\underline{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$*, we have equivalences:*

$$\mathsf{Eis}^{\mathrm{Coh}}_{\alpha_1+\alpha_2,\alpha_3} \circ \left(\mathsf{Eis}^{\mathrm{Coh}}_{\alpha_1,\alpha_2} \otimes \mathrm{Id}_{\alpha_3}\right) = \mathsf{Eis}^{\mathrm{Coh}}_{\alpha_1,\alpha_2+\alpha_3} \circ \left(\mathrm{Id}_{\alpha_1} \otimes \mathsf{Eis}^{\mathrm{Coh}}_{\alpha_2,\alpha_3}\right) = \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}$$

Similarly, the functor $\mathsf{CT}^{\mathrm{Coh}}$ *is co-associative, namely:*

$$\left(\mathsf{CT}^{\mathrm{Coh}}_{\alpha_1,\alpha_2} \otimes \mathrm{Id}_{\alpha_3}\right) \circ \mathsf{CT}^{\mathrm{Coh}}_{\alpha_1+\alpha_2,\alpha_3} = \left(\mathrm{Id}_{\alpha_1} \otimes \mathsf{CT}^{\mathrm{Coh}}_{\alpha_2,\alpha_3}\right) \circ \mathsf{CT}^{\mathrm{Coh}}_{\alpha_1,\alpha_2+\alpha_3} = \mathsf{CT}^{\mathrm{Coh}}_{\underline{\alpha}}$$

Proof. Let $\widetilde{\mathcal{E}}_{[\alpha_1,\alpha_2],\alpha_3}$ be the fiber product $\left(\widetilde{\mathbf{Coh}}_{\alpha_1,\alpha_2} \times \mathbf{Coh}_{\alpha_3}\right) \times_{\mathbf{Coh}_{\alpha_1+\alpha_2}\times\mathbf{Coh}_{\alpha_3}} \widetilde{\mathbf{Coh}}_{\alpha_1+\alpha_2,\alpha_3}$ and $\widetilde{\mathcal{E}}_{\alpha_1,[\alpha_2,\alpha_3]}$ be the fiber product $\left(\mathbf{Coh}_{\alpha_1} \times \widetilde{\mathbf{Coh}}_{\alpha_2,\alpha_3}\right) \times_{\mathbf{Coh}_{\alpha_1}\times\mathbf{Coh}_{\alpha_2+\alpha_3}} \widetilde{\mathbf{Coh}}_{\alpha_1,\alpha_2+\alpha_3}$. It is not hard to see:

Lemma 2.7. *We have a canonical isomorphism* $\widetilde{\mathcal{E}}_{[\alpha_1,\alpha_2],\alpha_3} \cong \widetilde{\mathcal{E}}_{\alpha_1,[\alpha_2,\alpha_3]} \cong \widetilde{\mathbf{Coh}}_{\alpha_1,\alpha_2,\alpha_3}$.

We can consider the following diagram:

$$\begin{tikzcd}
\mathbf{Coh}_{\alpha_1} \times \mathbf{Coh}_{\alpha_2} \times \mathbf{Coh}_{\alpha_3} & \widetilde{\mathbf{Coh}}_{\alpha_1,\alpha_2} \times \mathbf{Coh}_{\alpha_3} \arrow[l, "\mathsf{q}^{\mathrm{Coh}}_{\alpha_1,\alpha_2}\times\mathrm{id}_{\alpha_3}"'] \arrow[r, "\mathsf{p}^{\mathrm{Coh}}_{\alpha_1,\alpha_2}\times\mathrm{id}_{\alpha_3}"] & \mathbf{Coh}_{\alpha_1+\alpha_2} \times \mathbf{Coh}_{\alpha_3} \\
\mathbf{Coh}_{\alpha_1} \times \widetilde{\mathbf{Coh}}_{\alpha_2,\alpha_3} \arrow[u, "\mathrm{id}_{\alpha_1}\times\mathsf{q}^{\mathrm{Coh}}_{\alpha_2,\alpha_3}"] \arrow[d, "\mathrm{id}_{\alpha_1}\times\mathsf{p}^{\mathrm{Coh}}_{\alpha_2,\alpha_3}"'] & \widetilde{\mathbf{Coh}}_{\alpha_1,\alpha_2,\alpha_3} \arrow[l] \arrow[r] \arrow[u] \arrow[d] & \widetilde{\mathbf{Coh}}_{\alpha_1+\alpha_2,\alpha_3} \arrow[u, "\mathsf{q}^{\mathrm{Coh}}_{\alpha_1+\alpha_2,\alpha_3}"'] \arrow[d, "\mathsf{p}^{\mathrm{Coh}}_{\alpha_1+\alpha_2,\alpha_3}"] \\
\mathbf{Coh}_{\alpha_1} \times \mathbf{Coh}_{\alpha_2+\alpha_3} & \widetilde{\mathbf{Coh}}_{\alpha_1,\alpha_2+\alpha_3} \arrow[l, "\mathsf{q}^{\mathrm{Coh}}_{\alpha_1,\alpha_2+\alpha_3}"'] \arrow[r, "\mathsf{p}^{\mathrm{Coh}}_{\alpha_1,\alpha_2+\alpha_3}"] & \mathbf{Coh}_{\alpha_1+\alpha_2+\alpha_3}
\end{tikzcd} \tag{2.2.1}$$

all the squares are cartesian, then the associativity and coassociativity can be deduced from the base change along upper-right circuit and lower-left circuit. ■

In addition, there is a description of $\mathsf{Eis}^{\mathrm{Coh}}$ on charts, given in [SS20]:

Lemma 2.8. *For fixed* $\mathcal{L}'$*, there exists small enough* $N \ll 0$ *such that in* $\widetilde{\mathbf{Coh}}_{\alpha,\beta}$*, for any* $\deg(\mathcal{L}) \leqslant N$*, we have*

$$\left(\mathsf{p}^{\mathrm{Coh}}_{\alpha,\beta}\right)^{-1}\left(\mathbf{Coh}^{>\mathcal{L}'}_{\alpha+\beta}\right) \subseteq \left(\mathsf{q}^{\mathrm{Coh}}_{\alpha,\beta}\right)^{-1}\left(\mathbf{Coh}^{>\mathcal{L}}_{\alpha} \times \mathbf{Coh}^{>\mathcal{L}}_{\beta}\right)$$

The Lemma allows us to reformulate functor $\mathsf{Eis}^{\mathrm{Coh}}$ on charts. For a fixed pair of line bundles $(\mathcal{L}, \mathcal{L}')$ that satisfy the above lemma, we can start with $\mathbf{Coh}^{>\mathcal{L}}_{\alpha} \times \mathbf{Coh}^{>\mathcal{L}}_{\beta} \cong \left[Q^{\mathcal{L}}_{\alpha} \times Q^{\mathcal{L}}_{\beta} / \mathrm{G}^{\mathcal{L}}_{\alpha} \times \mathrm{G}^{\mathcal{L}}_{\beta}\right]$ and arrive at $\mathbf{Coh}^{>\mathcal{L}'}_{\alpha+\beta} \cong \left[Q^{\mathcal{L},\mathcal{L}'}_{\alpha+\beta} / \mathrm{G}^{\mathcal{L}}_{\alpha+\beta}\right]$. The group $\mathrm{G}^{\mathcal{L}}_{\alpha} \times \mathrm{G}^{\mathcal{L}}_{\beta}$ is a Levi in $\mathrm{G}^{\mathcal{L}}_{\alpha+\beta}$. We can consider

$$\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta} := \widetilde{Q}^{\mathcal{L}}_{\alpha,\beta} \bigcap Q^{\mathcal{L},\mathcal{L}'}_{\alpha+\beta}$$

where $\widetilde{Q}^{\mathcal{L}}_{\alpha,\beta}$ is a subscheme of $Q^{\mathcal{L}}_{\alpha+\beta}$ that classifies

$$\left\{\left[\phi : \mathcal{L} \otimes \mathbf{k}^{\langle c(\mathcal{L}),\alpha+\beta\rangle} \twoheadrightarrow \mathcal{F}\right] \in \mathrm{Quot}^{\mathcal{L}}_{\alpha} \,\middle|\, \mathrm{Im}\left(\phi\mid_{\mathcal{L}\otimes\mathbf{k}^{\langle c(\mathcal{L}),\beta\rangle}}\right) \text{ strongly generated by } \mathcal{L}\right\}$$

then $\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta}$ is closed in $Q^{\mathcal{L},\mathcal{L}'}_{\alpha+\beta}$ and carries an action of the parabolic $\mathrm{P}^{\mathcal{L}}_{\alpha,\beta} = \left(\mathrm{G}^{\mathcal{L}}_{\alpha} \times \mathrm{G}^{\mathcal{L}}_{\beta}\right) \times U^{\mathcal{L}}_{\alpha,\beta}$. We denote by $\widetilde{\mathbf{Coh}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta} := \left[\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta} / \mathrm{P}^{\mathcal{L}}_{\alpha,\beta}\right]$ the global quotient, then it is the stack classifying extensions

$$0 \to \mathcal{E}' \to \mathcal{E}'' \to \mathcal{E} \to 0; \quad \text{where } (\mathcal{E}, \mathcal{E}') \in \mathbf{Coh}^{>\mathcal{L}}_{\alpha} \times \mathbf{Coh}^{>\mathcal{L}}_{\beta}, \text{ and } \mathcal{E}'' \in \mathbf{Coh}^{>\mathcal{L}'}_{\alpha+\beta}$$

In conclusion, we have the following commutative diagram:

$$\begin{tikzcd}
\mathbf{Coh}_{\alpha}^{>\mathcal{L}} \times \mathbf{Coh}_{\beta}^{>\mathcal{L}} \arrow[d, hook] & \widetilde{\mathbf{Coh}}_{\alpha,\beta}^{\mathcal{L},\mathcal{L}'} \arrow[l, "\mathsf{q}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}"'] \arrow[r, "\mathsf{p}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}"] \arrow[d, hook] & \mathbf{Coh}_{\alpha+\beta}^{>\mathcal{L}'} \arrow[d, hook] \\
\mathbf{Coh}_{\alpha} \times \mathbf{Coh}_{\beta} & \widetilde{\mathbf{Coh}}_{\alpha,\beta} \arrow[l, "\mathsf{q}_{\alpha,\beta}^{\mathrm{Coh}}"'] \arrow[r, "\mathsf{p}_{\alpha,\beta}^{\mathrm{Coh}}"] & \mathbf{Coh}_{\alpha+\beta}
\end{tikzcd} \tag{2.2.2}$$

where the vertical arrows are open immersions. For $\mathbf{F} \in \mathrm{D}\left(\mathbf{Coh}_{\alpha} \times \mathbf{Coh}_{\beta}\right)$, we have

$$\mathsf{Eis}_{\alpha,\beta}(\mathbf{F})^{\mathrm{Coh}} = \mathrm{colim}_{\mathcal{L},\mathcal{L}'} \left(\mathsf{p}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}\right)_! \left(\mathsf{q}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}\right)^* \left(\mathbf{F}|_{\mathbf{Coh}_{\alpha}^{>\mathcal{L}} \times \mathbf{Coh}_{\beta}^{>\mathcal{L}}}\right)$$

2.3 The Hall categories of curves

2.3.1 Monoidal and comonoidal structures. Let us denote $\mathbf{Coh} = \coprod_{\alpha \in \mathbf{Z}^+} \mathbf{Coh}_{\alpha}$ the entire stack of coherent sheaves on X, then we can see that $\mathrm{D}(\mathbf{Coh}) = \prod_{\alpha \in \mathbf{Z}^+} \mathrm{D}(\mathbf{Coh}_{\alpha})$ can be regarded as the categorical version of $\mathbf{H}_{X_0}$ and the functors $\mathsf{Eis}^{\mathrm{Coh}}$, $\mathsf{CT}^{\mathrm{Coh}}$ play the role of multipilication and co-multiplication respectively. We refer the category $\mathrm{D}(\mathbf{Coh})$ equips with functor $\mathsf{Eis}^{\mathrm{Coh}}$ the **Hall category** associated with curve X.

In fact, the notion of Hall categories or categorified Hall algebras has been proposed in [PS22]. Let $\mathcal{X}$ be a smooth proper scheme of dimension either 1 or 2, put $\mathbf{Coh}(\mathcal{X})$ the derived stack of coherent sheaves on $\mathcal{X}$ and $\widetilde{\mathbf{Coh}}(\mathcal{X})$ the stack classifies the extensions, we have natural maps

$$\mathbf{Coh}(\mathcal{X}) \times \mathbf{Coh}(\mathcal{X}) \xleftarrow{\mathsf{q}_{\mathcal{X}}} \widetilde{\mathbf{Coh}}(\mathcal{X}) \xrightarrow{\mathsf{p}_{\mathcal{X}}} \mathbf{Coh}(\mathcal{X})$$

Theorem 2.9. *The composition of functors*

$$\mathsf{QCoh}(\mathbf{Coh}(\mathcal{X})) \otimes \mathsf{QCoh}(\mathbf{Coh}(\mathcal{X})) \xrightarrow{\boxtimes} \mathsf{QCoh}(\mathbf{Coh}(\mathcal{X}) \times \mathbf{Coh}(\mathcal{X})) \xrightarrow{(\mathsf{p}_{\mathcal{X}})_!(\mathsf{q}_{\mathcal{X}})^*} \mathsf{QCoh}(\mathbf{Coh}(\mathcal{X}))$$

endows $\mathbb{E}_1$-monoidal structure on $\mathsf{QCoh}(\mathbf{Coh}(\mathcal{X}))$, *namely we have* $\mathsf{QCoh}(\mathbf{Coh}(\mathcal{X})) \in \mathrm{Alg}_{\mathbb{E}_1}(\mathsf{DGCat}_{\mathrm{cont}})$.

In particular, take $\mathcal{X}$ to be the Dolbeault shape of curve X, the stack $\mathbf{Coh}(\mathcal{X})$ is the stack of Higgs sheaves **Higgs** (see Section 4 for more details). Roughly, the classical limit argument says we have

$$\mathcal{D}\text{-}\mathsf{mod}(\mathbf{Coh}) \overset{\hbar \to 0}{\rightsquigarrow} \mathsf{QCoh}\left(\mathbf{Higgs}\right)$$

Then it is natural to ask whether $\mathcal{D}\text{-}\mathsf{mod}\left(\mathbf{Coh}\right)$ is also an $\mathbb{E}_1$-monoidal, Proposition 2.6 tells us that

Proposition 2.10. *The category* $\mathrm{D}(\mathbf{Coh})$ *equips with* $\mathsf{Eis}^{\mathrm{Coh}}$ *is an $\mathbb{E}_1$-monoidal object in* $\mathsf{DGCat}_{\mathrm{cont}}$ *and* $\mathrm{D}(\mathbf{Coh})_{\mathrm{co}}$ *equips with* $\mathsf{CT}^{\mathrm{Coh}}$ *is an $\mathbb{E}_1$-co-monoidal object, namely* $\left(\mathrm{D}(\mathbf{Coh}), \mathsf{CT}^{\mathrm{Coh}}\right) \in \mathrm{coAlg}_{\mathbb{E}_1}(\mathsf{DGCat}_{\mathrm{cont}})$.

Proof. The proof is similar to the one of Theorem 2.9. For any $\mathcal{T}$ presentable ∞-category, it is proved in [DK19] that there exists a canonical functor $2\text{-}\mathsf{Segal}(\mathcal{T}) \to \mathrm{Alg}_{\mathbb{E}_1}\left(\mathsf{Corr}^{\times}(\mathcal{T})\right)$ from the category of 2-Segal objects in $\mathcal{T}$ to $\mathbb{E}_1$-monoidal objects in the $(\infty,2)$-category of correspondences in $\mathcal{T}$. Then, take $\mathcal{T}$ to be dGeom, the category of derived geometric stacks, different choices of sheaf theories give monoidal functor from subcategory of correspondence to ∞-category. For the case of quasi-coherent sheaves, QCoh defines a lax monoidal functor $\mathsf{Corr}^{\times}_{\mathrm{rep,all}}(\mathsf{dGeom}) \longrightarrow \mathsf{DGCat}_{\mathrm{cont}}$, for the case of $\mathcal{D}$-modules, $\mathcal{D}$-mod defines a lax monoidal functor $\mathsf{Corr}^{\times}_{\mathrm{sm,proper}}(\mathsf{dGeom}) \longrightarrow \mathsf{DGCat}_{\mathrm{cont}}$ and finally for constructible sheaves, $\mathsf{Shv} : \mathsf{Corr}^{\times}_{\mathrm{all}}(\mathsf{dGeom}) \to \mathsf{DGCat}_{\mathrm{cont}}$. We can consider $\mathcal{S}_\bullet\mathbf{Coh}(\mathcal{X}) : \Delta^{\mathrm{op}} \to \mathsf{dSt}$ the Waldhausen $\mathcal{S}_\bullet$-construction, which is a simplicial object in derived stacks, the contruction is given in [PS22, Section 4.1]. In particular, $\mathcal{S}_1\mathbf{Coh}(\mathcal{X}) = \mathbf{Coh}(\mathcal{X})$ and $\mathcal{S}_2\mathbf{Coh}(\mathcal{X}) = \widetilde{\mathbf{Coh}}(\mathcal{X})$. As is proved in [PS22, Lemma 4.1], $\mathcal{S}_\bullet\mathbf{Coh}(\mathcal{X})$ is a 2-Segal object in dGeom. In particular, when we take $\mathcal{X} = X$ to be a smooth projective curve, Proposition 2.6 tells us that the correspondence $\mathbf{Coh}\times\mathbf{Coh} \xleftarrow{\mathsf{q}^{\mathrm{Coh}}} \widetilde{\mathbf{Coh}} \xrightarrow{\mathsf{p}^{\mathrm{Coh}}} \mathbf{Coh}$ which is the image of $\mathcal{S}_\bullet\mathbf{Coh}(X)$, lies in $\mathsf{Corr}^{\times}_{\mathrm{sm,proper}}(\mathsf{dGeom})$. Applying each sheaf theory, we get the desired result. ■

2.3.2 *The Hopf pairing.* We then consider the categorical analog of Hopf pairing in Theorem 2.1(3). As we have seen, for $\mathrm{D}(\mathbf{Coh}_\alpha)$ or $\mathrm{D}(\mathbf{Bun}_G)$, the pairing itself is given by the counit

$$\langle -,-\rangle := \mathrm{ev}_{\mathrm{D}(\mathbf{Coh}_\alpha)} : \mathrm{D}(\mathbf{Coh}_\alpha)_{\mathrm{co}} \otimes \mathrm{D}(\mathbf{Coh}_\alpha) \to \mathbf{Vect}$$

We want to see the pairing is compatible with the monoidal and co-monoidal structure of the Hall category, in other words, the adjunction of operators Eis and CT with respect to the pairing.

Proposition 2.11. *Let* C *and* D *be compactly generated categories, with dual categories* $\mathsf{C}^\vee$ *and* $\mathsf{D}^\vee$. *Assume there is adjoint pair* $F : \mathsf{C} \underset{\perp}{\rightleftarrows} \mathsf{D} : G$ *that induces* $G^\vee : \mathsf{C}^\vee \underset{\perp}{\rightleftarrows} \mathsf{D}^\vee : F^\vee$ *, then we have*

$$\mathrm{ev}_{\mathsf{D}}\left(F(\mathbf{c})\otimes\mathbf{d}^\vee\right) = \mathrm{ev}_{\mathsf{C}}\left(\mathbf{c}\otimes F^\vee(\mathbf{d}^\vee)\right) \quad \text{and} \quad \mathrm{ev}_{\mathsf{D}}\left(G^\vee(\mathbf{c}^\vee)\otimes\mathbf{d}\right) = \mathrm{ev}_{\mathsf{C}}\left(\mathbf{c}^\vee\otimes G(\mathbf{d})\right)$$

Proof. By definition and adjunction we have

$$\mathrm{ev}_{\mathsf{C}}\left(\mathbf{c}\otimes F^\vee(\mathbf{d}^\vee)\right) = \mathrm{Hom}_{\mathsf{C}^\vee}\left(\mathbf{c}^\vee, F^\vee(\mathbf{d}^\vee)\right) = \mathrm{Hom}_{\mathsf{D}^\vee}\left(G^\vee\left((\mathbf{c})^\vee\right), \mathbf{d}^\vee\right) = \mathrm{ev}_{\mathsf{D}}\left(\left(G^\vee(\mathbf{c}^\vee)\right)^\vee\otimes\mathbf{d}^\vee\right)$$

So it remains to show that $(G^\vee(\mathbf{c}^\vee))^\vee = F(\mathbf{c})$, which is proved in [Gai16, Lemma 1.5.6]. ■

To apply Proposition 2.11, we need to consider the dual notions of functors Eis and CT. Notice that by Theorem 1.9, the adjunctions $\mathsf{Eis}^G_{M,!} \dashv \mathsf{CT}^M_{G,*}$ and $\mathsf{Eis}^{\mathrm{Coh}} \dashv \mathsf{CT}^{\mathrm{Coh}}_*$ are adjunctions between compactly generated categories. Taking the dual categories and dual functors, we obtain:

$$\mathsf{Eis}_{\mathrm{co},*} := (\mathsf{CT}_*)^\vee : \mathrm{D}(\mathbf{Bun}_M)_{\mathrm{co}} \underset{\perp}{\rightleftarrows} \mathrm{D}(\mathbf{Bun}_G)_{\mathrm{co}} : (\mathsf{Eis}_!)^\vee =: \mathsf{CT}_{\mathrm{co},?}$$

$$\mathsf{Eis}^{\mathrm{Coh}}_{\mathrm{co},*} := (\mathsf{CT}^{\mathrm{Coh}}_*)^\vee : \mathrm{D}(\mathbf{Coh}_{\alpha,\beta})_{\mathrm{co}} \underset{\perp}{\rightleftarrows} \mathrm{D}(\mathbf{Coh}_{\alpha+\beta})_{\mathrm{co}} : (\mathsf{Eis}^{\mathrm{Coh}})^\vee =: \mathsf{CT}^{\mathrm{Coh}}_{\mathrm{co},?}$$

where we identify $\mathrm{D}(\mathbf{Bun}_G)^\vee$ with $\mathrm{D}(\mathbf{Bun}_G)_{\mathrm{co}}$ and $\mathrm{D}(\mathbf{Coh}_\alpha)^\vee$ with $\mathrm{D}(\mathbf{Coh}_\alpha)_{\mathrm{co}}$. As a corollary of Proposition 2.11, we have the categorical version of Theorem 2.1(3):

Proposition 2.12. *For any reductive group* G *with pairing* $\langle -,-\rangle_{\mathrm{D}(\mathbf{Bun}_G)}$ *and* $\mathbf{P}\in\mathrm{D}(\mathbf{Bun}_M)$, $\mathbf{Q}\in\mathrm{D}(\mathbf{Bun}_G)_{\mathrm{co}}$, *we have*

$$\left\langle \mathsf{Eis}^G_{M,!}(\mathbf{P}), \mathbf{Q}\right\rangle_{\mathrm{D}(\mathbf{Bun}_G)} = \left\langle \mathbf{P}, \mathsf{CT}^M_{\mathrm{co},G,?}(\mathbf{Q})\right\rangle_{\mathrm{D}(\mathbf{Bun}_M)}$$

Similarly, under the pairing on $\mathrm{D}(\mathbf{Coh}_\alpha)$*, for* $\mathbf{P} \in \mathrm{D}(\mathbf{Coh}_\alpha \times \mathbf{Coh}_\beta)$*,* $\mathbf{Q} \in \mathrm{D}(\mathbf{Coh}_{\alpha+\beta})_{\mathrm{co}}$*, we have*

$$\left\langle \mathsf{Eis}^{\mathrm{Coh}}_{\alpha,\beta,!}(\mathbf{P}), \mathbf{Q} \right\rangle_{\mathrm{D}(\mathbf{Coh}_{\alpha+\beta})} = \left\langle \mathbf{P}, \mathsf{CT}^{\mathrm{Coh}}_{\mathrm{co},\alpha,\beta,?}(\mathbf{Q}) \right\rangle_{\mathrm{D}(\mathbf{Coh}_\alpha \times \mathbf{Coh}_\beta)}$$

2.4 The Langlands formula and adjunctions

2.4.1 *The coherent Langlands formula.* To see a categorical analog of Theorem 2.1(2) over the Hall category, namely the compatiblity between product and coproduct. We need to consider the composition of functors $\mathsf{CT}^{\mathrm{Coh}} \circ \mathsf{Eis}^{\mathrm{Coh}}$.

Recall the classical Langlands formula (see for example [MW95]) gives a re-interpretation of the constant term of Eisenstein series. Namely fix G be a reductive group, one can prove for an automorphic form f on G the following formula:

$$\mathsf{CT}^M_G \circ \mathsf{Eis}^G_L(f) = \sum_{w \in W_L \backslash W / W_M} \mathsf{Eis}^M_{M^w} \circ \mathcal{R}_w \circ \mathsf{CT}^{{}^wL}_L(f)$$

where L, M are Levi subgroups of G and W_L is the sub-Weyl group of W Weyl group of G. For each representative w in the above double coset, the groups M^w and ${}^wL$ are Levi subgroups of M and L respectively that are conjugate by w. The middle term $\mathcal{R}_w$ on the right hand side is the induced conjugation on automorphic forms by w.

For sheaves on $\mathbf{Bun}_G$ or $\mathbf{Coh}_\alpha$, we can also deduce a similar relation like above. For tuple $\underline{r} = (r_1, \dots, r_s) \in \mathbb{Z}^s$, let $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{r}}$ denote $\bigoplus_{(r_i,d_i)\in \mathbf{Z}^+} \mathsf{Eis}^{\mathrm{Coh}}_{(r_1,d_1),\dots,(r_s,d_s)}$ and also $\mathsf{CT}^{\mathrm{Coh}}_{\underline{r}} = \bigoplus_{(r_i,d_i)\in \mathbf{Z}^+} \mathsf{CT}^{\mathrm{Coh}}_{(r_1,d_1),\dots,(r_s,d_s)}$. And for two tuples $\underline{r} = (r_1, \dots, r_{s_1})$ and $\underline{r'} = (r'_1, \dots, r'_{s_2})$ with the same sum n, we say a matrix $\mathbb{M}$ of size $s_1 \times s_2$ is a *contingency matrix* from $\underline{r}$ to $\underline{r'}$ if its i-th row sums to r_i and its j-th column sums to r'_j.

Theorem 2.13 (Langlands formula for **Coh**)**.** *The composition of functor* $\mathsf{CT}^{\mathrm{Coh}}_{r,s} \circ \mathsf{Eis}^{\mathrm{Coh}}_{p,q}$ *admits a filtration with graded pieces are classified by contingency matrices from* $[r,s]$ *to* $[p,q]$*. In particular, let* $\mathbb{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ *be a contingency matrix from* $[p,q]$ *to* $[r,s]$*, the graded piece associated with* $\mathbb{M}$ *is given by*

$$\left(\mathsf{Eis}^{\mathrm{Coh}}_{a,c} \times \mathsf{Eis}^{\mathrm{Coh}}_{b,d}\right) \circ \mathcal{R}^c_b \circ \left(\mathsf{CT}^{\mathrm{Coh}}_{a,b} \times \mathsf{CT}^{\mathrm{Coh}}_{c,d}\right)$$

where $\mathcal{R}^c_b$ *is induced equivalence by the swapping isomorphism* $\mathbf{Coh}_{b,c} \xrightarrow{\cong} \mathbf{Coh}_{c,b} : (\mathcal{F}_1, \mathcal{F}_2) \mapsto (\mathcal{F}_2, \mathcal{F}_1)$.

Proof of Theorem 2.13. We let $\widetilde{\mathcal{E}}^{r,s}_{p,q}$ denote the fiber product $\widetilde{\mathbf{Coh}}_{p,q} \times_{\mathbf{Coh}_n} \widetilde{\mathbf{Coh}}_{r,s}$. Then for any affine scheme S, $\widetilde{\mathcal{E}}^{r,s}_{p,q}(S)$ classifies the crosses as the following left diagram Figure 1:

$$\begin{tikzcd}
 & & 0 \arrow[d] & & \\
 & & \mathcal{J} \arrow[d, "g"] & & \\
0 \arrow[r] & \mathcal{F} \arrow[r, "f"] & \mathcal{M} \arrow[r, "f'"] \arrow[d, "g'"] & \mathcal{G} \arrow[r] & 0 \\
 & & \mathcal{K} \arrow[d] & & \\
 & & 0 & &
\end{tikzcd}$$

Figure 1: The cross diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
0 \arrow[r] & \mathcal{J}_1 \arrow[r, "u"] \arrow[d, "u'"'] & \mathcal{J} \arrow[r, "v"] & \mathcal{J}_2 \arrow[r] \arrow[d, "x"] & 0 \\
 & \mathcal{F} \arrow[d, "v'"'] & & \mathcal{G} \arrow[d, "y"] & \\
0 \arrow[r] & \mathcal{K}_1 \arrow[r, "x'"'] \arrow[d] & \mathcal{K} \arrow[r, "y'"'] & \mathcal{K}_2 \arrow[r] \arrow[d] & 0 \\
 & 0 & & 0 &
\end{tikzcd}$$

Figure 2: The square diagram

We fix $\mathcal{F}, \mathcal{G}, \mathcal{J}, \mathcal{K} \in \mathsf{Coh}(X_S)$ fiberwise with ranks p, q, r, s respectively. Let $\mathbf{Cr}_{\mathcal{M}}$ be the set of all choices (f, f', g, g') of the cross with middle term being $\mathcal{M}$. For any element in $\mathbf{Cr}_{\mathcal{M}}$, we can take

$$\mathcal{K}_1 := \mathrm{Im}(g'f); \mathcal{K}_2 := \mathrm{Coker}(g'f); \mathcal{J}_1 := \mathrm{Ker}(f'g); \mathcal{J}_2 := \mathrm{Im}(f'g).$$

We can deduce from the short exactness of both arrows in the cross that $\mathrm{Ker}(g'f) \cong \mathrm{Ker}(f'g)$ and $\mathrm{Coker}(g'f) \cong \mathrm{Coker}(f'g)$, and thus starting from the cross, we get the right above square diagram of short exact sequences (Figure 2). Let $\mathbf{Sq}(\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2)$ be the set of square diagrams with vertices $\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2$. We have constructed a map $\Phi : \coprod \mathbf{Cr}_{\mathcal{M}} \to \coprod \mathbf{Sq}(\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2)$. Moreover, if we let $a = \mathrm{rk}(\mathcal{J}_1), b = \mathrm{rk}(\mathcal{K}_1), c = \mathrm{rk}(\mathcal{J}_2), d = \mathrm{rk}(\mathcal{K}_2)$, we can associate a cross diagram with a contingency matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ from $[p, q]$ to $[r, s]$ by matching the rank of vertices of its image under Φ.

Given a square diagram Sq in $\mathbf{Sq}(\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2)$, we can complete its middle: take $\mathcal{M}_1$ to be the pushout of (u, u') and $\mathcal{M}_2$ to be the fiber product $\mathcal{G} \times_{\mathcal{K}_2} \mathcal{K}$ given by y, y'. Also, consider the direct sum of two extensions $\sigma_1 = \mathcal{K}_1 \xrightarrow{x'} \mathcal{K} \xrightarrow{y'} \mathcal{K}_2$ and $\sigma_1 = \mathcal{J}_2 \xrightarrow{x} \mathcal{G} \xrightarrow{y} \mathcal{K}_2$, we get an extension $\sigma_1 \oplus \sigma_2 \in \mathrm{Ext}^1(\mathcal{K}_2, \mathcal{K}_1 \oplus \mathcal{J}_2)$, the corresponding middle term is exactly $\mathcal{M}_2$. Similarly, we can also obtain $\mathcal{J}_1 \longrightarrow \mathcal{M}_1 \longrightarrow \mathcal{K}_1 \oplus \mathcal{J}_2$. If a cross $\mathsf{Cr} \in \mathbf{Cr}_{\mathcal{M}}$ maps to Sq, then we obtain natural maps $\mathcal{M}_1 \to \mathcal{M}$ and $\mathcal{M} \to \mathcal{M}_2$, Moreover, they fit canonically into the following right octahedral diagram. Also, the inverse image $\Phi^{-1}(\mathsf{Sq}) \cap \mathbf{Cr}_{\mathcal{M}}$ are in bijection with the following left diagram, where the red arrows vary according to the choice of crosses in $\mathbf{Cr}_{\mathcal{M}}$.

$$\begin{tikzcd}
 & 0 & 0 & & \\
0 \arrow[r] & \mathcal{K}_1 \oplus \mathcal{J}_2 \arrow[u] \arrow[r] & \mathcal{M}_2 \arrow[u] \arrow[r] & \mathcal{K}_2 \arrow[r] & 0 \\
0 \arrow[r] & \mathcal{M}_1 \arrow[u] \arrow[r, red] & \mathcal{M} \arrow[u, red] \arrow[r, red] & \mathcal{K}_2 \arrow[u, equal] \arrow[r] & 0
\end{tikzcd}
\qquad
\begin{tikzcd}
 & 0 & 0 & & \\
0 \arrow[r] & \mathcal{K}_1 \oplus \mathcal{J}_2 \arrow[u] \arrow[r] & \mathcal{M}_2 \arrow[u] \arrow[r] & \mathcal{K}_2 \arrow[r] & 0 \\
0 \arrow[r] & \mathcal{M}_1 \arrow[u] \arrow[r, blue] & \mathcal{M} \arrow[u, blue] \arrow[r, blue] & \mathcal{K}_2 \arrow[u, equal] \arrow[r] & 0 \\
 & \mathcal{J}_1 \arrow[u] \arrow[r, equal] & \mathcal{J}_1 \arrow[u, blue] & & \\
 & 0 \arrow[u] & 0 \arrow[u] & &
\end{tikzcd} \tag{2.4.1}$$

For a contingency matrix $\mathbb{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ from $[p, q]$ to $[r, s]$, we let $\widetilde{\mathcal{E}}_{\mathbb{M}}$ be the stack that classifies the classes of square diagrams whose vertices have ranks correspond to the entries of $\mathbb{M}$, namely

$$\widetilde{\mathcal{E}}_{\mathbb{M}} : S \in \mathsf{Sch}_{\mathbf{k}} \mapsto \left\langle (\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2) \;\middle|\; \begin{array}{c} \mathcal{J}_i, \mathcal{K}_i \text{ flat over } S; \forall s : \operatorname{Spec} \mathbf{k} \to S, \\ \operatorname{rk}(\mathcal{J}_1|X_s) = a, \operatorname{rk}(\mathcal{J}_2|X_s) = b, \operatorname{rk}(\mathcal{K}_1|X_s) = c, \operatorname{rk}(\mathcal{K}_2|X_s) = d. \end{array} \right\rangle$$

We can see the map Φ induces a map sending any point in $\widetilde{\mathcal{E}}^{r,s}_{p,q}$ to one of $\widetilde{\mathcal{E}}_{\mathbb{M}}$, and the inverse construction induces a locally closed embedding $\iota_{\mathbb{M}} : \widetilde{\mathcal{E}}_{\mathbb{M}} \hookrightarrow \widetilde{\mathcal{E}}^{r,s}_{p,q}$. Eventually, it turns out that the stack $\widetilde{\mathcal{E}}^{r,s}_{p,q}$ can be stratified by $\bigcup_{\mathbb{M}} \widetilde{\mathcal{E}}_{\mathbb{M}}$, see the proof in [Sch09, Theorem 1.9] comparing the groupoids $\widetilde{\mathcal{E}}^{r,s}_{p,q}(\mathbf{k})$ and $\bigcup_{\mathbb{M}} \widetilde{\mathcal{E}}_{\mathbb{M}}(\mathbf{k})$.

We now turn to the composition of functors $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$, we have

$$\begin{aligned} \mathsf{CT}^{\mathrm{Coh}}_{r,s} \circ \mathsf{Eis}^{\mathrm{Coh}}_{p,q} &= (\mathsf{p}^{\mathrm{Coh}}_{r,s})_!(\mathsf{q}^{\mathrm{Coh}}_{r,s})^*(\mathsf{p}^{\mathrm{Coh}}_{p,q})_!(\mathsf{q}^{\mathrm{Coh}}_{p,q})^* \\ &\cong (\mathsf{p}^{\mathrm{Coh}}_{r,s})_!(\pi_2)_!(\pi_1)^*(\mathsf{q}^{\mathrm{Coh}}_{p,q})^* \end{aligned}$$

where π_i is the projection of $\tilde{\mathcal{E}}^{r,s}_{p,q}$ along the following diagram

$$\begin{tikzcd} \mathbf{Coh}_p \times \mathbf{Coh}_q & \widetilde{\mathbf{Coh}}_{p,q} \arrow[l, "\mathsf{q}^{\mathrm{Coh}}_{p,q}"'] \arrow[r, "\mathsf{p}^{\mathrm{Coh}}_{p,q}"] & \mathbf{Coh}_n \\ & \bigcup_{\mathbb{M}} \widetilde{\mathcal{E}}_{\mathbb{M}} = \widetilde{\mathcal{E}}^{r,s}_{p,q} \arrow[u, "\pi_1"] \arrow[r, "\pi_2"] & \widetilde{\mathbf{Coh}}_{r,s} \arrow[u, "\mathsf{q}^{\mathrm{Coh}}_{r,s}"'] \arrow[d, "\mathsf{p}^{\mathrm{Coh}}_{r,s}"] \\ & & \mathbf{Coh}_r \times \mathbf{Coh}_s \end{tikzcd} \tag{2.4.2}$$

For each contingency matrix $\mathbb{M}$, we can complete the above diagram to a big commutative diagram:

$$\begin{tikzcd} \mathbf{Coh}_p \times \mathbf{Coh}_q & \widetilde{\mathbf{Coh}}_{p,q} \arrow[l, "\mathsf{q}^{\mathrm{Coh}}_{p,q}"'] \arrow[r, "\mathsf{p}^{\mathrm{Coh}}_{p,q}"] & \mathbf{Coh}_n \\ \widetilde{\mathbf{Coh}}_{a,b} \times \widetilde{\mathbf{Coh}}_{c,d} \arrow[u, "\mathsf{p}^{\mathrm{Coh}}_{a,b} \times \mathsf{p}^{\mathrm{Coh}}_{c,d}"] \arrow[d, "\mathsf{q}^{\mathrm{Coh}}_{a,b} \times \mathsf{q}^{\mathrm{Coh}}_{c,d}"'] & \widetilde{\mathcal{E}}_{\mathbb{M}} \arrow[l, "\tilde{\pi}^{\mathbb{M}}_1"'] \arrow[u, "\pi^{\mathbb{M}}_1"] \arrow[r, "\pi^{\mathbb{M}}_2"] \arrow[d, "\tilde{\pi}^{\mathbb{M}}_2"] & \widetilde{\mathbf{Coh}}_{r,s} \arrow[u, "\mathsf{q}^{\mathrm{Coh}}_{r,s}"'] \arrow[d, "\mathsf{p}^{\mathrm{Coh}}_{r,s}"] \\ \mathbf{Coh}_{a,b,c,d} \cong \mathbf{Coh}_{a,c,b,d} & \widetilde{\mathbf{Coh}}_{a,c} \times \widetilde{\mathbf{Coh}}_{b,d} \arrow[l, "\mathsf{q}^{\mathrm{Coh}}_{a,c} \times \mathsf{q}^{\mathrm{Coh}}_{b,d}"] \arrow[r, "\mathsf{p}^{\mathrm{Coh}}_{a,c} \times \mathsf{p}^{\mathrm{Coh}}_{b,d}"'] & \mathbf{Coh}_r \times \mathbf{Coh}_s \end{tikzcd} \tag{2.4.3}$$

where $\pi^{\mathbb{M}}_i$ is the composition $\widetilde{\mathcal{E}}_{\mathbb{M}} \overset{\iota_{\mathbb{M}}}{\hookrightarrow} \widetilde{\mathcal{E}}^{r,s}_{p,q}$ with π_i. Note that by construction, $\tilde{\mathcal{E}}_{\mathbb{M}}$ is the fiber product $\left(\widetilde{\mathbf{Coh}}_{a,b} \times \widetilde{\mathbf{Coh}}_{c,d}\right) \times_{\mathbf{Coh}_{a,b,c,d}} \left(\widetilde{\mathbf{Coh}}_{a,c} \times \widetilde{\mathbf{Coh}}_{b,d}\right)$, where the morphism $\widetilde{\mathbf{Coh}}_{a,c} \times \widetilde{\mathbf{Coh}}_{b,d} \to \mathbf{Coh}_{a,b,c,d}$ is given by the composition

$$\widetilde{\mathbf{Coh}}_{a,c} \times \widetilde{\mathbf{Coh}}_{b,d} \xrightarrow{\mathsf{q}^{\mathrm{Coh}}_{a,c} \times \mathsf{q}^{\mathrm{Coh}}_{b,d}} \mathbf{Coh}_{a,b,c,d} \xrightarrow{\mathrm{Id}_a \times \tau^c_b \times \mathrm{Id}_d} \mathbf{Coh}_{a,b,c,d}$$

where τ^c_b is the swapping isomorphism.

For each contingency matrix $\mathbb{M}$, we can consider the following functor:

$$\begin{aligned}
&(\mathsf{Eis}^{\mathrm{Coh}}_{a,c}\times\mathsf{Eis}^{\mathrm{Coh}}_{b,d})\circ\mathcal{R}^c_b\circ(\mathsf{CT}^{\mathrm{Coh}}_{a,b}\times\mathsf{CT}^{\mathrm{Coh}}_{c,d})\\
=&\left(\mathsf{p}^{\mathrm{Coh}}_{a,c}\times\mathsf{p}^{\mathrm{Coh}}_{b,d}\right)_!\left(\mathsf{q}^{\mathrm{Coh}}_{a,c}\times\mathsf{q}^{\mathrm{Coh}}_{b,d}\right)^*\circ\mathcal{R}^c_b\circ\left(\mathsf{p}^{\mathrm{Coh}}_{a,b}\times\mathsf{p}^{\mathrm{Coh}}_{c,d}\right)_!\left(\mathsf{q}^{\mathrm{Coh}}_{a,b}\times\mathsf{q}^{\mathrm{Coh}}_{c,d}\right)^*\\
\cong&\left(\mathsf{p}^{\mathrm{Coh}}_{a,c}\times\mathsf{p}^{\mathrm{Coh}}_{b,d}\right)_!(\tilde{\pi}^{\mathbb{M}}_2)_!(\tilde{\pi}^{\mathbb{M}}_1)^*\left(\mathsf{q}^{\mathrm{Coh}}_{a,b}\times\mathsf{q}^{\mathrm{Coh}}_{c,d}\right)^*\\
\cong&(\mathsf{p}^{\mathrm{Coh}}_{r,s})_!(\pi^{\mathbb{M}}_2)_!(\pi^{\mathbb{M}}_1)^*(\mathsf{q}^{\mathrm{Coh}}_{p,q})^*
\end{aligned}$$

On the other hand, by adjunction we have $(\iota_{\mathbb{M}})_!(\iota_{\mathbb{M}})^! \longrightarrow \mathrm{Id}$ and thus we get

$$\begin{aligned}
&(\mathsf{p}^{\mathrm{Coh}}_{r,s})_!(\pi^{\mathbb{M}}_2)_!(\pi^{\mathbb{M}}_1)^*(\mathsf{q}^{\mathrm{Coh}}_{p,q})^*\cong(\mathsf{p}^{\mathrm{Coh}}_{r,s})_!(\pi_2)_!(\iota_{\mathbb{M}})_!(\iota_{\mathbb{M}})^!(\pi_1)^*(\mathsf{q}^{\mathrm{Coh}}_{p,q})^*\\
\longrightarrow&(\mathsf{p}^{\mathrm{Coh}}_{r,s})_!(\pi_2)_!(\pi_1)^*(\mathsf{q}^{\mathrm{Coh}}_{p,q})^*\cong(\mathsf{p}^{\mathrm{Coh}}_{r,s})_!(\mathsf{q}^{\mathrm{Coh}}_{r,s})^*(\mathsf{p}^{\mathrm{Coh}}_{p,q})_!(\mathsf{q}^{\mathrm{Coh}}_{p,q})^*=\mathsf{CT}^{\mathrm{Coh}}_{r,s}\circ\mathsf{Eis}^{\mathrm{Coh}}_{p,q}
\end{aligned}$$

this gives the desired graded piece appears in the filtration of $\mathsf{CT}^{\mathrm{Coh}}_{r,s}\circ\mathsf{Eis}^{\mathrm{Coh}}_{p,q}$. ■

Remark 2.14. To compare Theorem 2.13 with the classical Langlands formula above, we notice that $[p,q]$ and $[r,s]$ determines Levi subgroups $L=\mathrm{GL}_p\times\mathrm{GL}_q$ and $M=\mathrm{GL}_r\times\mathrm{GL}_s$ respectively. In this case, the set of contingency matrices from $[p,q]$ to $[r,s]$ are in bijection with the double coset $W_L\backslash W/W_M$ and thus admits an order given by the reduced length in the double coset.
Also, one should be cautious that due to the existence of rank 0 coherent sheaves, having 0 in the entry of contingency matrices does not mean the corresponding sheaf vanishes. In fact, choose $(\alpha_1,\ldots,\alpha_{s_1})$ and $(\beta_1,\ldots,\beta_{s_2})$ such that $\sum\alpha_i=\sum\beta_j$, we can consider the refined contingency matrices of size $s_1\times s_2$ with coeffcient in $\mathbf{Z}^+$, the above proof and consequence still hold.

3 Canonical bases associated with a curve

The part 3 is devoted to describe a categorification of the spherical Hall algebra $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$ and construct the *canonical bases* of $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$ via so-called Eisenstein sheaves. We work with $\mathsf{Shv}(\mathbf{Coh})$ in this part.

3.1 Lusztig perverse sheaves for Coh

3.1.1 *Lusztig sheaves and Laumon-Lusztig categories.* In this section, we consider categorical analog of $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$. We have seen that for each $\alpha\in\mathbf{Z}^+$, the stack $\mathbf{Coh}_\alpha$ can be exhibited by smooth quasi-compact open substacks $\mathbf{Coh}^{>\mathcal{L}}_\alpha\cong\left[\mathbf{Q}^{\mathcal{L}}_\alpha/\mathrm{G}^{\mathcal{L}}_\alpha\right]$. When X is over $\overline{\mathbb{F}}_q$, we denote by $\mathbf{1}_\alpha$ the colimit of perverse sheaves $\overline{\mathbb{Q}_\ell}|_{\left[\mathbf{Q}^{\mathcal{L}}_\alpha/\mathrm{G}_\alpha\right]}[\dim(\mathbf{Coh}_\alpha)]$, which lies in $\mathsf{Perv}\left(\mathbf{Coh}_\alpha\right)\subseteq\mathsf{Shv}(\mathbf{Coh}_\alpha)$. Similarly for $X/\mathbb{C}$, let $\mathbf{1}_\alpha$ be the colimit of $\mathbb{C}|_{\left[\mathbf{Q}^{\mathcal{L}}_\alpha/\mathrm{G}_\alpha\right]}[\dim(\mathbf{Coh}_\alpha)]$, lies in $\mathsf{Perv}\left(\mathbf{Coh}_\alpha\right)=\mathcal{D}\text{-}\mathsf{mod}_{\mathrm{reg.hol}}\left(\mathbf{Coh}_\alpha\right)^{\heartsuit}\subseteq\mathcal{D}\text{-}\mathsf{mod}\left(\mathbf{Coh}_\alpha\right)$.

Lemma 3.1. *Let $\underline{\alpha}=(\alpha_1,\ldots,\alpha_s)$ be a partition of α, then $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}$ preserves the category of perverse sheaves. Moreover, $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}$ sends semi-simple perverse sheaves to semi-simple ones.*

Proof. The first half is because $\mathsf{q}^{\mathrm{Coh}}_{\underline{\alpha}}$ is smooth and $\mathsf{p}^{\mathrm{Coh}}_{\underline{\alpha}}$ is proper, thus $\left(\mathsf{q}^{\mathrm{Coh}}_{\underline{\alpha}}\right)^*$ and $\left(\mathsf{p}^{\mathrm{Coh}}_{\underline{\alpha}}\right)_!$ are t-exact up to shift. The second half is because $\mathsf{q}^{\mathrm{Coh}}_{\underline{\alpha}}$ is moreover surjective thus $\left(\mathsf{q}^{\mathrm{Coh}}_{\underline{\alpha}}\right)^*$ preserves simple objects and there is decomposition theorem for $\mathsf{p}^{\mathrm{Coh}}_{\underline{\alpha}}$. ■

As a consequence, we have a semi-simple perverse sheaf $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbf{1}_{\alpha_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\alpha_s}\right)\in\mathsf{Perv}\left(\mathbf{Coh}_{\alpha}\right)$. We abbreviate the constant sheaf $\mathbf{1}_{\alpha_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\alpha_s}$ as $\mathbf{1}_{\underline{\alpha}}$. We refer the sheaf $\mathsf{Eis}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right)$ as Eisenstein sheaf, which has been studied extensively by Laumon [Lau90]. Starting from Eisenstein sheaves, we define:

Definition 3.2. By a **Lusztig perverse sheaf** on $\mathbf{Coh}_{\alpha}$ we mean a perverse sheaf whose shift is a simple subobject of Laumon's *Eisenstein sheaf* $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right)$, where $\sum_{i=1}^{k}\alpha_i=\alpha$ and $\mathrm{rk}(\alpha_i)\leqslant 1$.
We let $\mathsf{P}^{\mathrm{Coh}}_{\alpha}$ be the set of Lusztig perverse sheaves on $\mathbf{Coh}_{\alpha}$ and write $\mathsf{P}^{\mathrm{Coh}}_{r}=\bigsqcup_{\mathrm{rk}\,\alpha=r}\mathsf{P}^{\mathrm{Coh}}_{\alpha}$. When $r=0$, we also use $\mathsf{P}^{\mathrm{Tor}}$ to denote for $\mathsf{P}^{\mathrm{Coh}}_{0}=\bigsqcup_{d\geqslant 1}\mathsf{P}^{\mathrm{Coh}}_{\tau d}$.

Definition 3.3. For $\alpha\in\mathbf{Z}^+$, we denote by $\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}$ the category additively generated by $\mathsf{P}^{\mathrm{Coh}}_{\alpha}$, viewed as a DG-category. Its ind-completion $\mathsf{Ind}\left(\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}\right)$ is a subcategory of $\mathsf{Shv}\left(\mathbf{Coh}_{\alpha}\right)$. We use the notation $\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\mathrm{Kar}}$ for the DG-category which is the Karoubi completion of $\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}$ and call it **Laumon-Lusztig category**.

We recall some terminologies and properties about Karoubi completion:

Definition 3.4. An essentially small DG category C is said to be *Karoubi complete* or *Karoubian* in short, if its homotopy category is idempotent complete.

Let C be an essentially small DG category, the Karoubi completion of C is a Karoubian category $\mathsf{C}_{\mathrm{Kar}}$ with a functor $\mathsf{C}\hookrightarrow\mathsf{C}_{\mathrm{Kar}}$ that induces equivalence $\mathrm{Hom}_{\mathsf{DGCat}}\left(\mathsf{C}_{\mathrm{Kar}},\mathsf{D}\right)\cong\mathrm{Hom}_{\mathsf{DGCat}}\left(\mathsf{C},\mathsf{D}\right)$ for any Karoubian category D. We have the following Thomason-Trobaugh-Neeman localization Theorem:

Lemma 3.5. *For essentially small DG category* C *which is not cocomplete, the functor* $\mathsf{C}\longrightarrow\mathsf{Ind}(\mathsf{C})^c$ *realizes* $\mathsf{Ind}(\mathsf{C})^c$ *as Karoubi completion of* C*, which is unique under equivalence.*

For an essentially small DG category C with a subset of objects $\mathbf{S}$, we say $\mathbf{S}$ Karoubi generates C if $\mathrm{Ho}(\mathsf{C})$, the homotopy category of C, can be obtained by taking finite iterations of mapping cones and direct summand. We have the following Lemma.

Lemma 3.6. *[DG15, Corollary 1.4.6] Let* C *be cocomplete DG category with a subset of objects* $\mathbf{S}\subseteq\mathsf{C}^c$ *that generates* C*, then* $\mathbf{S}$ *Karoubi generates* C^c*.*

Remark 3.7. In our case, we have inclusions

$$\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}\subseteq\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\mathrm{Kar}}=\mathsf{Ind}\left(\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}\right)^c\subseteq\mathsf{Ind}\left(\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}\right)$$

However, one need to be cautious here that objects in $\mathsf{P}^{\mathrm{Coh}}_{\alpha}$ are compact in $\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}$ for tautological reason, but not compact in general in the ambient category $\mathsf{Shv}\left(\mathbf{Coh}_{\alpha}\right)$. Also, the set $\mathsf{P}^{\mathrm{Coh}}_{\alpha}$ Karoubi generates the Laumon-Lusztig category $\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\mathrm{Kar}}$, but Lusztig perverse sheaves are not among compact generators of $\mathsf{Shv}(\mathbf{Coh}_\alpha)$. We abbreviate $\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\mathrm{Kar}}$ as $\mathsf{Q}^{\mathrm{Coh}}_{\alpha}$ if there is no ambiguity. Now for a left adjoint functor $\mathsf{F}:\mathsf{C}\to\mathsf{D}$ and $\mathbf{c}\in\mathsf{C}$ compact object, if the right adjoint F^R is compatible with colimits, for example if F^R is also a left adjoint, then $\mathsf{F}(\mathbf{c})$ is again compact. Similar to the case of geometric constant term functor considered in [DG16], we have adjunctions $\mathsf{Eis}^{\mathrm{Coh}}\dashv\mathsf{CT}^{\mathrm{Coh}}_{*}\dashv\mathsf{Eis}^{\mathrm{Coh},-}$ (see Corollary A.10) over $\mathsf{Shv}(\mathbf{Coh}_\alpha)$. We can restrict the functor to $\mathsf{Q}^{\mathrm{Coh}}_{\alpha}$, in combining the following Proposition 3.8, we can show that $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$ preserves compactness in $\mathsf{Q}^{\mathrm{Coh}}_{\alpha}$.

Proposition 3.8. *$\mathsf{Eis}^{\mathrm{Coh}}_{\alpha,\beta}$ sends $\mathsf{Q}^{\mathrm{Coh}}_{\alpha}\times\mathsf{Q}^{\mathrm{Coh}}_{\beta}$ to $\mathsf{Q}^{\mathrm{Coh}}_{\alpha+\beta}$ and $\mathsf{CT}^{\mathrm{Coh}}_{\alpha,\beta}$ sends $\mathsf{Q}^{\mathrm{Coh}}_{\alpha+\beta}$ to $\mathsf{Q}^{\mathrm{Coh}}_{\alpha}\times\mathsf{Q}^{\mathrm{Coh}}_{\beta}$.*

Proof. Assume C,D two cocomplete DG category, $\mathbf{S}_1$ Karoubi generates C^c and $\mathbf{S}_2$ Karoubi generates D^c, $\mathsf{F}:\mathsf{C}\to\mathsf{D}$ is a functor that sends $\mathbf{S}_1$ to D^c, then F maps C^c to D^c. Thus the argument for $\mathsf{Eis}^{\mathrm{Coh}}$ follows by definition of $\mathsf{P}^{\mathrm{Coh}}$: For $\mathbf{P}\in\mathsf{P}^{\mathrm{Coh}}_{\alpha},\mathbf{Q}\in\mathsf{P}^{\mathrm{Coh}}_{\beta}$, there exists $\mathbf{P}'\in\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}$ and $\mathbf{Q}'\in\mathsf{Q}^{\mathrm{Coh}}_{\beta,\oplus}$ such that $\mathbf{P}\oplus\mathbf{P}'=\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}(\mathbf{1}_{\underline{\alpha}})$ and $\mathbf{Q}\oplus\mathbf{Q}'=\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}\left(\mathbf{1}_{\underline{\beta}}\right)$. Thus $\mathsf{Eis}^{\mathrm{Coh}}_{\alpha,\beta}(\mathbf{P}\boxtimes\mathbf{Q})\subseteq\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}\cup\underline{\beta}}\left(\mathbf{1}_{\underline{\alpha}\cup\underline{\beta}}\right)$ and combining Lemma 3.1, we know it is a semi-simple and compact object in $\mathsf{Q}^{\mathrm{Coh}}_{\alpha+\beta,\oplus}$.
For functor $\mathsf{CT}^{\mathrm{Coh}}$, we want to show the following refined Langlands formula:

Proposition 3.9. *Let $\underline{\gamma}=(\gamma_1,\ldots,\gamma_r)\in(\mathbf{Z}^+)^r$ and $\beta,\gamma\in\mathbf{Z}^+$ such that $\gamma=\sum\gamma_i=\alpha+\beta$, then*

$$\mathsf{CT}^{\mathrm{Coh}}_{\alpha,\beta}\left(\mathsf{Eis}^{\mathrm{Coh}}\left(\mathbf{1}_{\gamma_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\gamma_r}\right)\right)=\bigoplus_{\underline{\alpha},\underline{\beta}}\mathsf{Eis}^{\mathrm{Coh}}\left(\mathbf{1}_{\alpha_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\alpha_r}\right)\boxtimes\mathsf{Eis}^{\mathrm{Coh}}\left(\mathbf{1}_{\beta_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\beta_r}\right)$$

where $\underline{\beta},\underline{\gamma}$ runs over $(\beta_1,\ldots,\beta_r)\in(\mathbf{Z}^+)^r$ and $(\gamma_1,\ldots,\gamma_r)\in(\mathbf{Z}^+)^r$such that $\sum\beta_i=\beta,\sum\gamma_i=\gamma$.

Proof of Proposition 3.9. The graded pieces of the equivalence are given by Langlands formula 2.13. It remains to show that the extensions among them are trivial. Let us consider the composition $\mathsf{CT}^{\mathrm{Coh}}\circ\mathsf{Eis}^{\mathrm{Coh}}$ under local charts. Fix $\mathcal{L}'$, define $\widetilde{Q}^{\mathcal{L}'}_{\alpha,\beta}$ to be a subvariety of $Q^{\mathcal{L}'}_{\gamma}$ cut out by condition:

$$\widetilde{Q}^{\mathcal{L}'}_{\alpha,\beta}=\left\{\phi\in Q^{\mathcal{L}'}_{\gamma}\,\middle|\,\left[\phi\left(\mathcal{L}'\otimes\mathbf{k}^{\langle\alpha,c(\mathcal{L}')\rangle}\right)\right]=\alpha\right\}$$

We have diagram of varieties $Q^{\mathcal{L}'}_{\alpha}\times Q^{\mathcal{L}'}_{\beta}\xleftarrow{\widetilde{\mathsf{q}}^{\mathcal{L}'}_{\alpha,\beta}}\widetilde{Q}^{\mathcal{L}'}_{\alpha,\beta}\xrightarrow{\widetilde{\mathsf{q}}^{\mathcal{L}'}_{\alpha,\beta}}Q^{\mathcal{L}'}_{\gamma}$. On the other hand, pick appropriate $\mathcal{L}$ as in Lemma 2.8, $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\gamma}}$ locally factors through $Q^{\mathcal{L}}_{\underline{\gamma}}\xleftarrow{\widetilde{\mathsf{q}}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}}\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}\xrightarrow{\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}}Q^{\mathcal{L}'}_{\gamma}$, where

$$\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}:=\widetilde{Q}^{\mathcal{L}}_{\underline{\gamma}}\bigcap Q^{\mathcal{L},\mathcal{L}'}_{\gamma}=\left\{(\phi,\mathcal{V}_\bullet)\,\middle|\,\begin{array}{l}\phi\in Q^{\mathcal{L}'}_{\gamma},\mathcal{V}_\bullet=\mathcal{V}_1\subseteq\mathcal{V}_2\subseteq\ldots\mathcal{V}_r=\mathcal{L}\otimes\mathbf{k}^{\langle\gamma,c(\mathcal{L}')\rangle}\\ \mathcal{V}_i/\mathcal{V}_{i-1}\cong\mathcal{L}\otimes\mathbf{k}^{\langle\gamma_i,c(\mathcal{L}')\rangle},\left[\phi(\mathcal{V}_i)/\phi(\mathcal{V}_{i-1})\right]=\gamma_i\end{array}\right\}\bigcap Q^{\mathcal{L},\mathcal{L}'}_{\gamma}$$

and the map $\widetilde{\mathsf{q}}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}$ is smooth. Let ${}^{\alpha,\beta}\mathcal{E}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}$ be the fiber product $\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}\times_{Q^{\mathcal{L}'}_{\gamma}}\widetilde{Q}^{\mathcal{L}'}_{\alpha,\beta}$. We have the following diagram

$$\begin{tikzcd}
Q^{\mathcal{L}}_{\alpha}\times Q^{\mathcal{L}}_{\beta} & \widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta} \arrow[l, "\widetilde{\mathsf{q}}^{\mathcal{L}'}_{\alpha,\beta}"'] \arrow[r, "\widetilde{\mathsf{p}}^{\mathcal{L}'}_{\alpha,\beta}"] & Q^{\mathcal{L}'}_{\gamma} \\
 & {}^{\alpha,\beta}\mathcal{E}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}} \arrow[u, "\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta}"] \arrow[r, "\pi_{\underline{\gamma}}"] & \widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}} \arrow[u, "\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}"']
\end{tikzcd} \tag{3.1.1}$$

To calculate $\mathsf{CT}^{\mathrm{Coh}}_{\alpha,\beta}\left(\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\gamma}}\left(\mathbf{1}_{\underline{\gamma}}\right)\right)|_{\mathbf{Coh}^{>\mathcal{L}'}_{\alpha}\times\mathbf{Coh}^{>\mathcal{L}'}_{\beta}}$, it suffices to consider the complex

$$\left(\widetilde{\mathsf{q}}^{\mathcal{L}'}_{\alpha,\beta}\right)_!\left(\widetilde{\mathsf{p}}^{\mathcal{L}'}_{\alpha,\beta}\right)^*\left(\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}\right)_!\left(\overline{\mathbb{Q}_\ell}\right)\cong\left(\widetilde{\mathsf{q}}^{\mathcal{L}'}_{\alpha,\beta}\circ\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta}\right)_!(\pi_{\underline{\gamma}})^*\left(\overline{\mathbb{Q}_\ell}\right)\cong\left(\widetilde{\mathsf{q}}^{\mathcal{L}'}_{\alpha,\beta}\circ\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta}\right)_!\left(\overline{\mathbb{Q}_\ell}|_{\alpha,\beta}\mathcal{E}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}\right)$$

The same argument in the proof of Theorem 2.13 implies that it admits a stratification given by $\underline{\alpha},\underline{\beta}$, let us denote each stratum by $\mathbf{X}_{\underline{\alpha},\underline{\beta}}$. We have the following Lemma of Lusztig:

Lemma 3.10. *For $f:Y\longrightarrow Z$ a morphism between varieties such that there exists a finite stratification $Y=\bigsqcup_\sigma Y_\sigma$, restriction of f on Y_σ can be decomposed into $Y_\sigma\xrightarrow{f_\sigma}\tilde{Y}_\sigma\xrightarrow{g_\sigma}Z$ and that f_σ is vector bundle, g_σ is proper. Then we have $f_!\left(\overline{\mathbb{Q}_\ell}\right)=\bigoplus_\sigma(f|_{Y_\sigma})_!\left(\overline{\mathbb{Q}_\ell}\right)$.*

In our case, the restriction of $\widetilde{\mathsf{q}}^{\mathcal{L}'}_{\alpha,\beta}\circ\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta}$ onto each stratum $\mathbf{X}_{\underline{\alpha},\underline{\beta}}$ factors as

$$\mathbf{X}_{\underline{\alpha},\underline{\beta}}\xrightarrow{\pi_{\underline{\alpha},\underline{\beta}}}\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\underline{\alpha}}\times\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\underline{\beta}}\xrightarrow{\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\underline{\alpha},\underline{\beta}}}Q^{\mathcal{L}'}_{\alpha}\times Q^{\mathcal{L}'}_{\beta}$$

where $\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\underline{\alpha}}$ is defined similarly as $\widetilde{Q}^{\mathcal{L},\mathcal{L}'}_{\underline{\gamma}}$ by changing $\underline{\gamma}$ with $\underline{\alpha}$. And the map $\pi_{\underline{\alpha},\underline{\beta}}$ is a vector bundle map, the map $\widetilde{\mathsf{p}}^{\mathcal{L},\mathcal{L}'}_{\underline{\alpha},\underline{\beta}}$ is proper so that we can apply Lemma 3.10. Since we are dealing with constant sheaves, they descend to equivariant sheaves categories, i.e. constant sheaves on quotient stacks. Thus we get the desired formula as in the Proposition for $\mathsf{CT}^{\mathrm{Coh}}_{\alpha,\beta}\left(\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\gamma}}\left(\mathbf{1}_{\underline{\gamma}}\right)\right)|_{\mathbf{Coh}^{>\mathcal{L}'}_{\alpha}\times\mathbf{Coh}^{>\mathcal{L}'}_{\beta}}$, passing to the colimit, we prove the Proposition 3.9. ∎

Now for any $\mathbf{P}\in\mathsf{P}^{\mathrm{Coh}}_{\alpha+\beta}$, $\mathsf{CT}^{\mathrm{Coh}}_{\alpha,\beta}(\mathbf{P})$ is a subcomplex of certain $\mathsf{CT}^{\mathrm{Coh}}\circ\mathsf{Eis}^{\mathrm{Coh}}\left(\mathbf{1}_{\underline{\gamma}}\right)$, which is revealed to be semi-simple. Thus $\mathsf{CT}^{\mathrm{Coh}}_{\alpha,\beta}(\mathbf{P})$ lies in $\mathsf{Q}^{\mathrm{Coh}}_{\alpha,\oplus}\times\mathsf{Q}^{\mathrm{Coh}}_{\beta,\oplus}$, as we want. ∎

3.1.2 *rank 0 case.* The rank 0=torsion case of Lusztig sheaves is already interesting, in fact, it resembles the Springer theory. For fixed torsion degree $d\geqslant 1$, we have the following diagram of Abel-Jacobi maps and induction:

$$\begin{tikzcd}
\mathbf{Coh}_{\tau 1}\times\ldots\times\mathbf{Coh}_{\tau 1} \arrow[dr, "\mathrm{supp}_{\times d}"'] & \widetilde{\mathbf{Coh}}_{\underline{\tau 1}} \arrow[l, "\mathsf{q}^{\mathrm{Coh}}_{\underline{\tau 1}}"'] \arrow[r, "\mathsf{q}^{\mathrm{Coh}}_{\underline{\tau 1}}"] \arrow[d, shift left, "\widetilde{\mathrm{supp}}_d"] & \mathbf{Coh}_{\tau d} \arrow[d, shift left, "\mathrm{supp}_d"] \\
& X^d \arrow[u, shift left, "\widetilde{\mathrm{AJ}}^d"] \arrow[r, "\mathrm{Sym}"'] & X^{(d)} \arrow[u, shift left, "\mathrm{AJ}^d"]
\end{tikzcd} \tag{3.1.2}$$

where $\underline{\tau 1}$ is d-copies of $\tau 1=(0,1)$. Inside of X^d (resp. $X^{(d)}$), consider the open locus $X^d\backslash\Delta$ (resp. $X^{(d)}\backslash\Delta$) of points with multiplicities 1. Then the Abel Jacobi map gives rises to an open inclusion $\left(X^{(d)}\backslash\Delta\right)/(\mathbb{G}_m)^d=:\mathbf{Coh}^{\mathrm{rss}}_{\tau d}\subseteq\mathbf{Coh}_{\tau d}$, where the $(\mathbb{G}_m)^d$-action is trivial and the upper script rss refers to regular semi-simple. In fact, recall when we take X to be $\mathbb{A}^1$, $\mathbf{Coh}_{\tau d}\cong\left[\mathfrak{gl}_d/\mathrm{GL}_d\right]$, the map $\mathsf{q}^{\mathrm{Coh}}_{\underline{\tau 1}}$ is the Grothendieck-Springer resolution $\left[\widetilde{\mathfrak{gl}}_d/\mathrm{GL}_d\right]\longrightarrow\left[\mathfrak{gl}_d/\mathrm{GL}_d\right]$, the preimage of supp_d (resp. $\widetilde{\mathrm{supp}}_d$) corresponds to the regular semi-simple part $\left[\mathfrak{gl}^{\mathrm{rss}}_d/\mathrm{GL}_d\right]$ (resp. $\left[\widetilde{\mathfrak{gl}}^{\mathrm{rss}}_d/\mathrm{GL}_d\right]$). The restriction of $\mathsf{q}^{\mathrm{Coh}}_{\underline{\tau 1}}$ to $\left[\widetilde{\mathfrak{gl}}^{\mathrm{rss}}_d/\mathrm{GL}_d\right]$ is an étale $\mathfrak{S}_d$-cover. In general, for X a smooth projective curve, the map $X^d\backslash\Delta\longrightarrow X^{(d)}\backslash\Delta$ is also a $\mathfrak{S}_d$-cover. A representation σ of $\mathfrak{S}_d$ thus gives rises to a local system $\mathbf{L}_\sigma$ on $\mathbf{Coh}^{\mathrm{rss}}_{\tau d}$.

Proposition 3.11. *For X smooth projective curve, the simple objects in $\mathsf{P}^{\mathrm{Coh}}_{\tau d}$ are in bijection with*

$$\left\{\mathbf{IC}\left(\mathbf{Coh}^{\mathrm{rss}}_{\tau d}, \mathbf{L}_\sigma\right) \mid \sigma : \textit{irreducible representation of } \mathfrak{S}_d\right\}.$$

In particular, we know irreducible representations of $\mathfrak{S}_d$ are classified by Young diagrams of size d. We use $\mathrm{Part}(d)$ to denote the set of Young diagrams of size d and σ_λ for the irreducible representation that is associated with $\lambda \in \mathrm{Part}(d)$. We abbreviate $\mathbf{L}_\lambda$ the object $\mathbf{IC}\left(\mathbf{Coh}^{\mathrm{rss}}_{\tau d}, \mathbf{L}_{\sigma_\lambda}\right)$ that corresponds to irreducible representation σ_λ. Now for $d_1, d_2 \geqslant 1$ and $d = d_1 + d_2$, we have the following relation

$$\mathsf{Eis}^{\mathrm{Coh}}_{\tau d_1, \tau d_2}\left(\mathbf{L}_{\lambda_1} \boxtimes \mathbf{L}_{\lambda_2}\right) \cong \bigoplus_{\lambda \in \mathrm{Part}(d)} \mathbb{V}_\lambda \otimes \mathbf{L}_\lambda[\mathbf{shift}]$$

where $\mathbb{V}_\lambda$ is vector space $\mathbb{C}^{c^\lambda_{\lambda_1,\lambda_2}}$ of dimension $c^\lambda_{\lambda_1,\lambda_2}$, which equals the multiplicity of σ_λ that appears in the induced representation $\mathrm{Ind}^{\mathfrak{S}_d}_{\mathfrak{S}_{d_1}\times\mathfrak{S}_{d_2}}\left(\sigma_{\lambda_1} \boxtimes \sigma_{\lambda_2}\right)$.

3.2 Compactified Eisenstein and Lusztig perverse sheaves for Bun

3.2.1 *Compatified Eisenstein series.* We have seen a description of Lusztig perverse sheaves for torsion case, one may wonder if there is similar construction of Lusztig perverse sheaves for vector bundle part, namely over the stacks $\mathbf{Bun}_{\mathrm{GL}_r}$. And it is natural to ask if there is a splitting of $\mathsf{P}^{\mathrm{Coh}}_r$ into vector bundle part and torsion part. Unfotunately, unlike Proposition 1.6, the morphism $\mathsf{p}^{\mathrm{Bun}}_{\alpha,\beta}$ or more generally p_P, is not proper, which means $\mathsf{Eis}_!$ does not commute with Verdier duality. As a result, $\mathsf{Eis}_!$ does not preserve perversity. To fix, we need to modify the Eisenstein series functor. The method is given by Braverman and Gaitsgory in [BG02], via so-called Drinfeld compactification.
For $\mathring{\mathcal{Y}} \overset{\text{open}}{\subseteq} \mathcal{Y}$ an open substack, set $\mathbf{Maps}_{\mathrm{gen}}(X, \mathcal{Y} \supseteq \mathring{\mathcal{Y}})$ to be a subsheaf in groupoids of $\mathbf{Maps}(X, \mathcal{Y})$:

$$\mathbf{Maps}_{\mathrm{gen}}(X, \mathcal{Y} \supseteq \mathring{\mathcal{Y}}) : \mathrm{Sch}_{\mathbf{k}} \longrightarrow \mathrm{Gpd};$$

$$S \mapsto \left\langle f \in \mathbf{Maps}(X, \mathcal{Y})(S) \,\middle|\, \begin{array}{l} \text{for } \forall \bar{s} \to S \text{ geometric point, there exists} \\ U \overset{\text{open}}{\subseteq} X \times \bar{s} \text{ so that } f\mid_U \text{ factors through } \mathring{\mathcal{Y}} \end{array} \right\rangle$$

Definition 3.12. Let Y be a scheme over $\mathbf{k}$, we say that Y is *strongly quasi-affine* if its global section is a finitely generated $\mathbf{k}$-algebra and there is an open embedding $Y \hookrightarrow \mathrm{Spec}\left(\Gamma(Y, \mathcal{O}_Y)\right) =: \overline{Y}$. Call $\overline{Y}$ the *affine closure* of Y.

Lemma 3.13. *[BG02] Both G/U and $G/[P,P]$ are strongly quasi-affine.*

We define the (variations of) compactification of Bun_P to be

$$\widetilde{\mathrm{Bun}}_P := \mathbf{Maps}_{\mathrm{gen}}\left(X, G\backslash\overline{G/U}/M \supseteq \mathbf{B}P\right) \quad \textit{Drinfeld's Compactification}$$

$$\overline{\mathrm{Bun}}_P := \mathbf{Maps}_{\mathrm{gen}}\left(X, G\backslash\overline{G/[P,P]}/(P/[P,P]) \supseteq \mathbf{B}P\right) \quad \textit{Naive Compactification}$$

For $S \in \mathrm{Sch}_{\mathbf{k}}$, we know that $\mathrm{Bun}_P(S)$ is the groupoid of triples $(\mathcal{F}_G, \mathcal{F}_M, \kappa)$, where κ is a family of subbundles $\kappa^{\mathcal{V}} : (\mathcal{V}^{U(P)})_{\mathcal{F}_M} \hookrightarrow \mathcal{V}_{\mathcal{F}_G}$ associated with $\mathcal{V} \in \mathrm{Rep}\, G$ (finite dimensional G-representations).

And the *Plucker relation* (i.e. the following commutative diagram)

$$\begin{tikzcd} (\mathcal{V}_1^U)_{\mathcal{F}_M} \otimes (\mathcal{V}_2^U)_{\mathcal{F}_M} \arrow[r, "\kappa^{\mathcal{V}_1} \otimes \kappa^{\mathcal{V}_2}"] \arrow[d] & (\mathcal{V}_1 \otimes \mathcal{V}_2)_{\mathcal{F}_G} \arrow[d] \\ (\mathcal{V}_3^U)_{\mathcal{F}_M} \arrow[r, "\kappa^{\mathcal{V}_3}"'] & (\mathcal{V}_3)_{\mathcal{F}_G} \end{tikzcd}$$

holds for every G-module map $\mathcal{V}_1 \otimes \mathcal{V}_2 \to \mathcal{V}_3$. In fact, by definition $\mathrm{Bun}_P(S)$ classifies the data of

$$\left(\mathcal{F}_G, \mathcal{F}_M, \kappa : \mathcal{F}_G = \mathcal{F}_M \overset{M}{\times} G \longrightarrow \mathcal{F}_M \overset{M}{\times} G/U(P)\right)$$

We can thus easily see by Tannakian the equivalence of two definitions of Bun_P.

Now let H be either U or $[P, P]$, then $\mathbf{Maps}_{\text{gen}}\left(G\backslash\overline{G/H}/(P/H) \supseteq \mathbf{B}P\right)(S)$ is the groupoid of triples:

$$(\mathcal{F}_G, \mathcal{F}_{P/H}, \kappa)$$

where as above, κ is a family of maps of coherent sheaves

$$\kappa^{\mathcal{V}} : (\mathcal{V}^H)_{\mathcal{F}_{P/H}} \to \mathcal{V}_{\mathcal{F}_G}$$

so that the restriction of $\kappa^{\mathcal{V}}$ on $X \times \bar{s}$ is injective for any geometric point $\bar{s} \to S$.

Theorem 3.14. *The natural maps $\widetilde{\mathsf{p}}_P : \widetilde{\mathrm{Bun}}_P \to \mathrm{Bun}_G$ and $\overline{\mathsf{p}}_P : \overline{\mathrm{Bun}}_P \to \mathrm{Bun}_G$ are schematic and proper.*

Definition 3.15. We define the **compactified Eisenstein series functor** $\widetilde{\mathsf{Eis}}_M^G : \mathsf{Shv}(\mathrm{Bun}_M) \longrightarrow \mathsf{Shv}(\mathrm{Bun}_G)$

$$\widetilde{\mathsf{Eis}}_M^G : \mathbf{F} \mapsto (\widetilde{\mathsf{p}}_P)_* \left(\mathbf{IC}_{\widetilde{\mathrm{Bun}}_P} \otimes \widetilde{\mathsf{q}}_P^*(\mathbf{F}) \otimes \left(\overline{\mathbb{Q}_\ell}[1]\right)^{-\otimes \dim(\mathrm{Bun}_M)}\right)$$

Remark 3.16. Notice that when $P = B$ is Borel, we have $\widetilde{\mathrm{Bun}}_B = \overline{\mathrm{Bun}}_B$. We will call the associated Eisenstein series functor the *principal series* and denote it by $\overline{\mathsf{Eis}}_T^G$. For convenience, we will abbreviate compactified Eisenstein series as $\widetilde{\mathsf{Eis}}$ or $\overline{\mathsf{Eis}}$ when there is no ambiguity in the choice of groups. The notion of principal series goes also for $\mathsf{Eis}_{\underline{\alpha}}^{\mathrm{Coh}}$, we say it is *principal series* if $\mathrm{rk}(\alpha_i) \leqslant 1$ for any α_i appears in $\underline{\alpha}$.

It is easy to see that $\mathsf{Eis}_M^G \circ \mathsf{Eis}_L^M = \mathsf{Eis}_L^G$ for Levis $L \subseteq M \subseteq G$, however it is highly non-trivial that

Theorem 3.17. *[BG02, Theorem 2.3.10] For any Levis $L \subseteq M \subseteq G$, we have a canonical isomorphism*

$$\widetilde{\mathsf{Eis}}_M^G \circ \widetilde{\mathsf{Eis}}_L^M \cong \widetilde{\mathsf{Eis}}_L^G$$

The Theorem is proved in [BG02] for $\widetilde{\mathsf{Eis}} \circ \overline{\mathsf{Eis}} \cong \widetilde{\mathsf{Eis}}$. One can mimic the same method there to get proof for general parabolic subgroups. In particular, we define Lusztig perverse sheaves to be

Definition 3.18. A **Lusztig perverse sheaf** on $\mathbf{Bun}_\alpha$ for $\alpha \in \mathbf{Z}^{++}$ is a perverse sheaf whose shift is a simple subquotient of $\overline{\mathsf{Eis}}_{\underline{\alpha}}\left(\mathbf{1}_{\alpha_1} \boxtimes \ldots \boxtimes \mathbf{1}_{\alpha_k}\right)$, where $\sum_{i=1}^{l} \alpha_i = \alpha$ and $\mathrm{rk}(\alpha_i) = 1$. We denote by $\mathsf{P}_\alpha^{\mathrm{Bun}}$ the set of Luszitg perverse sheaves on $\mathbf{Bun}_\alpha$ and write $\mathsf{P}_r^{\mathrm{Bun}} = \bigsqcup_{\mathrm{rk}\,\alpha = r} \mathsf{P}_\alpha^{\mathrm{Bun}}$.

Similarly, we denote by $\mathsf{Q}_\alpha^{\mathrm{Bun}}$ the DG-category which is the Karoubi completion of $\mathsf{Q}_{\alpha,\oplus}^{\mathrm{Bun}}$, the category additively generated by $\mathsf{P}_\alpha^{\mathrm{Bun}}$.

Remark 3.19. The sheaf $\overline{\mathsf{Eis}}_{\underline{\alpha}}(\mathbf{1}_{\underline{\alpha}})$ is a priori not a perverse sheaf. However, since we are in the situation of group GL_n, a theorem (see [BG02, Conjecture 2.1.10] and [BG08, Theorem 10.2]) says that is indeed true and moreover $\overline{\mathsf{Eis}}_{\underline{\alpha}}(\mathbf{1}_{\underline{\alpha}})$ is semi-simple. We can deduce the following Proposition:

Proposition 3.20. *The compactified Eisenstein series $\widetilde{\mathsf{Eis}}_{\alpha,\beta}$ sends $\mathsf{Q}^{\mathrm{Bun}}_{\alpha} \times \mathsf{Q}^{\mathrm{Bun}}_{\beta}$ to $\mathsf{Q}^{\mathrm{Bun}}_{\alpha+\beta}$.*

Proof. Theorem 3.17 implies $\widetilde{\mathsf{Eis}}_{\mathrm{rk}(\alpha),\mathrm{rk}(\beta)} \circ \left(\overline{\mathsf{Eis}}_{\underline{\alpha}} \boxtimes \overline{\mathsf{Eis}}_{\underline{\beta}}\right) = \overline{\mathsf{Eis}}_{\underline{\alpha}\cup\underline{\beta}}$. Also as is stated in the Remark 3.19, $\overline{\mathsf{Eis}}_{\underline{\alpha}}(\mathbf{1}_{\underline{\alpha}})$ is semi-simple, then the Proposition follows by construction. ∎

Proposition 3.21. *The compactified principal Eisenstein series $\overline{\mathsf{Eis}}$ for $\mathrm{GL}_r (r \geqslant 1)$ is compatible with $\mathsf{Eis}^{\mathrm{Coh}}$. More concretely, recall $\jmath_\alpha : \mathbf{Bun}_\alpha \hookrightarrow \mathbf{Coh}_\alpha$, then for any $\underline{\alpha} = (\alpha_1, \dots, \alpha_r)$ with $\mathrm{rk}(\alpha_i) = 1$, we have*

$$\jmath_\alpha^* \circ \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}} \circ (\jmath_{\underline{\alpha}})_{!*} \cong \overline{\mathsf{Eis}}_{\underline{\alpha}} \Longleftrightarrow \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}} \circ (\jmath_{\underline{\alpha}})_{!*} \cong (\jmath_\alpha)_{!*} \circ \overline{\mathsf{Eis}}_{\underline{\alpha}}$$

Proof. There is another compactification of $\mathbf{Bun}_B$ in the case of GL_r, given by Laumon [Lau90]. Let $\overline{\mathbf{Bun}}^L_B$ be the Laumon's compactification that classifies flags $\mathcal{E}_1 \subseteq \mathcal{E}_2 \subseteq \dots \subseteq \mathcal{E}_r$ such that $\mathcal{E}_{i-1} \subseteq \mathcal{E}_i$ is a subsheaf. We can see thus $\overline{\mathbf{Bun}}^L_B$ fits into the cartesian diagram

$$\begin{tikzcd}
\overline{\mathbf{Bun}}^L_B \arrow[r] \arrow[d] & \mathbf{Bun}_T \arrow[d, hook] \\
\widetilde{\mathbf{Coh}} \arrow[r, "\mathsf{p}^{\mathrm{Coh}}"] & \prod_{\mathrm{rk}\,T} \mathbf{Coh}_1
\end{tikzcd} \tag{3.2.1}$$

It is proved in [Kuz97] that $\overline{\mathbf{Bun}}^L_B \to \overline{\mathbf{Bun}}_B$ is a small resolution, thus the functor $\overline{\mathsf{Eis}}^L := (\bar{\mathsf{p}}^L)_!(\bar{\mathsf{q}}^L)^*$ along $\mathbf{Bun}_T \xleftarrow{\bar{\mathsf{q}}^L} \overline{\mathbf{Bun}}^L_B \xrightarrow{\bar{\mathsf{p}}^L} \mathbf{Bun}_G$ is equivalent to $\overline{\mathsf{Eis}}$ and the Proposition follows from a diagram chase. ∎

Recall we use $\mathsf{P}^{\mathrm{Tor}}$ denote $\mathsf{P}^{\mathrm{Coh}}_0 = \bigsqcup_{d \geqslant 1} \mathsf{P}^{\mathrm{Coh}}_{\tau d}$, then we have the following separation Theorem:

Theorem 3.22. *The set of Lusztig perverse sheaves $\mathsf{P}^{\mathrm{Coh}}_r$ on $\mathbf{Coh}_r$ is in bijection with $\mathsf{P}^{\mathrm{Bun}}_r \times \mathsf{P}^{\mathrm{Tor}}$.*

Proof. Begin with $\mathbf{P} \boxtimes \mathbf{T} \in \mathsf{P}^{\mathrm{Bun}}_r \times \mathsf{P}^{\mathrm{Tor}}$, we can consider the following diagram:

$$\begin{tikzcd}
\mathbf{Bun}_\alpha \times \mathbf{Coh}_{\tau l} \arrow[d, hook, "\jmath_\alpha \times \mathrm{Id}"] & \mathcal{Y} \arrow[l] \arrow[r, hook, "\iota^l"] \arrow[d] \arrow[dl, phantom, "\lrcorner"] & \mathbf{Coh}^{\mathrm{tor.deg} \leqslant l}_{\alpha+\tau l} \arrow[d, hook, "\jmath^{\leqslant l}_{\alpha+\tau l}"] \\
\mathbf{Coh}_\alpha \times \mathbf{Coh}_{\tau l} & \widetilde{\mathbf{Coh}}_{\alpha,\tau l} \arrow[l, "\mathsf{q}^{\mathrm{Coh}}"'] \arrow[r, "\mathsf{p}^{\mathrm{Coh}}"] & \mathbf{Coh}_{\alpha+\tau l}
\end{tikzcd} \tag{3.2.2}$$

where $\mathcal{Y}$ is the fiber product $(\mathbf{Bun}_\alpha \times \mathbf{Coh}_{\tau l}) \underset{\mathbf{Coh}_\alpha \times \mathbf{Coh}_{\tau l}}{\times} \widetilde{\mathbf{Coh}}_{\alpha,\tau l}$.

Lemma 3.23. *We have $\mathcal{Y} \cong \mathbf{Coh}^{\mathrm{tor.deg}=l}$, and the right square in the above diagram commutes.*

We send $\mathbf{P} \boxtimes \mathbf{T}$ to

$$(\mathbf{P} \boxtimes \mathbf{T})^{\mathrm{Coh}} := \mathsf{Eis}^{\mathrm{Coh}}\left((\jmath_\alpha \times \mathrm{Id})_{!*}(\mathbf{P} \boxtimes \mathbf{T})\right) \in \mathsf{Shv}(\mathbf{Bun}_{\alpha+\tau l})$$

and we would like to see this is a simple perverse sheaf on $\mathbf{Coh}_{\alpha+\tau l}$. As $\mathsf{Eis}^{\mathrm{Coh}}$ preserves perversity and we take the intermediate extension along open immersion $\mathbf{Bun}_\alpha \hookrightarrow \mathbf{Coh}_\alpha$, we can see immediately that $(\mathbf{P}\boxtimes\mathbf{T})^{\mathrm{Coh}}$ is a perverse sheaf. Note that $\mathcal{Y}\to\mathbf{Bun}_\alpha\times\mathbf{Coh}_{\tau l}$ is a unipotent gerbe, then by smooth and proper base change, we have

$$(\mathbf{P}\boxtimes\mathbf{T})^{\mathrm{Coh}}\cong\left(\jmath^{\leqslant l}_{\alpha+\tau l}\right)_{!*}(\iota^l)_!(\mathbf{P}\boxtimes\mathbf{T})\,[\mathbf{shift}]$$

By smoothness and the fact that $\mathbf{Coh}^{\mathrm{tor.deg}\leqslant l}_{\alpha+\tau l}$ is open in $\mathbf{Coh}_{\alpha+\tau l}$, it suffices to check that $(\iota^l)_!(\mathbf{P}\boxtimes\mathbf{T})$ is irreducible. But ι^l is proper, by decomposition theorem, it suffices to check the irreducibility of $\mathbf{P}\boxtimes\mathbf{T}$, which is automatic. We then verify that $(\mathbf{P}\boxtimes\mathbf{T})^{\mathrm{Coh}}$ is in principal series, i.e. a component of some $\mathsf{Eis}^{\mathrm{Coh}}\left(\mathbf{1}_{\alpha_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\alpha_k}\right)$. By assumption, $\mathbf{P}\in\mathsf{P}^{\mathrm{Bun}}_r$, then there exists $\underline{\alpha'}=\left(\alpha'_1,\ldots,\alpha'_m\right)$ such that $\mathbf{P}\subseteq\overline{\mathsf{Eis}}\left(\mathbf{1}_{\alpha'_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\alpha'_m}\right)$. By compatibility between $\overline{\mathsf{Eis}}$ and $\mathsf{Eis}^{\mathrm{Coh}}$, we have

$$\begin{aligned}\mathsf{Eis}^{\mathrm{Coh}}_{m,\tau l}\left((\jmath_m)_{!*}\left(\overline{\mathsf{Eis}}(\mathbf{1}_{\alpha'_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\alpha'_m})\right)\boxtimes\mathbf{T}\right)&=\mathsf{Eis}^{\mathrm{Coh}}_{m,\tau l}\left(\mathsf{Eis}^{\mathrm{Coh}}\left((\jmath_{\alpha_1})_{!*}(\mathbf{1}_{\alpha'_1})\boxtimes\ldots\boxtimes(\jmath_{\alpha_1})_{!*}(\mathbf{1}_{\alpha'_m})\right)\boxtimes\mathbf{T}\right)\\&=\mathsf{Eis}^{\mathrm{Coh}}\left(\mathbf{1}_{\alpha'_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\alpha'_m}\boxtimes\mathbf{T}\right)\end{aligned}$$

We can see $(\mathbf{P}\boxtimes\mathbf{T})^{\mathrm{Coh}}\in\mathsf{P}^{\mathrm{Coh}}_\alpha$ by definition. Conversely, let $\mathbf{P}\in\mathsf{P}^{\mathrm{Coh}}_r$, then there exists a minimal integer $d\gg 0$ such that $\mathrm{Supp}(\mathbf{P})\bigcap\mathbf{Coh}^{\mathrm{tor.deg}\leqslant d}\neq\emptyset$. We claim that there exists $\widetilde{\mathbf{P}}$ such that

$$(\iota^d)_!(\widetilde{\mathbf{P}})\cong(\jmath^{\leqslant d}_\alpha)^*(\mathbf{P})\in\mathsf{Perv}\left(\mathbf{Coh}^{\mathrm{tor.deg}\leqslant d}\right)$$

In fact, since $\mathbf{P}$ is simple, its restriction to open locus $(\jmath^{\leqslant d}_\alpha)^*(\mathbf{P})$ is also simple, so it must be in the form of $\mathbf{IC}_Z(\mathcal{L})$, where $Z\subseteq\mathbf{Coh}^{\mathrm{tor.deg}\leqslant d}$ locally closed. There exists a minimal $0\leqslant d'\leqslant d$ such that $Z\bigcap\mathbf{Coh}^{\mathrm{tor.deg}=d'}\neq\emptyset$. If $d'<d$, then it contradicts to our assumption on d, this forces $d'=d$, but this means Z is included in $\mathbf{Coh}^{\mathrm{tor.deg}=d}$, then we find the $\widetilde{\mathbf{P}}$ we want, which is simple. Also use the fact that $\pi^{\mathrm{tor.deg}=d}:\mathbf{Coh}^{\mathrm{tor.deg}=d}_\alpha\longrightarrow\mathbf{Bun}_{\alpha-\tau d}\times\mathbf{Coh}_{\tau d}$ is a unipotent gerbe, we can deduce that $(\pi^{\mathrm{tor.deg}=d})_![\mathbf{shift}]$ induces an equivalence $\mathsf{Perv}\left(\mathbf{Coh}^{\mathrm{tor.deg}=d}_\alpha\right)\cong\mathsf{Perv}\left(\mathbf{Bun}_{\alpha-\tau d}\right)\times\mathsf{Perv}\left(\mathbf{Coh}_{\tau d}\right)$, so we obtain from $\widetilde{\mathbf{P}}$ a pair of simple perverse sheaves $(\mathbf{P}',\mathbf{T})\in\mathsf{Perv}\left(\mathbf{Bun}_{\alpha-\tau d}\right)\times\mathsf{Perv}\left(\mathbf{Coh}_{\tau d}\right)$. Moreover, since $\mathbf{P}\cong(\jmath^{\leqslant d}_\alpha)_{!*}(\jmath^{\leqslant d}_\alpha)^*(\mathbf{P})$, we have $(\mathbf{P}'\boxtimes\mathbf{T})^{\mathrm{Coh}}\cong\mathbf{P}$. It remains to show that $\mathbf{P}'$ and $\mathbf{T}$ are in principal series:

Lemma 3.24. *For any $\mathbf{P}_1\boxtimes\mathbf{T}_1\in\mathbf{Coh}_\beta\times\mathbf{Coh}_\gamma$, we have $\mathbf{P}_1\boxtimes\mathbf{T}_1\subseteq\mathsf{CT}^{\mathrm{Coh}}_{\beta,\gamma}\circ\mathsf{Eis}^{\mathrm{Coh}}_{\beta,\gamma}(\mathbf{P}_1\boxtimes\mathbf{T}_1)$.*

Proof. This is an easy consequence of Langlands formula, cf. Theorem 2.13. ∎

Apply to $(\jmath_{\alpha-\tau d})_{!*}(\mathbf{P}')\boxtimes\mathbf{T}$, we only need to show $\mathsf{CT}^{\mathrm{Coh}}_{\alpha-\tau d,\tau d}(\mathbf{P})$ is in principal series. By our assumption, $\mathbf{P}$ is in principal series, thus $\mathsf{CT}^{\mathrm{Coh}}_{\alpha-\tau d,\tau d}(\mathbf{P})\subseteq\mathsf{CT}^{\mathrm{Coh}}_{\alpha-\tau d,\tau d}\left(\mathsf{Eis}^{\mathrm{Coh}}\left(\mathbf{1}_{\alpha_1}\boxtimes\ldots\boxtimes\mathbf{1}_{\alpha_r}\right)\right)$, we can conclude the proof of Theorem 3.22 with the above Lemma. ∎

3.3 Trace maps and canonical bases associated with curves

In this section, we are going to relate algebras $\mathbf{H}^{\mathrm{sph}}_{X_0}$ (respectively $\widehat{\mathbf{H}}^{\mathrm{sph,Bun}}_{X_0}$, $\widehat{\mathbf{H}}^{\mathrm{sph,Tor}}_{X_0}$) and the Grothendieck group of categories $\mathsf{Q}^{\mathrm{Coh}}$ (respectively $\mathsf{Q}^{\mathrm{Bun}}$, $\mathsf{Q}^{\mathrm{Tor}}$) via trace map and we will state the construction of canonical bases of the spherical Hall algebra of X_0, in the spirit of Lusztig.

Throughout the whole section, unless otherwise specified, we assume X_0 a smooth projective curve over $\mathbb{F}_q$ and we write $X = X_0 \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$.

3.3.1 Trace of Lusztig sheaves. As before, we write ${}^0\mathbf{Coh}_\alpha$ the stack of coherent sheaves of class α on X_0 and $\mathbf{Coh}_\alpha$ is the notation for the stack of coherent sheaves on X, then ${}^0\mathbf{Coh}_\alpha \times_{\mathrm{Spec}(\mathbb{F}_q)} \mathrm{Spec}(\overline{\mathbb{F}}_q) \cong \mathbf{Coh}_\alpha$. We can consider the geometric Frobenius action on $\mathbf{Coh}_\alpha$. In addition, over curve X_0, for any $\mathcal{L} \in \mathrm{Pic}(X_0)$, there is a representable Quot scheme ${}^0\,\mathrm{Quot}_\alpha^{\mathcal{L}}$ and its open ${}^0\mathbf{Q}_\alpha^{\mathcal{L}}$ whose underlying functors are similarly defined as $\mathrm{Quot}_\alpha^{\mathcal{L}}$ and $\mathbf{Q}_\alpha^{\mathcal{L}}$, so that we can see the quasi-compact open covers $\mathbf{Coh}_\alpha^{\mathcal{L}}$ are defined over $\mathbb{F}_q$, namely we have $\mathcal{Q}_\alpha^{\mathcal{L}} \cong {}^0\mathcal{Q}_\alpha^{\mathcal{L}} \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$.
Over each quasi-compact open substack $\mathbf{Coh}_\alpha^{>\mathcal{L}} \cong \left[\mathcal{Q}_\alpha^{\mathcal{L}} / \mathrm{G}_\alpha^{\mathcal{L}}\right]$, for $\mathbf{F} \in \mathsf{Shv}_{\mathrm{constr}}\left[\mathcal{Q}_\alpha^{\mathcal{L}} / \mathrm{G}_\alpha^{\mathcal{L}}\right]$, by faisceaux-fonctions correspondence, its trace of Frobenius is defined to be a function over $\left[{}^0\mathcal{Q}_\alpha^{\mathcal{L}} / {}^0\mathrm{G}_\alpha^{\mathcal{L}}\right](\mathbb{F}_q)$:

$$\mathrm{Tr}\,(\mathrm{Frob}, \mathbf{F}) : x \in \left[{}^0\mathcal{Q}_\alpha^{\mathcal{L}} / {}^0\mathrm{G}_\alpha^{\mathcal{L}}\right](\mathbb{F}_q) \mapsto \mathrm{Tr}\,(\mathrm{Frob}_x, H^\bullet(\mathbf{F}))$$

More concretely, assmue the point x is represented by $\mathcal{F} \in \mathsf{Coh}(X_0)$, and the value of $\mathrm{Tr}\,(\mathrm{Frob}, \mathbf{F})$ at x is

$$-q^{\frac{1}{2}\dim(\mathrm{G}_\alpha^{\mathcal{L}})} \sum_i (-1)^i \,\mathrm{Tr}(\mathrm{Frob}, H^i(\mathbf{F}_{\mathcal{F}}))$$

We abbreviate $\mathrm{Tr}\,(\mathrm{Frob}, \mathbf{F})$ as $f_{\mathbf{F}}$. Now for $\mathbf{F} \in \mathsf{Shv}_{\mathrm{constr}}\,(\mathbf{Coh}_\alpha)$ which is presented as compatible $\mathbf{F}^{\mathcal{L}} \in \mathsf{Shv}_{\mathrm{constr}}\left(\mathbf{Coh}_\alpha^{>\mathcal{L}}\right)$, we have for $\mathcal{L}' > \mathcal{L}$, $f_{\mathbf{F}^{\mathcal{L}}}|_{\left[{}^0\mathcal{Q}_\alpha^{\mathcal{L}'} / {}^0\mathrm{G}_\alpha^{\mathcal{L}'}\right](\mathbb{F}_q)} = f_{\mathbf{F}^{\mathcal{L}'}} \in \mathrm{Funct}\left(\left[{}^0\mathcal{Q}_\alpha^{\mathcal{L}} / {}^0\mathrm{G}_\alpha^{\mathcal{L}}\right](\mathbb{F}_q), \overline{\mathbb{Q}_\ell}\right)$. We take the limit of $f_{\mathbf{F}^{\mathcal{L}}}$ to be the trace of Frobenius of $\mathbf{F}$, which is in $\mathrm{Funct}\left({}^0\mathbf{Coh}_\alpha(\mathbb{F}_q), \overline{\mathbb{Q}_\ell}\right)$, it lives in the h-adic formal completion $\widehat{\mathbf{H}}_{X_0}$ of the Hall algebra $\mathbf{H}_{X_0}$ (see section 2.1). Since $\mathsf{Eis}_{\underline{\alpha}}^{\mathrm{Coh}}\left(\mathbf{1}_{\underline{\alpha}}\right) \in \mathsf{Shv}_{\mathrm{constr}}\,(\mathbf{Coh}_\alpha)$, thus for any $\mathbf{P} \in \mathsf{P}_\alpha^{\mathrm{Coh}}$, it lies automatically in $\mathsf{Shv}_{\mathrm{constr}}\,(\mathbf{Coh}_\alpha)$ and we have:

$$\mathrm{Tr}\,(\mathrm{Frob}, -) : \mathsf{Obj}(\mathsf{Q}_{\alpha,\oplus}^{\mathrm{Coh}}) \longrightarrow \widehat{\mathbf{H}}_{X_0}[\alpha] \qquad \mathbf{P} \mapsto f_{\mathbf{P}}$$

Also, since $\mathsf{Q}_{\alpha,\oplus}^{\mathrm{Coh}} \subseteq \mathsf{Shv}_{\mathrm{constr}}\,(\mathbf{Coh}_\alpha)$, its Karoubi completion

$$\mathsf{Q}_\alpha^{\mathrm{Coh}} = \left(\mathsf{Ind}\left(\mathsf{Q}_{\alpha,\oplus}^{\mathrm{Coh}}\right)\right)^c \subseteq \left(\mathsf{Shv}\,(\mathbf{Coh}_\alpha)^{\mathrm{ren}}\right)^c = \mathsf{Ind}\,(\mathsf{Shv}_{\mathrm{constr}}\,(\mathbf{Coh}_\alpha))^c = \mathsf{Shv}_{\mathrm{constr}}\,(\mathbf{Coh}_\alpha)$$

We can see that the above trace map extends to $\mathrm{Tr}\,(\mathrm{Frob}, -) : \mathsf{Obj}(\mathsf{Q}_\alpha^{\mathrm{Coh}}) \longrightarrow \widehat{\mathbf{H}}_{X_0}[\alpha]$. Since the homotopy category of $\mathsf{Q}_\alpha^{\mathrm{Coh}}$ is also Karoubi complete, we can take the Grothendieck group over it. Let us denote $K_0\left(\mathsf{Q}_\alpha^{\mathrm{Coh}}\right) = K_0\left(\mathsf{Ho}(\mathsf{Q}_\alpha^{\mathrm{Coh}})\right)$, then trace map of Frobenius can be upgraded into algebra morphism

$$\mathrm{Tr}\,(\mathrm{Frob}, -) : K_0\left(\mathsf{Q}_\alpha^{\mathrm{Coh}}\right) \longrightarrow \widehat{\mathbf{H}}_{X_0}[\alpha]$$

We define the following subspace of $\widehat{\mathbf{H}}_{X_0}$:

$$\hat{\mathfrak{U}}_X^+ = \bigoplus_{\alpha \in \mathbf{Z}^+} \mathrm{Tr}\left(\mathrm{Frob}, K_0\left(\mathsf{Q}_\alpha^{\mathrm{Coh}}\right)\right) \subseteq \widehat{\mathbf{H}}_{X_0}$$

Proposition 3.25. *The trace of Frobenius on* $\mathsf{Q}^{\mathrm{Coh}}$ *commutes with* $\mathsf{Eis}^{\mathrm{Coh}}$ *and* $\mathsf{CT}^{\mathrm{Coh}}$.

Proof. For $\mathsf{Eis}^{\mathrm{Coh}}$, this is because trace of Frobenius is compatible with smooth pullback and proper pushforward, for $\mathsf{CT}^{\mathrm{Coh}}$, we can apply Proposition 3.9. ∎

As a corollary, the functors $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$ endows $\hat{\mathfrak{U}}_X^+$ with structures of multiplication and comultiplication respectively. We use the same notation $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$ for operations in $\hat{\mathfrak{U}}^+_X$. Moreover, the pairing on $\mathsf{Shv}(\mathbf{Coh}_\alpha)$ (see section 2.3) restricts to $\mathsf{Q}^{\mathrm{Coh}}_\alpha$ and the pseudo-identity is just the Verdier duality on $\mathsf{Q}^{\mathrm{Coh}}$. Thus, they endow $\hat{\mathfrak{U}}^+_X$ a non-degenerate pairing such that $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$ are adjoint operators.

Remark 3.26. There is a notion of categorical trace, for dualizable C with endo-functor F, the trace $\mathrm{Tr}(\mathsf{F},\mathsf{C})$ is defined to be the composition $\mathbf{1} \xrightarrow{\text{counit}} \mathsf{C}\otimes\mathsf{C}^\vee \xrightarrow{F\otimes \mathrm{Id}_{\mathsf{C}^\vee}} \mathsf{C}\otimes\mathsf{C}^\vee \xrightarrow{\text{unit}} \mathbf{1}$, one may wonder of $\hat{\mathfrak{U}}^+_X$ can be recovered from categorical trace of Frobenius over $\mathsf{Q}^{\mathrm{Coh}}$. In fact, this can be done with the *local term* map: for stack $\mathcal{X}$ with correspondence $\mathcal{X} \xleftarrow{c_1} C \xrightarrow{c_2} \mathcal{X}$, we get an endo-functor $[\mathbf{c}] := (c_1)_*(c_2)^!$: $\mathsf{Shv}(\mathcal{X})^{\mathrm{ren}} \to \mathsf{Shv}(\mathcal{X})^{\mathrm{ren}}$. Let $\mathcal{X}^{\mathbf{c}\text{-fix}}$ be the fiber product $\mathcal{X}\times_{\mathcal{X}\times\mathcal{X}} C$ with respect to diagonal embedding and $C \xrightarrow{c_1\times c_2} \mathcal{X}\times\mathcal{X}$. In [GV25], they constructed $\mathrm{LT}^{\mathrm{true}}_{[\mathbf{c}]} : \mathrm{Tr}\left([\mathbf{c}], \mathsf{Shv}(\mathcal{X})^{\mathrm{ren}}\right) \longrightarrow \Gamma\left(\mathcal{X}^{\mathbf{c}\text{-fix}}, \omega_{\mathcal{X}^{\mathbf{c}\text{-fix}}}\right)$. In particular, taking the correspondence to be $(c_1, c_2) = (\mathrm{Id}, \mathrm{Frob})$ gives rises to

$$\mathrm{LT}^{\mathrm{true}}_{\mathrm{Frob}} : \mathrm{Tr}\left(\mathrm{Frob}, \mathsf{Shv}(\mathcal{X})^{\mathrm{ren}}\right) \longrightarrow \Gamma\left(\mathcal{X}^{\mathrm{Frob}}, \omega_{\mathcal{X}^{\mathrm{Frob}}}\right).$$

We can take $\mathcal{X} = \mathbf{Coh}_\alpha$ and restrict to $\mathsf{Q}^{\mathrm{Coh}}$, we can show

Proposition 3.27. *The image of the following composition identifies with* $\hat{\mathfrak{U}}^+_X$*:*

$$\mathrm{Tr}\left(\mathrm{Frob}, \mathsf{Q}^{\mathrm{Coh}}\right) \hookrightarrow \mathrm{Tr}\left(\mathrm{Frob}, \mathsf{Shv}(\mathbf{Coh}_\alpha)^{\mathrm{ren}}\right) \xrightarrow{\mathrm{LT}^{\mathrm{true}}_{\mathrm{Frob}}} \Gamma\left({}^0\mathbf{Coh}_\alpha(\mathbb{F}_q), \omega_{{}^0\mathbf{Coh}_\alpha}\right)$$

3.3.2 The Verdier duality and purity. The Verdier duality induces $\mathsf{Shv}_{\mathrm{constr}}(\mathbf{Coh}_\alpha)^\vee \longrightarrow \mathsf{Shv}_{\mathrm{constr}}(\mathbf{Coh}_\alpha)$, it induces the duality on $\mathsf{Q}^{\mathrm{Coh}}_\alpha$. Since Verdier duality commutes with smooth pullback and proper pushforward, $\mathbb{D}\left(\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}(\mathbf{1}_{\underline{\alpha}})\right) \cong \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbb{D}(\mathbf{1}_{\underline{\alpha}})\right)$, the constant sheaf $\mathbf{1}_{\underline{\alpha}}$ is Verdier self-dual, we can deduce the Eisenstein sheaf $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}(\mathbf{1}_{\underline{\alpha}})$ is also Verdier self-dual. This implies that the Verdier duality induces a permutation on $\mathsf{P}^{\mathrm{Coh}}_\alpha$. We formulate the following conjecture, which is proven in [Sch06] for $g_X \leqslant 1$.

Conjecture 3.28. *For arbitrary genus, every simple component appears in* $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}(\mathbf{1}_{\underline{\alpha}})$ *is Verdier self-dual.*

For $\mathrm{rk}(\alpha) = 0$, the above Conjeture is a consequence of Proposition 3.11. Thus, for $\alpha \in \mathbf{Z}^{++}$, above Conjecture is equivalent to: every elements $\mathsf{P}^{\mathrm{Bun}}_\alpha$ is self-dual. In fact, recall for $\mathbf{P} \in \mathsf{P}^{\mathrm{Coh}}$, we can associate it with $(\mathbf{P}^{\mathrm{Bun}}, \mathbf{T}) \in \mathsf{P}^{\mathrm{Bun}} \times \mathsf{P}^{\mathrm{Tor}}$ and in the proof of Theorem 3.22, we showed that $\mathbf{P} \cong \left(\mathbf{P}^{\mathrm{Bun}} \boxtimes \mathbf{T}\right)^{\mathrm{Coh}}$. Thus if $\mathbf{P}^{\mathrm{Bun}}$ is self-dual, we can calculate the Verdier dual of $\mathbf{P}$:

$$\begin{aligned}\mathbb{D}\left(\left(\mathbf{P}^{\mathrm{Bun}}\boxtimes\mathbf{T}\right)^{\mathrm{Coh}}\right) &= \mathbb{D}\left(\mathsf{Eis}^{\mathrm{Coh}}\left((\jmath_\alpha\times\mathrm{Id})_{!*}\left(\mathbf{P}^{\mathrm{Bun}}\boxtimes\mathbf{T}\right)\right)\right)\\ &= \mathsf{Eis}^{\mathrm{Coh}}\left(\mathbb{D}\left((\jmath_\alpha\times\mathrm{Id})_{!*}\left(\mathbf{P}^{\mathrm{Bun}}\boxtimes\mathbf{T}\right)\right)\right)\\ &= \mathsf{Eis}^{\mathrm{Coh}}\left((\jmath_\alpha\times\mathrm{Id})_{!*}\left(\mathbb{D}\left(\mathbf{P}^{\mathrm{Bun}}\right)\boxtimes\mathbb{D}(\mathbf{T})\right)\right) = \left(\mathbf{P}^{\mathrm{Bun}}\boxtimes\mathbf{T}\right)^{\mathrm{Coh}}\end{aligned}$$

Let us consider also the purity of Lusztig perverse sheaves. Any object $\mathbf{P}$ in $\mathsf{Shv}_{\mathrm{constr}}(\mathbf{Coh}_\alpha)$ is called a Weil sheaf if there is an isomorphism $\mathbf{P} \xrightarrow{\cong} (\mathrm{Frob})^*(\mathbf{P})$, it is further pointwise pure of weight zero if for every point $x \in \mathbf{Coh}_\alpha$, any eigenvalue of Frobenius on the stalks $H^i(\mathbf{P})|_x$ has complex norm $q^{\frac{i}{2}}$ (for fixed isomorphism $\overline{\mathbb{Q}_\ell} \cong \mathbb{C}$).

Proposition 3.29. *[Sch06], [Sch12] When genus of X is less or equal than 1, any Eisenstein sheaf $\mathsf{Eis}_{\underline{\alpha}}^{\mathrm{Coh}}(\mathbf{1}_{\underline{\alpha}})$ is pointwise pure of weight zero. As a corollary, in this case, any $\mathbf{P} \in \mathsf{P}_{\alpha}^{\mathrm{Coh}}$ is a Weil sheaf that pointwise pure of weight zero.*

Proof. We refer to the proof in [Sch06]. In fact, assume $\mathsf{Eis}_{\alpha}^{\mathrm{Coh}}(\mathbf{1}_{\underline{\alpha}})$ is pointwise pure of weight zero, due the semisimplicity of $\mathsf{Eis}_{\alpha}^{\mathrm{Coh}}(\mathbf{1}_{\underline{\alpha}})$, it can be written as $\bigoplus_{\mathbf{P}\in\mathsf{P}_{\alpha}^{\mathrm{Coh}}}(\mathbb{V}_{\mathbf{P}} \oplus \mathbf{P})$, where for each $\mathbf{P}$ where the complex $\mathbb{V}_{\mathbf{P}} = \oplus_j \operatorname{Hom}\left(\mathbf{P}, {}^{p}H^{j}\left(\mathsf{Eis}_{\alpha}^{\mathrm{Coh}}(\mathbf{1}_{\underline{\alpha}})\right)\right)[j]$ is of weight zero. Moreover, we have pointwise

$$H^{k}\left(\mathsf{Eis}_{\alpha}^{\mathrm{Coh}}(\mathbf{1}_{\underline{\alpha}})\right)|_{\mathcal{F}} = \bigoplus_{\mathbf{P}} \bigoplus_{i+j=k} H^{i}(\mathbb{V}_{\mathbf{P}}) \otimes H^{j}(\mathbf{P})|_{\mathcal{F}}$$

thus we can deduce $\mathbf{P}$ is pointwise pure of weight zero from the assertion. ∎

Conjecture 3.30. *We conjecture that $\mathsf{Eis}_{\underline{\alpha}}^{\mathrm{Coh}}(\mathbf{1}_{\underline{\alpha}})$ is pure pointwise of weight zero for any genus.*

We introduce the main Theorem of this section, the proof will occupy the rest of the section.

Theorem 3.31. *The inclusion $\hat{\mathfrak{U}}_X^{+} \subseteq \widehat{\mathbf{H}}_{X_0}$ identifies $\hat{\mathfrak{U}}_X^{+}$ with the subalgebra $\widehat{\mathbf{H}}_{X_0}^{\mathrm{sph}}$.*

3.3.3 The canonical bases of spherical Hall algebras. Based on Theorem 3.31, we introduce:

Definition 3.32. We use the notation $\mathbf{B}_X := \{\mathbf{b}_{\mathbf{P}}\}_{\mathbf{P}\in\mathsf{P}^{\mathrm{Coh}}}$ for the subset in $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right)$ that labelled by simple Eisenstein sheaves in $\mathsf{P}^{\mathrm{Coh}}$, we refer $\mathbf{B}_X$ as the set of **canonical bases** associated with X.

For any degree $e \geqslant 1$, the trace of Frobenius (with e-th power) induces a map $\mathbf{B}_X \mapsto \left\{f_{\mathbf{P}}^{e}\right\}_{\mathbf{P}\in\mathsf{P}^{\mathrm{Coh}}}$, where $f_{\mathbf{P}}^{e}$ is a function in $\mathbf{H}_{X/\mathbb{F}_q^e}$: the Hall algebra associated with $X_0 \times_{\mathbb{F}_q} \mathbb{F}_{q^e}$. We have the following results of Schiffmann [Sch06] [Sch12] concerning the set $\mathbf{B}_X$ for the case genus 0 and 1.

Theorem 3.33. *(i) When $g_X = 0$ i.e. $X_0 = \mathbb{P}^1_{\mathbb{F}_q}$, the (trace of) set $\mathbf{B}_X$ gives canonical bases for $\mathbf{U}_v^{+}\left(\hat{\mathfrak{sl}}_2\right)$, $v = q^{-\frac{1}{2}}$.*
(ii) When $g_X = 1$, the set $\mathbf{B}_X$ gives an $\mathcal{R}$-basis of universal Elliptic Hall algebra $\boldsymbol{\mathcal{E}}_{\mathcal{R}}^{+}$, where $\mathcal{R} = \mathbb{C}\left[\sigma^{\pm\frac{1}{2}}, \sigma'^{\pm\frac{1}{2}}\right]$. It means for each elliptic curve E, spcializing $\boldsymbol{\mathcal{E}}_{\mathcal{R}}^{+}$ at σ, σ' the Weil numbers of E, we get $\widehat{\mathbf{H}}_{E}^{\mathrm{sph}}$.
Moreover, in above cases, the set $\mathbf{B}_X$ shares the following properties with canonical bases of quantum algebra:

- *(Convolution) The Verdier duality induces an isomorphism $\overline{\mathbf{b}_{\mathbf{P}}} := \mathbf{b}_{\mathbb{D}(\mathbf{P})} \cong \mathbf{b}_{\mathbf{P}}$, for any $\mathbf{P} \in \mathsf{P}^{\mathrm{Coh}}$.*
- *(Positivity) For $\mathbf{P}, \mathbf{P}' \in \mathsf{P}^{\mathrm{Coh}}$, we have $\mathbf{b}_{\mathbf{P}} \cdot \mathbf{b}_{\mathbf{P}'} \in \prod_{\mathbf{P}''} \mathbb{N}[v^{\pm 1}]\mathbf{b}_{\mathbf{P}''}$.*

As a corollary of Theorem 3.31, we can generalize the result of Schiffmann to curve of higher genus.

Corollary 3.34. *Let X_0 be of any genus, then for each degree $e \geqslant 1$, the set $\mathbf{B}_X$ gives rises to a basis of spherical Hall algebra $\mathbf{H}_{X/\mathbb{F}_q^e}$ by taking $\operatorname{Tr}(\mathrm{Frob}^{e}, -)$, the trace of e-th power of Frobenius. Moreover, $\mathbf{B}_X$ satisfies the same positivity as in Theorem 3.33 and the Verdier duality induces an auto-equivalence on $\mathbf{B}_X$.*

Also, we can consider the space of principal series automorphic forms $\widehat{\mathbf{H}}^{\mathrm{sph,Bun}}_{X_0} \subseteq \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$. It is shown in [NSS25] (see also [Neg23]) that when X_0 is taken to be a *generic curve*, namely when its Weil numbers are distinct, the bundle part of spherical Hall algebra $\widehat{\mathbf{H}}^{\mathrm{sph,Bun}}_{X_0}$ is isomorphic to the (small) shuffle algebra associated with quiver of g_X-loops, which can be regarded as a quantum loop algebra. Combine with Theorem 3.22, we obtain the canonical bases for quantum loop algebra $\widehat{\mathbf{H}}^{\mathrm{sph,Bun}}_{X_0}$ via compactified Eisenstein series:

Corollary 3.35. *The set* $\{\mathbf{b}_{\mathbf{P}}\}_{\mathbf{P}\in\bigsqcup_r \mathsf{P}^{\mathrm{Bun}}_r}$ *forms a canonical basis of* $\widehat{\mathbf{H}}^{\mathrm{sph,Bun}}_{X_0}$.

3.4 Proof of Theorem 3.31

By definition and Proposition 3.11, for $i = 0, 1$, we have isomorphisms

$$\hat{\mathfrak{U}}^{+}_{X}[i] := \mathrm{Tr}\left(\mathrm{Frob}, \bigoplus_{\mathrm{rk}(\alpha)=i} \mathsf{Q}^{\mathrm{Coh}}_{\alpha}\right) \cong \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}[i]$$

We would use Proposition 3.25 to translate sheaves to functions. For higher ranks, it is clear that $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}[\alpha] \subseteq \hat{\mathfrak{U}}^{+}_{X}[\alpha]$, we want to show the inclusion in the reverse way. Assume first that $\mathbf{P} \in \mathsf{P}^{\mathrm{Coh}}_{\alpha}$ is a simple object, we want to show when $\mathbf{P}$ is supported only on very unstable part, the corresponding function $f_{\mathbf{P}} \in \mathbf{H}^{\mathrm{sph}}_{X_0}[\alpha]$. The following Lemma is crucial, which gives a bound to the support for Eisenstein sheaves.

Lemma 3.36. *Let* $\underline{\beta} = (\beta_1, \ldots, \beta_m)$ *be a tuple of HN-type, then for any* $\mathcal{F} \in \mathbf{Coh}_{\underline{\beta}}$, *we have*

$$\mathsf{p}^{\mathrm{Coh}}_{\underline{\beta}}\left(\mathsf{q}^{\mathrm{Coh}}_{\underline{\beta}}\right)^{-1}(\mathcal{F}) \subseteq \bigcup_{\underline{\gamma}\leq\underline{\beta}} \mathbf{HN}_{\underline{\gamma}}$$

as a consequence, the support of Eisenstein series $\mathrm{supp}\left(\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{Q})\right) \subseteq \bigcup_{\underline{\gamma}\leq\underline{\beta}} \mathbf{HN}_{\underline{\gamma}}$ *for* $\mathbf{Q} \in \mathsf{Q}^{\mathrm{Coh}}_{\underline{\beta}}$.

By our assumption, $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{HN}_{\underline{\alpha}} \neq \emptyset$ for some very unstable $\underline{\alpha}$, we can take $\underline{\beta}$ to be the unique very unstable HN-type among those $\underline{\alpha}$ such that $\mathrm{supp}(\mathbf{P}) \subseteq \overline{\mathbf{HN}_{\underline{\gamma}}}$.

Lemma 3.37. *For the above very unstable* $\underline{\beta}$ *associated with* $\mathbf{P}$, *we have*

$$\mathbf{P}\Big|_{\mathbf{HN}_{\underline{\gamma}}} = \left(\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}} \circ \mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P})\right)\Big|_{\mathbf{HN}_{\underline{\gamma}}}$$

By induction hypothesis and the fact that $\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}} : \mathsf{Q}^{\mathrm{Coh}}_{\alpha} \to \mathsf{Q}^{\mathrm{Coh}}_{\underline{\beta}}$ preserves Lusztig sheaf categories, we can deduce $f_{\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P})}$ lies in $\widehat{\mathbf{H}}^{\mathrm{sph}}\left[\underline{\beta}\right]$, and thus

$$\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}\left(f_{\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P})}\right) = f_{\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}} \circ \mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P})} \in \widehat{\mathbf{H}}^{\mathrm{sph}}$$

On the other hand, abbreviate $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}} \circ \mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P})$ as $\mathbf{R}$, by assumption $\mathbf{P} \subseteq \mathbf{R}$ is a simple direct summand and there exists $\mathbf{P}'$ (not necessarily simple) such that $\mathbf{P} \oplus \mathbf{P}' = \mathbf{R}$ and that $\mathrm{supp}(\mathbf{P}') \subseteq \bigcup_{\underline{\beta}<\underline{\alpha}} \mathbf{HN}_{\underline{\beta}}$. Using the induction hypothesis, we have $f_{\mathbf{P}'} \in \mathbf{H}^{\mathrm{sph}}_{X_0}[\alpha]$, so we can deduce $f_{\mathbf{P}} = f_{\mathbf{R}} - f_{\mathbf{P}'} \in \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}[\alpha]$. As a consequence, the space of traces of objects in $\mathsf{Q}^{\mathrm{Coh}}_{\alpha}$ that having very unstable supports is included in $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}[\alpha]$, denote it by $\widehat{\mathbf{H}}^{\mathrm{sph,vus}}_{X_0}[\alpha]$.

Next we deal with the case where $\mathrm{supp}(\mathbf{P}) \bigcap \left(\mathbf{Coh}^{\mathsf{nvu}}_{\alpha} \cup \mathbf{Coh}^{\mathsf{as}}_{\alpha}\right) \neq \emptyset$. We impose the induction hypothesis that $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}[r] \cong \hat{\mathfrak{U}}^{+}_{X}[r]$ holds for $r < \mathrm{rk}(\alpha)$, the pieces of rank strictly smaller than α. We look at the case when $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{Coh}^{\mathsf{as}}_{\alpha} \neq \emptyset$. Using Lemma 3.36 and similar argument as above, we can find $\underline{\beta}$ not very unstable such that $\mathrm{supp}(\mathbf{P}) \subseteq \overline{\mathbf{HN}_{\underline{\beta}}}$. We assume $\underline{\beta} = (\beta_1, ..., \beta_s)$, then by assumption there exists $1 \leqslant i \leqslant s-1$ such that $\mu(\beta_{i+1}) - \mu(\beta_i) > 2g_X - 2$. Apart from i, we separate $\underline{\beta}$ into two parts $\underline{\beta^{\leqslant i}}$ and $\underline{\beta^{>i}}$ with $\beta^{\leqslant i} = \sum_{j=1}^{i} \beta_j, \beta^{>i} = \sum_{j=i+1}^{s} \beta_j$. We write $\mathsf{CT}^{\mathrm{Coh}}_{\beta^{\leqslant i}, \beta^{>i}}(\mathbf{P})$ as $\mathbf{P}_1 \boxtimes \mathbf{P}_2$, where $\mathbf{P}_1$ and $\mathbf{P}_2$ are supported in the stacks $\mathbf{Coh}_{\beta^{\leqslant i}}$ and $\mathbf{Coh}_{\beta^{>i}}$ respectively. By induction hypothesis, $f_{\mathbf{P}_i} \in \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$ for $i = 1, 2$. Note that $\mu\left(\beta^{>i}\right) - \mu\left(\beta^{\leqslant i}\right) \geqslant \mu\left(\beta_{i+1}\right) - \mu\left(\beta_i\right)$, thus the HN-type given by $\left(\beta^{\leqslant i}, \beta^{>i}\right)$ is very unstable. We can apply Lemma 3.37 to deduce that $\mathbf{P} \subseteq \mathbf{R} := \mathsf{Eis}^{\mathrm{Coh}}_{\beta^{\leqslant i}, \beta^{>i}}\left(\mathbf{P}_1 \boxtimes \mathbf{P}_2\right)$ and the HN-types of simple components in the complementary of $\mathbf{P}$ in $\mathbf{R}$ are $\prec \underline{\beta}$. For another simple component $\mathbf{P}' \subseteq \mathbf{R}$, if its HN-type is very unstable, then we already know that $f_{\mathbf{P}'} \in \mathbf{H}^{\mathrm{sph,vus}}_{X_0}$. Otherwise, we apply the previous steps for $\mathbf{P}$ to $\mathbf{P}'$, after several repeats, all the simple components created locate in very unstable locus and we obtain $f_{\mathbf{R}} - f_{\mathbf{P}} \in \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$. Since we have also $\mathsf{Eis}^{\mathrm{Coh}}\left(f_{\mathbf{P}_1} \boxtimes f_{\mathbf{P}_2}\right) \in \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$ by induction hypothesis, we can deduce that $f_{\mathbf{P}} \in \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$.

Finally, we deal with the case where $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{Coh}^{\mathsf{as}}_{\alpha} \neq \emptyset$. We denote by $\widehat{\mathbf{H}}^{\mathrm{sph,vus}}_{X_0}[\alpha] \subseteq \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}[\alpha]$ the subspace spanned by $f_{\mathbf{P}}$ such that $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{Coh}^{\mathsf{as}}_{\alpha} = \emptyset$, which includes the functions considered in the above two cases. Now we want to show

$$\hat{\mathfrak{U}}^{+}_{X}[\alpha] \Big/ \widehat{\mathbf{H}}^{\mathrm{sph,vus}}_{X_0}[\alpha] \cong \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}[\alpha] \Big/ \widehat{\mathbf{H}}^{\mathrm{sph,vus}}_{X_0}[\alpha]$$

We know $\hat{\mathfrak{U}}^{+}_{X}[\alpha] \Big/ \widehat{\mathbf{H}}^{\mathrm{sph,vus}}_{X_0}[\alpha]$ is finite dimensional and admits a non-degenerate inner product inherit from $\hat{\mathfrak{U}}^{+}_{X}[\alpha]$, it thus suffices to show for any $\bar{f} \in \hat{\mathfrak{U}}^{+}_{X}[\alpha] \Big/ \widehat{\mathbf{H}}^{\mathrm{sph,vus}}_{X_0}[\alpha]$, there exists $\bar{g} \in \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}[\alpha] \Big/ \widehat{\mathbf{H}}^{\mathrm{sph,vus}}_{X_0}[\alpha]$ such that $\langle f, g \rangle \neq 0$.

To begin with, it is true that the non-vanishing of pairing holds when $f = f_{\mathbf{P}}$ is the function corresponds some simple Eisenstein sheaf $\mathbf{P} \in \mathsf{P}^{\mathrm{Coh}}$. By assumption, there exists $\mathbf{P}'$ such that $\mathbf{P} \oplus \mathbf{P}' = \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right)$ for some principal $\underline{\alpha}$, which is not necessarily HN-type. Since $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{HN}_{\underline{\beta}} \neq \emptyset$, the constant term $\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P}) \neq 0$. On the other hand, we have Langlands formula that

$$\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}} \circ \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right) = \bigoplus_{\underline{\gamma}} \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\gamma}} \circ \mathsf{CT}^{\mathrm{Coh}}_{\underline{\gamma}}\left(\mathbf{1}_{\underline{\alpha}}\right)$$

where $\underline{\gamma}$ runs over the contingency matrices between $\underline{\alpha}$ and $\underline{\beta}$. Thus $\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P})$ contains several components in $\bigoplus_{\underline{\gamma}} \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\gamma}} \circ \mathsf{CT}^{\mathrm{Coh}}_{\underline{\gamma}}\left(\mathbf{1}_{\underline{\alpha}}\right)$ For any $g' \in \mathbf{H}^{\mathrm{sph}}_{X_0}\left[\underline{\beta}\right]$, we have

$$\left\langle f_{\mathbf{P}}, \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}(g')\right\rangle = \left\langle \mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}\left(f_{\mathbf{P}}\right), g'\right\rangle = \left\langle \sum_{\underline{\gamma'}} \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\gamma'}} \circ \mathsf{CT}^{\mathrm{Coh}}_{\underline{\gamma'}}\left(\mathbf{1}_{\underline{\alpha}}\right), g'\right\rangle$$

where $\underline{\gamma'}$ appears in the sum corresponds to the component in $\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P})$. Since $\underline{\alpha}$ is principal, its subpartition $\underline{\gamma}$ is again principal. We know $\mathsf{CT}^{\mathrm{Coh}}_{\underline{\gamma}} : \mathbf{H}^{\mathrm{sph}}_{X_0}\left[\underline{\alpha}\right] \longrightarrow \mathbf{H}^{\mathrm{sph}}_{X_0}\left[\underline{\gamma}\right]$ preserves spherical part, thus the left side $\sum_{\underline{\gamma'}} \mathsf{Eis}^{\underline{\beta},\mathrm{Coh}}_{\underline{\gamma'}} \circ \mathsf{CT}^{\underline{\gamma'},\mathrm{Coh}}_{\underline{\alpha}}(\mathbf{1}_{\underline{\alpha}})$ in the pairing appears in $\mathbf{H}^{\mathrm{sph}}_{X_0}\left[\underline{\beta}\right]$, we can take g' to be exactly the same function, this will give a non-vanishing pairing $\left\langle f_{\mathbf{P}}, \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}(g')\right\rangle$ and clearly $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}(g')$ is the function in $\mathbf{H}^{\mathrm{sph}}_{X_0}[\alpha]$ we want.

To generalize the case of simple Eisenstein sheaves, for any $\mathbf{P} \in \mathsf{Q}^{\mathrm{Coh}}_{\alpha}$ such that $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{HN}_{\underline{\beta}} \neq \emptyset$ for some almost semi-stable HN-type $\underline{\beta}$, we want to construct g such that $\langle f_{\mathbf{P}}, g\rangle \neq 0$, it suffices to have:

Proposition 3.38. *The restriction of $\mathsf{Q}^{\mathrm{Coh}}_{\alpha}$ to the intersection $\bigcap_{\underline{\beta}} \mathrm{Ker}\left(\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}\right)$ of all principal series partition is zero, in other words, $\mathsf{Q}^{\mathrm{Coh}}_{\alpha}$ has trivial intersection with*

$$\mathsf{Shv}\left(\mathbf{Coh}_{\alpha}\right)^{\mathrm{Bun}}_{\mathrm{cusp}} := \mathsf{Shv}\left(\mathbf{Coh}_{\alpha}\right) \otimes_{\mathsf{Shv}\left(\mathbf{Bun}_{\mathrm{rk}(\alpha)}\right)} \mathsf{Shv}\left(\mathbf{Bun}_{\mathrm{rk}(\alpha)}\right)_{\mathrm{cusp}}$$

Admitting the Proposition 3.38 and use Proposition 3.25, we can deduce that for any $\mathbf{P} \in \mathsf{Q}^{\mathrm{Coh}}_{\alpha}$, there exists some principal $\underline{\beta}$ such that $\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}\left(f_{\mathbf{P}}\right) = f_{\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}(\mathbf{P})} \neq 0$. And also there exists $g_0 \in \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}\left[\underline{\beta}\right]$ such that the pairing $\left\langle g_0, \mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}\left(f_{\mathbf{P}}\right)\right\rangle \neq 0$. Take $g = \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}\left(g_0\right)$, we have

$$\langle f_{\mathbf{P}}, g\rangle = \left\langle f_{\mathbf{P}}, \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}\left(g_0\right)\right\rangle = \left\langle \mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}\left(f_{\mathbf{P}}\right), g_0\right\rangle \neq 0$$

By induction hypothesis, the function $\mathsf{CT}^{\mathrm{Coh}}_{\underline{\beta}}\left(f_{\mathbf{P}}\right) \in \widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}\left[\underline{\beta}\right]$ and we know $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$ is stable under $\mathsf{Eis}^{\mathrm{Coh}}$, thus g is also a function in $\widehat{\mathbf{H}}^{\mathrm{sph}}_{X_0}$, this encloses the proof.

3.5 Proof of Proposition 3.38

So it remains to show Proposition 3.38. Since $\mathsf{Q}^{\mathrm{Coh}}$ is Karoubi generated by $\mathsf{P}^{\mathrm{Coh}}$, which according to Theorem 3.22, can be decomposed into $\mathsf{P}^{\mathrm{Bun}}$ and $\mathsf{P}^{\mathrm{Tor}}$. We only need to show $\mathsf{CT}^{\mathrm{Coh}}$ is conservative on $\overline{\mathsf{Eis}}$-generated part and since those $\overline{\mathsf{Eis}}$-generated sheaves are supported on **Bun**, $\mathsf{CT}^{\mathrm{Coh}}$ reduces to $\mathsf{CT}^{\mathrm{Bun}}$. In our case, we can further reduce to the conservativity of $\mathsf{CT}^{\mathrm{Bun}}$ on $\mathsf{Eis}_!$-generated sheaves. We explain how to apply restricted geometric Langlands into this case.

Let $\mathbf{LocSys}^{\mathrm{restr}}_{\check{G}}$ be the prestack of $\check{G}$-local system over X, defined in [Ari+22b]. For scheme S,

$$\mathbf{LocSys}^{\mathrm{restr}}_{\check{G}}(S) = \left\langle \text{right t-exact symmetric monoidal functors } \mathrm{Rep}(\check{G}) \to \mathsf{QCoh}(S) \otimes \mathsf{qLisse}(X)\right\rangle$$

where $\mathsf{qLisse}(X)$ is the category of quasi-Lisse sheaves on X, namely whose objects have cohomologies being filtered colimit of lisse sheaves. The restricted version of geometric Langlands predicts there is an equivalence of categories

$$\mathbb{L}^{\mathrm{restr}}_{G} : \mathsf{Shv}_{\mathsf{Nilp}}\left(\mathbf{Bun}_{G}\right) \xrightarrow{\cong} \mathsf{IndCoh}_{\mathsf{Nilp}}\left(\mathbf{LocSys}^{\mathrm{restr}}_{\check{G}}\right)$$

where on the left hand side $\mathsf{Shv}_{\mathsf{Nilp}}\left(\mathbf{Bun}_{G}\right)$ is subcategory of $\mathsf{Shv}\left(\mathbf{Bun}_{G}\right)$ whose objects are constructible sheaves with singular support contained in global nilpotent cone $\mathsf{Nilp} = \mathbf{Maps}(X, \frac{\mathcal{N}}{G})$. The equivalence should be compatible with Eisenstein series and constant term functors on both sides. Also, as a consequence of the equivalence, the categorical trace of two sides should match

$$\mathrm{Tr}\left(\mathrm{Frob}, \mathsf{Shv}_{\mathsf{Nilp}}\left(\mathbf{Bun}_G\right)\right) \cong \mathrm{Tr}\left(\mathrm{Frob}, \mathsf{IndCoh}_{\mathsf{Nilp}}\left(\mathbf{LocSys}_{\check{G}}^{\mathrm{restr}}\right)\right)$$

The following results have been established in a series works concern restricted Langlands:

Theorem 3.39. *(1) [GR25] The restricted geometric Langlands holds for $G = \check{G} = \mathrm{GL}_n, \forall n \geqslant 1$.*
(2) [Ari+22a] The categorical trace of Frobenius over $\mathsf{Shv}_{\mathsf{Nilp}}(\mathbf{Bun}_G)$ identifies with the space of automorphic functions $\mathrm{Funct}_c\left(\mathbf{Bun}_G(\mathbb{F}_q), \mathbb{C}\right)$. The categorical trace of Frobenius on restricted local systems is given by

$$\mathrm{Tr}\left(\mathrm{Frob}, \mathsf{IndCoh}_{\mathsf{Nilp}}\left(\mathbf{LocSys}_{\check{G}}^{\mathrm{restr}}\right)\right) \cong \mathrm{Tr}\left(\mathrm{Frob}, \mathsf{IndCoh}\left(\mathbf{LocSys}_{\check{G}}^{\mathrm{restr}}\right)\right) = \Gamma\left(\mathbf{LocSys}_{\check{G}}^{\mathrm{arith}}, \omega_{\mathbf{LocSys}_{\check{G}}^{\mathrm{arith}}}\right)$$

The Frobenius action on $\mathsf{qLisse}(X)$ gives rises to Frobenius action on the stack $\mathbf{LocSys}_{\check{G}}^{\mathrm{restr}}$, here $\mathbf{LocSys}_{\check{G}}^{\mathrm{arith}}$ is defined to be the stack of Frobenius fixed points of $\mathbf{LocSys}_{\check{G}}^{\mathrm{restr}}$, it is shown in [Ari+22b] that $\mathbf{LocSys}_{\check{G}}^{\mathrm{arith}}$ is an algebraic stack whose irreducible components are labelled by semisimple $\check{G}$-local systems. For our purpose, we concentrate the component corresponds to trivial local system:
Let $\mathbf{LocSys}_{\check{G}}^{\mathrm{restr},\circ} \subseteq \mathbf{LocSys}_{\check{G}}^{\mathrm{restr}}$ be the irreducible component that corresponds to the trivial local system, according to [Ari+22b, Proposition 3.7.2], its field-valued points are $\check{G}$-local systems whose semi-simplification is the trivial $\check{G}$-local system. On the other hand, let $\mathsf{Shv}\left(\mathbf{Bun}_n\right)_{\mathrm{ps}} \subseteq \mathsf{Shv}\left(\mathbf{Bun}_n\right)$ be the category generated by $\mathsf{Eis}_!\left(\mathbf{1}_{\underline{\alpha}}\right)$ under colimits, for $\underline{\alpha} = (\alpha_1, ..., \alpha_n)$ with $\mathrm{rk}(\alpha_i) = 1$. As a special case of Theorem 3.39, we have the following equivalence:

$$\mathsf{Shv}\left(\mathbf{Bun}_T\right)_{\mathrm{ps}} \xrightarrow{\cong} \mathsf{QCoh}\left(\mathbf{LocSys}_{\check{T}}^{\mathrm{restr},\circ}\right)$$

for torus T of any rank. In particular, $\mathsf{Shv}\left(\mathbf{Bun}_T\right)_{\mathrm{ps}} \subseteq \mathsf{qLisse}\left(\mathbf{Bun}_T\right) \cong \mathsf{Shv}_{\mathsf{Nilp}}\left(\mathbf{Bun}_T\right)$. Combining with the following Eisenstein compatibility:

$$\begin{tikzcd}
\mathsf{qLisse}\left(\mathbf{Bun}_T\right) \arrow[r, "\mathsf{Eis}_{T,!}"] \arrow[d] & \mathsf{Shv}_{\mathsf{Nilp}}\left(\mathbf{Bun}_G\right) \arrow[d] \\
\mathsf{QCoh}\left(\mathbf{LocSys}_{\check{G}}^{\mathrm{restr}}\right) \arrow[r, "\mathsf{Eis}_{\check{T}}^{\mathrm{spec}}"] & \mathsf{IndCoh}_{\mathsf{Nilp}}\left(\mathbf{LocSys}_{\check{G}}^{\mathrm{restr}}\right)
\end{tikzcd} \tag{3.5.1}$$

We can deduce an equivalence $\mathsf{Shv}\left(\mathbf{Bun}_n\right)_{\mathrm{ps}} \cong \mathsf{IndCoh}_{\mathsf{Nilp}}\left(\mathbf{LocSys}_{\check{G}}^{\mathrm{restr},\circ}\right)$, $\check{G} = \mathrm{GL}_n$, since the right hand side $\mathsf{IndCoh}_{\mathsf{Nilp}}\left(\mathbf{LocSys}_{\check{G}}^{\mathrm{restr},\circ}\right)$ is generated under colimits by image of $\mathsf{Eis}_{\check{T}}^{\mathrm{spec}}$ over $\mathsf{QCoh}\left(\mathbf{LocSys}_{\check{T}}^{\mathrm{restr},\circ}\right)$. A variation of Proposition 3.38 is the following result due to L. Lafforgue and reformed in [Ari+22b]:

Proposition 3.40. *The intersection $\mathsf{Shv}\left(\mathbf{Bun}_n\right)_{\mathrm{ps}} \bigcap \mathsf{Shv}\left(\mathbf{Bun}_n\right)_{\mathrm{cusp}}$ is trivial.*

Proof. In fact, under the restricted Langlands correspondence, we have

$$\mathbb{L}_{\mathrm{GL}_n}^{\mathrm{restr}} : \mathsf{Shv}\left(\mathbf{Bun}_n\right)_{\mathrm{cusp}} \xrightarrow{\cong} \mathsf{QCoh}\left(\mathbf{LocSys}_n^{\mathrm{restr,irred}}\right)$$

However, the intersection $\mathbf{LocSys}_n^{\mathrm{restr,irred}} \bigcap \mathbf{LocSys}_n^{\mathrm{restr},\circ}$ is trivial. ■

It remains to show that Proposition 3.40 implies Proposition 3.38. First, the singular support of constant sheaf $\mathbf{1}_{\underline{\alpha}}$ is just the zero section and the functor $\mathsf{Eis}^{\mathsf{Coh}}$ preserves nilpotent singular support. Thus we have $\mathsf{Q}_n^{\mathsf{Coh}} \subseteq \mathsf{Shv}_{\mathsf{Nilp}}(\mathbf{Coh}_n)$ and the $*$-restriction of $\mathsf{Q}_n^{\mathsf{Coh}}$ to $\mathbf{Bun}_n$ satisfies:

$$\mathsf{Q}_n^{\mathsf{Coh}} \otimes_{\mathsf{Shv}(\mathbf{Coh}_n)} \mathsf{Shv}(\mathbf{Bun}_n) \subseteq \mathsf{Shv}_{\mathsf{Nilp}}(\mathbf{Bun}_n)$$

Due to Theorem 3.22, the left hand side category is nothing but $\mathsf{Q}_n^{\mathsf{Bun}}$. We obtain following diagram:

$$\begin{tikzcd} \mathsf{Q}_1^{\mathsf{Bun}} \times \ldots \times \mathsf{Q}_1^{\mathsf{Bun}} \arrow[r, "\overline{\mathsf{Eis}}"] \arrow[d, "\cong"'] & \mathsf{Q}_n^{\mathsf{Bun}} \arrow[d, hook] \\ \mathsf{qLisse}(\mathbf{Bun}_T)_{\mathsf{ps}} \arrow[r, "\overline{\mathsf{Eis}}"] & \mathsf{Shv}_{\mathsf{Nilp}}(\mathbf{Bun}_n) \end{tikzcd} \tag{3.5.2}$$

We need to compare $\mathsf{Q}_n^{\mathsf{Bun}}$ with $\mathsf{Shv}(\mathbf{Bun}_n)_{\mathsf{ps}}$ for $n \geqslant 2$. We claim that one can replace $\mathsf{Shv}_{\mathsf{Nilp}}(\mathbf{Bun}_n)$ in the above diagram with $\mathsf{Shv}(\mathbf{Bun}_n)_{\mathsf{ps}}$. It suffices to have that the category generated under colimits by $\overline{\mathsf{Eis}}(\mathbf{1}_{\underline{\alpha}})$ is included in the category generated by $\mathsf{Eis}_!(\mathbf{1}_{\underline{\alpha}})$.

Via the open embedding $\mathbf{Bun}_B \hookrightarrow \overline{\mathbf{Bun}}_B$, we can extend the parabolic induction diagram into the following:

$$\begin{tikzcd} \mathbf{Bun}_B \arrow[r, hook, "\bar{\jmath}"] & \overline{\mathbf{Bun}}_B \arrow[dl, "\bar{\mathsf{p}}"'] \arrow[dr, "\bar{\mathsf{q}}"] & \\ \mathbf{Bun}_T & & \mathbf{Bun}_G \end{tikzcd}$$

such that $\mathsf{p}_B = \bar{\mathsf{p}} \circ \bar{\jmath}$ and $\mathsf{q}_B = \bar{\mathsf{q}} \circ \bar{\jmath}$. To compare $\mathsf{Eis}_!$ and $\overline{\mathsf{Eis}}$, we need the following result

Theorem 3.41. *The complex $\bar{\jmath}_!(\mathbf{IC}_{\mathbf{Bun}_B})$ and $\mathbf{IC}_{\widetilde{\mathbf{Bun}}_B}$ are universal locally acyclic.*

As a consequence of the above Theorem, we can write $\mathsf{Eis}_!(\mathcal{F})$ as $\bar{\mathsf{q}}_!\left((\bar{\mathsf{p}})^!(\mathcal{F}) \overset{!}{\otimes} \bar{\jmath}_!(\mathbf{IC}_{\mathbf{Bun}_B})\right)$ and $\overline{\mathsf{Eis}}(\mathcal{F})$ as $\bar{\mathsf{q}}_!\left((\bar{\mathsf{p}})^!(\mathcal{F}) \overset{!}{\otimes} \mathbf{IC}_{\widetilde{\mathbf{Bun}}_B}\right)$, there is a natural transformation $\mathsf{Eis}_! \longrightarrow \overline{\mathsf{Eis}}$ induced by $\bar{\jmath}_!(\mathbf{IC}_{\mathbf{Bun}_B}) \to \mathbf{IC}_{\widetilde{\mathbf{Bun}}_B}$. To compare $\mathsf{Eis}_!(\mathbf{1}_{\underline{\alpha}})$ and $\overline{\mathsf{Eis}}(\mathbf{1}_{\underline{\alpha}})$, we first consider a stratification of $\overline{\mathbf{Bun}}_B$:

Denote by $\check{\Lambda}_G$ the coweight lattice of G and $\check{\Lambda}_G^{\mathrm{pos}}$ the positive coweights. Let $\check{\alpha}_i$ be the coroot where $i \in \mathcal{I}_G$ corresponds to vertices of Dynkin diagram. For $\check{\lambda} \in \check{\Lambda}_G^{\mathrm{pos}}$, which can be written as $\sum_{i \in \mathcal{I}_G} n_i \check{\alpha}_i$, we can associate it with $X^{\check{\lambda}} = \prod_{i \in \mathcal{I}_G} X^{(n_i)}$. There is a locally closed embedding $\iota_{\check{\lambda}} : X^{\check{\lambda}} \times \mathbf{Bun}_B \hookrightarrow \overline{\mathbf{Bun}}_B$[BG02, Proposition 1.2.7], given by $\left(\prod_i \underline{x}_i, \mathcal{F}_G, \mathcal{F}_T, \kappa\right) \mapsto \left(\mathcal{F}_G, \mathcal{F}_T', \kappa'\right)$, let $\mathcal{V}_{\mathcal{F}_G}^{\theta}$ (resp. $\mathcal{L}_{\mathcal{F}_G}^{\theta}$) be the associated vector bundle (resp. line bundle) of character θ, then $\mathcal{F}_T', \kappa'$ in the image of embedding are

$$\mathcal{F}_T' = \mathcal{F}_T\left(\sum_i -\check{\alpha}_i \cdot \underline{x}_i\right); \quad \kappa'^{,\theta} : \mathcal{L}_{\mathcal{F}_T'}^{\theta} = \mathcal{L}_{\mathcal{F}_T}^{\theta}\left(-\sum_i \langle \check{\alpha}_i, \theta \rangle \cdot \underline{x}_i\right) \to \mathcal{L}_{\mathcal{F}_T}^{\theta} \xrightarrow{\kappa^{\theta}} \mathcal{V}_{\mathcal{F}_G}^{\theta}$$

Denote $\overline{\mathbf{Bun}}_B^{\check{\lambda}}$ the image of $\jmath_{\check{\lambda}}$, then $\overline{\mathbf{Bun}}_B$ is stratified by $\check{\Lambda}_G^{\mathrm{pos}}$ such that $\overline{\mathbf{Bun}}_B^{\leqslant \check{\lambda}} = \bigcup_{\check{\theta} \leqslant \check{\lambda}} \overline{\mathbf{Bun}}_B^{\check{\theta}}$ is an open substack. We denote $\mathrm{Ran}\left(X, \check{\Lambda}_G^{\mathrm{pos}}\right) := \mathrm{colim}_{\check{\lambda} \in \check{\Lambda}_G^{\mathrm{pos}}} X^{\check{\lambda}}$ the $\check{\Lambda}_G^{\mathrm{pos}}$-colored Ran space of X. It carries a structure of addition, given by $\mathrm{add}_{\check{\lambda}_1, \check{\lambda}_2} : X^{\check{\lambda}_1} \times X^{\check{\lambda}_2} \to X^{\check{\lambda}_1 + \check{\lambda}_2}$, whose associated $*$-pushforward endows the category $\mathsf{Shv}\left(\mathrm{Ran}\left(X, \check{\Lambda}_G^{\mathrm{pos}}\right)\right)$ a monoidal structure $\star$: for $\mathcal{F}_1, \mathcal{F}_2 \in \mathsf{Shv}\left(\mathrm{Ran}\left(X, \check{\Lambda}_G^{\mathrm{pos}}\right)\right)$ represented by $\left(\mathcal{F}_i^{\check\lambda} \in \mathsf{Shv}\left(X^{\check\lambda}\right)\right)_{\check\Lambda_G^{\mathrm{pos}}}$ respectively, we can put $\mathcal{F}_1 \star \mathcal{F}_2 = \left(\bigoplus_{\check\lambda_1+\check\lambda_2=\check\lambda} (\mathrm{add}_{\check\lambda_1,\check\lambda_2})_* \left(\mathcal{F}_1^{\check\lambda_1} \boxtimes \mathcal{F}_2^{\check\lambda_2}\right)\right)$. Then $\mathsf{Shv}\left(\mathrm{Ran}\left(X, \check\Lambda_G^{\mathrm{pos}}\right)\right)$ acts on $\mathsf{Shv}(\mathbf{Bun}_T)$ by $\mathsf{Shv}\left(\mathrm{Ran}\left(X, \check\Lambda_G^{\mathrm{pos}}\right)\right) \otimes \mathsf{Shv}(\mathbf{Bun}_T) \to \mathsf{Shv}(\mathbf{Bun}_T)$:

$$\left(\mathcal{F} = \left(\mathcal{F}^{\check\lambda}\right), \mathcal{S}\right) \mapsto \mathcal{F} \star \mathcal{S} := \bigoplus_{\check\lambda} (\mathrm{pr}_{\mathbf{Bun}_T})_* \left(\mathrm{pr}_{X^{\check\lambda}}^! (\mathcal{F}^{\check\lambda}) \overset{!}{\otimes} \mathrm{mult}_{\check\lambda}^! (\mathcal{S})\right)$$

where $\mathrm{mult}_{\check\lambda}$ is the composition $X^{\check\lambda} \times \mathbf{Bun}_T \xrightarrow{\mathrm{AJ}^{\check\lambda} \times \mathrm{Id}} \mathbf{Bun}_T \times \mathbf{Bun}_T \xrightarrow{\mathrm{mult}} \mathbf{Bun}_T$ and $\mathrm{pr}_{X^{\check\lambda}}, \mathrm{pr}_{\mathbf{Bun}_T}$ are projections from $X^{\check\lambda} \times \mathbf{Bun}_T$. And the action on $\mathsf{Shv}\left(\overline{\mathbf{Bun}}_B\right)$ is given by:

$$\left(\mathcal{F} = \left(\mathcal{F}^{\check\lambda}\right), \mathcal{S}\right) \mapsto \mathcal{F} \star \mathcal{S} := \bigoplus_{\check\lambda} \bar\iota_*^{\check\lambda} \left(\mathcal{F}^{\check\lambda} \boxtimes \mathcal{S}\right)$$

where $\bar\iota^{\check\lambda}$ is the map $X^{\check\lambda} \times \overline{\mathbf{Bun}}_B \to \overline{\mathbf{Bun}}_B$ such that its composition with the inclusion $X^{\check\lambda} \times \mathbf{Bun}_B \hookrightarrow X^{\check\lambda} \times \overline{\mathbf{Bun}}_B$ is the strata $\iota^{\check\lambda} : X^{\check\lambda} \times \mathbf{Bun}_B \hookrightarrow \overline{\mathbf{Bun}}_B$.

We consider a commutative algebra object in $\mathsf{Shv}\left(\mathrm{Ran}\left(X, \check\Lambda_G^{\mathrm{pos}}\right)\right)$, constructed in [Gai]: consider $\mathsf{Shv}(X)^{\check\Lambda^{\mathrm{pos}}}$ the category of $\check\Lambda_G^{\mathrm{pos}}$-graded objects in $\mathsf{Shv}(X)$, the collection of $\Delta_*^{\check\lambda}$ along $\Delta^{\check\lambda} : X \mapsto X^{\check\lambda}$ gives a functor $\Delta_*^{\mathrm{Ran}} : \mathsf{Shv}(X)^{\check\Lambda^{\mathrm{pos}}} \to \mathsf{Shv}\left(\mathrm{Ran}\left(X, \check\Lambda_G^{\mathrm{pos}}\right)\right)$, which sends Lie co-algebra objects to Lie co-algebra objects (with respect to $\star$). Denote by $\check{\mathfrak{n}}_X^- = \check{\mathfrak{n}}^- \otimes \overline{\mathbb{Q}_\ell}_X$ be $\check\Lambda^{\mathrm{neg}}$-graded Lie $\star$-algebra in $\mathsf{Shv}(X)$, where $\check{\mathfrak{n}}^-$ is the Lie algebra of unipotent of dual group. The Verdier dual $\mathbb{D}\left(\check{\mathfrak{n}}_X^-\right) \cong (\check{\mathfrak{n}}^-)^\vee \otimes \overline{\mathbb{Q}_\ell}[2]$ is a Lie co-algebra. Take $\Omega\left(\check{\mathfrak{n}}_X^-\right)$ to be the associated Chevalley complex (or Koszul dual) of $\Delta_*^{\mathrm{Ran}}\left(\mathbb{D}\left(\check{\mathfrak{n}}_X^-\right)\right)$, this is a commutative factorization algebra in $\mathsf{Shv}\left(\mathrm{Ran}\left(X, \check\Lambda_G^{\mathrm{pos}}\right)\right)$. According to previous discussion on the strata, the sheaf $\bar{j}_!\left(\mathbf{IC}_{\mathbf{Bun}_B}\right)$ is naturally $\Omega\left(\check{\mathfrak{n}}_X^-\right)$-module. We have:

Theorem 3.42. *The complex $\mathbf{IC}_{\widetilde{\mathbf{Bun}}_B}$ is Bar resolution to $\widetilde{j}_!\left(\mathbf{IC}_{\mathbf{Bun}_B}\right)$. Stratumwise, for each $\check\lambda \in \check\Lambda_G^{\mathrm{pos}}$, we have following isomorphism in Grothendieck group*

$$\left[\bar{j}_!\left(\mathbf{IC}_{\mathbf{Bun}_B^{\check\lambda}}\right)\right] = \sum_{\check\theta \in \check\Lambda_G^{\mathrm{pos}}} \left[\Omega\left(\check{\mathfrak{n}}_X^-\right)^{\check\theta} \star \mathbf{IC}_{\overline{\mathbf{Bun}}_B^{\check\theta+\check\lambda}}\right] \in K_0\left(\overline{\mathbf{Bun}}_B\right)$$

Now for $\mathcal{S} \in \mathsf{Shv}\left(\mathrm{Ran}\left(X, \check\Lambda_G^{\mathrm{pos}}\right)\right)$, its actions on $\mathcal{D}\text{-mod}(\mathbf{Bun}_T)$ and $\mathcal{D}\text{-mod}\left(\overline{\mathbf{Bun}}_B\right)$ are compatible in the following way: for $(\mathcal{F}_1, \mathcal{F}_2) \in \mathcal{D}\text{-mod}(\mathbf{Bun}_T) \otimes \mathcal{D}\text{-mod}\left(\overline{\mathbf{Bun}}_B\right)$, we have

$$\bar{\mathsf{p}}_!\left(\bar{\mathsf{q}}^!(\mathcal{S} \star \mathcal{F}_1) \overset{!}{\otimes} \mathcal{F}_2\right) \cong \bar{\mathsf{p}}_!\left(\bar{\mathsf{q}}^!(\mathcal{F}_1) \overset{!}{\otimes} (\mathcal{S} \star \mathcal{F}_2)\right)$$

Combine with Theorem 3.42, we get the desired comparison result of $\mathsf{Eis}_!$ and $\overline{\mathsf{Eis}}$ over constant sheaves/functions.

4 Singular support of Eisenstein sheaves

Throughout the part 4, unless otherwise specified, the curve X is taken to be smooth projective over $\mathbb{C}$. We are going to study the microlocal behaviour of Eisenstein sheaves.

4.1 Stacks of Higgs sheaves and global nilpotent cones

4.1.1 *Stacks of Higgs sheaves and global nilpotent cones.* Let us denote by $\mathbf{Higgs}_\alpha$ the (underived) moduli stack of Higgs sheaves on $\mathbf{Coh}_\alpha$, i.e. the functor given by

$$S \mapsto \left\langle (\mathcal{F}, \varphi) \middle| \mathcal{F} \in \mathbf{Coh}_\alpha(S), \varphi \in H^0\left(X_S, \mathrm{End}(\mathcal{F}) \otimes \Omega^1_{X_S}\right). \right\rangle$$

The map φ in above definition is called a Higgs field. Let $\mathbb{T}_{\mathbf{Coh}_\alpha}$ be the tangent complex of $\mathbf{Coh}_\alpha$, we can see that $\mathbf{Higgs}_\alpha$ is classical truncation of the derived cotangent space of $\mathbf{Coh}_\alpha$:

$$\mathbf{Higgs}_\alpha = {}^{\mathsf{cl}}\mathbf{T}^*\mathbf{Coh}_\alpha = {}^{\mathsf{cl}}\left(\mathrm{Spec}_{\mathbf{Coh}_\alpha}\left(\mathrm{Sym}\left(\mathbb{T}_{\mathbf{Coh}_\alpha}\right)\right)\right)$$

In using the Dolbeault shape of X, see [PS22] and Appendix B.1, the derived stack $\mathbf{T}^*\mathbf{Coh} = \coprod_\alpha \mathbf{T}^*\mathbf{Coh}_\alpha$ can be identified with $\mathbf{Coh}\left(X_{\mathsf{Dol}}\right)$. Let $\mathsf{r}_\alpha : \mathbf{Higgs}_\alpha \to \mathbf{Coh}_\alpha$ be the projection map, then the local chart $\mathbf{Coh}_\alpha^{>\mathcal{L}}$ gives rise to a local chart of $\mathbf{Higgs}_\alpha$. Let $\mathbf{Higgs}_\alpha^{>\mathcal{L}}$ be the following fiber product

$$\begin{tikzcd}
\mathbf{Higgs}_\alpha^{>\mathcal{L}} \arrow[r] \arrow[d] & \mathbf{Higgs}_\alpha \arrow[d] \\
\mathbf{Coh}_\alpha^{>\mathcal{L}} \arrow[r] & \mathbf{Coh}_\alpha
\end{tikzcd}$$

Then $\mathbf{Higgs}_\alpha^{>\mathcal{L}}$ is exactly ${}^{\mathsf{cl}}\left(\mathrm{Spec}_{\mathbf{Coh}_\alpha}\left(\mathrm{Sym}\left(\mathbb{T}_{\mathbf{Coh}_\alpha^{>\mathcal{L}}}\right)\right)\right)$, where $\mathbb{T}_{\mathbf{Coh}_\alpha^{>\mathcal{L}}}$ is given by $\left[\mathfrak{g}_\alpha^{\mathcal{L}} \otimes \mathcal{O}_{Q_\alpha^{\mathcal{L}}} \to \mathcal{T}_{Q_\alpha^{\mathcal{L}}}\right]$, with $\mathfrak{g}_\alpha^{\mathcal{L}} = \mathrm{Lie}\left(\mathrm{G}_\alpha^{\mathcal{L}}\right)$ and $\mathcal{T}_{Q_\alpha^{\mathcal{L}}}$ the tangent sheaf. Thus $\mathbf{Higgs}_\alpha^{>\mathcal{L}}$ is a quotient stack $\left[T^*_{\mathrm{G}_\alpha^{\mathcal{L}}} Q_\alpha^{\mathcal{L}} \middle/ \mathrm{G}_\alpha^{\mathcal{L}}\right]$, where for affine group scheme H acting on Y, we use $T^*_H Y \subseteq T^*Y$ denote the zero of moment map with respect to Hamiltonian action of H on T^*Y. If $\mathcal{L}'$ is strongly generated by $\mathcal{L}$, then the open inclusion $j^{\mathcal{L},\mathcal{L}'} : \mathbf{Coh}_\alpha^{>\mathcal{L}'} \hookrightarrow \mathbf{Coh}_\alpha^{>\mathcal{L}}$ induces an open inclusion $j_H^{\mathcal{L},\mathcal{L}'} : \mathbf{Higgs}_\alpha^{>\mathcal{L}'} \hookrightarrow \mathbf{Higgs}_\alpha^{>\mathcal{L}}$, see [SS20, (2.11)]. We get an open exhaustion of $\mathbf{Higgs}_\alpha = \mathrm{colim}_{\mathcal{L}}\, \mathbf{Higgs}_\alpha^{>\mathcal{L}}$.

For a Higgs field $\varphi : \mathcal{F} \to \mathcal{F} \otimes \Omega^1_X$, we can consider its k-th power

$$\varphi^k : \mathcal{F} \xrightarrow{\varphi} \mathcal{F} \otimes \Omega^1_X \xrightarrow{\varphi \otimes \mathrm{id}_{\Omega^1_X}} \mathcal{F} \otimes (\Omega^1_X)^{\otimes 2} \dots \xrightarrow{\varphi^{k-1} \otimes \mathrm{id}_{\Omega^1_X}} \mathcal{F} \otimes (\Omega^1_X)^{\otimes k}$$

And we say φ is nilpotent if there exists some $k \geqslant 1$ such that φ^k vanishes. Restrict to the locus where φ is nilpotent, we get a closed substack $\mathbf{\Lambda}_\alpha \subseteq \mathbf{Higgs}_\alpha$, called the global nilpotent cone, which is first proposed by Laumon in [Lau88]. Moreover, it is known that $\mathbf{\Lambda}_\alpha$ is a Lagrangian in $\mathbf{Higgs}_\alpha$. The irreducible stack $\mathbf{Coh}_\alpha$ can be regard as the locus in $\mathbf{Higgs}_\alpha$ such that φ is zero. However, the Lagrangian $\mathbf{\Lambda}_\alpha$ is not irreducible in general (for example $\mathbf{Coh}_\alpha$ is one of the irreducible components). All irreducible components of $\mathbf{\Lambda}_\alpha$ can be arranged combinatorically via Jordan stratification, which is described explicitly in [Boz22], for later usage, we give a quick review here:

4.1.2 *Irreducible components of global nilpotent cones.* Let $(\mathcal{F}, \varphi) \in \mathbf{\Lambda}_\alpha$, then there exists $s \in \mathbb{N}$ such that $\varphi^s = 0$ and $\varphi^{s-1} \neq 0$. Write $\Omega^i_X := (\Omega^1_X)^{\otimes i}$, let $\mathcal{F}_i := \mathrm{Im}(\varphi^i) \otimes \Omega^{-i}_X$ and put $\mathcal{F}_0 := \mathcal{F}$, then we get two series:

$$\mathcal{F}_0 \twoheadrightarrow \mathcal{F}_1 \otimes \Omega_X^1 \twoheadrightarrow \mathcal{F}_2 \otimes \Omega_X^2 \twoheadrightarrow \ldots \twoheadrightarrow \mathcal{F}_s \otimes \Omega_X^s = 0$$

$$\mathcal{F}_0 \hookleftarrow \mathcal{F}_1 \hookleftarrow \mathcal{F}_2 \hookleftarrow \ldots \hookleftarrow \mathcal{F}_s = 0$$

The second series allows us to define $\mathcal{F}_k'' := \mathcal{F}_k / \mathcal{F}_{k+1}$ and the first series induces

$$\mathcal{F}_0'' \twoheadrightarrow \mathcal{F}_1'' \otimes \Omega_X^1 \twoheadrightarrow \mathcal{F}_2'' \otimes \Omega_X^2 \twoheadrightarrow \ldots \twoheadrightarrow \mathcal{F}_s'' \otimes \Omega_X^s = 0$$

We abbreviate $\mathcal{F}_k'' \otimes \Omega_X^k$ as $\mathcal{F}_k''(k\Omega_X)$ and denote by α_k the class of $\operatorname{Ker}\left(\mathcal{F}_{k-1}''((k-1)\Omega_X) \twoheadrightarrow \mathcal{F}_k''(k\Omega_X)\right)$, we use $\alpha_k(m\Omega)$ to denote the class of $\operatorname{Ker}\left(\mathcal{F}_{k-1}''((k-1)\Omega) \twoheadrightarrow \mathcal{F}_k''(k\Omega_X)\right)(m\Omega_X)$, the series $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_s)$ is called the *Jordan type* of $(\mathcal{F}, \varphi)$, corresponds to the outer blocks of the following Young tableau:

$$\begin{array}{|c|c|c|c|c|}
\hline
\alpha_s \\
\cline{1-2}
\alpha_s(-\Omega) & \alpha_{s-1} \\
\cline{1-4}
\cdots & \cdots & \cdots & \ddots \\
\hline
\alpha_s((1-s)\Omega) & \alpha_{s-1}((2-s)\Omega) & \ldots & \alpha_2(-\Omega) & \alpha_1 \\
\hline
\end{array}$$

All blocks in the Young tableau sums to α, thus tuple $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_s)$ corresponds to $(\mathcal{F}, \varphi)$ should satisfy the following relations:

$$\sum_{k=1}^{s} k \cdot \operatorname{rk}(\alpha_k) = \operatorname{rk}(\alpha); \quad \sum_{k=1}^{s} k \cdot \deg(\alpha_k) - \operatorname{rk}(\alpha_k)\left(\sum_{j=1}^{k-1} j \cdot (2g-2)\right) = \deg(\alpha)$$

For any tuple with coordinates in $\mathbf{Z}^+$ which is potentially a Jordan type of nilpotent Higgs sheaf of class α should satisfy the above condition. If $\boldsymbol{\alpha}$ is such, we say $\boldsymbol{\alpha}$ is a Jordan type of α and use the notation $\boldsymbol{\alpha} \dashv \alpha$ in [Boz22]. We have a map $\mathrm{JT} : \mathbf{\Lambda}_\alpha \longrightarrow \{\text{Jordan types of } \alpha\}$.

Theorem 4.1. *[Boz22, Corollary 2.5] For $\boldsymbol{\alpha} \dashv \alpha$, we denote by $\mathbf{\Lambda}_{\boldsymbol{\alpha}} \subseteq \mathbf{\Lambda}_\alpha$ the preimage $\mathrm{JT}^{-1}(\boldsymbol{\alpha})$. Then this gives a cell decomposition $\mathbf{\Lambda}_\alpha = \bigsqcup_{\boldsymbol{\alpha} \dashv \alpha} \mathbf{\Lambda}_{\boldsymbol{\alpha}}$. Moreover, the set of irreducible components of $\mathbf{\Lambda}_\alpha$ is given by*

$$\left\{ \overline{\mathbf{\Lambda}_{\boldsymbol{\alpha}}} \,\middle|\, \boldsymbol{\alpha} \dashv \alpha \right\}, \quad \textit{where } \overline{\mathbf{\Lambda}_{\boldsymbol{\alpha}}} \textit{ is the Zariski closure of } \mathbf{\Lambda}_{\boldsymbol{\alpha}}$$

For Jordan type $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_s)$, there is a projection $\pi_{\boldsymbol{\alpha}} : \mathbf{\Lambda}_{\boldsymbol{\alpha}} \longrightarrow \prod_{k=1}^{s} \mathbf{Coh}_{\alpha_s}$ given by

$$\pi_{\boldsymbol{\alpha}} : (\mathcal{F}, \varphi) \mapsto \left(\operatorname{Ker}\left(\mathcal{F}_{k-1}''((k-1)\Omega_X) \twoheadrightarrow \mathcal{F}_k''(k\Omega_X)\right)\right)$$

which is an iterated vector bundle, hence $\mathbf{\Lambda}_{\boldsymbol{\alpha}}$ is irreducible, to show that $\overline{\mathbf{\Lambda}_{\boldsymbol{\alpha}}}$ is exactly the irreducible component of $\mathbf{\Lambda}_\alpha$, it is enough to calculate its dimension, which is done in [Boz22]. Moreover, we can describe the closure $\overline{\mathbf{\Lambda}_{\boldsymbol{\alpha}}}$ via a partial order on Jordan type of α. For two Jordan types $\boldsymbol{\alpha} = (a_1, \ldots, a_s)$ and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_t)$ of α, we say $\boldsymbol{\beta} \preceq \boldsymbol{\alpha}$ if the following holds:

$$\sum_{h=1}^{k} \sum_{j=h}^{s} \alpha_j \left((h-j)(2g-2)\right) \preceq \sum_{h=1}^{k} \sum_{j=h}^{t} \beta_j \left((h-j)(2g-2)\right)$$

where for each k, the sum is over blocks in the first k rows (from bottom to top) in the Young tableau and the relation $\preceq$ is the one in $\mathbf{Z}^+$. Thus the minimal Jordan type of α is itself under this order, we denote this Jordan type by (α), in which case $(\mathcal{F},\varphi)\in\mathbf{\Lambda}_{(\alpha)}$ is the zero section $\mathbf{Coh}_\alpha\subseteq\mathbf{Higgs}_\alpha$. In general, the closed irreducible components of $\mathbf{\Lambda}_\alpha$ can be described as $\overline{\mathbf{\Lambda}_{\boldsymbol{\alpha}}}=\bigsqcup_{\boldsymbol{\beta}\preceq\boldsymbol{\alpha}}\mathbf{\Lambda}_{\boldsymbol{\beta}}$.

The global nilpotent cone $\mathbf{Higgs}$ can also be truncated from a derived substack of $\mathbf{T}^*\mathbf{Coh}_\alpha$. Let $X_{\mathsf{Dol}}^{\mathsf{nil}}$ be the relative classifying space $\mathbf{B}_X TX$ with respect to TX-action on X. Then $\mathbf{\Lambda}=\bigsqcup_\alpha\mathbf{\Lambda}_\alpha={}^{\mathsf{cl}}\mathbf{Coh}\left(X_{\mathsf{Dol}}^{\mathsf{nil}}\right)$.

4.1.3 *Parabolic induction/restriction for Higgs.* Similar to the diagram over $\mathbf{Coh}$ to construct $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathsf{CT}^{\mathrm{Coh}}$, we have induction diagram for $\mathbf{Higgs}$ and $\mathbf{\Lambda}$:

$$\begin{tikzcd}
\mathbf{Higgs}_\alpha\times\mathbf{Higgs}_\beta & \widetilde{\mathbf{Higgs}}_{\alpha,\beta} \arrow[l, "\mathsf{q}^H_{\alpha,\beta}"'] \arrow[r, "\mathsf{p}^H_{\alpha,\beta}"] & \mathbf{Higgs}_{\alpha+\beta} \\
\mathbf{\Lambda}_\alpha\times\mathbf{\Lambda}_\beta \arrow[u, hook] & \widetilde{\mathbf{\Lambda}}_{\alpha,\beta} \arrow[l, "\mathsf{q}^\Lambda_{\alpha,\beta}"'] \arrow[r, "\mathsf{p}^\Lambda_{\alpha,\beta}"] \arrow[u, hook] & \mathbf{\Lambda}_{\alpha+\beta} \arrow[u, hook]
\end{tikzcd} \tag{4.1.1}$$

where $\widetilde{\mathbf{Higgs}}_{\alpha,\beta}$ is the derived stack classifying following commutative diagrams:

$$\left\langle \begin{tikzcd}
\mathcal{E}' \arrow[r, "f"] \arrow[d, "\varphi'"] & \mathcal{E}'' \arrow[r, "g"] \arrow[d, "\varphi''"'] & \mathcal{E} \arrow[d, "\varphi"] \\
\mathcal{E}'\otimes\Omega_X^1 \arrow[r, "f\otimes\Omega_X^1"] & \mathcal{E}''\otimes\Omega^1_X \arrow[r, "g\otimes\Omega_X^1"] & \mathcal{E}\otimes\Omega_X^1
\end{tikzcd} \;\middle|\; \begin{array}{l} (\mathcal{E},\varphi)\in\mathbf{Higgs}_\alpha;\,(\mathcal{E}',\varphi')\in\mathbf{Higgs}_\beta; \\ \\ (f,g)\in\mathrm{Ext}^1(\mathcal{E},\mathcal{E}')\end{array}\right\rangle$$

and the maps $\mathsf{q}^H_{\alpha,\beta}$ (resp. $\mathsf{p}^H_{\alpha,\beta}$) are given by projection to the third and the first column (resp. the middle column). The stack $\widetilde{\mathbf{\Lambda}}_{\alpha,\beta}$ in the bottom is the substack of $\widetilde{\mathbf{Higgs}}_{\alpha,\beta}$ by adding the condition that one (hence all) of $\varphi',\varphi'',\varphi$ is nilpotent.

Proposition 4.2. *The maps $\mathsf{q}^H_{\alpha,\beta}$ and $\mathsf{q}^\Lambda_{\alpha,\beta}$ are quasi-smooth, the maps $\mathsf{p}^H_{\alpha,\beta}$ and $\mathsf{p}^\Lambda_{\alpha,\beta}$ are schematic and proper.*

Proof. For the first half, we need the compute the tangent complex with respect to q^H or q^Λ. Since the moduli stack $\mathbf{Higgs}_\alpha$ can also be regarded as stack of coherent sheaves on T^*X with proper supports, it suffices to compute the tangent complex for extension diagram on $\mathbf{Coh}(T^*X)$. In general, for S smooth quasi-projective algebraic surface, we denote by $\overline{C}_S$ the universal coherent sheaf on $\mathbf{Coh}\,(S)\times S$. Then the relative cotangent complex is given by

$$\mathbb{L}_{\mathsf{q}^H}=\left(\mathsf{q}^H\right)^*\left((p_{12})_*\mathcal{H}\mathrm{om}\left((p_{13})^*\overline{C}_S,(p_{23})^*\overline{C_S}\right)[-1]\right)$$

where the morphisms are:

$$\begin{tikzcd}
 & \mathbf{Coh}(S)\times\mathbf{Coh}(S)\times S \arrow[r, "p_{12}"] \arrow[dl, "p_{13}"'] \arrow[dr, "p_{23}"] & \mathbf{Coh}(S)\times\mathbf{Coh}(S) & \widetilde{\mathbf{Coh}}(S) \arrow[l, "\mathsf{q}^H"'] \\
\mathbf{Coh}(S)\times\mathbf{Coh}(S) & & \mathbf{Coh}(S)\times S &
\end{tikzcd}$$

In particular, for a point $x\in\mathbf{Coh}(S)$ represented by $\mathcal{F}'\to\mathcal{F}''\to\mathcal{F}$, the restriction of $x^*\mathbb{L}_{\mathsf{q}^H}$ is $(\mathcal{H}\mathrm{om}\,(\mathcal{F},\mathcal{F}'')\,[1])^\vee$. Thus, we can conclude fiberwise the Tor-amplitude of $\mathbb{L}_{\mathsf{q}^H}$ within $[-1,1]$, which implies that q^H is quasi-smooth. The properness is a consequence that the Quot scheme is proper. ∎

4.2 Cohomological Hall algebras

4.2.1 Borel-Moore homologies. We constructed the Hall algebras for curve X_0 over $\mathbb{F}_q$, now for a curve X over $\mathbb{C}$, the replacement for point-counting of $\mathbf{Coh}_\alpha$ is the Borel-Moore homology of $\mathbf{Higgs}$.

For scheme $Y \in \mathsf{Sch}_{\mathrm{ft}}$ of finite type over $\mathbf{k} = \mathbb{C}$, consider the complex $C_\bullet^{\mathrm{BM}}(Y) = R\Gamma\left(Y, \pi^!\mathbb{C}\right)$ with $\pi : Y \to \mathrm{pt}$. We denote by $H_\bullet^{\mathrm{BM}}(Y)$ the cohomology of the complex $C_\bullet^{\mathrm{BM}}(Y)$ and refer it as the Borel-Moore homology of Y. It has following properties:

(i) For $f : Y \to Z$ proper, the unit map $f_* f^! \to \mathrm{Id}$ induces proper pushforward on Borel-Moore homology: $f_* : H_\bullet^{\mathrm{BM}}(Y) \to H_\bullet^{\mathrm{BM}}(Z)$.

(ii) For $f : Y \to Z$ locally complete intersection (l.c.i in short) of relative dimension d, then the counit $\mathrm{Id} \to f_* f^!$ induces pullback on Borel-Moore homology: $f^! : H_\bullet(Z) \to H_{\bullet+2d}(Y)$.

Now let $\mathcal{X}$ be an Artin stack and $\mathsf{Sm}_{\mathcal{X}}$ be smooth cover of $\mathcal{X}$ with $(S, s) \in \mathsf{Sm}_{\mathcal{X}}$ contains $S \in \mathsf{Sch}_{\mathrm{ft}}$, $s : S \to X$ smooth of relative dimension d_s. Then the Borel-Moore complex for $\mathcal{X}$ is defined to be $\lim_{(S,s)\in \mathsf{Sm}_{\mathcal{X}}} C_\bullet^{\mathrm{BM}}(S)[-d_s]$, the limit is taken over $*$-pullback among smooth translations. Write $H_\bullet^{\mathrm{BM}}(\mathcal{X})$ the corresponding homology with same properties as schemes.

4.2.2 CoHA for Higgs sheaves. Now we turn to $H_\bullet^{\mathrm{BM}}(\mathbf{Higgs}) := \bigoplus_\alpha H_\bullet^{\mathrm{BM}}(\mathbf{Higgs}_\alpha)$ and we want to endow it with a multiplication structure given by pull-push along the extension diagram, analogous to that of the Hall algebra $\mathbf{H}_{X_0}$. The problem is that q^H is only quasi-smooth, but not l.c.i, so it is not clear that the pull-push functor $\mathsf{Eis}^H_{\alpha,\beta} := \left(\mathsf{p}^H_{\alpha,\beta}\right)_! \left(\mathsf{q}^H_{\alpha,\beta}\right)^*$ would induce a product on Borel-Moore homology. There are several approaches to define an associative algebra structure on $H_\bullet^{\mathrm{BM}}(\mathbf{Higgs})$, for example [KV22], [PS22] and they turn out to be equivalent. We adapt here the method in [Min20] and [SS20] by defining the product on local charts. Recall we have $\mathbf{Higgs}_\alpha^{>\mathcal{L}} \cong \left[T^*_{\mathrm{G}_\alpha^{\mathcal{L}}} \mathcal{Q}_\alpha^{\mathcal{L}} / \mathrm{G}_\alpha^{\mathcal{L}}\right]$. Let $\widetilde{\mathbf{Higgs}}_{\alpha,\beta}^{\mathcal{L},\mathcal{L}'} \subseteq \widetilde{\mathbf{Higgs}}_{\alpha,\beta}$ be the substack that $\mathsf{p}^H_{\alpha,\beta}\left((\mathcal{E}, \mathcal{E}', \mathcal{E}'', \phi, \phi', \phi'')\right) \in \mathbf{Higgs}_{\alpha+\beta}^{>\mathcal{L}'}$ and $\mathsf{q}^H_{\alpha,\beta}\left((\mathcal{E}, \mathcal{E}', \mathcal{E}'', \phi, \phi', \phi'')\right) \in \mathbf{Higgs}_{\alpha,\beta}^{>\mathcal{L}}$. The stacks appear in the extension diagram $\mathbf{Higgs}_\alpha^{>\mathcal{L}} \times \mathbf{Higgs}_\beta^{>\mathcal{L}} \xleftarrow{\left(\mathsf{q}^H_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}} \widetilde{\mathbf{Higgs}}_{\alpha,\beta}^{\mathcal{L},\mathcal{L}'} \xrightarrow{\left(\mathsf{p}^H_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}} \mathbf{Higgs}_{\alpha+\beta}^{>\mathcal{L}'}$ are in fact quotient stacks. We use the notation in [SS20]:

$$Y = \mathcal{Q}_\alpha^{\mathcal{L}} \times \mathcal{Q}_\beta^{\mathcal{L}}; \quad M = \mathrm{G}_\alpha^{\mathcal{L}} \times \mathrm{G}_\beta^{\mathcal{L}} \subseteq P = P_{\alpha,\beta}^{\mathcal{L}} \subseteq G = G_{\alpha+\beta}^{\mathcal{L}}; \quad X' = Y \overset{P}{\times} G; \quad X'' = \mathcal{Q}_{\alpha+\beta}^{\mathcal{L},\mathcal{L}'}; \quad W = \widetilde{\mathcal{Q}}_{\alpha,\beta}^{\mathcal{L},\mathcal{L}'}$$

Then $\mathbf{Higgs}_\alpha^{\mathcal{L}} \times \mathbf{Higgs}_\beta^{>\mathcal{L}} \cong \left[T^*_G X'/G\right]$, $\mathbf{Higgs}_{\alpha+\beta}^{>\mathcal{L}'} \cong \left[T^*_G X''/G\right]$ and $\widetilde{\mathbf{Higgs}}_{\alpha,\beta}^{\mathcal{L},\mathcal{L}'}$ is isomorphic to

$$\left[T^*_W(X' \times X'') \cap \left(T^*_G X' \times T^*_G X''\right)/G\right] =: [Z/G]$$

where $T^*_W(X' \times X'')$ is the conormal bundle of smooth subscheme W in $X' \times X''$. Thus the local extension diagram of Higgs sheaves fits into the G-equivariant version following correspondence:

$$\begin{tikzcd}
T^*_G X' \arrow[d] & Z \arrow[l, "\overline{\Phi}"'] \arrow[r, "\overline{\Psi}"] \arrow[d] & T^*_G X'' \arrow[d] \\
T^* X' & T^*_W(X' \times X'') \arrow[l, "\Phi"'] \arrow[r, "\Psi"] & T^* X''
\end{tikzcd}$$

where the left square in the diagram is cartesian. Combine with the properties of (equivariant) Borel-Moore homology, we have the following Theorem in [SS20].

Theorem 4.3. *[SS20, Theorem 3.3] (1) Let $\mathcal{L}, \mathcal{L}'$ be as above, the functor $\left(\left(\mathsf{p}^H_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}\right)_* \left(\left(\mathsf{q}^H_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}\right)^!$ induces a map $\mathsf{m}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta} : H^{\mathrm{BM}}_\bullet\left(\mathbf{Higgs}^{\mathcal{L}}_\alpha \times \mathbf{Higgs}^{>\mathcal{L}}_\beta\right) \longrightarrow H^{\mathrm{BM}}_\bullet\left(\mathbf{Higgs}^{>\mathcal{L}'}_{\alpha+\beta}\right)$.*
(2) The collection of maps $\mathsf{m}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta}$ for various $\mathcal{L}, \mathcal{L}'$ gives rise to a map

$$\mathsf{Eis}^H_{\alpha,\beta} : H^{\mathrm{BM}}_\bullet\left(\mathbf{Higgs}_\alpha \times \mathbf{Higgs}_\beta\right) \longrightarrow H^{\mathrm{BM}}_\bullet\left(\mathbf{Higgs}_{\alpha+\beta}\right)$$

Moreover. $\mathsf{Eis}^H_{\alpha,\beta}$ endows $H^{\mathrm{BM}}_\bullet(\mathbf{Higgs})$ structure of an associative algebra.

For Theorem 4.3(1), one can show that restrict to the local chart, $\left(\mathsf{q}^H_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}$ is l.c.i and thus one has refined Gysin pullback. For(2), one need to check that if there is tuple $(\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3)$ such that $\mathcal{L}_{i+1}$ is strongly generated by $\mathcal{L}_i$, then $\mathsf{m}^{\mathcal{L}_1,\mathcal{L}_3}_{\alpha,\beta} = \mathsf{m}^{\mathcal{L}_2,\mathcal{L}_3}_{\alpha,\beta} \circ \left(j^{\mathcal{L}_1,\mathcal{L}_2}_H\right)^!$ and $\mathsf{m}^{\mathcal{L}_1,\mathcal{L}_3}_{\alpha,\beta} = \left(j^{\mathcal{L}_2,\mathcal{L}_3}_H\right)^! \circ \mathsf{m}^{\mathcal{L}_1,\mathcal{L}_2}_{\alpha,\beta}$ hold. For the associativity, it suffices to show $\mathsf{m}^{\mathcal{L},\mathcal{L}'}$ does not depend on the order of multiplication. The associativity is proved in [Min20, Theorem 2.2]. Also, let $\mathbf{\Lambda}^{>\mathcal{L}}_\alpha := \mathbf{\Lambda}_\alpha \times_{\mathbf{Higgs}_\alpha} \mathbf{Higgs}^{>\mathcal{L}}_\alpha$ and $\widetilde{\mathbf{\Lambda}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta} \subseteq \widetilde{\mathbf{Higgs}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta}$ such that ϕ, ϕ' nilpotent. Then we have a commutative diagram:

$$\begin{tikzcd}
\mathbf{Higgs}^{>\mathcal{L}}_\alpha \times \mathbf{Higgs}^{>\mathcal{L}}_\beta & \widetilde{\mathbf{Higgs}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta} \arrow[l, "\left(\mathsf{q}^H_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}"'] \arrow[r, "\left(\mathsf{p}^H_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}"] & \mathbf{Higgs}^{>\mathcal{L}'}_{\alpha+\beta} \\
\mathbf{\Lambda}^{>\mathcal{L}}_\alpha \times \mathbf{\Lambda}^{>\mathcal{L}}_\beta \arrow[u, hook, "i^{\mathcal{L}}_{\alpha,\beta}"] & \widetilde{\mathbf{\Lambda}}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta} \arrow[l, "\left(\mathsf{q}^\Lambda_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}"'] \arrow[r, "\left(\mathsf{p}^\Lambda_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}"] \arrow[u, hook, "\tilde{i}^{\mathcal{L},\mathcal{L}'}_{\alpha,\beta}"'] & \mathbf{\Lambda}^{>\mathcal{L}'}_{\alpha+\beta} \arrow[u, hook, "i^{\mathcal{L}'}_{\alpha+\beta}"']
\end{tikzcd} \tag{4.2.1}$$

such that the left square in the diagram 4.2.1 is cartesian. Apply the similar arguments in Theorem 4.3 gives the morphism for Borel-Moore homology on global nilpotent cone:

$$\mathsf{Eis}^\Lambda_{\alpha,\beta} : H^{\mathrm{BM}}_\bullet(\mathbf{\Lambda}_\alpha \times \mathbf{\Lambda}_\beta) \longrightarrow H^{\mathrm{BM}}_\bullet(\mathbf{\Lambda}_{\alpha+\beta})$$

There is also an associative algebra structure on $H^{\mathrm{BM}}_\bullet(\mathbf{\Lambda}) := \bigoplus_\alpha H^{\mathrm{BM}}_\bullet(\mathbf{\Lambda}_\alpha)$, and the proper pushforward with respect to $i_\alpha : \mathbf{\Lambda}_\alpha \hookrightarrow \mathbf{Higgs}_\alpha$ induces algebra morphism $H^{\mathrm{BM}}_\bullet(\mathbf{\Lambda}) \longrightarrow H^{\mathrm{BM}}_\bullet(\mathbf{Higgs})$.

Definition 4.4. We call the algebras $H^{\mathrm{BM}}_\bullet(\mathbf{\Lambda})$ resp. $H^{\mathrm{BM}}_\bullet(\mathbf{Higgs})$ the **cohomological Hall algebra**(in short=CoHA) of Higgs sheaves (resp. nilpotent Higgs sheaves).

The irreducible component $\mathbf{\Lambda}_{\boldsymbol{\alpha}}$ of $\mathbf{\Lambda}_\alpha$ corresponds to an element $[\mathbf{\Lambda}_{\boldsymbol{\alpha}}] \in H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda}_\alpha) = H^{\mathrm{BM}}_{2\dim(\mathbf{\Lambda}_\alpha)}(\mathbf{\Lambda}_\alpha)$. Also, we can calculate the relative dimension of $\left(\mathsf{q}^\Lambda_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}$, which is the same as $\left(\mathsf{q}^H_{\alpha,\beta}\right)^{\mathcal{L},\mathcal{L}'}$. Use the quotient stack expression, it remains to look at the map $\overline{\Phi} : Z \to T^*_G X'$, since the square $(\overline{\Phi}, \Phi)$ is cartesian, we have

$$\begin{aligned}
\mathrm{rel.}\dim(\overline{\Phi}) = \mathrm{rel.}\dim(\Phi) &= \dim\left(T^*_W(X' \times X'')\right) - \dim(T^*X') = \dim(X' \times X'') - \dim(T^*X') \\
&= \dim\left(\mathbf{Coh}^{\mathcal{L},\mathcal{L}''}_{\alpha+\beta}\right) - \dim\left(\mathbf{Coh}^{>\mathcal{L}}_\alpha \times \mathbf{Coh}^{>\mathcal{L}}_\beta\right) = \dim\left(\mathbf{Coh}_{\alpha+\beta}\right) - \dim\left(\mathbf{Coh}_\alpha \times \mathbf{Coh}_\beta\right)
\end{aligned}$$

Thus we can deduce that $\mathsf{Eis}^\Lambda_{\alpha+\beta}$ sends $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda}_\alpha \times \mathbf{\Lambda}_\beta) \longrightarrow H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda}_{\alpha+\beta})$ and $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda}) = \bigoplus_\alpha H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda}_\alpha)$ is a subalgebra of $H^{\mathrm{BM}}_\bullet(\mathbf{\Lambda})$.

4.3 Dimensional reduction and coproducts on CoHA

4.3.1 Dimensional reduction. In this section we are going to construct a coproduct on the Borel-Moore homology of Higgs sheaves. First, note that the construction of Borel-Moore homology can be generalized to derived Artin stacks, see [Kha19], also it is proved *loc. cit* that the Borel-Moore homology is insensitive of the derived structure:

Lemma 4.5. *[Kha19, Theorem 2.19] Let $\mathcal{X}$ be a derived stack and ${}^{\mathrm{cl}}\mathcal{X}$ the classical truncation, then there is an isomorphism $H_\bullet^{\mathrm{BM}}\left({}^{\mathrm{cl}}\mathcal{X}\right) \xrightarrow{\cong} H_\bullet^{\mathrm{BM}}(\mathcal{X})$ induced by $*$-pushforward along ${}^{\mathrm{cl}}\mathcal{X} \hookrightarrow \mathcal{X}$.*

Also, from the construction of analytification functor (see Appendix B.1), we can detect an equivalence for derived Artin stack $\mathcal{X}$: $H_\bullet^{\mathrm{BM}}(\mathcal{X}) \cong H_\bullet(\mathcal{X}^{\mathrm{an}})$. Combine Theorem B.10, we can deduce the following sequences of isomorphisms:

$$H_\bullet^{\mathrm{BM}}\left(\mathbf{Higgs}_\alpha\right) \cong H_\bullet^{\mathrm{BM}}\left(\mathbf{Coh}_\alpha\left(X_{\mathrm{Dol}}\right)\right) \cong H_\bullet\left(\left(\mathbf{Coh}_\alpha\left(X_{\mathrm{Dol}}\right)\right)^{\mathrm{an}}\right) \cong H_\bullet\left(\mathbf{AnCoh}\left(X_{\mathrm{Dol}}^{\mathrm{an}}\right)\right) \cong H_\bullet\left(\mathbf{Higgs}_\alpha^{\mathrm{an}}\right)$$
$$H_\bullet^{\mathrm{BM}}\left(\mathbf{\Lambda}_\alpha\right) \cong H_\bullet^{\mathrm{BM}}\left(\mathbf{Coh}_\alpha\left(X_{\mathrm{Dol}}^{\mathrm{nil}}\right)\right) \cong H_\bullet\left(\left(\mathbf{Coh}_\alpha\left(X_{\mathrm{Dol}}^{\mathrm{nil}}\right)\right)^{\mathrm{an}}\right) \cong H_\bullet\left(\mathbf{AnCoh}\left(\left(X_{\mathrm{Dol}}^{\mathrm{nil}}\right)^{\mathrm{an}}\right)\right) \cong H_\bullet\left(\mathbf{\Lambda}_\alpha^{\mathrm{an}}\right)$$

On the other hand, we have an isomorphism $\omega_{T^*X} \cong \mathcal{O}_{T^*X}$ between the canonical sheaf and the structure sheaf. So the total space $\mathrm{Tot}_{T^*X}\left(\omega_{T^*X}\right) \cong T^*X \times \mathbb{A}^1$, which is a Calabi-Yau threefold and we denote it by Y. Also, we use $\mathfrak{M}_Y, \mathfrak{M}_{T^*X}$ to denote the derived moduli stack of coherent sheaves on Y (resp. on T^*X) with proper support. It is known in [Pan+13] that the moduli stack $\mathfrak{M}_Y$ admits a (-1)-shifted symplectic structure and $\mathfrak{M}_{T^*X}$ is 0-shifted symplectic. For integer n, we can consider the n-shifted cotangent stack $\mathbf{T}^*[n]\mathfrak{M}_{T^*X} := \mathrm{Spec}_{\mathfrak{M}_{T^*X}}\left(\mathrm{Sym}\left(\mathbb{T}_{\mathfrak{M}_{T^*X}}[-n]\right)\right)$. For $n = -1$, the (-1)-shifted symplectic stacks as above, there is an associated $\boldsymbol{\varphi}$ monodromic mixed Hodge module.
Consider also an analytic version, $\mathfrak{M}_Y^{\mathrm{an}}$ is the substack of $\mathbf{AnCoh}\left(T^*X^{\mathrm{an}} \times \mathbb{C}\right)$ with compact support condition. For $U \subseteq \mathbb{C}$ convex open subset of $\mathbb{C}$, the restriction gives a substack

$$\mathfrak{M}_{T^*X^{\mathrm{an}} \times U}^{\mathrm{an}} \subseteq \mathfrak{M}_{T^*X^{\mathrm{an}} \times \mathbb{C}}^{\mathrm{an}} = \mathfrak{M}_Y^{\mathrm{an}}$$

we abbreviate it as $\mathfrak{M}_U^{\mathrm{an}}$. Denote by $\boldsymbol{\varphi}^{\mathrm{an}}$ the monodromic mixed Hodge module on $\mathfrak{M}_Y^{\mathrm{an}}$ and for $U \subseteq \mathbb{C}$ convex open, let $\boldsymbol{\varphi}_U^{\mathrm{an}}$ be the restriction along the inclusion. We have the following result:

Theorem 4.6. *(1) [Kin22, Theorem 4.14] There is an isomorphism between (-1)-shifted symplectic derived stacks $\mathfrak{M}_Y \cong \mathbf{T}^*[-1]\mathfrak{M}_{T^*X}$ such that $\pi_{T^*X} : \mathbf{T}^*[-1]\mathfrak{M}_{T^*X} \to \mathfrak{M}_{T^*X}$ induces a natural isomorphism:*

$$\left(\pi_{T^*X}\right)_*\left(\boldsymbol{\varphi}\right) \cong \mathbb{D}\mathbb{C}_{\mathfrak{M}_{T^*X}}\left[-\mathrm{vdim}\left({}^{\mathrm{cl}}\mathfrak{M}_{T^*X}\right)\right]$$

(2) Let U be convex open in $\mathbb{C}$, then the substack $\mathfrak{M}_{T^*X^{\mathrm{an}} \times U}^{\mathrm{an}} = \mathfrak{M}_U^{\mathrm{an}}$ is open in $\mathfrak{M}_Y$ and the restriction induces an isomorphism $R\Gamma\left(\mathfrak{M}_Y^{\mathrm{an}}, \boldsymbol{\varphi}^{\mathrm{an}}\right) \cong R\Gamma\left(\mathfrak{M}_U^{\mathrm{an}}, \boldsymbol{\varphi}_U^{\mathrm{an}}\right)$.

4.3.2 The co-factorization structure. As a consequence of Theorem 4.6(1), take the cohomology on both sides, we have an isomorphism $H^{\mathrm{vdim}-\bullet}\left(\mathfrak{M}_Y, \boldsymbol{\varphi}\right) \cong H_\bullet^{\mathrm{BM}}\left({}^{\mathrm{cl}}\mathfrak{M}_{T^*X}\right)$, which is nothing but $H_\bullet^{\mathrm{BM}}\left(\mathbf{Higgs}\right)$. Also for $U_1, U_2, \ldots, U_s$ disjoint convex open subsets in $\mathbb{C}$, we have identification

$$\mathfrak{M}_{U_1 \sqcup \ldots \sqcup U_s}^{\mathrm{an}} \cong \mathfrak{M}_{U_1}^{\mathrm{an}} \times \ldots \times \mathfrak{M}_{U_s}^{\mathrm{an}}$$

under this identification we have the decomposition of DT-sheaf $\varphi_{U_1\sqcup\ldots\sqcup U_s} \cong \varphi_{U_1} \boxtimes \ldots \boxtimes \varphi_{U_s}$. For any convex open U that contains $U_1 \sqcup \ldots \sqcup U_s$, the restriction of $\mathfrak{M}_U^{\mathrm{an}}$ to $\mathfrak{M}^{\mathrm{an}}_{U_1\sqcup\ldots\sqcup U_s}$ gives a morphism

$$\Delta_{U,U_1\sqcup\ldots\sqcup U_s} : R\Gamma\left(\mathfrak{M}_U^{\mathrm{an}}, \varphi_U^{\mathrm{an}}\right) \longrightarrow R\Gamma\left(\mathfrak{M}_{U_1}^{\mathrm{an}}, \varphi_{U_1}^{\mathrm{an}}\right) \otimes \ldots \otimes R\Gamma\left(\mathfrak{M}_{U_s}^{\mathrm{an}}, \varphi_{U_s}^{\mathrm{an}}\right)$$

In combining with Theorem 4.6(2), we can see the above morphism gives a map $H_\bullet^{\mathrm{BM}}(\mathbf{Higgs}) \longrightarrow H_\bullet^{\mathrm{BM}}(\mathbf{Higgs}) \otimes \ldots \otimes H_\bullet^{\mathrm{BM}}(\mathbf{Higgs})$. To show it indeed defines an coassociative coproduct, we need to show the above morphism $\Delta_{U,U_1\sqcup\ldots\sqcup U_s}$ is independent of the choice of $(U_1 \sqcup \ldots \sqcup U_s \subseteq U)$.

First, for $V \subseteq U$ that contains $U_1 \sqcup \ldots \sqcup U_s$, we know the restriction $\Delta_{U,V}$ is an isomorphism by Theorem 4.6 and the map $\Delta_{U,U_1\sqcup\ldots\sqcup U_s}$ factors as $\Delta_{V,U_1\sqcup\ldots\sqcup U_s} \circ \Delta_{U,V}$, thus $\Delta_{U,U_1\sqcup\ldots\sqcup U_s}$ is invariant under shrinking U. It is also invariant under shrinking U_i for the same reason. Next, for two open subsets $V, W \subseteq U$ that V, W disjoint, then $\Delta_{U,V} \cong \Delta_{U,W}$, thus for two collections of disjoint open subsets $U_1 \sqcup \ldots \sqcup U_s \subseteq U$ and $V_1 \sqcup \ldots \sqcup V_s \subseteq U$ such that $V_i \cap U_j = \emptyset$, we have $\Delta_{U,U_1\sqcup\ldots\sqcup U_s} \cong \Delta_{U,V_1\sqcup\ldots\sqcup V_s}$. Finally, if $V_i \cap U_j$ is non-empty, we can find $W \subseteq V_i \cap U_j$ open convex and we have $\Delta_{U,U_1\sqcup\ldots\sqcup U_s} \xrightarrow{\cong} \Delta_{U,U_1\sqcup\ldots\sqcup W\sqcup\ldots\sqcup U_s} \xleftarrow{\cong} \Delta_{U,V_1\sqcup\ldots\sqcup W\sqcup\ldots\sqcup V_s}$ by replacing U_i and V_j by W respectively. Since there are finitely many intersection, we can do the above replacement in finite steps and this shows Δ does not depend on choice of U_i's.

Now we construct a coproduct over the nilpotent CoHA (i.e. over $H_\bullet^{\mathrm{BM}}(\mathbf{\Lambda})$). Consider the derived Artin stack of nilpotent Higgs sheaves $\mathbf{Coh}\left(X_{\mathsf{Dol}}^{\mathsf{nil}}\right)$, its associated shifted cotangent stack $\mathbf{T}^*[-1]\mathbf{Coh}\left(X_{\mathsf{Dol}}^{\mathsf{nil}}\right)$ is also -1-shifted symplectic, we denote the corresponding projection map by $\mathbf{T}^*[-1]\mathfrak{M}_{T^*X}^{\mathsf{nil}} \xrightarrow{\pi_{T^*X}^{\mathsf{nil}}} \mathfrak{M}_{T^*X}^{\mathsf{nil}}$.

Lemma 4.7. *Let $i : \mathcal{X} \hookrightarrow \mathcal{Y}$ be a quasi-smooth closed embedding of derived Artin stacks, consider the following correspondence induced by i:*

$$\begin{tikzcd}
\mathbf{T}^*[-1]\mathcal{Y} \arrow[d, "\pi_{\mathcal{Y}}"'] & \mathbf{T}^*[-1]\mathcal{Y} \times_{\mathcal{Y}} \mathcal{X} \arrow[l, "\mathrm{pr}_1"'] \arrow[r, "\mathrm{d}i"] & \mathbf{T}^*[-1]\mathcal{X} \arrow[d, "\pi_{\mathcal{X}}"] \\
\mathcal{Y} & & \mathcal{X} \arrow[ll, hook', "i"']
\end{tikzcd}$$

then we have

$$(\pi_{\mathcal{X}})_*(\mathrm{d}i)_*(\mathrm{pr}_1)^!\left(\varphi_{\mathbf{T}^*[-1]\mathcal{Y}}\right) \cong i^!(\pi_{\mathcal{Y}})_*\left(\varphi_{\mathbf{T}^*[-1]\mathcal{Y}}\right)[-\mathrm{codim}_{\mathcal{X}}(\mathcal{Y})] \cong i^!\mathbb{DC}_{\mathcal{Y}}[\dim(\mathcal{X})] \cong \mathbb{DC}_{\mathcal{X}}[\dim(\mathcal{X})]$$

Now we can consider the following diagram and its analytic version:

$$\begin{tikzcd}
\mathbf{T}^*[-1]\mathfrak{M}_{T^*X} \arrow[d, "\pi_{T^*X}"] & \mathbf{T}^*[-1]\mathfrak{M}_{T^*X} \times_{\mathfrak{M}_{T^*X}} \mathfrak{M}_{T^*X}^{\mathsf{nil}} \arrow[l, "\mathrm{pr}_1"'] \arrow[r, "\mathrm{d}i"] \arrow[d, "\mathrm{pr}_2"] & \mathbf{T}^*[-1]\mathfrak{M}_{T^*X}^{\mathsf{nil}} \arrow[dl, "\pi_{T^*X}^{\mathsf{nil}}"] \\
\mathfrak{M}_{T^*X} & \mathfrak{M}_{T^*X}^{\mathsf{nil}} \arrow[l, hook', "i"'] &
\end{tikzcd} \tag{4.3.1}$$

Define $\varphi^{\mathsf{nil}} := (\mathrm{d}i)_*(\mathrm{pr}_1)^!\left(\varphi_{\mathbf{T}^*[-1]\mathfrak{M}_{T^*X}}\right)$. We have $\mathrm{pr}_2 = \pi_{T^*X}^{\mathsf{nil}} \circ \mathrm{d}i$ and thus by Lemma 4.7,

$$(\mathrm{pr}_2)_*(\mathrm{pr}_1)^!\left(\varphi_{\mathbf{T}^*[-1]\mathfrak{M}_{T^*X}}\right) \cong \left(\pi_{T^*X}^{\mathsf{nil}}\right)_*(\mathrm{d}i)_*(\mathrm{pr}_1)^!\left(\varphi_{\mathbf{T}^*[-1]\mathfrak{M}_{T^*X}}\right) \cong \mathbb{DC}_{\mathfrak{M}_{T^*X}^{\mathsf{nil}}}[\mathbf{shift}]$$

For $U \subseteq \mathbb{C}$ analytic open convex subset, recall we have inclusion $\mathfrak{M}_U^{\mathrm{an}} \subseteq \mathfrak{M}_{T^*X} \times \mathbb{C}$, for the same U, we denote by $\mathfrak{M}_U^{\mathsf{nil,an}}$ the image of $\mathfrak{M}_U^{\mathrm{an}} \times_{\mathfrak{M}_{T^*X}^{\mathrm{an}}} \mathfrak{M}_{T^*X}^{\mathsf{nil,an}}$ under $(\mathrm{d}i)^{\mathrm{an}} : \mathfrak{M}_{T^*X^{\mathrm{an}}\times\mathbb{C}}^{\mathrm{an}} \times_{\mathfrak{M}_{T^*X}^{\mathrm{an}}} \mathfrak{M}_{T^*X}^{\mathsf{nil,an}} \to \mathbf{T}^*[-1]\mathfrak{M}_{T^*X}^{\mathsf{nil,an}}$.

Similarly, for $(U_1,...,U_s)$ disjoint open convex subsets, we can define $\mathfrak{M}^{\mathrm{nil,an}}_{U_1\sqcup...\sqcup U_s}$. We have the following diagram of analytic stacks, where the two upper squares are cartesian:

$$\begin{tikzcd}
\mathfrak{M}^{\mathrm{an}}_U \arrow[d, hook, "r_U"] & \mathfrak{M}^{\mathrm{an}}_U \times_{\mathfrak{M}^{\mathrm{an}}_{T^*X}} \mathfrak{M}^{\mathrm{nil,an}}_{T^*X} \arrow[l, "\mathrm{pr}_{1,U}"'] \arrow[r, "\mathrm{d}i_U"] \arrow[d, hook, "\widetilde{r}_U"] & \mathfrak{M}^{\mathrm{nil,an}}_U \arrow[d, hook, "r^{\mathrm{nil}}_U"] \\
\mathbf{T}^*[-1]\mathfrak{M}^{\mathrm{an}}_{T^*X} \arrow[d, "\pi_{T^*X}"] & \mathbf{T}^*[-1]\mathfrak{M}^{\mathrm{an}}_{T^*X} \times_{\mathfrak{M}^{\mathrm{an}}_{T^*X}} \mathfrak{M}^{\mathrm{nil,an}}_{T^*X} \arrow[l, "\mathrm{pr}_1"'] \arrow[r, "\mathrm{d}i"] \arrow[d, "\mathrm{pr}_2"] & \mathbf{T}^*[-1]\mathfrak{M}^{\mathrm{nil,an}}_{T^*X} \arrow[dl, "\pi^{\mathrm{nil}}_{T^*X}"] \\
\mathfrak{M}^{\mathrm{an}}_{T^*X} & \mathfrak{M}^{\mathrm{nil,an}}_{T^*X} \arrow[l, hook', "i"'] &
\end{tikzcd} \tag{4.3.2}$$

We define the U-local DT-sheaf φ^{nil}_U on $\mathfrak{M}^{\mathrm{nil,an}}_U$ to be $\left(r^{\mathrm{nil}}_U\right)^* \varphi^{\mathrm{nil}}$, then by above diagram 4.3.2

$$\varphi^{\mathrm{nil}}_U \cong \left(r^{\mathrm{nil}}_U\right)^* (\mathrm{d}i)_* \left(\mathrm{pr}_1\right)^! (\varphi) \cong (\mathrm{d}i_U)_* (\widetilde{r}_U)^! (\mathrm{pr}_1)^! \varphi \cong (\mathrm{d}i_U)_* (\mathrm{pr}_{1,U})^! \varphi_U$$

The unit $\mathrm{Id} \to \left(r^{\mathrm{nil}}_U\right)_* \left(r^{\mathrm{nil}}_U\right)^*$ induces a map $R\Gamma\left(\mathbf{T}^*[-1]\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, \varphi^{\mathrm{nil}}\right) \longrightarrow R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_U, \varphi^{\mathrm{nil}}_U\right)$, we claim it is an isomorphism. In fact, by Theorem 4.6(2), we can deduce the unit $\mathrm{Id} \to (r_U)_*(r_U)^*$ induces

$$R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{T^*X\times\mathbb{A}^1}, (r_U)_*\varphi_U\right) = R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{T^*X\times\mathbb{A}^1}, (r_U)_*(r_U)^*\varphi\right) \cong R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{T^*X\times\mathbb{A}^1}, \varphi\right)$$

On the side of the global nilpotent cone,

$$\begin{aligned}
&R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, (\pi^{\mathrm{nil}}_{T^*X})_*(r^{\mathrm{nil}}_U)_*\left(\varphi^{\mathrm{nil}}_U\right)\right) = R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, (\pi^{\mathrm{nil}}_{T^*X})_*(r^{\mathrm{nil}}_U)_*(r^{\mathrm{nil}}_U)^*(\mathrm{d}i)_*(\mathrm{pr}_1)^!\left(\varphi\right)\right)\\
=&R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, (\pi^{\mathrm{nil}}_{T^*X})_*(r^{\mathrm{nil}}_U)_*(\mathrm{d}i_U)_*(\widetilde{r}_U)^*(\mathrm{pr}_1)^!\left(\varphi\right)\right) = R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, (\pi^{\mathrm{nil}}_{T^*X})_*(\mathrm{d}i)_*(\widetilde{r}_U)_*(\widetilde{r}_U)^*(\mathrm{pr}_1)^!\left(\varphi\right)\right)\\
=&R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, (\pi^{\mathrm{nil}}_{T^*X})_*(\mathrm{d}i)_*(\widetilde{r}_U)_*(\mathrm{pr}_{1,U})^!(r_U)^*\left(\varphi\right)\right) = R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, (\pi^{\mathrm{nil}}_{T^*X})_*(\mathrm{d}i)_*(\mathrm{pr}_1)^!(r_U)_*(r_U)^*\left(\varphi\right)\right)\\
=&R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, (\pi^{\mathrm{nil}}_{T^*X})_*(\mathrm{d}i)_*(\mathrm{pr}_1)^!\left(\varphi\right)\right) = R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}, i^!(\pi_{\mathfrak{M}^{\mathrm{an}}_{T^*X}})_*\varphi\right) \textbf{[shift]}
\end{aligned}$$

which is the Borel-Moore complex on $\mathfrak{M}^{\mathrm{nil,an}}_{T^*X}$ according to Lemma 4.7, the claim is proved. In particular, taking the cohomology of the complex we recover $H^{\mathrm{BM}}_{\bullet}(\mathbf{\Lambda})$.

Proposition 4.8. *The nilpotent CoHA $H^{\mathrm{BM}}_{\bullet}(\mathbf{\Lambda})$ admits a coproduct and the algebra morphism $H^{\mathrm{BM}}_{\bullet}(\mathbf{\Lambda}) \longrightarrow H^{\mathrm{BM}}_{\bullet}(\mathbf{Higgs})$ is also a co-algebra morphism. Moreover, the coproduct on $H^{\mathrm{BM}}_{\bullet}(\mathbf{\Lambda})$ preserves the top homology, which makes $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$ also a co-algebra.*

Proof. The same argument for $\mathfrak{M}^{\mathrm{an}}_U$ also works for $\mathfrak{M}^{\mathrm{nil,an}}_U$ and φ^{nil}_U, namely we have $\mathfrak{M}^{\mathrm{nil,an}}_{U_1\sqcup...\sqcup U_s} \cong \mathfrak{M}^{\mathrm{nil,an}}_{U_1}\times ...\times\mathfrak{M}^{\mathrm{nil,an}}_{U_s}$ and $\varphi^{\mathrm{nil}}_{U_1\sqcup...\sqcup U_s} \cong \varphi^{\mathrm{nil}}_{U_1}\boxtimes...\boxtimes\varphi^{\mathrm{nil}}_{U_s}$. The coproduct is thus induced by the factorization structure of φ^{nil} and the above diagram shows the compatibility between the factorization structure on φ and φ^{nil}. Finally, for $U_1 \sqcup U_2 \subseteq U$, the map $R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_U \varphi^{\mathrm{nil}}_U\right) \to R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{U_1} \varphi^{\mathrm{nil}}_{U_1}\right) \otimes R\Gamma\left(\mathfrak{M}^{\mathrm{nil,an}}_{U_2}, \varphi^{\mathrm{nil}}_{U_2}\right)$ is of cohomological degree 0, thus preserves the induced top homology. ■

Proposition 4.9. *There is a cohomological version of Langlands formula for all algebras $H^{\mathrm{BM}}_{\bullet}(\mathbf{Higgs})$, $H^{\mathrm{BM}}_{\bullet}(\mathbf{\Lambda})$ and $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$ above, which integrate the products and coproducts into bialgebra structures.*

Proof. We consider the case of $H^{\mathrm{BM}}_{\bullet}(\mathbf{Higgs})$. Pick $U_1, U_2, U \subseteq \mathbb{C}$ open convex subsets with $U_1 \cup U_2 \subseteq U$, unwinding the definition, we want to prove the commutativity of the following diagram:

$$\begin{tikzcd}
R\Gamma\left(\mathfrak{M}_U^{\mathrm{an}},\varphi_U\right)\otimes R\Gamma\left(\mathfrak{M}_U^{\mathrm{an}},\varphi_U\right) \arrow[rrr, "\mathsf{m}_U"] \arrow[d, "\Delta\otimes\Delta"] & & & R\Gamma\left(\mathfrak{M}_U^{\mathrm{an}},\varphi_U\right) \arrow[d, "\Delta"] \\
\begin{matrix} R\Gamma\left(\mathfrak{M}_{U_1}^{\mathrm{an}},\varphi_{U_1}\right)\otimes R\Gamma\left(\mathfrak{M}_{U_2}^{\mathrm{an}},\varphi_{U_2}\right) \\ \otimes \\ R\Gamma\left(\mathfrak{M}_{U_1}^{\mathrm{an}},\varphi_{U_1}\right)\otimes R\Gamma\left(\mathfrak{M}_{U_2}^{\mathrm{an}},\varphi_{U_2}\right)\end{matrix} \arrow[r, "\mathrm{id}_{U_1}\circ\mathrm{swap}\circ\mathrm{id}_{U_2}"] & \begin{matrix} R\Gamma\left(\mathfrak{M}_{U_1}^{\mathrm{an}},\varphi_{U_1}\right)\otimes R\Gamma\left(\mathfrak{M}_{U_1}^{\mathrm{an}},\varphi_{U_1}\right) \\ \otimes \\ R\Gamma\left(\mathfrak{M}_{U_2}^{\mathrm{an}},\varphi_{U_2}\right)\otimes R\Gamma\left(\mathfrak{M}_{U_2}^{\mathrm{an}},\varphi_{U_2}\right)\end{matrix} \arrow[rr, "\mathsf{m}_{U_1}\otimes\mathsf{m}_{U_2}"] & & R\Gamma\left(\mathfrak{M}_{U_1}^{\mathrm{an}},\varphi_{U_1}\right)\otimes R\Gamma\left(\mathfrak{M}_{U_2}^{\mathrm{an}},\varphi_{U_2}\right)
\end{tikzcd}$$

It reveals that $\Delta\circ\mathsf{m}_U = \left(\mathsf{m}_{U_1}\otimes\mathsf{m}_{U_2}\right)\circ\left(\mathrm{id}_{U_1}\circ\mathrm{swap}\circ\mathrm{id}_{U_2}\right)\circ(\Delta\otimes\Delta)$ upto a sign, see [Dav+, Proposition 6.7] for details. ■

4.4 Characteristic cycles on stacks

In the previous section, we have construct the following bialgebras

$$H_{\mathrm{top}}^{\mathrm{BM}}(\mathbf{\Lambda})\longrightarrow H_{\bullet}^{\mathrm{BM}}(\mathbf{\Lambda})\longrightarrow H_{\bullet}^{\mathrm{BM}}(\mathbf{Higgs})$$

where the first arrow is a sub-bialgebra and the second one is a bialgebra morphism. On the other hand, similar to positive characteristic case, we can construct a bialgebra $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right)$ from Eisenstein sheaves. We are going to relate them in considering the characteristic cycle of Eisenstein sheaves.

4.4.1 *Characteristic cycles.* Let first consider X a smooth scheme, with cotangent bundle T^*X. We say a Zariski closed subset $C\subseteq T^*X$ is a conical subset if it is invariant under $\mathbb{G}_m$-action. Let $f: W\longrightarrow X$ be a morphism of smooth schemes and $C\subseteq T^*X$ be a conical subset, we say f is C-transversal (resp. at a point w) if $(W\times_X C)\cap df^{-1}\left(T_W^*W\right)$ is included in the zero section $W\times_X T_X^*X\subseteq W\times_X T^*X$ (resp. the intersection with point w: $\left((W\times_X C)\cap df^{-1}\left(T_W^*W\right)\right)\times_W w$ is included in $W\times_X T_X^*X$).
In particular, when f is C-transversal, the image $df\left((W\times_X C)\cap df^{-1}\left(T_W^*W\right)\right)$ is a conical subset in T^*W. Likewise, for $h: X\longrightarrow Y$ a morphism from smooth scheme X and $C\subseteq T^*X$ conical subset, we say h is C-transversal if $df^{-1}(C)\subseteq X\times_Y T_Y^*Y$. The pair of morphisms $X\xleftarrow{f}W\xrightarrow{h}Y$ with X,W smooth is said to be C-transversal if f is C-transversal and h is $df\left((W\times_X C)\cap df^{-1}\left(T_W^*W\right)\right)$-transversal. We adapt to the following result of Saito as definition of characteristic cycle:

Theorem 4.10. *For $\mathcal{F}\in\mathsf{Shv}_{\mathrm{constr}}(X)$ over smooth irreducible scheme X and conical Lagrangian subset $C\subseteq T^*X$ there exists $\mathbb{Z}$-linear combination $\mathrm{CC}_C(\mathcal{F})=\sum_a m_a[C_a]$ over irreducible components of C such that for any C-transversal pair $X\xleftarrow{f}W\xrightarrow{h}Y$ that h is étale and Y a smooth curve, we have the following Milnor formula*

$$-\dim\Phi_u\left(f^*\mathcal{F},h\right)=\left(\mathrm{CC}_C(\mathcal{F}),dh\right)_{T^*W,u}$$

for every at most C-isolated point $u\in W$. Moreover $\mathrm{CC}_C(\mathcal{F})$ is independent of conical subset $\mathcal{F}$ has micro-support on, we thus simplify as $\mathrm{CC}(\mathcal{F})$, The support $\mathrm{supp}\left(\mathrm{CC}(\mathcal{F})\right)=\bigcup_{m_a\neq 0}C_a$ is exactly $\mathrm{SS}(\mathcal{F})$.

Lemma 4.11. *Let $j: X_0\hookrightarrow X_1$ be an open immersion between smooth irreducible schemes and let $\mathcal{F}\in$*

$\mathsf{Shv}_{\mathrm{constr}}(X_0)$ *with characteristic cycle* $\mathrm{CC}(\mathcal{F}) = \sum_a m_a[\Lambda_a^0]$, *then*

$$\mathrm{CC}\left((j)_{!*}(\mathcal{F})\right) = \sum_a m_a[\Lambda_a^1], \text{ where } \Lambda_a^1 \times_{X_1} X_0 = \Lambda_a^0$$

In other words, $(dj)^{-1}(\mathrm{CC}(\mathcal{F})) \cong \mathrm{CC}\left((j)_{!*}(\mathcal{F})\right)$.

4.4.2 Equivariant and inductive characteristic cycles. We next consider the characteristic cycles for global quotient stack of the form $\left[\mathcal{X}/G\right]$, where $\mathcal{X}$ is a smooth scheme over $\mathbf{k}$ and G is a reductive group over $\mathbf{k}$. Suppose we have diagram $\mathcal{Y} \xleftarrow{q} \mathcal{V} \xrightarrow{p} \mathcal{X}$ where $\mathcal{Y}, \mathcal{V}$ admits action of $P \subseteq G$ parabolic subgroup, $\mathcal{X}$ admits an action of G and that q is P-equivariant smooth map, p is closed P-equivariant. Then we have naturally the diagram $\mathcal{Y} \overset{P}{\times} G \longleftarrow \mathcal{V} \overset{P}{\times} G \longrightarrow \mathcal{X}$. Quotienting G of the diagram, we get $[\mathcal{Y}/P] \xleftarrow{\bar{q}} [\mathcal{V}/P] \xrightarrow{\bar{p}} [\mathcal{X}/G]$, compose $\bar{q}$ with $U(P)$-gerbe $[\mathcal{Y}/P] \to [\mathcal{Y}/L]$, where L is the Levi subgroup of P and $U(P)$ is the unipotent radical of P, we get a correspondence diagram:

$$[\mathcal{Y}/L] \xleftarrow{\tilde{q}} [\mathcal{V}/P] \xrightarrow{\bar{p}} [\mathcal{X}/G]$$

where $\tilde{q}$ is smooth and $\bar{p}$ is proper, we denote $\mathrm{Ind}_{[\mathcal{Y}/L]}^{[\mathcal{X}/G]} := (\bar{p})_!(\tilde{q})^*[\mathbf{shift}] : \mathrm{D}([\mathcal{Y}/L]) \longrightarrow \mathrm{D}([\mathcal{X}/G])$, the equivariant induction, where the shift is given by the relative dimension $\dim[\mathcal{V}/P] - \dim[\mathcal{Y}/L]$. The $\mathcal{D}$-modules or constructible sheaves on $[\mathcal{Y}/L]$ are L-equivariant ones on Y. We define the equivariant characteristic cycle on $[\mathcal{Y}/L]$ to be the same as the characteristic cycle of the sheaf over underlying Y. In other words, put $\mathrm{Fgt} : \mathrm{D}([\mathcal{Y}/L]) \longrightarrow \mathrm{D}(\mathcal{Y})$, then $\mathrm{CC}(\mathcal{F}) := \mathrm{CC}\left(\mathrm{Fgt}(\mathcal{F})\right)$ for $\mathcal{F} \in \mathrm{D}([\mathcal{Y}/L])$.

On the other hand, we have the following (equivariant) induction of cotangent spaces: assume we have a correspondence $\mathcal{Y}' \xleftarrow{f} \mathcal{V}' \xrightarrow{g'} \mathcal{X}'$ such that f is smooth, g is proper and $\mathcal{V}' \xrightarrow{(f,g)} \mathcal{Y}' \times \mathcal{X}'$ is closed. Then the conormal bundle $T^*_{\mathcal{V}'}(\mathcal{X}' \times \mathcal{Y}')$ fits into the following diagram:

$$\begin{tikzcd}
T^*_{\mathcal{V}'}(\mathcal{X}' \times \mathcal{Y}') \arrow[r, "\psi'"'] \arrow[d, "\phi'"] \arrow[rr, bend left, "\psi"] \arrow[dd, bend right, "\phi"'] & T^*\mathcal{X}' \times_{\mathcal{X}'} \mathcal{V}' \arrow[r, "\mathrm{pr}^{\mathcal{X}'}"'] \arrow[d, "\mathrm{d}g"] & T^*\mathcal{X}' \\
T^*\mathcal{Y}' \times_{\mathcal{Y}'} \mathcal{V}' \arrow[r, "\mathrm{d}f"'] \arrow[d, "\mathrm{pr}^{\mathcal{Y}'}"] & T^*\mathcal{V}' & \\
T^*\mathcal{Y}' & &
\end{tikzcd} \tag{4.4.1}$$

where $\phi = \mathrm{pr}^{\mathcal{Y}'} \circ \phi'$ is the natural projection which is l.c.i. and $\psi = \mathrm{pr}^{\mathcal{X}'} \circ \psi'$ is also natural projection which is proper. Moreover, [Hen24, Lemma 3.3] shows that both ϕ' and $\mathrm{d}g$ are l.c.i. For $\Lambda \subseteq T^*\mathcal{Y}'$ and $\Lambda' \subseteq T^*\mathcal{X}'$ such that $\psi\left(\phi^{-1}(\Lambda)\right) \subseteq \Lambda'$, then $(\psi)_*(\phi)^!$ induces a map $H_\bullet^{\mathrm{BM}}(\Lambda) \longrightarrow H^{\mathrm{BM}}_{\bullet+2(\dim \mathcal{X}' - \dim \mathcal{Y}')}(\Lambda')$. Apply the above model to the correspondence diagram $\mathcal{Y} \xleftarrow{q} \mathcal{V} \xrightarrow{p} \mathcal{X}$ as above, put $\mathcal{Y}' = \mathcal{Y} \overset{P}{\times} G$ and suppose $\Lambda_0 \subseteq T^*\mathcal{Y}$ which induces $\Lambda = \Lambda \overset{P}{\times} G \subseteq T^*\mathcal{Y}'$ such that $\psi\left(\phi^{-1}(\Lambda)\right) \subseteq \Lambda'$, we get a map

$$\mathrm{Ind}_{[\mathcal{Y}/L]}^{[\mathcal{X}/G]} : H^{\mathrm{BM}}_{2\dim \mathcal{Y}}(\Lambda) \longrightarrow H^{\mathrm{BM}}_{2\dim \mathcal{X}'}(\Lambda')$$

There is compatibility between the above cohomological induction and characteristic cycle map:

Proposition 4.12. *[Hen24, Theorem 6.3] The cohomological induction is compatible with equivariant characteristic cycle, namely*

$$\mathrm{CC}\left(\mathrm{Ind}_{[\mathcal{Y}/L]}^{[\mathcal{X}/G]}(\mathcal{F})\right) \cong \mathrm{Ind}_{[\mathcal{Y}/L]}^{[\mathcal{X}/G]}(\mathrm{CC}(\mathcal{F}))$$

We can generalize the above result to stacks that locally as quotient stack. For $j : Y_0 \hookrightarrow Y_1$ is G-equivariant open embedding of smooth irreducible schemes which induces $\bar{j} : [Y_0/G] \hookrightarrow [Y_1/G]$, then Lemma 4.11 gives $(dj)^{-1}(\mathrm{CC}(\mathcal{F})) \cong \mathrm{CC}\left((\bar{j})_{!*}(\mathcal{F})\right)$. Now we assume that the stack $\mathcal{X}$ is irreducible, locally of finite type and can be written as colimit of quotient stacks $\mathcal{X} := \mathrm{colim}_i[\mathcal{X}_i/G_i]$, such that the colimit is taken along the arrows $[\mathcal{X}_i/G_i] \hookrightarrow [\mathcal{X}_j/G_j]$ whenever $[\mathcal{X}_j/G_j] \cong [\mathcal{X}_j^i/G_i]$ where $\mathcal{X}_i \hookrightarrow \mathcal{X}_j^i$. Then, for $\mathcal{F} := \mathrm{colim}_i \mathcal{F}_i$ as a colimit of intermediate extension along open inclusions as above, we can define the following *inductive characteristic cycle* of $\mathcal{F}$ on $\mathcal{X}$:

$$\mathrm{CC}^{\mathsf{ind}}(\mathcal{F}) := \sum_a m_a[\Lambda_a], \quad \text{where } \Lambda_a \text{ runs over } \mathrm{Lagr}(T^*\mathcal{X})$$

In fact, under the above assumption, we can restrict to any of $\mathcal{F}_i$, then $\mathrm{CC}(\mathcal{F}_i)$ is of the form $\sum_a m_a^i[\Lambda_a^i]$, where $\Lambda_a^i = \Lambda_a \times_{\mathcal{X}} [\mathcal{X}_i/G_i]$ is the restriction of the irreducible Lagrangian Λ_a.

Remark 4.13. For varieties over $\mathbb{C}$, Saito's construction of characteristic cycles matches with the definition in the litterature, also it works for ℓ-adic setting. Since the definition of CC only depends on local behaviour, one can check if we have a smooth covering of stack $\mathcal{X}$ by compatible system of global quotient as above, our construction of $\mathrm{CC}^{\mathsf{ind}}$ for $\mathcal{X}$ is well defined for both ℓ-adic setting of $\mathcal{X}$ in positive characteristic and $\mathcal{D}$-mod setting for $\mathcal{X}$ over $\mathbb{C}$.

We then consider the compatibility between $\mathsf{Eis}^{\mathrm{Coh}}$ and $\mathrm{CC}^{\mathsf{ind}}$. As $\mathbf{Coh}_\alpha$ can be exhibited as colimit $\mathrm{colim}_{\mathcal{L}}\, \mathbf{Coh}_\alpha^{>\mathcal{L}}$. For two classes α and β, we can consider $\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}$, which is isomorphic to $\left[\mathsf{Q}_\alpha^{\mathcal{L}}/\mathsf{G}_\alpha^{\mathcal{L}}\right] \times \left[\mathsf{Q}_\beta^{\mathcal{L}}/\mathsf{G}_\beta^{\mathcal{L}}\right]$. We consider $\mathsf{Eis}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'} = \left(\mathsf{p}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}\right)_! \left(\mathsf{q}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}\right)^*$, then by Proposition 4.12, we have:

$$\mathrm{CC}\left(\mathsf{Eis}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}(\mathcal{F})\right) \cong \mathrm{Ind}_{\left[\mathsf{Q}_\alpha^{\mathcal{L}}\times\mathsf{Q}_\beta^{\mathcal{L}}/\mathsf{G}_\alpha^{\mathcal{L}}\times\mathsf{G}_\beta^{\mathcal{L}}\right]}^{\left[\mathsf{Q}_{\alpha+\beta}^{\mathcal{L},\mathcal{L}'}/\mathsf{G}_{\alpha+\beta}^{\mathcal{L}}\right]}(\mathrm{CC}(\mathcal{F}))$$

We then move to whole stack $\mathbf{Coh}_\alpha \times \mathbf{Coh}_\beta$, we can conclude:

Proposition 4.14. *The inductive characteristic cycle $\mathrm{CC}^{\mathsf{ind}}$ induces a map $K_0\left(\mathsf{Q}_\alpha^{\mathrm{Coh}}\right) \longrightarrow H_{\mathrm{top}}^{\mathrm{BM}}(\mathbf{\Lambda}_\alpha)$, which commutes with $\mathsf{Eis}^{\mathrm{Coh}}$, namely we have the following commutative diagram:*

$$\begin{tikzcd} K_0(\mathsf{Q}_\alpha^{\mathrm{Coh}})\hat{\otimes}K_0(\mathsf{Q}_\beta^{\mathrm{Coh}}) \arrow[r, "\mathrm{CC}^{\mathsf{ind}}"] \arrow[d, "\mathsf{Eis}_{\alpha,\beta}^{\mathrm{Coh}}"'] & H_{\mathrm{top}}^{\mathrm{BM}}\left(\mathbf{\Lambda}_\alpha \times \mathbf{\Lambda}_\beta\right) \arrow[d, "\mathsf{Eis}_{\alpha,\beta}^{\Lambda}"] \\ K_0(\mathsf{Q}_{\alpha+\beta}^{\mathrm{Coh}}) \arrow[r, "\mathrm{CC}^{\mathsf{ind}}"] & H_{\mathrm{top}}^{\mathrm{BM}}\left(\mathbf{\Lambda}_{\alpha+\beta}\right) \end{tikzcd} \tag{4.4.2}$$

In other words, $\mathrm{CC}^{\mathsf{ind}}$ is an algebra morphism between $K_0(\mathsf{Q}^{\mathrm{Coh}})$ and $H_{\mathrm{top}}^{\mathrm{BM}}(\mathbf{\Lambda})$.

Proof. First, notice that the objects in $\mathsf{Q}^{\mathrm{Coh}}$ has nilpotent singular support, then for $(\mathcal{F}_1, \mathcal{F}_2) \in \mathsf{Q}_\alpha^{\mathrm{Coh}} \times \mathsf{Q}_\beta^{\mathrm{Coh}}$, the restricted singular support $\mathrm{SS}\left((\mathcal{F}_1 \boxtimes \mathcal{F}_2)|_{\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}}\right) \subseteq \mathbf{\Lambda}_\alpha^{>\mathcal{L}} \times \mathbf{\Lambda}_\beta^{>\mathcal{L}}$. Thus the restriction of cohomological induction $\mathrm{Ind}_{\left[\mathsf{Q}_\alpha^{\mathcal{L}} \times \mathsf{Q}_\beta^{\mathcal{L}} / \mathsf{G}_\alpha^{\mathcal{L}} \times \mathsf{G}_\beta^{\mathcal{L}}\right]}^{\left[\mathsf{Q}_{\alpha+\beta}^{\mathcal{L},\mathcal{L}'} / \mathsf{G}_{\alpha+\beta}^{\mathcal{L}}\right]}$ to $\mathrm{CC}\left((\mathcal{F}_1 \boxtimes \mathcal{F}_2)|_{\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}}\right)$ coincides with

$$\mathsf{m}_{\alpha,\beta}^{\mathcal{L},\mathcal{L}'} : H_{\mathrm{top}}^{\mathrm{BM}}\left(\mathbf{\Lambda}_\alpha^{>\mathcal{L}} \times \mathbf{\Lambda}_\beta^{>\mathcal{L}}\right) \longrightarrow H_{\mathrm{top}}^{\mathrm{BM}}\left(\mathbf{\Lambda}_{\alpha+\beta}^{>\mathcal{L}'}\right)$$

The image of $\mathcal{F}_1 \boxtimes \mathcal{F}_2$ through the lower circuit of the diagram is given by

$$\begin{aligned}
&\mathrm{CC}^{\mathrm{ind}}\left(\mathrm{colim}_{\mathcal{L}'} \mathsf{Eis}_{\alpha,\beta}^{\mathrm{Coh}}(\mathcal{F}_1 \boxtimes \mathcal{F}_2)|_{\mathbf{Coh}_{\alpha+\beta}^{>\mathcal{L}'}}\right) = \mathrm{CC}^{\mathrm{ind}}\left(\mathrm{colim}_{\mathcal{L}'}\left(\mathrm{colim}_{\mathcal{L}} \mathsf{Eis}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}\left((\mathcal{F}_1 \boxtimes \mathcal{F}_2)|_{\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}}\right)\right)\right) \\
&= \mathrm{colim}_{\mathcal{L}'}\left(\mathrm{CC}^{\mathrm{ind}}\left(\mathrm{colim}_{\mathcal{L}}\left(\mathsf{Eis}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}\left((\mathcal{F}_1 \boxtimes \mathcal{F}_2)|_{\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}}\right)\right)\right)\right) \\
&= \mathrm{colim}_{\mathcal{L},\mathcal{L}'}\left(\mathrm{CC}\left(\mathsf{Eis}_{\alpha,\beta}^{\mathrm{Coh},\mathcal{L},\mathcal{L}'}\left((\mathcal{F}_1 \boxtimes \mathcal{F}_2)|_{\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}}\right)\right)\right) = \mathrm{colim}_{\mathcal{L},\mathcal{L}'}\left(\mathsf{m}_{\alpha,\beta}^{\mathcal{L},\mathcal{L}'}\left(\mathrm{CC}\left((\mathcal{F}_1 \boxtimes \mathcal{F}_2)|_{\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}}\right)\right)\right) \\
&= \mathrm{colim}_{\mathcal{L}'}\left(\mathsf{m}_{\alpha,\beta}^{\mathcal{L}'}\left(\mathrm{colim}_{\mathcal{L}}\left(\mathrm{CC}\left((\mathcal{F}_1 \boxtimes \mathcal{F}_2)|_{\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}}\right)\right)\right)\right) = \mathrm{colim}_{\mathcal{L}'}\left(\mathsf{m}_{\alpha,\beta}^{\mathcal{L}'}\left(\mathrm{CC}^{\mathrm{ind}}(\mathcal{F}_1 \boxtimes \mathcal{F}_2)\right)\right) \\
&= \mathsf{Eis}_{\alpha,\beta}^{\Lambda}\left(\mathrm{CC}^{\mathrm{ind}}(\mathcal{F}_1 \boxtimes \mathcal{F}_2)\right).
\end{aligned}$$

which matches with the upper circuit, as we want. ■

Moreover, it reveals that $\mathrm{CC}^{\mathrm{ind}}$ is not only an algebra morphism, it is furthur an isomorphism:

Theorem 4.15. *The inductive characteristic cycle $\mathrm{CC}^{\mathrm{ind}}$ is a co-algebra morphism from $K_0(\mathsf{Q}^{\mathrm{Coh}})$ to $H_\bullet^{\mathrm{BM}}(\mathbf{Higgs})$, namely we have the following commutative diagram:*

$$\begin{tikzcd}
K_0(\mathsf{Q}_\alpha^{\mathrm{Coh}}) \arrow[r, "\mathrm{CC}^{\mathrm{ind}}"] \arrow[d, "\mathsf{CT}_{\beta,\gamma}^{\mathrm{Coh}}"'] & H_\bullet^{\mathrm{BM}}(\mathbf{Higgs}_\alpha) \arrow[d, "\Delta_{\beta,\gamma}"] \\
K_0(\mathsf{Q}_\beta^{\mathrm{Coh}}) \hat{\otimes} K_0(\mathsf{Q}_\gamma^{\mathrm{Coh}}) \arrow[r, "\mathrm{CC}^{\mathrm{ind}}"] & H_\bullet^{\mathrm{BM}}\left(\mathbf{Higgs}_\beta \times \mathbf{Higgs}_\gamma\right)
\end{tikzcd} \tag{4.4.3}$$

In particular, $\mathrm{CC}^{\mathrm{ind}}$ induces an isomorphism of bialgebras $K_0(\mathsf{Q}^{\mathrm{Coh}}) \cong H_{\mathrm{top}}^{\mathrm{BM}}(\mathbf{\Lambda})$.

4.5 Proof of Theorem 4.15

The proof of Theorem 4.15 will occupy this section. First, analogous to Theorem 3.31, we can deduce that $K_0\left(\mathsf{Q}_\alpha^{\mathrm{Coh}}\right)$ is spanned by the classes of $\mathsf{Eis}_{\underline{\alpha}}^{\mathrm{Coh}}\left(\mathbf{1}_{\underline{\alpha}}\right)$ for all partitions $\underline{\alpha}$ (not necessarily principal). It is enough to show for any simple $\mathbf{P} \in \mathsf{P}_\alpha^{\mathrm{Coh}}$, the corresponding $[\mathbf{P}] \in K_0\left(\mathsf{Q}_\alpha^{\mathrm{Coh}}\right)$ is in span $\left(\left[\mathsf{Eis}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right)\right]\right)$. As in the proof of Theorem 3.31, we divide into two cases: when $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{Coh}_\alpha^{\mathrm{as}} = \emptyset$, we can use the induction hypothesis to conclude. If $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{Coh}_\alpha^{\mathrm{as}} \neq \emptyset$, we can find $\underline{\beta} \in \mathrm{HN}(\alpha)$ such that $\mathrm{supp}(\mathbf{P}) \bigcap \mathbf{HN}_{\underline{\beta}} \neq \emptyset$ and $\underline{\alpha}$ some principal partition of α such that $\mathbf{P}$ appears in principal series $\mathsf{Eis}_{\underline{\alpha}}^{\mathrm{Coh}}\left(\mathbf{1}_{\underline{\alpha}}\right)$. We then conclude with the Langlands formula and the induction hypothesis applied to $\mathsf{CT}_{\underline{\beta}}^{\mathrm{Coh}}$.

Let $[\mathbf{Coh}_\alpha]$ be the fundamental class in $H_{\mathrm{top}}^{\mathrm{BM}}(\mathbf{Coh}_\alpha)$. As there is an embedding $h_\alpha : \mathbf{Coh}_\alpha \hookrightarrow \mathbf{\Lambda}_\alpha$ as an irreducible component $\mathbf{\Lambda}_{(\alpha)}$, we can identify $(h_\alpha)_*[\mathbf{Coh}_\alpha]$ as $[\mathbf{\Lambda}_{(\alpha)}] \in H_{\mathrm{top}}^{\mathrm{BM}}(\mathbf{\Lambda}_\alpha)$. Also, consider the closed embedding $i : \mathfrak{M}^{\mathrm{nil}}_{T^*X} \hookrightarrow \mathfrak{M}_{T^*X}$, whose classical truncation induces $i_\alpha : \mathbf{\Lambda}_\alpha \hookrightarrow \mathbf{Higgs}_\alpha$, we write also $\left[\mathbf{\Lambda}_{(\alpha)}\right]$ the cycle class corresponding to the zero section $(i_\alpha \circ h_\alpha)_* [\mathbf{Coh}_\alpha] \in H^{\mathrm{BM}}_{\mathrm{mid}}\left(\mathbf{Higgs}_\alpha\right)$.

Proposition 4.16. *On the side of $H^{\mathrm{BM}}_\bullet\left(\mathbf{Higgs}_\alpha\right)$, we have $\Delta_{\beta,\gamma}\left(\left[\mathbf{\Lambda}_{(\alpha)}\right]\right) = \left[\mathbf{\Lambda}_{(\beta)}\right] \otimes \left[\mathbf{\Lambda}_{(\gamma)}\right]$.*

Proof. The coproduct on $H^{\mathrm{BM}}_\bullet\left(\mathbf{Higgs}\right)$ is given by the cofactorization of $\boldsymbol{\varphi}$ on $T^*[-1]\mathfrak{M}_{T^*X} \cong \mathfrak{M}_{T^*X\times\mathbb{A}^1}$. On the other hand, denote by $\mathfrak{M}_{X\times\mathbb{A}^1}$ the derived moduli stack of coherent sheaves with proper support on $X\times\mathbb{A}^1$, then we have $T^*[-1]\mathfrak{M}_{X\times\mathbb{A}^1} \cong \mathfrak{M}_{T^*X\times\mathbb{A}^1}$, so pushing forward along $\pi_{X\times\mathbb{A}^1} : \mathfrak{M}_{T^*X\times\mathbb{A}^1} \to \mathfrak{M}_{X\times\mathbb{A}^1}$ gives $(\pi_{X\times\mathbb{A}^1})_*\left(\boldsymbol{\varphi}\right) \cong \mathbb{DC}_{\mathfrak{M}_{X\times\mathbb{A}^1}}\left[\mathrm{vdim}\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}\right)\right]$. Both $\mathfrak{M}_{T^*X}$ and $\mathfrak{M}_{X\times\mathbb{A}^1}$ map down to $\mathbf{Coh}(X)$, we have

$$\begin{tikzcd}
 & \mathfrak{M}_{T^*X\times\mathbb{A}^1} \arrow[dl, "\pi_{X\times\mathbb{A}^1}"'] \arrow[dr, "\pi_{T^*X}"] & \\
\mathfrak{M}_{X\times\mathbb{A}^1} \arrow[dr, "\mathrm{pr}^\vee[-1]"'] & & \mathfrak{M}_{T^*X} \arrow[dl, "\mathrm{pr}"] \\
 & \mathbf{Coh}(X) &
\end{tikzcd} \tag{4.5.1}$$

In fact, a point in $\mathfrak{M}_{T^*X\times\mathbb{A}^1}$ classifies $(\mathcal{F},\varphi,\phi)$, where $(\mathcal{F},\varphi)$ is a point in $\mathfrak{M}_{T^*X}$, which corresponds to a Higgs sheaf and $\phi : (\mathcal{F},\varphi) \to (\mathcal{F},\varphi)$ is an endomorphism between Higgs sheaves. The map $\pi_{T^*X} : \mathfrak{M}_{T^*X\times\mathbb{A}^1} \to \mathfrak{M}_{T^*X}$ sends $(\mathcal{F},\varphi,\phi) \mapsto (\mathcal{F},\varphi)$ while $\pi_{X\times\mathbb{A}^1} : (\mathcal{F},\varphi,\phi) \mapsto (\mathcal{F},\phi)$ forgets the Higgs field. While a point in $\mathfrak{M}_{X\times\mathbb{A}^1}$ classifies $(\mathcal{F},\psi)$ where ψ is an endomorphism of coherent sheaves $\mathcal{F}$ and $\mathfrak{M}_{X\times\mathbb{A}^1} \to \mathbf{Coh}$ is the forgetful map, we write this map as $\mathrm{pr}^\vee[-1]$ since the Serre duality shows:

$$\mathcal{H}om\left(\mathcal{E},\mathcal{E}\otimes\omega_X\right) \cong \mathcal{H}om\left(\mathcal{E},\mathcal{E}\right)^\vee[-1], \qquad \forall\mathcal{E}\in\mathbf{Coh}$$

Thus $\mathfrak{M}_{X\times\mathbb{A}^1} \cong \mathbf{T}[-1]\mathbf{Coh}$ is (-1)-shifted dual vector bundle of $\mathfrak{M}_{T^*X}$ over $\mathbf{Coh}$. The irreducible components of $\mathfrak{M}_{X\times\mathbb{A}^1}$ are thus also parametrized by Euler classes of X and $\mathrm{vdim}\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\alpha}\right) = 0, \forall\alpha\in\mathbf{Z}^+$. The diagram 4.5.1 can be combined with diagram 4.3.1 and we get the following complete diagram:

$$\begin{tikzcd}
\mathfrak{M}_{X\times\mathbb{A}^1} \arrow[r, "\widetilde{h}"] \arrow[d, "\mathrm{pr}^\vee[-1]"'] & \mathfrak{M}_{T^*X\times\mathbb{A}^1}\times_{\mathfrak{M}_{T^*X}}\mathfrak{M}^{\mathrm{nil}}_{T^*X} \arrow[r, "\mathrm{pr}_1"] \arrow[d, "\mathrm{pr}_2"] & \mathfrak{M}_{T^*X\times\mathbb{A}^1} \arrow[d, "\pi_{T^*X}"] \arrow[ll, bend right, "\pi_{X\times\mathbb{A}^1}"'] \\
\mathbf{Coh} \arrow[r, "h"] & \mathfrak{M}^{\mathrm{nil}}_{T^*X} \arrow[r, "i"] & \mathfrak{M}_{T^*X} \arrow[ll, bend left, "\mathrm{pr}"]
\end{tikzcd} \tag{4.5.2}$$

where $i\circ h$ is the section of $\mathrm{pr} : \mathbf{T}^*[0]\mathbf{Coh} \to \mathbf{Coh}$ and $\mathrm{pr}_1\circ\widetilde{h}$ is the section of $\pi_{X\times\mathbb{A}^1} : \mathbf{T}^*[-1]\mathfrak{M}_{X\times\mathbb{A}^1} \to \mathfrak{M}_{X\times\mathbb{A}^1}$. We have the following Fourier transform result established in [Kha+25, Theorem 4.23]:

Lemma 4.17. *Let B be a derived Artin stack, and $E \to B$ is a perfect complex, then there is a natural isomorphism between Borel-Moore homology groups:*

$$\mathrm{FT}_{E\mapsto E^\vee} : H^{\mathrm{BM}}_\bullet(E) \cong H^{\mathrm{BM}}_{\bullet-2\,\mathrm{rk}(E)}\left(E^\vee[-1]\right)$$

Moreover, the cycle class $[B] \in H^{\mathrm{BM}}_\bullet(E)$ is sent to $[E^\vee[-1]]^{\mathrm{vir}}$ the virtual fundamental class of $E^\vee[-1]$.

Apply the Lemma 4.17 to our case, we can get an isomorphism

$$H^{\mathrm{BM}}_{\bullet+(2g_X-2)(\mathrm{rk}(\alpha))^2}\left(\mathbf{Higgs}_\alpha\right) \xrightarrow{\cong} H^{\mathrm{BM}}_\bullet\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\alpha}\right)$$

and that the cycle class $\left[\mathbf{\Lambda}_{(\alpha)}\right]$ is sent to $\left[{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\alpha}\right]^{\mathrm{vir}}$ (see also [Kha+25, Corollary 4.27]). This isomorphism can be upgraded to an isomorphism of co-algebras. The co-algebra structure on $H^{\mathrm{BM}}_\bullet(\mathfrak{M}_{X\times\mathbb{A}^1})$ is also inherited from the cofactorisation structure of the DT-sheaf $\varphi_{\mathfrak{M}_{T^*X\times\mathbb{A}^1}}$:

For $\underline{\alpha} = (\beta,\gamma)$ a partition of α, we have the following commutative diagram of extensions where the right square is cartesian:

$$\begin{tikzcd}
\mathfrak{M}_{T^*X\times\mathbb{A}^1,\beta}\times\mathfrak{M}_{T^*X\times\mathbb{A}^1,\gamma} \arrow[d, "\pi_{X\times\mathbb{A}^1}\times\pi_{X\times\mathbb{A}^1}"'] & \widetilde{\mathfrak{M}}_{T^*X\times\mathbb{A}^1,\underline{\alpha}} \arrow[l, "\mathsf{q}_{T^*X\times\mathbb{A}^1}"'] \arrow[r, "\mathsf{p}_{T^*X\times\mathbb{A}^1}"] \arrow[d, "\tilde{\pi}_{X\times\mathbb{A}^1}"'] & \mathfrak{M}_{T^*X\times\mathbb{A}^1,\alpha} \arrow[d, "\pi_{X\times\mathbb{A}^1}"] \\
\mathfrak{M}_{X\times\mathbb{A}^1,\beta}\times\mathfrak{M}_{X\times\mathbb{A}^1,\gamma} & \widetilde{\mathfrak{M}}_{X\times\mathbb{A}^1,\underline{\alpha}} \arrow[l, "\mathsf{q}_{X\times\mathbb{A}^1}"'] \arrow[r, "\mathsf{p}_{X\times\mathbb{A}^1}"] & \mathfrak{M}_{X\times\mathbb{A}^1,\alpha}
\end{tikzcd} \tag{4.5.3}$$

Denote by $\varphi_{\underline{\alpha}}$ the restriction of DT sheaf $\varphi\boxtimes\varphi$ on $\mathfrak{M}_{T^*X\times\mathbb{A}^1,\beta}\times\mathfrak{M}_{T^*X\times\mathbb{A}^1,\gamma}$. It is proved in [Dav+, Theorem 2.16] that $\varphi_{\underline{\alpha}} = (\mathsf{q}_{T^*X\times\mathbb{A}^1})_*(\mathsf{p}_{T^*X\times\mathbb{A}^1})^!\left(\varphi_\alpha\right)$. Take the $*$-pushforward long $\pi_{X\times\mathbb{A}^1}$, we get

$$\mathbb{DC}_{\mathfrak{M}_{X\times\mathbb{A}^1,\underline{\alpha}}} = (\mathsf{q}_{X\times\mathbb{A}^1})_*(\mathsf{p}_{X\times\mathbb{A}^1})^!\,\mathbb{DC}_{\mathfrak{M}_{X\times\mathbb{A}^1,\alpha}}\,[\mathbf{shift}]$$

where the shift is the difference of virtual dimension $\mathrm{vdim}\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\underline{\alpha}}\right) - \mathrm{vdim}\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\alpha}\right)$, thus vanishes. We obtain the coproduct $H^{\mathrm{BM}}_\bullet\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\alpha}\right) \xrightarrow{\Delta_{\underline{\alpha}}} H^{\mathrm{BM}}_\bullet\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\beta}\right)\otimes H^{\mathrm{BM}}_\bullet\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\gamma}\right)$. Restrict to degree 0, we get $H^{\mathrm{BM}}_0\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\alpha}\right) \xrightarrow{\Delta_{\underline{\alpha}}} H^{\mathrm{BM}}_0\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\beta}\right)\otimes H^{\mathrm{BM}}_0\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\gamma}\right)$. The coproduct is compatible with Fourier transform:

$$\begin{tikzcd}
H^{\mathrm{BM}}_{(2g_X-2)\,\mathrm{rk}(\alpha)^2}\left(\mathbf{Higgs}_\alpha\right) \arrow[r, "\Delta_{\underline{\alpha}}"] \arrow[d, "\cong"] & H^{\mathrm{BM}}_{(2g_X-2)\,\mathrm{rk}(\beta)^2}\left(\mathbf{Higgs}_\beta\right)\otimes H^{\mathrm{BM}}_{(2g_X-2)\,\mathrm{rk}(\gamma)^2}\left(\mathbf{Higgs}_\gamma\right) \arrow[d, "\cong"'] \\
H^{\mathrm{BM}}_0\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\alpha}\right) \arrow[r, "\Delta_{\underline{\alpha}}"] & H^{\mathrm{BM}}_0\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\beta}\right)\otimes H^{\mathrm{BM}}_0\left({}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1,\gamma}\right)
\end{tikzcd}$$

Thus, we reduce to show $\Delta\left(\left[{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}\right]^{\mathrm{vir}}\right) = \left[{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}\right]^{\mathrm{vir}}\otimes\left[{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}\right]^{\mathrm{vir}}$. Since $\mathfrak{M}_{X\times\mathbb{A}^1}$ is quasi-smooth, its classical truncation ${}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}$ carries a perfect obstruction given by $\mathbb{L}_{\mathfrak{M}_{X\times\mathbb{A}^1}|{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}} \to \mathbb{L}_{{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}}$. And the virtual fundamental class of ${}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}$ is given by

$$\left[{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}\right] = s^!_{{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}}\left[\mathfrak{C}_{{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}}\right]$$

the Gynsin pullback of the intrinsic normal cone along zero section $s_{{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}} : {}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1} \hookrightarrow \mathfrak{C}_{{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}}$, which is the same as derived Gynsin pull back $f^![\mathrm{pt}]$ along $f : \mathfrak{M}_{X\times\mathbb{A}^1} \to \mathrm{pt}$. Applying the analytification functor, we get perfect obstruction $\mathbb{L}^{\mathrm{an}}_{\mathfrak{M}_{X\times\mathbb{A}^1}|{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}} \to \mathbb{L}^{\mathrm{an}}_{{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}}$, this gives the analytic intrinsic normal cone $\left[\mathfrak{C}_{{}^{\mathrm{cl}}\mathfrak{M}^{\mathrm{an}}_{X\times\mathbb{A}^1}}\right]$, then the same construction gives $\left[{}^{\mathrm{cl}}\mathfrak{M}^{\mathrm{an}}_{X\times\mathbb{A}^1}\right]^{\mathrm{vir}}$, which corresponds to $\left[{}^{\mathrm{cl}}\mathfrak{M}_{X\times\mathbb{A}^1}\right]^{\mathrm{vir}}$ under the isomorphism of homologies $H^{\mathrm{BM}}_\bullet(\mathfrak{M}_{X\times\mathbb{A}^1}) \cong H_\bullet\left(\mathfrak{M}^{\mathrm{an}}_{X\times\mathbb{A}^1}\right)$.

Moreover, the cofactorisation structure on $\mathfrak{M}^{\mathrm{an}}_{T^*X\times\mathbb{A}^1}$ descends to $\mathfrak{M}^{\mathrm{an}}_{X\times\mathbb{A}^1}$: for $U_1 \sqcup U_2 \subseteq U$ convex open subsets, we have $j_{U_1,U_2} : \mathfrak{M}^{\mathrm{an}}_{X\times(U_1\sqcup U_2)} \cong \mathfrak{M}^{\mathrm{an}}_{X\times U_1} \times \mathfrak{M}^{\mathrm{an}}_{X\times U_2} \subseteq \mathfrak{M}^{\mathrm{an}}_{X\times U}$ an open inclusion. Denote by $\pi_{X\times U} : \mathfrak{M}^{\mathrm{an}}_{T^*X\times U} \to \mathfrak{M}^{\mathrm{an}}_{X\times U}$, then we have

$$\begin{tikzcd} R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{T^*X\times U}, \varphi^{\mathrm{an}}_U\right) \arrow[r, "\Delta"] \arrow[d, "(\pi_{X\times U})_*"'] & R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{T^*X\times U_1}, \varphi^{\mathrm{an}}_{U_1}\right) \otimes R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{T^*X\times U_2}, \varphi^{\mathrm{an}}_{U_2}\right) \arrow[d, "(\pi_{X\times(U_1\times U_2)})_*"] \\ R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{X\times U}, (\pi_{X\times U})_*(\varphi^{\mathrm{an}}_U)\right) \arrow[r, "\Delta"] & R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{X\times U_1}, (\pi_{X\times U_1})_*(\varphi^{\mathrm{an}}_{U_1})\right) \otimes R\Gamma\left(\mathfrak{M}^{\mathrm{an}}_{X\times U_2}, (\pi_{X\times U_2})_*(\varphi^{\mathrm{an}}_{U_2})\right) \end{tikzcd}$$

Thus to prove the desired coproduct formula, we can pass to verification on the analytic side. It suffices to show that $(j_{U_1,U_2})^! \left[\mathfrak{M}^{\mathrm{an}}_{X\times U}\right]^{\mathrm{vir}} \cong \left[\mathfrak{M}^{\mathrm{an}}_{X\times(U_1\sqcup U_2)}\right]^{\mathrm{vir}}$, which follows from the fact that j_{U_1,U_2} is open. ■

We then calculate the upper circuit of the diagram 4.4.3 for $\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right)$, with $\underline{\alpha} = (\alpha_1, \ldots, \alpha_s)$:

$$\Delta_{\beta,\gamma}\left(\mathrm{CC}^{\mathrm{ind}}\left(\mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right)\right)\right) = \Delta_{\beta,\gamma}\left(\mathsf{Eis}^{\Lambda}_{\underline{\alpha}}\left(\mathrm{CC}^{\mathrm{ind}}\left(\mathbf{1}_{\underline{\alpha}}\right)\right)\right)$$

where the equivalence is given by Proposition 4.14. We notice that $\mathrm{CC}^{\mathrm{ind}}\left(\mathbf{1}_{\underline{\alpha}}\right) = \left[\mathbf{\Lambda}_{(\alpha_1)}\right] \otimes \ldots \otimes \left[\mathbf{\Lambda}_{(\alpha_s)}\right]$. Thus the top circuit of the diagram (applying to $\mathsf{Eis}^{\mathrm{Coh}}$) is $\Delta_{\beta,\gamma}\left(\mathsf{Eis}^{\Lambda}_{\underline{\alpha}}\left(\left[\mathbf{\Lambda}_{(\alpha_1)}\right] \otimes \ldots \otimes \left[\mathbf{\Lambda}_{(\alpha_s)}\right]\right)\right)$. On the other hand, the lower circuit gives

$$\begin{aligned}\mathrm{CC}^{\mathrm{ind}}\left(\mathsf{CT}^{\mathrm{Coh}}_{\beta,\gamma} \circ \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\alpha}}\left(\mathbf{1}_{\underline{\alpha}}\right)\right) &= \mathrm{CC}^{\mathrm{ind}}\left(\bigoplus_{\underline{\beta},\underline{\gamma}} \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\beta}}\left(\mathbf{1}_{\underline{\beta}}\right) \boxtimes \mathsf{Eis}^{\mathrm{Coh}}_{\underline{\gamma}}\left(\mathbf{1}_{\underline{\gamma}}\right)\right) \\ &= \bigoplus_{\underline{\beta},\underline{\gamma}} \left(\mathrm{Ind}_{\underline{\beta}}\left(\left[\mathbf{\Lambda}_{(\beta_1)}\right] \otimes \ldots \otimes \left[\mathbf{\Lambda}_{(\beta_k)}\right]\right)\right) \otimes \left(\mathrm{Ind}_{\underline{\gamma}}\left(\left[\mathbf{\Lambda}_{(\gamma_1)}\right] \otimes \ldots \otimes \left[\mathbf{\Lambda}_{(\gamma_l)}\right]\right)\right)\end{aligned}$$

Thus the compatibility of $\mathrm{CC}^{\mathrm{ind}}$ and coproducts is a consequence of the Langlands formula (Proposition 4.9) for $H^{\mathrm{BM}}_{\bullet}(\mathbf{Higgs})$. Next we want to show $\mathrm{CC}^{\mathrm{ind}}$ induces an isomorphism between $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right)$ and $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$. First, we note that $\mathrm{CC}^{\mathrm{ind}}$ factors as $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right) \xrightarrow{\mathrm{CC}^{\mathrm{ind}}} H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda}) \xrightarrow{i_*} H^{\mathrm{BM}}_{\bullet}(\mathbf{Higgs})$. Thus, to show the injectivity of $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right) \xrightarrow{\mathrm{CC}^{\mathrm{ind}}} H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$, it is enough to show the composition $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right) \xrightarrow{\mathrm{CC}^{\mathrm{ind}}} H^{\mathrm{BM}}_{\bullet}(\mathbf{Higgs})$ is injective. Back to the diagram 4.4.3, we have $\Delta_{\beta,\gamma} \circ \mathrm{CC}^{\mathrm{ind}} = \mathrm{CC}^{\mathrm{ind}} \circ \mathsf{CT}^{\mathrm{Coh}}_{\beta,\gamma}$. The map $\mathsf{CT}^{\mathrm{Coh}}_{\beta,\gamma}$ is injective since the functor $\mathsf{CT}^{\mathrm{Coh}}$ is conservative on $\mathsf{Q}^{\mathrm{Coh}}$, and the bottom $\mathrm{CC}^{\mathrm{ind}}$ is injective by induction hypothesis. Thus the composition $\Delta_{\beta,\gamma} \circ \mathrm{CC}^{\mathrm{ind}}$ is injective, which implies $K_0\left(\mathsf{Q}^{\mathrm{Coh}}_{\alpha}\right) \to H^{\mathrm{BM}}_{\bullet}\left(\mathbf{Higgs}_{\alpha}\right)$ is injective. On the other hand, $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right) \xrightarrow{\mathrm{CC}^{\mathrm{ind}}} H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$ is surjective due to the explicit description of irreducible components of $\mathbf{\Lambda}_{\alpha}$.

Finally, since the coproduct on $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$ is induced by the cofactorization of $(\mathrm{pr}_1)^!(\varphi)$ as in 4.5.2, then using diagram chasing, we can find the proof of Proposition 4.16 shows also that $\Delta_{\beta,\gamma}\left(\left[\mathbf{\Lambda}_{(\alpha)}\right]\right) = \left[\mathbf{\Lambda}_{(\beta)}\right] \otimes \left[\mathbf{\Lambda}_{(\gamma)}\right]$ holds in $H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$. Hence, $K_0\left(\mathsf{Q}^{\mathrm{Coh}}\right) \xrightarrow{\mathrm{CC}^{\mathrm{ind}}} H^{\mathrm{BM}}_{\mathrm{top}}(\mathbf{\Lambda})$ is a bialgebra isomorphism.

A Category toolkits

A.1 Recollements and truncatabilities

A.1.1 *Recollements and right adjoints.* We say a recollement of (stable ∞) category C is a collection of categories $(\mathsf{C}_0, \mathsf{C}, \mathsf{C}_1)$ accompany with the following diagram of adjunctions:

$$
\begin{tikzcd}
\mathsf{C}_0 \arrow[r, bend left, "j_!"] \arrow[r, bend right, "j_*"'] & \mathsf{C} \arrow[l, "j^*=j^!"'] \arrow[r, bend left, "i^*"] \arrow[r, bend right, "i^!"'] & \mathsf{C}_1 \arrow[l, "i_*=i_!"']
\end{tikzcd}
$$

such that the functors $j_!, j_*, i_*$ are fully-faithful and $\mathrm{Ker}(i^*) = \mathrm{Im}(j_!), \mathrm{Ker}(i^!) = \mathrm{Im}(j_*)$. Assume there is another recollement $(\mathsf{D}_0, \mathsf{D}, \mathsf{D}_1)$ for D, then a morphism of recollement data from C to D is given by functors $(\mathsf{F}_0, \mathsf{F}, \mathsf{F}_1)$ where $\mathsf{F}_i : \mathsf{C}_i \to \mathsf{D}_i$ and $\mathsf{F} : \mathsf{C} \to \mathsf{D}$ making the following diagram commute:

$$
\begin{tikzcd}
\mathsf{C}_0 \arrow[d, "\mathsf{F}_0"] & \mathsf{C} \arrow[l, "j^*=j^!"'] \arrow[d, "\mathsf{F}"] & \mathsf{C}_1 \arrow[l, "i_*=i_!"'] \arrow[d, "\mathsf{F}_1"] \\
\mathsf{D}_0 & \mathsf{D} \arrow[l, "\tilde{j}^*=\tilde{j}^!"'] & \mathsf{D}_1 \arrow[l, "\tilde{i}_*=\tilde{i}_!"']
\end{tikzcd}
\tag{A.1.1}
$$

Proposition A.1. *Assume* $(\mathsf{F}_0, \mathsf{F}, \mathsf{F}_1)$ *is a morphism of recollement data and that both functors F_0 and F_1 admit right adjoints (denoted by* G_0, G_1 *respectively), then the functor* F *is also right adjointable.*

Proof. We can construct the right adjoint of F by gluing $j_* \circ \mathsf{G}_0$ and $i_* \circ \mathsf{G}_1$. More precisely, for any object $y \in \mathsf{D}$, the image of right adjoint G of F on y is the following fiber product:

$$
\begin{tikzcd}
\mathsf{G}(y) \arrow[r] \arrow[d] & i_*\mathsf{G}_1\tilde{i}^*(y) \arrow[d] \\
j_*\mathsf{G}_0\tilde{j}^*(y) \arrow[r] & i_*\mathsf{G}_1\tilde{i}^*\tilde{j}_*\tilde{j}^*(y)
\end{tikzcd}
\tag{A.1.2}
$$

The map $i_*\mathsf{G}_1\tilde{i}^*(y) \to i_*\mathsf{G}_1\tilde{i}^*\tilde{j}_*\tilde{j}^*(y)$ is induced from the unit $\mathrm{id}_{\mathsf{D}} \to \tilde{j}_*\tilde{j}^*$. Note that we have

$$
\mathrm{Hom}_{\mathsf{C}}\left(j_*\mathsf{G}_0\tilde{j}^*(y), i_*\mathsf{G}_1\tilde{i}^*\tilde{j}_*\tilde{j}^*(y)\right) = \mathrm{Hom}_{\mathsf{D}_1}\left(\mathsf{F}_1 i^* j_*\mathsf{G}_0\tilde{j}^*(y), \tilde{i}^*\tilde{j}_*\tilde{j}^*(y)\right) = \mathrm{Hom}_{\mathsf{D}_1}\left(\tilde{i}^*\tilde{j}_*\mathsf{F}_0\mathsf{G}_0\tilde{j}^*(y), \tilde{i}^*\tilde{j}_*\tilde{j}^*(y)\right)
$$

The map $j_*\mathsf{G}_0\tilde{j}^*(y) \to i_*\mathsf{G}_1\tilde{i}^*\tilde{j}_*\tilde{j}^*(y)$ in diagram A.1.2 is induced by counit $\mathsf{F}_0\mathsf{G}_0$ applied to $\tilde{j}^*(y)$. For the sanity check, apply $\mathrm{Hom}_{\mathsf{C}}(x, -)$ to diagram A.1.2, we get

$$
\begin{aligned}
\mathrm{Hom}_{\mathsf{C}}\left(x, \mathsf{G}(y)\right) &= \mathrm{Hom}\left(x, i_*\mathsf{G}_1\tilde{i}^*(y)\right) \times_{\mathrm{Hom}(x, i_*\mathsf{G}_1\tilde{i}^*\tilde{j}_*\tilde{j}^*(y))} \mathrm{Hom}\left(x, j_*\mathsf{G}_0\tilde{j}^*(y)\right) \\
&= \mathrm{Hom}\left(\tilde{i}_!\mathsf{F}_1 i^* x, y\right) \times_{\mathrm{Hom}(\tilde{j}_!\tilde{j}^*\tilde{i}_!\mathsf{F}_1 i^* x, y)} \mathrm{Hom}\left(\tilde{j}_!\mathsf{F}_0 j^* x, y\right)
\end{aligned}
$$

Now by assumption of recollement of D, the element $F(x)$ is the fiber product as the following

$$
\begin{tikzcd}
\mathsf{F}(x) \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner", very near start] & \tilde{i}_*\tilde{i}^*\mathsf{F}(x) \arrow[d] \\
\tilde{j}_*\tilde{j}^*\mathsf{F}(x) \arrow[r] & \tilde{i}_*\tilde{i}^*\tilde{j}_*\tilde{j}^*\mathsf{F}(x)
\end{tikzcd}
\cong
\begin{tikzcd}
\mathsf{F}(x) \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner", very near start] & \tilde{i}_!\mathsf{F}_1 i^*(x) \arrow[d] \\
\tilde{j}_!\mathsf{F}_0 j^*(x) \arrow[r] & \tilde{j}_!\tilde{j}^*\tilde{i}_!\mathsf{F}_1 i^*(x)
\end{tikzcd}
$$

Hence we get $\mathrm{Hom}_{\mathsf{C}}\left(x, \mathsf{G}(y)\right) = \mathrm{Hom}_{\mathsf{D}}\left(\mathsf{F}(x), y\right)$. ■

Lemma A.2. *[Lur17, Proposition A.8.14] Assume* $(\mathsf{F}_0, \mathsf{F}, \mathsf{F}_1)$ *is a morphism of recollement data and that both functors* $\mathsf{F}_0, \mathsf{F}_1$ *are equivalences, suppose also* F *is left-exact, then* F *is an equivalence itself.*

A.1.2 *Truncatabilities.* For an open inclusion $j : \mathcal{U} \hookrightarrow \mathcal{X}$ with closed complement $i : \mathcal{Z} \hookrightarrow \mathcal{X}$, it not always true that we have recollement of $\mathcal{D}\text{-mod}(\mathcal{X})$ via $\mathcal{D}\text{-mod}(\mathcal{U})$ and $\mathcal{D}\text{-mod}(\mathcal{Z})$.

Lemma A.3. *In [DG15], it is proved that the following conditions are equivalent:*
(i) The left adjoint functor $j_!$ *of* j^* *is well-defined on whole of* $\mathcal{D}\text{-mod}(\mathcal{U})$.
(ii) The left adjoint functor i^* *to* i_* *is well-defined on whole of* $\mathcal{D}\text{-mod}(\mathcal{X})$.

In the situation that the conditions in the above Lemma holds, we say $\mathcal{Z}$ is *truncative* and $\mathcal{U}$ is *co-truncative* in $\mathcal{X}$. And $\mathcal{X}$ is *truncatable* if it can be covered by co-truncative open quasi-compact substacks.

A typical example of truncative closed substack is given by contraction: suppose $p : W \to S$ is an affine morphism with section $\iota : S \to W$ and there is an action of $\mathbb{G}_m$ that contracts W to $\iota(S)$. Let $\mathcal{X} := W/\mathbb{G}_m$ and $\mathcal{Z} := S/\mathbb{G}_m$, then [DG15, Proposition 5.1.2] shows the induced closed embedding $\mathcal{Z} \hookrightarrow \mathcal{X}$ is truncative. In general, we say a locally closed substack $\mathcal{Z} \subseteq \mathcal{X}$ is *contractive* if the inclusion is locally of the form $S/\mathbb{G}_m \hookrightarrow W/\mathbb{G}_m$ as above. In such situation, we have

Lemma A.4. *[DG15, Corollary 5.2.3] A contractive substack is truncative.*

Proposition A.5. *The stack* $\mathbf{Coh}_\alpha$ *is truncatable.*

Proof. The proof for $\mathbf{Coh}_\alpha$ is almost identical to the one in [DG15]. Note that any quasicompact open in $\mathbf{Coh}_\alpha$ in contained in one of $\bigsqcup_{\underline{\beta} \geq \underline{\alpha}} \mathbf{HN}_{\underline{\alpha}}$, for $\underline{\alpha}$ very unstable, it suffices to prove that the locally closed substack $\mathbf{HN}_{\underline{\alpha}}$ for every unstable HN-partition $\underline{\alpha}$ is truncative. We adapt [DG15, Proposition 9.2.2, Lemma 11.3.4] to the following diagram, where the right vertical arrow is the isomorphism in 1.2.1:

$$\begin{tikzcd}
\mathbf{Coh}^{\mathrm{ss}}_{\underline{\alpha}} \arrow[r, "\iota_{\underline{\alpha}}"] \arrow[d, "\iota_{\underline{\alpha}}^-"'] & \widetilde{\mathbf{Coh}}_{\underline{\alpha}} \times_{\mathbf{Coh}_{\underline{\alpha}}} \mathbf{Coh}^{\mathrm{ss}}_{\underline{\alpha}} \arrow[d, "\cong"] \\
\widetilde{\mathbf{Coh}}^-_{\underline{\alpha}} \times_{\mathbf{Coh}_{\underline{\alpha}}} \mathbf{Coh}^{\mathrm{ss}}_{\underline{\alpha}} \arrow[r] & \mathbf{HN}_{\underline{\alpha}}
\end{tikzcd}$$

The upper horizontal arrow $\iota_{\underline{\alpha}}$ is smooth and surjective thanks to Proposition 1.6 and the very unstable condition. Also there is an action of $\mathbb{A}^1$ on $\widetilde{\mathbf{Coh}}^-_{\underline{\alpha}} \times_{\mathbf{Coh}_{\underline{\alpha}}} \mathbf{Coh}^{\mathrm{ss}}_{\underline{\alpha}}$ via the fibers $\widetilde{\mathbf{Coh}}^-_{\underline{\alpha}} \to \mathbf{Coh}_{\underline{\alpha}}$, and the image $\mathrm{Im}\left(\iota_{\underline{\alpha}}^-\right)$ is contractive with respect to the induced $\mathbb{G}_m$-action. ∎

We have seen that the truncatability of a closed substack gives rise to a recollement of $\mathcal{D}$-modules, in combine with Proposition A.1, we get the following Lemma, which is useful for the second adjunction introduced in the sequel.

Lemma A.6. *Let* $\mathcal{U} \subseteq \mathcal{X}$ *and* $\mathcal{U}' \subseteq \mathcal{Y}$ *be open and co-truncative substack in* $\mathcal{X}$ *and* $\mathcal{Y}$ *respectively and consider* $\mathcal{Z} = \mathcal{X} \backslash \mathcal{U}$, $\mathcal{Z}' = \mathcal{Y} \backslash \mathcal{U}'$. *Suppose we have a functor* $\mathsf{F} : \mathcal{D}\text{-mod}(\mathcal{X}) \to \mathcal{D}\text{-mod}(\mathcal{Y})$ *such that restricts to right adjointable functors* $\mathcal{D}\text{-mod}(U) \to \mathcal{D}\text{-mod}(U')$, $\mathcal{D}\text{-mod}(\mathcal{Z}) \to \mathcal{D}\text{-mod}(\mathcal{Z}')$, *then* F *admit a right adjoint.*

A.2 Hyperbolic localization and the second adjunction

In [DG16], it was proved that for a fixed coweight $\mu \in \pi_1(G)$ and $P = MU(P)$ parabolic subgroup, there is an equivalence of functors

$$\mathsf{CT}_*^{\mu} \cong \mathsf{CT}_!^{\mu,-} : \mathcal{D}\text{-mod}\left(\mathbf{Bun}_G^{\mu}\right) \longrightarrow \mathcal{D}\text{-mod}\left(\mathbf{Bun}_M^{\mu}\right)$$

where $\mathsf{CT}_* = (\mathsf{q}_P)_*(\mathsf{p}_P)^!$ and $\mathsf{CT}_!^- = (\mathsf{q}_{P^-})_!(\mathsf{p}_{P^-})^*$ is the !-constant term functor associated with the opposite parabolic. The equivalence gives a right adjoint of CT_*^{μ}, which is $\mathsf{Eis}_*^{\mu,-}$, this adjunction is referred to be the **second adjunction** in [DG16]. Notice that $\mathsf{CT}_!^-$ is a priori not well-defined, however the proof uses the following Braden's theorem about hyperbolic localization to show this functor exists and the above equivalence. In general, let $\mathcal{Z}$ be an algebraic stack, equipped with an action of $\mathbb{G}_m$. Denote

$$\mathcal{Z}^0 = \mathbf{Maps}^{\mathbb{G}_m}\left(\mathrm{pt}, \mathcal{Z}\right); \quad \mathcal{Z}^+ = \mathbf{Maps}^{\mathbb{G}_m}\left(\mathbb{A}^1_+, \mathcal{Z}\right); \quad \mathcal{Z}^- = \mathbf{Maps}^{\mathbb{G}_m}\left(\mathbb{A}^1_-, \mathcal{Z}\right)$$

the *fixed points, attractor, repeller* of $\mathcal{Z}$ respectively (with respect to the $\mathbb{G}_m$-action), where for for $T, \mathcal{Z}$ with $\mathbb{G}_m$-action and any test scheme S, $\mathrm{Maps}\left(S, \mathbf{Maps}^{\mathbb{G}_m}(T, \mathcal{Z})\right) = \mathrm{Maps}\left(S \times T, \mathcal{Z}\right)^{\mathbb{G}_m}$. Here $\mathbb{A}^1_+$ is $\mathbb{A}^1$ equipped with dilation action of $\mathbb{G}_m$ and $\mathbb{A}^1_-$ is $\mathbb{A}^1$ equipped with opposite $\mathbb{G}_m$-action via $t \mapsto \frac{1}{t}$. We have the following diagram

$$\begin{tikzcd}
\mathcal{Z}^0 \arrow[rrd, shift left, "\mathsf{i}^-"] \arrow[rdd, shift left, "\mathsf{i}^+"] & & \\
 & & \mathcal{Z}^- \arrow[llu, shift left, "\mathsf{q}^-"'] \arrow[d, "\mathsf{p}^-"] \\
 & \mathcal{Z}^+ \arrow[luu, shift left, "\mathsf{q}^+"] \arrow[r, "\mathsf{p}^+"] & \mathcal{Z}
\end{tikzcd} \tag{A.2.1}$$

where $\mathsf{p}^{\pm}$ is induced by $\mathbb{A}^1_{\pm} \to \{1\}$, $\mathsf{q}^{\pm}$ is induced by $\mathbb{G}_m$-equivariant map $0 : \mathrm{pt} \to \mathbb{A}^1_{\pm}$ and $\mathsf{i}^{\pm}$ is induced by $\mathbb{A}^1_{\pm} \to \mathrm{pt}$. Let $\mathcal{D}\text{-mod}(\mathcal{Z})^{\mathbb{G}_m-\mathrm{mon}} \subseteq \mathcal{D}\text{-mod}(\mathcal{Z})$ be the subcategory of $\mathbb{G}_m$-monodromic objects that generated by pullback $\mathcal{D}\text{-mod}(\mathcal{Z}/\mathbb{G}_m) \to \mathcal{D}\text{-mod}(\mathcal{Z})$.

Proposition A.7. *[DG14] (1) Let $\mathcal{Z}$ be an algebraic space and $\mathcal{Z}^0, \mathcal{Z}^{\pm}$ as above, restrict to $\mathcal{D}\text{-mod}(\mathcal{Z})^{\mathbb{G}_m-\mathrm{mon}}$, we have an adjunction*

$$\left(\mathsf{q}^+\right)_* \circ \left(\mathsf{p}^+\right)^! |_{\mathcal{D}\text{-mod}(\mathcal{Z})^{\mathbb{G}_m-\mathrm{mon}}} \dashv \left(\mathsf{p}^-\right)_* \circ \left(\mathsf{q}^-\right)^!$$

Also, there is natural transformation $(\mathsf{i}^+)^* \circ (\mathsf{p}^+)^! \to (\mathsf{i}^-)_! \circ (\mathsf{p}^-)^* : \mathcal{D}\text{-mod}\left(\mathcal{Z}\right)^{\mathbb{G}_m-\mathrm{mon}} \to \mathrm{Pro}\left(\mathcal{D}\text{-mod}\left(\mathcal{Z}^0\right)\right)$ which takes value in $\mathcal{D}\text{-mod}\left(\mathcal{Z}^0\right) = \mathcal{D}\text{-mod}\left(\mathcal{Z}^0\right)^{\mathbb{G}_m-\mathrm{mon}}$ and is an isomorphism.

(2) Assume further for algebraic stacks $\mathcal{Y}, \mathcal{Y}^0, \mathcal{Y}^{\pm}$, the diagram A.2.1 extends to a commutative cube

$$\begin{tikzcd}
\mathcal{Z}^0 \arrow[rr, "\psi^0"] \arrow[d, "\mathsf{i}^+"'] \arrow[ddr, "\mathsf{i}^-"] & & \mathcal{Y}^0 \arrow[d, "\widetilde{\mathsf{i}}^+"'] \arrow[ddr, "\widetilde{\mathsf{i}}^-"] & \\
\mathcal{Z}^+ \arrow[rr, "\psi^+"] \arrow[ddr, "\mathsf{p}^+"'] & & \mathcal{Y}^+ \arrow[ddr, "\widetilde{\mathsf{p}}^+"'] & \\
 & \mathcal{Z}^- \arrow[rr, "\psi^-"'] \arrow[d, "\mathsf{p}^-"] & & \mathcal{Y}^- \arrow[d, "\widetilde{\mathsf{p}}^-"] \\
 & \mathcal{Z} \arrow[rr, "\psi"'] & & \mathcal{Y}
\end{tikzcd} \tag{A.2.2}$$

such that $\mathcal{Y}^0$ open embeds to $\mathcal{Y}^+ \times_{\mathcal{Y}} \mathcal{Y}^-$, $\psi, \psi^{\pm}, \psi^0$ are smooth, ψ^0 surjective and ψ is $\mathbb{G}_m$-equivariant with respect to trivial $\mathbb{G}_m$-action on $\mathcal{Y}$, then $\left(\widetilde{\mathsf{i}^-}\right)^* \left(\widetilde{\mathsf{p}^-}\right)^! : \mathcal{D}\text{-mod}(\mathcal{Y}) \to \mathsf{Pro}\left(\mathcal{D}\text{-mod}\left(\mathcal{Y}^0\right)\right)$ takes value in $\mathcal{D}\text{-mod}\left(\mathcal{Y}^0\right)$. Moreover, there is a natrual isomorphism $\left(\widetilde{\mathsf{i}^-}\right)^* \left(\widetilde{\mathsf{p}^-}\right)^! \xrightarrow{\cong} \left(\widetilde{\mathsf{i}^+}\right)^! \left(\widetilde{\mathsf{p}^+}\right)^*$.

In the situation of Proposition A.7(2), the following diagram is said to be a *hyperbolic diagram*.

$$\begin{tikzcd}
\mathcal{Y}^0 \arrow[r, "\widetilde{\mathsf{i}^-}"'] \arrow[d, "\widetilde{\mathsf{i}^+}"'] & \mathcal{Y}^- \arrow[d, "\widetilde{\mathsf{p}^-}"] \\
\mathcal{Y}^+ \arrow[r, "\widetilde{\mathsf{p}^+}"'] & \mathcal{Y}
\end{tikzcd}$$

Now we want to prove that there is an analogous second adjunction for $\mathsf{CT}^{\mathrm{Coh}}$. Unfortunately, the method used in [DG16] cannot be applied directly into the case of $\mathbf{Coh}_\alpha$ when $\mathrm{rk}(\alpha) \geqslant 1$, since there is no unified $\mathbb{G}_m$-action on the torsion part and on locally free part. We first treat separately the case of torsion sheaves, namely the constant term to $\mathcal{D}\text{-mod}(\mathbf{Coh}_{\tau d})$. Recall $\mathbf{Coh}_{\tau d}$ is the quotient stack $\left[Q_{\tau d}/\mathrm{GL}_d\right]$, where for test scheme S,

$$Q_{\tau d}(S) \subseteq \mathrm{Quot}_{\tau d}(S) = \left\{\phi : \mathbf{k}^d \otimes \mathcal{O}_{X_S} \twoheadrightarrow \mathcal{T} \mid \mathcal{T} \in \mathsf{Coh}(X \times S), \mathcal{T}_s \in \mathsf{Tor}^d(X), \forall s \in S\right\}$$

such that $H^0(\phi_s)$ induces an isomorphism for any $s \in S$. The Levi subgroups of GL_d are of the form $\mathrm{GL}_{\underline{d}} = \mathrm{GL}_{d_1} \times ... \times \mathrm{GL}_{d_l}$, parametrized by partition $\underline{d} = (d_1, ..., d_l)$ of d. For each partition $\underline{d}$, can find a coweight $\mathbb{G}_m \to T$ such that $\mathrm{GL}_d^{\mathbb{G}_m}$ is $\mathrm{GL}_{\underline{d}}$ and the attractor $\mathrm{Attr}(\mathrm{GL}_d) = \left\{g \in \mathrm{GL}_d \mid \lim_{t \to 0} t \cdot g \text{ exists}\right\}$ with respect to this $\mathbb{G}_m$-action is $P_{\underline{d}}$. This coweight induces $\mathbb{G}_m$-action on $\mathrm{Quot}_{\tau d}$, via GL_d-action on $\mathbf{k}^d$. And this action preserves $Q_{\tau d}$, with fixed point locus equals to $Q_{\tau d}^{\mathbb{G}_m} \cong Q_{\tau d_1} \times ... \times Q_{\tau d_l}$ and the space of attractors is $\widetilde{Q}_{\tau \underline{d}} = \left\{(\phi, \mathcal{T}_\bullet = \mathcal{T}_1 \subseteq ... \subseteq \mathcal{T}_{l-1} \subseteq \mathcal{T}_l = \mathcal{T}) \mid (\phi, \mathcal{T}) \in Q_{\tau d}, \phi(\mathbf{k}^{d_i} \otimes \mathcal{O}_X) = \mathcal{T}_i/\mathcal{T}_{i-1}\right\}$. Moreover, the space of repellers of $\mathbb{G}_m$-action of GL_d is the opposite parabolic $P^-_{\underline{d}}$, the repellers of $Q_{\tau d}$ is

$$\widetilde{Q}^-_{\tau \underline{d}} = \left\{\left(\phi, \mathcal{T} = \mathcal{T}_1 \twoheadrightarrow \mathcal{T}_2 \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{T}_l\right) \mid \phi(\mathbf{k}^{d_i} \otimes \mathcal{O}_X) = \mathrm{Ker}\left(\mathcal{T}_i/\mathcal{T}_{i+1}\right)\right\}$$

Passing to the quotient stack $\mathbf{Coh}_{\tau d} \cong \left[Q_{\tau d}/\mathrm{GL}_d\right]$, we have the following diagram:

$$\begin{tikzcd}
\left[Q_{\tau \underline{d}}/\mathrm{GL}_{\underline{d}}\right] \arrow[r, shift right, "\mathsf{i}^{\mathrm{Coh}}_{\tau \underline{d}}"'] \arrow[d, shift right, "\mathsf{i}^{\mathrm{Coh},-}_{\tau \underline{d}}"'] & \left[\widetilde{Q}_{\tau \underline{d}}/P_{\underline{d}}\right] \arrow[l, shift right, "\mathsf{q}^{\mathrm{Coh}}_{\tau \underline{d}}"'] \arrow[d, "\mathsf{p}^{\mathrm{Coh}}_{\tau \underline{d}}"] \\
\left[\widetilde{Q}^-_{\tau \underline{d}}/P^-_{\underline{d}}\right] \arrow[u, shift right, "\mathsf{q}^{\mathrm{Coh},-}_{\tau \underline{d}}"'] \arrow[r, "\mathsf{p}^{\mathrm{Coh},-}_{\tau \underline{d}}"] & \left[Q_{\tau d}/\mathrm{GL}_d\right]
\end{tikzcd} \tag{A.2.3}$$

Proposition A.8. *For any $d \geqslant 2$ and partitions $\underline{d} = (d_1, ..., d_l)$ of d, the coherent constant term functor $\mathsf{CT}^{\mathrm{Coh}}_{\underline{d},*}$ is isomorphic to $\left(\mathsf{i}^{\mathrm{Coh}}_{\tau \underline{d}}\right)^* \left(\mathsf{q}^{\mathrm{Coh},-}_{\tau \underline{d}}\right)^!$. The above diagram A.2.3 is hyperbolic, let $\mathsf{CT}^{\mathrm{Coh},-}_{\underline{d},!} := \left(\mathsf{p}^{\mathrm{Coh},-}_{\tau \underline{d}}\right)_! \left(\mathsf{q}^{\mathrm{Coh},-}_{\tau \underline{d}}\right)^*$, it is isomorphic to $\left(\mathsf{i}^{\mathrm{Coh},-}_{\tau \underline{d}}\right)^! \left(\mathsf{q}^{\mathrm{Coh},-}_{\tau \underline{d}}\right)^*$. There is a natural isomorphism $\mathsf{CT}^{\mathrm{Coh}}_{\underline{d},*} \cong \mathsf{CT}^{\mathrm{Coh},-}_{\underline{d},!}$ that both functors take value in $\mathcal{D}\text{-mod}\left(\mathbf{Coh}_{\tau d_1} \times ... \times \mathbf{Coh}_{\tau d_l}\right)$. We have the second adjunction*

$$\mathsf{Eis}^{\mathrm{Coh}}_{\underline{d},!} \dashv \mathsf{CT}^{\mathrm{Coh}}_{\underline{d},*} \cong \mathsf{CT}^{\mathrm{Coh},-}_{\underline{d},!} \dashv \mathsf{Eis}^{\mathrm{Coh},-}_{\underline{d},*} \tag{A.2.4}$$

Proof. By Proposition 1.6(1), the pair $\left[Q_{\tau\underline{d}}/\mathrm{GL}_{\underline{d}}\right] \underset{\mathsf{i}^{\mathrm{Coh}}_{\tau\underline{d}}}{\overset{\mathsf{q}^{\mathrm{Coh}}_{\tau\underline{d}}}{\rightleftarrows}} \left[\widetilde{Q}_{\tau\underline{d}}/P_{\underline{d}}\right]$ forms a contraction. According to [DG16, Proposition 4.1.5], we have $\left(\mathsf{i}^{\mathrm{Coh},\pm}_{\tau\underline{d}}\right)^* \cong \left(\mathsf{q}^{\mathrm{Coh},\pm}_{\tau\underline{d}}\right)_*$ and $\left(\mathsf{i}^{\mathrm{Coh},\pm}_{\tau\underline{d}}\right)^! \cong \left(\mathsf{q}^{\mathrm{Coh},\pm}_{\tau\underline{d}}\right)_!$ when restricted to $\mathcal{D}\text{-mod}\left(\left[\widetilde{Q}^{\pm}_{\tau d}/P^{\pm}_{\underline{d}}\right]\right)^{\mathbb{G}_m-\mathrm{mon}}$, which is isomorphic to $\mathcal{D}\text{-mod}\left(\left[\widetilde{Q}^{\pm}_{\tau d}/P^{\pm}_{\underline{d}}\right]\right)$.
Take $\mathcal{Z} = Q_{\tau d}, \mathcal{Y} = \mathbf{Coh}_{\tau d}$, the quotient $\psi : \mathcal{Z} \to \mathcal{Y}$ fits the above diagram into commutative cube as in Proposition A.7(2), one can check the conditions of Proposition A.7(2) are satisfied. ■

A.2.1 *Sketch of an alternative proof of Proposition A.8.* Let us first do the case when $d = 2$ and the only non-trivial partition is $\underline{d} = (1,1)$. The stack $\mathbf{Coh}_{\tau 2}$ can be decomposed into two parts $\mathbf{Coh}^{\mathrm{rss}}_{\tau 2}$ and $\mathbf{Coh}^{\Delta}_{\tau 2} = \mathbf{Coh}_{\tau 2} - \mathbf{Coh}^{\mathrm{rss}}_{\tau 2}$, recall $\mathbf{Coh}^{\mathrm{rss}}_{\tau 2} \cong \left(X^{(2)}\backslash\Delta\right)/(\mathbb{G}_m)^2$ has support on distinct points. Apply Lemma A.6, we reduce to construct the right adjoint to $\mathsf{CT}^{\mathrm{Coh}}_{\tau\underline{d},*}$ on $\mathbf{Coh}^{\mathrm{rss}}_{\tau 2}$ and $\mathbf{Coh}^{\Delta}_{\tau 2}$ separately. Restricting to the regular semi-simple locus, the constant term map is pull-push along $\left(X^{(2)}\backslash\Delta\right)/(\mathbb{G}_m)^2 \longleftarrow \widetilde{\mathbf{Coh}}_{\tau\underline{d}} \times_{X^2} \left(X^2\backslash\Delta\right) \longrightarrow \left(X^2\backslash\Delta\right)/(\mathbb{G}_m)^2$. Note that for any torsion sheaves $\mathcal{T}_1, \mathcal{T}_2$ with distinct support, $\mathrm{Hom}(\mathcal{T}_1,\mathcal{T}_2) = \mathrm{Ext}^1(\mathcal{T}_1,\mathcal{T}_2) = 0$, so we identify $\widetilde{\mathbf{Coh}}_{\tau\underline{d}} \times_{X^2} \left(X^2\backslash\Delta\right) \cong \left(X^2\backslash\Delta\right)/(\mathbb{G}_m)^2$. The actions of $(\mathbb{G}_m)^2$ on both $X^{(2)}\backslash\Delta$ and $X^2\backslash\Delta$ are trivial, thus over the disjoint locus, $\mathsf{CT}^{\mathrm{Coh}}_{\underline{2},*}$ and $\mathsf{CT}^{\mathrm{Coh},-}_{\underline{2},!}$ are given by $(\mathrm{Sym})^!$ and $(\mathrm{Sym}\circ\mathrm{swap})^*$ respectively. The two functor thus coincide since Sym is an étale covering on the regular semi-simple locus. In fact, we can do the same for any $d \geqslant 1$ and $\underline{d} = (1,\ldots,1)$ as $X^d\backslash\Delta \to X^{(d)}\backslash\Delta$ is an étale covering. This proves $\mathsf{CT}^{\mathrm{Coh}}_{\underline{d},*}|_{\mathbf{Coh}^{\mathrm{rss}}_{\tau d}} \cong \mathsf{CT}^{\mathrm{Coh},-}_{\underline{d},!}|_{\mathbf{Coh}^{\mathrm{rss}}_{\tau d}}$.
Next for $\mathbf{Coh}^{\Delta}_{\tau 2}$, we consider an affine cover U_i of X, such that there exists $U_i \to \mathbb{A}^1$ étale for each i. The étale map $f_i : U_i \to \mathbb{A}^1$ induces $g_i : \mathbf{Coh}_{\tau 2}(U_i) \to \mathbf{Coh}_{\tau 2}(\mathbb{A}^1)$, over the locus $\mathbf{Coh}^{\Delta}_{\tau 2}$, we have

$$\begin{tikzcd}
\Delta(U_i)/(\mathbb{G}_m)^2 \arrow[d, "\Delta(f_i)"'] & \widetilde{\mathbf{Coh}}^{\Delta}_{\tau\underline{2}}(\mathbb{A}^1)\times_{\Delta(\mathbb{A}^1)}\Delta(U_i) \arrow[l, "{'\mathsf{q}^{\Delta,\mathrm{Coh}}_{U_i}}"'] \arrow[r, "{'\mathsf{p}^{\Delta,\mathrm{Coh}}_{U_i}}"] \arrow[d, "{'\widetilde{f}_i}"] & \mathbf{Coh}^{\Delta}_{\tau 2}(\mathbb{A}^1)\times_{\mathbb{A}^1} U_i \arrow[d, "{'g_i}"] \\
\Delta(\mathbb{A}^1)/(\mathbb{G}_m)^2 & \widetilde{\mathbf{Coh}}^{\Delta}_{\tau\underline{2}}(\mathbb{A}^1) \arrow[l, "\mathsf{q}^{\Delta,\mathrm{Coh}}_{\mathbb{A}^1}"'] \arrow[r, "\mathsf{p}^{\Delta,\mathrm{Coh}}_{\mathbb{A}^1}"] & \mathbf{Coh}^{\Delta}_{\tau 2}(\mathbb{A}^1) \\
\Delta(U_i)/(\mathbb{G}_m)^2 \arrow[u, "\Delta(f_i)"] & \widetilde{\mathbf{Coh}}^{\Delta}_{\tau\underline{2}}(U_i) \arrow[l, "\mathsf{q}^{\Delta,\mathrm{Coh}}_{U_i}"'] \arrow[r, "\mathsf{p}^{\Delta,\mathrm{Coh}}_{U_i}"] \arrow[u, "\widetilde{f}_i"] & \mathbf{Coh}^{\Delta}_{\tau 2}(U_i) \arrow[u, "g_i"']
\end{tikzcd} \tag{A.2.5}$$

where $\Delta(U_i)$ (resp. $\Delta(\mathbb{A}^1)$) is the image of diagonal $U_i \to U_i \times U_i$ (resp. $\mathbb{A}^1 \to \mathbb{A}^2$) that $\widetilde{\mathbf{Coh}}^{\Delta}_{\tau\underline{2}}$ supported on, carries with a trivial $(\mathbb{G}_m)^2$-action. The functor $\left(\mathsf{q}^{\Delta,\mathrm{Coh}}_{\mathbb{A}^1}\right)_! \left(\mathsf{p}^{\Delta,\mathrm{Coh}}_{\mathbb{A}^1}\right)^*$ along the middle row is well-defined and isomorphic to $\left(\mathsf{q}^{\Delta,\mathrm{Coh},-}_{\mathbb{A}^1}\right)_* \left(\mathsf{p}^{\Delta,\mathrm{Coh},-}_{\mathbb{A}^1}\right)^!$ due the next Proposition A.9 stating the second adjunction for Lie algebras. Now the maps $\Delta(f_i), '\widetilde{f}_i, 'g_i$ are all étale, which implies $\left('\mathsf{q}^{\Delta,\mathrm{Coh}}_{U_i}\right)_! \left('\mathsf{p}^{\Delta,\mathrm{Coh}}_{U_i}\right)^*$ is well-defined over $('g_i)^*\left(\mathcal{D}\text{-mod}\left(\mathbf{Coh}^{\Delta}_{\tau 2}(\mathbb{A}^1)\right)\right)$, however, this is whole $\mathcal{D}\text{-mod}\left(\mathbf{Coh}^{\Delta}_{\tau 2}(\mathbb{A}^1)\times_{(\mathbb{A}^1)^{(2)}} U_i^{(2)}\right)$, so the functor along the first row is also well-defined. In comparison with the diagram for $\widetilde{\mathbf{Coh}}^{\Delta,-}$, we can further deduce $\left('\mathsf{q}^{\Delta,\mathrm{Coh}}_{U_i}\right)_! \left('\mathsf{p}^{\Delta,\mathrm{Coh}}_{U_i}\right)^* \cong \left('\mathsf{q}^{\Delta,\mathrm{Coh},-}_{U_i}\right)_* \left('\mathsf{p}^{\Delta,\mathrm{Coh},-}_{U_i}\right)^!$. Finally, consider the stack of length 2-torsion sheaves supported on one point $\mathbf{Coh}_{\tau 2}^{\Delta,\circ} := \mathrm{supp}^{-1}(0)$, we have $\mathbf{Coh}_{\tau 2}^{\Delta}(\mathbb{A}^1) \cong \mathbf{Coh}_{\tau 2}^{\Delta,\circ} \times \mathbb{A}^1$ and that $\mathbf{Coh}_{\tau 2}^{\Delta}(U_i) \cong \mathbf{Coh}_{\tau 2}^{\Delta}(\mathbb{A}^1) \times_{\mathbb{A}^1} U_i$ and $\widetilde{\mathbf{Coh}}_{\tau \underline{2}}^{\Delta}(U_i) \cong \widetilde{\mathbf{Coh}}_{\tau \underline{2}}^{\Delta}(\mathbb{A}^1) \times_{\Delta(\mathbb{A}^1)} \Delta(U_i)$. Hence, the bottom and the top lines are equivalent and it shows that $\mathsf{CT}_{\underline{2},*}^{\mathrm{Coh}}|_{\mathbf{Coh}_{\tau 2}^{\Delta}(U_i)} \cong \mathsf{CT}_{\underline{2},!}^{\mathrm{Coh},-}|_{\mathbf{Coh}_{\tau 2}^{\Delta}(U_i)}$.

For general partition $\underline{d}$, we shall stratify the space by number of distinct supports, and repeat the above argument, see the proof of Theorem 1.12 in the next section.

Proposition A.9. *For G a reductive group and with Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$, let $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{n}$ be a parabolic subalgebra accompany with $\mathfrak{l}$ the Levi subalgebra. Let $\mathfrak{p}^-$ be the opposite parabolic to $\mathfrak{p}$, then we have*

$$\mathsf{CT}_{\mathfrak{g},*}^{\mathfrak{p}} \cong \mathsf{CT}_{\mathfrak{g},!}^{\mathfrak{p}^-} : \mathcal{D}\text{-mod}\left([\mathfrak{g}/G]\right) \longrightarrow \mathcal{D}\text{-mod}\left([\mathfrak{l}/L]\right)$$

Proof. There is $\mathbb{G}_m$-action on $\mathfrak{g}$ via coweight and adjoint action, then Braden's theorem applies, see [Gun18, Theorem 3.7]. One can also deduce this by considering $\mathbf{Bun}_G(\mathbb{P}^1)$ with level structure at ∞, see [DG16, Remark 0.5.7]. ■

As a Corollary of Proposition A.8, we have the second adjunction for $\mathbf{Coh}_\alpha$:

Corollary A.10. *For any $\alpha \in \mathbf{Z}^+$ and partition $\underline{\alpha}$, the !-constant term functor $\mathsf{CT}_{\underline{\alpha},!}^{\mathrm{Coh}}$ is well defined and the second adjunction $\mathsf{CT}_{\underline{\alpha},*}^{\mathrm{Coh}} \cong \mathsf{CT}_{\underline{\alpha},!}^{\mathrm{Coh},-}$ holds.*

Proof. We have the following diagram:

$$\begin{tikzcd}
 & \mathbf{Bun}_{\underline{\alpha}} \arrow[dl, "\mathsf{q}_{\underline{\alpha}}^{\mathrm{Bun}}"'] \arrow[d, hook, "\jmath_{\underline{\alpha}}^L"] \arrow[dr, "\mathsf{p}_{\underline{\alpha}}^{\mathrm{Bun}}"] & \\
\mathbf{Bun}_{\underline{\alpha}} \arrow[d, hook, "\jmath_{\underline{\alpha}}"'] & \overline{\mathbf{Bun}}_{\underline{\alpha}}^L \arrow[dl, "\mathsf{q}_{\underline{\alpha}}^L"'] \arrow[d, hook, "\widetilde{\jmath}_{\underline{\alpha}}"] \arrow[r, "\mathsf{p}_{\underline{\alpha}}^L"] \arrow[dr] & \mathbf{Bun}_\alpha \arrow[d, hook, "\jmath_\alpha"] \\
\mathbf{Coh}_{\underline{\alpha}} & \widetilde{\mathbf{Coh}}_{\underline{\alpha}} \arrow[l, "\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}"] \arrow[r, "\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}"'] & \mathbf{Coh}_\alpha
\end{tikzcd} \tag{A.2.6}$$

where $\overline{\mathbf{Bun}}_{\underline{\alpha}}^L$ is Laumon's compactification and left rhombus in the diagram is cartesian. The map $\mathsf{p}_{\underline{\alpha}}^{\mathrm{Bun}}$ factors as $\mathsf{p}_{\underline{\alpha}}^L \circ \jmath_{\underline{\alpha}}^L$. Let us use again the notation $\mathsf{p}_{\underline{\alpha}}^L$ for the composition $\jmath_\alpha \circ \mathsf{p}_{\underline{\alpha}}^L$, then we have

$$\begin{aligned}
(\jmath_{\underline{\alpha}})^! \left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_* \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Coh}}\right)^! (\jmath_\alpha)_* &= (\jmath_{\underline{\alpha}})^! \left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Coh}}\right)_* \left(\widetilde{\jmath}_{\underline{\alpha}}\right)_* \left(\mathsf{p}_{\underline{\alpha}}^L\right)^! = (\jmath_{\underline{\alpha}})^! \left(\mathsf{q}_{\underline{\alpha}}^L\right)_* \left(\mathsf{p}_{\underline{\alpha}}^L\right)^! \\
&= \left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Bun}}\right)_* \left(\jmath_{\underline{\alpha}}^L\right)^! \left(\mathsf{p}_{\underline{\alpha}}^L\right)^! = \left(\mathsf{q}_{\underline{\alpha}}^{\mathrm{Bun}}\right)_* \left(\mathsf{p}_{\underline{\alpha}}^{\mathrm{Bun}}\right)^! = \mathsf{CT}_{\underline{\alpha},*}^{\mathrm{Bun}}
\end{aligned}$$

We then show the Proposition by recurrence on the torsion degree. It suffices to show $\mathsf{CT}_{\underline{\alpha},*}^{\mathrm{Coh}}$ and $\mathsf{CT}_{\underline{\alpha},!}^{\mathrm{Coh},-}$ matches on $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant l}$ for any l. For $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant l} = \mathbf{Bun}_\alpha \bigcup \mathbf{Coh}_\alpha^{\mathrm{tor.deg}\in[1,l]}$, we want to show the second adjunction result via recollement. We first need to show that $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}=l}$ is truncative in $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant l}$. This can be deduced from the following

Lemma A.11. *Let $\mathcal{Z} \subseteq \mathcal{X}$ be a locally closed substack, if for any field valued point $z \in \mathcal{Z}(\mathbf{k})$, there is a map $\mathbb{G}_m \to \mathrm{Aut}(z)$ such that the corresponding weights of $\mathbb{G}_m$ on the stalks of conormal bundle $\left(\mathbf{N}^*_{\mathcal{Z}/\mathcal{X}}\right)_z$ are strictly positive and on the stalks of cotangent bundle $(\mathbf{T}^*\mathcal{Z})_z$ are non-negative, then $\mathcal{Z}$ is truncative in $\mathcal{X}$.*

The Lemma is a pointwise version of contractiveness. To apply the Lemma, we can take $\mathcal{Z} \subseteq \mathcal{X}$ as $\mathbf{Coh}_\alpha^{\text{tor.deg}=l} \subseteq \mathbf{Coh}_\alpha^{\text{tor.deg}\leqslant l}$. Then for any point $\mathcal{F} \in \mathbf{Coh}_\alpha^{\text{tor.deg}=l}$, it admits a non-canonical splitting $\mathcal{F}^{\text{Tor}} \oplus \mathcal{F}^{\text{Bun}}$, we can calculate the fiber of corresponding conormal and cotangent bundles:

$$\left(\mathbf{N}^*_{\mathcal{Z}/\mathcal{X}}\right)_{\mathcal{F}} \cong \operatorname{Ext}^1\left(\mathcal{F}^{\text{Tor}}, \mathcal{F}^{\text{Bun}}\right)^\vee; \qquad (\mathbf{T}^*\mathcal{Z})_{\mathcal{F}} = \operatorname{Ext}^1\left(\mathcal{F}^{\text{Tor}}, \mathcal{F}^{\text{Tor}}\right) \oplus \operatorname{Ext}^1\left(\mathcal{F}^{\text{Bun}}, \mathcal{F}^{\text{Bun}}\right)$$

We can take the $\mathbb{G}_m$-action on $\mathbf{Coh}_\alpha^{\text{tor.deg}=l}$ so that it has positive weight on $\mathcal{F}^{\text{Tor}}$ and zero on $\mathcal{F}^{\text{Bun}}$.

Back to the proof of Corollary A.10, we have shown $(\jmath_{\underline{\alpha}})^! \left(\mathsf{CT}^{\text{Coh}}_{\underline{\alpha},*}\right)|_{\mathbf{Coh}_\alpha^{\text{tor.deg}\leqslant l}}(\jmath_\alpha)_*$ is just $\mathsf{CT}^{\text{Bun}}_{\underline{\alpha},*}$, thus is right adjointable and isomorphic to $\mathsf{CT}^{\text{Bun},-}_{\underline{\alpha},!}$ according to [DG16]. For the complement of $\mathbf{Bun}_\alpha$ in $\mathbf{Coh}_\alpha^{\text{tor.deg}\leqslant l}$, using the diagram 3.2.2, we can consider:

$$\begin{tikzcd}
\mathbf{Bun}_{\alpha-\tau l} \times \mathbf{Coh}_{\tau l} \arrow[d, "\jmath_{\alpha-\tau l}\times\mathrm{id}"'] & \mathbf{Coh}_\alpha^{\text{tor.deg}=l} \arrow[l, "{'\mathsf{q}^{\text{Coh}}}"'] \arrow[d, "\tilde{\iota}^l"'] \arrow[dr, hook, "\iota^l"] & \\
\mathbf{Coh}_{\alpha-\tau l} \times \mathbf{Coh}_{\tau l} & \widetilde{\mathbf{Coh}}_{\alpha-\tau l,\tau l} \arrow[l, "\mathsf{q}^{\text{Coh}}"'] \arrow[r, "\mathsf{p}^{\text{Coh}}_{\alpha-\tau l,\tau l}"'] & \mathbf{Coh}_\alpha \\
 & \mathcal{E}^{\underline{\alpha}}_{\alpha-\tau l,\tau l} \arrow[u, "\pi_1"] \arrow[r, "\pi_2"'] & \widetilde{\mathbf{Coh}}_{\underline{\alpha}} \arrow[u, "\mathsf{p}^{\text{Coh}}_{\underline{\alpha}}"'] \arrow[d, "\mathsf{q}^{\text{Coh}}_{\underline{\alpha}}"] \\
\mathcal{E}_{\mathbb{M}} \arrow[uu, "\pi_{1,\mathbb{M}}"] \arrow[ur, hook, "\iota_{\mathbb{M}}"] \arrow[rr, "\pi_{2,\mathbb{M}}"] & & \mathbf{Coh}_{\underline{\alpha}}
\end{tikzcd} \tag{A.2.7}$$

The functor $\mathsf{CT}^{\text{Coh}}_{\underline{\alpha},*} \circ (\iota^l)_*$ is thus $\left(\mathsf{q}^{\text{Coh}}_{\underline{\alpha}}\right)_* \left(\mathsf{p}^{\text{Coh}}_{\underline{\alpha}}\right)^! (\iota^l)_* = \left(\mathsf{q}^{\text{Coh}}_{\underline{\alpha}}\right)_* (\pi_2)_* (\pi_1)^! \left(\tilde{\iota}^l\right)_*$. We have seen the fiber product $\mathcal{E}^{\underline{\alpha}}_{\alpha-\tau l,\tau l}$ can be stratified by contingency matrices, the functor $\mathsf{CT}^{\text{Coh}}_{\underline{\alpha},*} \circ (\iota^l)_*$ is filtered also by contingency matrices. For each contingency matrix $\mathbb{M} = \begin{bmatrix} \alpha_1 - \tau l_1 & \alpha_2 - \tau l_2 & \dots & \alpha_s - \tau l_s \\ \tau l_1 & \tau l_2 & \dots & \tau l_s \end{bmatrix}$, denote by $\underline{\alpha'} = (\alpha_1 - \tau l_1, \dots, \alpha_s - \tau l_s)$, $\tau \underline{l} = (\tau l_1, \dots, \tau l_s)$ corresponding partitions, then the map $\pi_{1,\mathbb{M}}$ factors through $\widetilde{\mathbf{Coh}}_{\underline{\alpha'}} \times \widetilde{\mathbf{Coh}}_{\tau\underline{l}}$. By diagram chase, we can deduce the associated graded piece of the functor is given by

$$\begin{aligned}
&\left(\pi_{2,\mathbb{M}}\right)_* \left(\pi_{1,\mathbb{M}}\right)^! \left(\mathsf{q}^{\text{Coh}}\right)_* (\iota^l)_* \cong \left(\pi_{2,\mathbb{M}}\right)_* \left(\pi_{1,\mathbb{M}}\right)^! \left(\jmath_{\alpha-\tau l} \times \mathrm{id}\right)_* \left('\mathsf{q}^{\text{Coh}}\right)_* \\
\cong &\left(\mathsf{CT}^{\text{Coh}}_{\underline{\alpha'},*} \times \mathsf{CT}^{\text{Coh}}_{\tau\underline{l},*}\right) \left(\jmath_{\alpha-\tau l} \times \mathrm{id}\right)_* \left('\mathsf{q}^{\text{Coh}}\right)_* \cong \left(\jmath_{\underline{\alpha'}} \times \mathrm{id}\right)_* \left(\mathsf{CT}^{\text{Bun}}_{\underline{\alpha'}} \times \mathsf{CT}^{\text{Coh}}_{\tau\underline{l}}\right) \left('\mathsf{q}^{\text{Coh}}\right)_*
\end{aligned}$$

where $\left('\mathsf{q}^{\text{Coh}}\right)_*$ is an equivalence and $\mathsf{CT}^{\text{Coh}}_{\underline{\alpha'},*} \times \mathsf{CT}^{\text{Coh}}_{\tau\underline{l},*} \cong \mathsf{CT}^{\text{Coh},-}_{\underline{\alpha'},!} \times \mathsf{CT}^{\text{Coh},-}_{\tau\underline{l},!}$ by induction hypothesis. When $l = 1$, the desired second adjunction of $\mathsf{CT}^{\text{Coh}}_{\underline{\alpha},*}|_{\mathbf{Coh}_\alpha^{\text{tor.deg}\leqslant 1}}$ can be glued from $\mathsf{CT}^{\text{Bun}}_{\underline{\alpha},*}$ and $\mathsf{CT}^{\text{Coh}}_{\underline{\alpha},*}|_{\mathbf{Coh}_\alpha^{\text{tor.deg}=1}}$ from Lemma A.6. Hence, for general $l \geqslant 1$, the second adjunction of $\mathsf{CT}^{\text{Coh}}_{\underline{\alpha},*}|_{\mathbf{Coh}_\alpha^{\text{tor.deg}\leqslant l}}$ can be glued from $\mathsf{CT}^{\text{Coh}}_{\underline{\alpha},*}|_{\mathbf{Coh}_\alpha^{\text{tor.deg}\leqslant l-1}}$ by induction and $\mathsf{CT}^{\text{Coh}}_{\underline{\alpha},*}|_{\mathbf{Coh}_\alpha^{\text{tor.deg}=l}}$, as we want. ∎

A.3 Miraculous duality

A.3.1 *Some properties of pseudo-identity functor.* We recall $\mathcal{X}$ is said to be miraculous if $\text{Ps-Id}_{\mathcal{X},!}$: $\mathcal{D}\text{-mod}(\mathcal{X})_{\text{co}} \to \mathcal{D}\text{-mod}(\mathcal{X})$ induced by kernel $(\Delta_X)_!(\mathbf{e}_X)$ is an equivalence. It is proved in [Gai16, Corollary 6.7.2]

Proposition A.12. *If $\mathcal{X}$ is a QCA stack, then it is miraculous if and only if* $\mathrm{Ps\text{-}Id}_{\mathcal{X},!}$ *preserves compact objects.*

Now we assume that $\mathcal{X}$ is locally QCA. Under this assumption, every quasi-compact open in $\mathcal{X}$ is QCA. If we suppose that $\mathcal{X}$ is truncable, then $\mathrm{Ctrnk}(\mathcal{X})$ the poset of co-truncative open substacks in $\mathcal{X}$ is cofinal, and we have

$$\mathcal{D}\text{-mod}(\mathcal{X}) \cong \lim_{U \in \mathrm{Ctrnk}(\mathcal{X})^{\mathrm{op}}} \mathcal{D}\text{-mod}(U)$$

Let $(\mathcal{U} \subseteq \mathcal{X}) \in \mathrm{Ctrnk}(\mathcal{X})$ be a co-truncative open substack and $\mathcal{Z}$ the closed complement of $\mathcal{X}$ in $\mathcal{X}$. Then the definition of truncativeness gives us the following diagram of duality maps:

$$\begin{tikzcd}[column sep=huge, row sep=huge]
\mathcal{D}\text{-mod}(\mathcal{Z})_{\mathrm{co}} \arrow[r, bend left=50, "(j^?)^\vee = j_{\mathrm{co},!}"] \arrow[r, phantom, shift left=4, "\perp"] \arrow[r, leftarrow, shift left=2, "(j_*)^\vee = j^*_{\mathrm{co}}"] \arrow[r, phantom, "\perp"] \arrow[r, shift right=2, "(j^!)^\vee = j_{\mathrm{co},*}"'] \arrow[r, phantom, shift right=4, "\perp"] \arrow[r, leftarrow, bend right=50, "(j_!)^\vee = j^?_{\mathrm{co}}"'] \arrow[d, "\mathrm{Ps\text{-}Id}_{\mathcal{Z},!}"'] & \mathcal{D}\text{-mod}(\mathcal{X})_{\mathrm{co}} \arrow[r, bend left=50, "(i_?)^\vee = i^*_{\mathrm{co}}"] \arrow[r, phantom, shift left=4, "\perp"] \arrow[r, leftarrow, shift left=2, "(i^!)^\vee = i_{\mathrm{co},*}"] \arrow[r, phantom, "\perp"] \arrow[r, shift right=2, "(i_!)^\vee = i^!_{\mathrm{co}}"'] \arrow[r, phantom, shift right=4, "\perp"] \arrow[r, leftarrow, bend right=50, "(i^*)^\vee = i_{\mathrm{co},?}"'] \arrow[d, "\mathrm{Ps\text{-}Id}_{\mathcal{X},!}"] & \mathcal{D}\text{-mod}(\mathcal{U})_{\mathrm{co}} \arrow[d, "\mathrm{Ps\text{-}Id}_{\mathcal{U},!}"] \\
\mathcal{D}\text{-mod}(\mathcal{Z}) \arrow[r, bend left=50, "j_!"] \arrow[r, phantom, shift left=4, "\perp"] \arrow[r, leftarrow, shift left=2, "j^* = j^!"] \arrow[r, phantom, "\perp"] \arrow[r, shift right=2, "j_*"'] \arrow[r, phantom, shift right=4, "\perp"] \arrow[r, leftarrow, bend right=50, "j^?"'] & \mathcal{D}\text{-mod}(\mathcal{X}) \arrow[r, bend left=50, "i^*"] \arrow[r, phantom, shift left=4, "\perp"] \arrow[r, leftarrow, shift left=2, "i_* = i_!"] \arrow[r, phantom, "\perp"] \arrow[r, shift right=2, "i^!"'] \arrow[r, phantom, shift right=4, "\perp"] \arrow[r, leftarrow, bend right=50, "i_?"'] & \mathcal{D}\text{-mod}(\mathcal{U})
\end{tikzcd} \tag{A.3.1}$$

where the functors on the bottom line are well-defined, thanks to (co)truncativeness. As is indicated in the diagram, the functors in the top line are dual functors to those in the bottom. Moreover, we have the following properties:

Lemma A.13. *[DG15] The functors $i_?, j_!$ are fully faithful. The functors $i^!, j^*$ preserves compactness.*

Remark A.14. One should be cautious here: choosing three out of four functors makes the top or bottom a recollement diagram, but the pseudo-identity functors do not give a morphism between recollement data. In fact, we have the following strange compatibility result in [Gai16]:

Proposition A.15. *Assume $\mathcal{U} \subseteq \mathcal{X}$ is co-truncative, then in diagram A.3.1, we have*

$$\mathrm{Ps\text{-}Id}_{\mathcal{X},!} \circ j_{\mathrm{co},*} \cong j_! \circ \mathrm{Ps\text{-}Id}_{\mathcal{U},!} \qquad \mathrm{Ps\text{-}Id}_{\mathcal{U},!} \circ j^?_{\mathrm{co}} \cong j^* \circ \mathrm{Ps\text{-}Id}_{\mathcal{X},!}$$
$$i^* \circ \mathrm{Ps\text{-}Id}_{\mathcal{X},!} \cong \mathrm{Ps\text{-}Id}_{\mathcal{Z},!} \circ i^!_{\mathrm{co}} \qquad \mathrm{Ps\text{-}Id}_{\mathcal{X},!} \circ i_{\mathrm{co},?} \cong i_* \circ \mathrm{Ps\text{-}Id}_{\mathcal{Z},!}$$

Proposition A.16. *In the setting of diagram A.3.1, assume* $\mathrm{Ps\text{-}Id}_{\mathcal{Z},!}$ *and* $\mathrm{Ps\text{-}Id}_{\mathcal{U},!}$ *are equivalences, then* $\mathrm{Ps\text{-}Id}_{\mathcal{X},!}$ *preserves compactness.*

Proof. Let $\mathbf{F} \in \mathcal{D}\text{-mod}(\mathcal{X})^c_{\mathrm{co}}$, we have recollement $i_* i^* \left(\mathrm{Ps\text{-}Id}_{\mathcal{X},!}(\mathbf{F})\right) \to \mathrm{Ps\text{-}Id}_{\mathcal{X},!}(\mathbf{F}) \to j_! j^* \left(\mathrm{Ps\text{-}Id}_{\mathcal{X},!}(\mathbf{F})\right)$, thus to show $\mathrm{Ps\text{-}Id}_{\mathcal{X},!}(\mathbf{F}) \in \mathcal{D}\text{-mod}(\mathcal{X})^c$, it suffices to have $j^* \left(\mathrm{Ps\text{-}Id}_{\mathcal{X},!}(\mathbf{F})\right) \in \mathcal{D}\text{-mod}(\mathcal{U})^c$ and $i^* \left(\mathrm{Ps\text{-}Id}_{\mathcal{X},!}(\mathbf{F})\right) \in \mathcal{D}\text{-mod}(\mathcal{Z})^c$, as i_* and $j_!$ already preserve compactness.
Now $j^* \left(\mathrm{Ps\text{-}Id}_{\mathcal{X},!}(\mathbf{F})\right) = \mathrm{Ps\text{-}Id}_{\mathcal{U},!}\left(j^?_{\mathrm{co}}(\mathbf{F})\right)$. Since j^* preserves compactness, its dual functor $j_{\mathrm{co},*}$ is continuous and thus the right adjoint $j^?_{\mathrm{co}} \dashv j_{\mathrm{co},*}$ preserves compactness. Similarly, we can deduce $i^!_{\mathrm{co}}$ preserves compactness because $i^!$ does according to Lemma A.13. ∎

As an immediate consequence of Proposition A.16 and Proposition A.12, if $\mathcal{X}$ is a QCA stack with co-truncative open $\mathcal{U} \subseteq \mathcal{X}$ such that $\mathcal{U}$ and its complement $\mathcal{Z}$ are miraculous, then $\mathcal{X}$ is also miraculous. More generally, we have

Proposition A.17. *If $\mathcal{X}$ is a locally QCA truncatable stack, then* $\text{Ps-Id}_{\mathcal{X},!}$ *is an equivalence if and only if it preserves compactness. Thus $\mathcal{X}$ is miraculous if some co-truncative $\mathcal{U} \subseteq \mathcal{X}$ and $\mathcal{Z} = \mathcal{X} \backslash \mathcal{U}$ are.*

Proof. By [Gai16, Proposition 7.6.4], $\mathcal{X}$ is miraculous if and only if every co-truncative quasi-compact open is miraculous. Take $(j : \mathcal{U}_i \subseteq \mathcal{X}) \in \text{Ctrnk}(\mathcal{X})$ with $\mathcal{U}_i$ quasicompact, then $\mathcal{U}_i$ is automatically QCA, by Proposition A.16, it is enough to show $\text{Ps-Id}_{\mathcal{U}_i,!}$ preserves compactness. For $\mathbf{F} \in \mathcal{D}\text{-mod}(U_i)^c_{\text{co}}$, we can consider $\text{Ps-Id}_{U_i,!}(\mathbf{F}) \cong j^! \circ j_! \circ \text{Ps-Id}_{U_i,!}(\mathbf{F}) \cong j^! \circ \text{Ps-Id}_{\mathcal{X},!} \circ j_{\text{co},*}(\mathbf{F})$. Since $\mathcal{X}$ is truncatable, by [DG15, Corollary 1.9.4], $j_{\text{co},*}(\mathbf{F}) \in \mathcal{D}\text{-mod}(\mathcal{X})_{\text{co}}$ is compact. Moreover, by assumption there exists $\mathcal{U}$ and $\mathcal{Z} = \mathcal{X} \backslash \mathcal{U}$ are miraculous, thus by Proposition A.16, $\text{Ps-Id}_{\mathcal{X},!}$ preserves compactness, and we can deduce $\text{Ps-Id}_{\mathcal{U}_i,!}(\mathbf{F}) \in \mathcal{D}\text{-mod}(\mathcal{U}_i)^c$, as we want. ∎

A.3.2 Proof of Theorem 1.11. Since $\mathbf{Coh}_{\tau d}$ is a QCA stack, according to [Gai16, Theorem 6.7.2], it suffices to show that the pseudo-identity functor on $\mathbf{Coh}_{\tau d}$ preserves compactness, or equivalently, it admits a continuous right adjoint. We reduce the case to the following Proposition:
Back to the proof of Theorem. For smooth curve X, we can cover it by affine U_i maps étalely to $\mathbb{A}^1$, this gives a covering of $\mathbf{Coh}_{\tau d}(X)$ by $\mathbf{Coh}_{\tau d}(U_i)$, it suffices to show each $\mathbf{Coh}_{\tau d}(U)$ is miraculous. Let $U \to \mathbb{A}^1$ be étale, then the induced map $\mathcal{Q}_{\tau d}(U) \to \mathfrak{gl}_d$ is étale, which descends to an étale map $\mathbf{Coh}_{\tau d}(U) \to \mathbf{Coh}_{\tau d}(\mathbb{A}^1)$ by quotienting GL_d. We claim the miraculousity for $\mathbf{Coh}_{\tau d}(\mathbb{A}^1) \cong \left[\mathfrak{gl}_d / \text{GL}_d\right]$ holds, which will be proved in the next section. It remains to show it implies the miraculousity of $\mathbf{Coh}_{\tau d}(U)$. We can obtain a stratification of $\mathbf{Coh}_{\tau d}(U)$ by considering the number of distinct points of the support. For $\underline{d} = (d_1, ..., d_2)$ a partition of d, write $\mathbf{Coh}^{\text{disj}}_{\tau \underline{d}}(U)$ the substack of torsion sheaves whose supports contain r-distinct points, with multiplicity d_i respectively, then $\mathbf{Coh}^{\text{disj}}_{\tau \underline{d}}(U) \cong \left[\prod_i \mathbf{Coh}^{\Delta}_{\tau d_i}(U) \times_{U^r} U^r_{\text{disj}} / \Gamma_{\underline{d}}\right]$, where $\Gamma_{\underline{d}} \subseteq \mathfrak{S}_r$ is the subgroup acts on the support of $\mathbf{Coh}^{\text{disj}}_{\tau \underline{d}}(U)$ that preserves the ordered partition. Denote also $\mathbf{Coh}^{(=r)}_{\tau d}(U)$ the union of $\mathbf{Coh}^{\text{disj}}_{\tau \underline{d}}(U)$ that the length of partition $\underline{d}$ is r. We obtain an increasing filtration $\mathbf{Coh}^{(\leqslant 1)}_{\tau d}(U) \subseteq ... \subseteq \mathbf{Coh}^{(\leqslant d)}_{\tau d}(U) = \mathbf{Coh}_{\tau d}(U)$. The following consequences are important:

Lemma A.18. *(i) Let $\mathcal{U} \subseteq \mathcal{X}$ be a co-truncative open substack of $\mathcal{X}$, with closed complement $\mathcal{Z}$, if $\mathcal{X}$ is miraculous then $\mathcal{U}$ and $\mathcal{Z}$ are also.*
(ii) Assume $\mathcal{X}$ admits a closed or open truncative exhausion $\mathcal{X}_1 \subseteq \mathcal{X}_2 \subseteq ... \subseteq \mathcal{X}_r = \mathcal{X}$ such that each $\mathcal{X}_{i+1} \backslash \mathcal{X}_i$ is miraculous, then $\mathcal{X}$ is miraculous.
(iii) Let $\mathcal{X}$ and $\mathcal{Y}$ be miraculous, then $\mathcal{X} \times \mathcal{Y}$ is also. The quotient of $\mathcal{X}$ by a finite group is also miraculous.

In our case, we have seen that $\mathbf{Coh}^{\Delta}_{\tau d_i}(U) \cong \mathbf{Coh}^{\Delta,\circ}_{\tau d_i} \times U$. Using Lemma A.18(i) and the fact (Theorem A.20) that $\mathbf{Coh}_{\tau d}(\mathbb{A}^1)$ is miraculous, we can deduce $\mathbf{Coh}^{\Delta}_{\tau d_i}(\mathbb{A}^1)$ is miraculous, thus $\mathbf{Coh}^{\Delta,\circ}_{\tau d_i}$ and $\mathbf{Coh}^{\Delta}_{\tau d_i}(U)$

are also. Using Lemma A.18(iii), we can deduce $\mathbf{Coh}^{\mathrm{disj}}_{\tau\underline{d}}(U)$ is also miraculous. Combine the filtration $\mathbf{Coh}^{(\leqslant\bullet)}_{\tau d}(U)$ with Lemma A.18(ii), we can conclude that $\mathbf{Coh}_{\tau d}(U)$ is also miraculous. Finally, by gluing the étale cover, we get $\mathbf{Coh}_{\tau d}(X)$ is miraculous.

A.3.3 Proof of Theorem 1.12. The miraculousity of $\mathbf{Coh}_\alpha$ can be obtained by gluing **Bun** and $\mathbf{Coh}_{\tau d}$. Let us assume $\mathrm{rk}(\alpha) \geqslant 1$, then $\mathbf{Coh}_\alpha = \mathrm{colim}_{l\geqslant 0}\, \mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant l}$, by Lemma A.18, it is enough to prove that each $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant l}$ is miraculous. For each torsion degree l, we have

$$\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant l-1} \xhookrightarrow{j^l_\alpha} \mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant l} \xhookleftarrow{\iota^{=l}_\alpha} \mathbf{Coh}_\alpha^{\mathrm{tor.deg}=l}$$

Then $\mathcal{D}\text{-mod}\left(\mathbf{Coh}_\alpha^{\mathrm{tor.deg}=l}\right) \cong \mathcal{D}\text{-mod}\left(\mathbf{Coh}_{\alpha-\tau l}\times \mathbf{Coh}_{\tau l}\right)$, which is known to be miraculous by Lemma A.18 and Theorem 1.11. Also, $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}\leqslant l-1}$ is miraculous by induction, since we have shown $\mathbf{Coh}_\alpha^{\mathrm{tor.deg}=l}$ is truncative, we can use Proposition A.17 to conclude.

A.4 Deligne-Lusztig involution for Lie algebras

Let G be a reductive Lie group and $\mathfrak{g} = \mathrm{Lie}(G)$ be its Lie algebra. We would like to see that the pseudo-identity functor on $\mathcal{D}\text{-mod}\left([\mathfrak{g}/G]\right)$ or $\mathsf{Shv}\left([\mathfrak{g}/G]\right)$ is (a) an equivalence and, (b) related to Deligne-Lusztig involution. Let us use $\mathrm{D}\left([\mathfrak{g}/G]\right)$ for either two sheaf setups and use $\mathbb{C}$ for coefficient.

***A.4.1** The wonderful compactification and logarithmic bundles.* We consider $\overline{G}$ the De Concini-Procesi wonderful compactification of G, which has a geometric realization: let $\mathfrak{g}_\Delta$ denote the image of diagonal embedding $\mathfrak{g} \hookrightarrow \mathfrak{g}\times\mathfrak{g}$, then $\overline{G}$ is the closure of $(G\times G)$-orbit of $\mathfrak{g}_\Delta$ in $\mathrm{Gr}(\dim(\mathfrak{g}), \mathfrak{g}\times\mathfrak{g})$. Then $\overline{G}$ contains G as a open subset. We denote by $D_G := \overline{G}\backslash G$ the divisor in $\overline{G}$, also let $T_{D_G}\overline{G}$ and $T^*_{D_G}\overline{G}$ be the logarithmic tangent bundle and the logarithmic cotangent bundle of $\overline{G}$ with respect to the divisor D_G respectively. From the construction we can see the compactification $\overline{G}$, as well as D_G inherit a $(G\times G)$-action. This also gives rises to $(G\times G)$-action on $T_{D_G}\overline{G}$. Moreover the composition

$$T^*_{D_G}\overline{G} \hookrightarrow \overline{G}\times\mathfrak{g}\times\mathfrak{g} \twoheadrightarrow T_{D_G}\overline{G}$$

gives the duality between logarithmic cotangent and tangent bundles. Notice that by construction, the restriction of $T^*_{D_G}\overline{G}$ to $G \subseteq \overline{G}$ has pointwise fiber isomorphic $\mathfrak{g}$. Next, we consider the fibers of $T^*_{D_G}\overline{G}$ over D_G. As is depicted in [Băl23], one can describe the irreducible component of the simple normal crossing divisor D_G explicitly, they are labelled by C_i for $i \in \{1, \ldots, \mathrm{rk}(G)\}$, also the $(G\times G)$-orbits in D_G are in bijection with proper subsets of $\{1, \ldots, \mathrm{rk}(G)\}$. In particualr, the $(G\times G)$-orbit corresponds to I is given by $\mathcal{O}_I = \bigcup_{i\notin I} C_i$. We denote by $T^*_{D_G,I}\overline{G}$ the restriction of $T^*_{D_G}\overline{G}$ onto $\mathcal{O}_I$. Then we have the following result [Băl23, Section 3.3]:

Proposition A.19. *The stratification of D_G by $(G\times G)$-orbits gives rises to a stratification of $T^*_{D_G}\overline{G}$ over the divisor D_G. In particular, for any $z_I \in \mathcal{O}_I$, the $G\times G$ equivariant vector bundle $T^*_{D_G}\overline{G}$ has fiber $\mathfrak{p}_I \times_{\mathfrak{l}_I} \mathfrak{p}_I^-$ over z_I, where $\mathfrak{l}_I$ is the levi subalgebra of the parabolic $\mathfrak{p}_I$.*

Let $\mathfrak{a}_I = \mathfrak{p}_I/[\mathfrak{p}_I, \mathfrak{p}_I]$, the orbit O_I fibers over $G/P_I \times G/P_I^-$, thus any point $a \in O_I$ can be associated with a parabolic subalgebra $\mathfrak{p}_a$ such that $\mathfrak{p}_a \xrightarrow{\mathfrak{c}_a} \mathfrak{p}_a/[\mathfrak{p}_a, \mathfrak{p}_a] \cong \mathfrak{a}_I$. Then there is a smooth morphism $\nu_I : T^*_{D_G,I}\overline{G} \to \mathfrak{a}_I$ given by $(a, x, x^-) \mapsto \mathfrak{c}_a(x)$.

The diagonal embedding $[\mathfrak{g}/G] \hookrightarrow [\mathfrak{g}/G] \times [\mathfrak{g}/G]$ factors as the following:

$$\begin{tikzcd}
G\times G\backslash T^*G \cong G\times G\backslash G\times\mathfrak{g} \arrow[r, "j_G"] \arrow[d, "\Delta"'] & G\times G\backslash \overline{G}\times\mathfrak{g}\times\mathfrak{g} \arrow[d, "\mathrm{pr}"] \\
G\times G\backslash \mathfrak{g}\times\mathfrak{g} & G\times G\backslash T^*_{D_G}\overline{G} = G\times G\backslash \left(\bigcup_I T^*_{D_G,I}\overline{G}\cup T^*G\right) \arrow[l, "\overline{\mu}_G"']
\end{tikzcd} \tag{A.4.1}$$

where we identify $T_{D_G}\overline{G}$ with $T^*_{D_G}\overline{G}$ to obtain pr and $T^*_{D_G}\overline{G} \to \mathfrak{g}\times\mathfrak{g}$ is the *generalized moment map*, given by $(g, x, y) \mapsto (x, y)$. The category $\mathrm{D}\left([\mathfrak{g}/G]\right)$ of the QCA stack $[\mathfrak{g}/G]$ is self-dual and pseudo-identity functor for $\mathrm{D}([\mathfrak{g}/G])$ can be viewed as an endofunctor, which induced by the kernel $\Delta_!\left(\mathbb{C}_{[\mathfrak{g}/G]}\right)$, according to diagram A.4.1, it can be written as $(\overline{\mu}_G \circ \mathrm{pr} \circ j_G)_!\left(\mathbb{C}_{[\mathfrak{g}/G]}\right)$.

***A.4.2** Deligne-Lusztig duality for Lie algebras.* We are going to see the duality result for Lie algebras:

Theorem A.20. *The stack $[\mathfrak{g}/G]$ is miraculous, moreover, the pseudo-identity functor on $[\mathfrak{g}/G]$ can be regarded as Deligne-Lusztig duality functor.*

Proof. According to [Gai16, Theorem 6.7.5], it is enough to prove that the functor $\text{Ps-Id}_{\mathfrak{g},!}$ induced by the kernel $\Delta_!\left(\mathbb{C}_{[\mathfrak{g}/G]}\right)$ admits a continuous right adjoint. We prove this by showing that there is a recollement on the kernel such that each graded piece admits a continuous right adjoint. Write $\mathfrak{s}_I = \mathfrak{p}_I \times_{\mathfrak{l}_I} \mathfrak{p}_I^-$ with $\sigma_I : G\times G\backslash\mathfrak{s}_I \hookrightarrow G\times G\backslash T^*_{D_G,I}\overline{G}$ and ι_I the inclusion $G\times G\backslash T^*_{D_G,I}\overline{G} \hookrightarrow G\times G\backslash T^*_{D_G}\overline{G}$. We have the following diagram:

$$\begin{tikzcd}
 & G\times G\backslash\mathfrak{g}\times\mathfrak{g} & & \\
G\times G\backslash T^*G \arrow[r, "\mathrm{pr}\circ j_G"] \arrow[d, "\nu_\emptyset"'] & G\times G\backslash T^*_{D_G}\overline{G} \arrow[u, "\overline{\mu}_G"] \arrow[d, "\nu"] & G\times G\backslash T^*_{D_G,I}\overline{G} \arrow[l, hook', "\iota_I"'] \arrow[d, "\nu_I"'] & G\times G\backslash\mathfrak{s}_I \arrow[l, hook', "\sigma_I"'] \arrow[d, "\nu_{a_I}"] \\
\mathfrak{h} \arrow[r, "\Delta_\mathfrak{h}"] & \mathfrak{h}\times\mathfrak{h} & \mathfrak{a}_I \arrow[l, "\tilde{\iota}_I"'] & a_I \arrow[l, "\tilde{\sigma}_I"']
\end{tikzcd} \tag{A.4.2}$$

where $\mathfrak{h}$ is the universal Cartan subalgebra of $\mathfrak{g}$ and $\nu : T^*_{D_G}\overline{G} \to \mathfrak{h}\times\mathfrak{h}$ is the composition $T^*_{D_G}\overline{G} \to \mathfrak{g}\times\mathfrak{g} \to \mathfrak{h}\times\mathfrak{h}$, where the first map is the generalized moment map describe before and the second is the projection. The maps $\nu_I, \nu_\emptyset$ have been described above and the last map ν_{a_I} is just the restriction of ν_I to a point. The functor $\Delta_!(\mathbb{C}) = (\overline{\mu}_G \circ \mathrm{pr} \circ j_G)_!(\mathbb{C})$ is glued from $(\overline{\mu}_G)_!(\iota_I)_!(\iota_I)^*(\mathrm{pr}\circ j_G)_!(\mathbb{C})$. We have

$$(\overline{\mu}_G)_!(\iota_I)_*(\iota_I)^*(\mathrm{pr}\circ j_G)_!(\mathbb{C}) = (\overline{\mu}_G)_!(\iota_I)_*(\iota_I)^*(\mathrm{pr}\circ j_G)_!\nu_\emptyset^*(\mathbb{C})$$
$$=(\overline{\mu}_G)_!(\iota_I)_*(\iota_I)^*\nu^*(\Delta_\mathfrak{h})_!(\mathbb{C}) = (\overline{\mu}_G)_!(\iota_I)_*(\nu_I)^*(\tilde{\iota}_I)^*(\Delta_\mathfrak{h})_!(\mathbb{C})$$

By contraction principle, $(\tilde{\iota}_I)^*(\Delta_{\mathfrak{h}})_!(\mathbb{C}) = (\tilde{\sigma}_I)_!(\mathbb{C})$. The above becomes:

$$(\overline{\mu}_G)_!(\iota_I)_!(\sigma_I)_!(\nu_{a_I})^*(\mathbb{C}) = (\overline{\mu}_G \circ \iota_I \circ \sigma_I)_!(\mathbb{C})$$

Let us consider the diagram:

$$\begin{tikzcd}
 & & \mathfrak{p}_I \times_{\mathfrak{l}_I} \mathfrak{p}_I^- \arrow[dl, "\mathrm{pr}_I"] \arrow[dr, "\mathrm{pr}_I^-"'] \arrow[dll, bend right, "\iota_{I,1}"'] \arrow[drr, bend left, "\iota_{I,2}"] & & \\
 & \mathfrak{p}_I \arrow[dl, "\mathsf{p}_I"'] \arrow[dr, "\mathsf{q}_I"] & & \mathfrak{p}_I^- \arrow[dl, "\mathsf{q}_I^-"'] \arrow[dr, "\mathsf{p}_I^-"] & \\
\mathfrak{g} & & \mathfrak{l}_I & & \mathfrak{g}
\end{tikzcd}$$

The endfunctor on $\mathrm{D}([\mathfrak{g}/G])$ given by kernel $(\overline{\mu}_G \circ \iota_I \circ \sigma_I)_!(\mathbb{C})$ is exactly $\left(\iota_{I,2}\right)_! \circ \left(\iota_{I,1}\right)^*$, that equals to

$$(\mathsf{p}_I^-)_!(\mathrm{pr}_I^-)_!(\mathrm{pr}_I)^*(\mathsf{p}_I)^* \cong (\mathsf{p}_I^-)_!(\mathsf{q}_I^-)^*(\mathsf{q}_I)_!(\mathsf{p}_I)^* = \mathsf{Eis}^{\mathfrak{g}}_{\mathfrak{p}_I^-,!} \circ \mathsf{CT}^{\mathfrak{p}_I}_{\mathfrak{g},!}$$

who carries a continuous right adjoint. ∎

B Analytic stacks

B.1 Derived analytic stacks and analytification functor

In the construction of comultiplication of CoHA, we use the analytification of (derived) stack in the sense of [Lur], we recall some terminologies and properties there. Roughly speaking, we would define a derived analytic space to be a pair $X = (\mathcal{X}, \mathcal{O}_X)$ with extra conditions, where $\mathcal{X}$ is an ∞-topos and $\mathcal{O}_X$ is a $\mathcal{T}_{\mathrm{an}}$-structure for the pregeometry $\mathcal{T}_{\mathrm{an}}$. Let us unpack the definitions in the sequel:

Definition B.1. For an ∞-category $\mathcal{G}$, its admissibility structure is
(1) A subcategory $\mathcal{G}^{\mathrm{ad}} \subseteq \mathcal{G}$ contains every object of $\mathcal{G}$ and with morphisms said to be admissible that satisfy the following properties:

- Pullback of admissible morphism along any morphism is admissible.
- For any morphisms f, g, if g and $g \circ f$ are admissible, then f is admissible.
- Any retract of admissible morphism is admissible.

(2) A Grothendieck topology on $\mathcal{G}$ that generated by admissible morphisms.
A *pregeometry* $\mathcal{T}$ is an ∞-category accompany with an admissibility structure and with finite products.

Definition B.2. For a pregeometry $\mathcal{T}$ and an ∞-topos $\mathcal{X}$, a $\mathcal{T}$*-structure on* $\mathcal{X}$ is a finite-product preserving functor $\mathcal{O} : \mathcal{T} \to \mathcal{X}$ such that:

(1) For the following left pullback square in $\mathcal{T}$ with admissible f

$$\begin{tikzcd} U' \arrow[r] \arrow[d] & U \arrow[d, "f"] \\ X' \arrow[r] & X \end{tikzcd} \qquad \begin{tikzcd} \mathcal{O}(U') \arrow[r] \arrow[d] & \mathcal{O}(U) \arrow[d, "\mathcal{O}(f)"] \\ \mathcal{O}(X') \arrow[r] & \mathcal{O}(X) \end{tikzcd}$$

the induced diagram in right-above is a pullback square in $\mathcal{X}$.
(2) For admissible covering $\{U_\alpha \to X\}$, the induced map $\coprod \mathcal{O}(U_\alpha) \to \mathcal{O}(X)$ is an effective epimorphism.

For our usage, we consider the following pregeometries $\mathcal{T}_{\text{disc}}, \mathcal{T}_{\text{ét}}, \mathcal{T}_{\text{an}}$. The pregeometry $\mathcal{T}_{\text{disc}}$ is defined to be $\mathrm{N}\left(\mathrm{Poly}_{\mathbb{C}}\right)^{\mathrm{op}}$, where $\mathrm{Poly}_{\mathbb{C}}$ is the category of polynomials over $\mathbb{C}$ and N denotes for the nerve. The admissible morphisms in $\mathcal{T}_{\text{disc}}$ are those isomorphisms and the Grothendieck topology is taken so that $U_\alpha \to X$ is a covering if and only if it is non-empty. The pregeometry $\mathcal{T}_{\text{ét}}$ has underlying ∞-category to be $\mathrm{N}(\mathsf{SmSch}_{\mathbb{C}})$, the category of smooth schemes over $\mathbb{C}$ and a morphism is admissible if it is étale. The Grothendieck topology is taken to be the étale topology. Finally $\mathcal{T}_{\text{an}}$ has the underlying ∞-category to be (the nerve of) complex manifolds and a morphism is admissible if it is étale (namely locally biholomorphic). The Grothendieck topology is also taken to be the étale topology.
For any $\mathcal{T}_{\text{an}}$-structured topos $(\mathcal{X}, \mathcal{O}_X)$, we can associate $\mathcal{O}_X^{\text{alg}}$ sheaf of connective $\mathbb{E}_\infty$-algebras, see [Lur, Construction 11.7]. In particular, a complex analytic space X can be regarded as a pair $\left(\mathcal{X}, \mathcal{O}_X^{\text{alg}}\right)$, where $\mathcal{X} = \mathsf{Sh}\left(X^{\text{top}}\right)$ is the ∞-category of sheaves on the underlying topological space X^{top} and $\mathcal{T}_{\text{an}}$-structure on $\mathcal{O}_X^{\text{alg}}$ is a functor that sends $V \in \mathcal{T}_{\text{an}}$ to sheaf $\mathcal{O}_X^{\text{alg}}(V) \in \mathcal{X}$ such that $\mathcal{O}_X^{\text{alg}}(V)(U) := \mathrm{Hom}\left(\mathcal{O}_V, \mathcal{O}_X^{\text{alg}}|_U\right)$ for any open $U \subseteq X$ and here $\mathcal{O}_V$ is sheaf of germs of holomorphic functions on V.

Definition B.3. A *derived compelx analytic space* is a pair $(\mathcal{X}, \mathcal{O}_X)$, where $\mathcal{X}$ is an ∞-topos and $\mathcal{O}_X$ is a $\mathcal{T}_{\text{an}}$-structure on $\mathcal{X}$ such that there exists an effective epimorphism $\coprod U_i \to \mathbf{1}_{\mathcal{X}}$ with properties:
(1) Locally on U_i, the ∞-topos $\mathcal{X}_{/U_i}$ is category of sheaves on a topological space X_i.
(2) Locally on U_i, the pair $\left(X_i, \pi_0\left(\mathcal{O}_X^{\text{alg}}|_{U_i}\right)\right)$ is a complex analytic space.
(3) For each $k \geqslant 0$, the sheaf $\pi_k\left(\mathcal{O}_X^{\text{alg}}|_{U_i}\right)$ is a coherent sheaf of $\pi_0\left(\mathcal{O}_X^{\text{alg}}|_{U_i}\right)$-module.

By definition we can regard the usual complex analytic spaces as 0-truncated derived complex analytic spaces, but it is unclear this induces an embedding of categories since the category of derived complex analytic spaces is not defined. In fact, for any pregeometry $\mathcal{T}$, we can consider ${}^R\mathsf{Top}(\mathcal{T})$ the ∞-category of $\mathcal{T}$-structured ∞-topoi, then it can be shown that derived complex analytic spaces form a subcategory in ${}^R\mathsf{Top}(\mathcal{T})$ and we use $\mathsf{dAn}_{\mathbb{C}}$ to denote the ∞-category of derived analytic spaces. Moreover, we have the following Theorem of Lurie.

Theorem B.4. *[Lur, Theorem 12.8] The above construction sending complex analytic space X to $\left(\mathsf{Sh}\left(X^{\text{top}}\right), \mathcal{O}_X^{\text{alg}}\right)$ gives a fully-faithful embedding $\mathrm{N}(\mathsf{An}_{\mathbb{C}}) \hookrightarrow \mathsf{dAn}_{\mathbb{C}}$. A derived complex analytic stack $X = (\mathcal{X}, \mathcal{O}_X)$ is in $\mathsf{An}_{\mathbb{C}}$ if and only if $\mathcal{X}$ is 0-localic and $\mathcal{O}_X$ is 0-truncated.*

For $X = (\mathcal{X}, \mathcal{O}_X) \in \mathsf{dAn}_{\mathbb{C}}$, we use $t_0(X)$ to denote $(\mathcal{X}, \pi_0(\mathcal{O}_X))$. We say $X \in \mathsf{dAn}_{\mathbb{C}}$ is *derived affinoid* if $t_0(X)$ is a complex affinoid space, we denote by $\mathsf{dAfd}_{\mathbb{C}}$ the category of derived affinoid spaces.

Definition B.5. The ∞-category of derived analytic stacks is defined to be $\mathsf{Sh}\left(\mathsf{dAfd}_{\mathbb{C}}, \tau_{\text{ét}}\right)^{\wedge}$ the hypercomplete sheaves on derived affinoid spaces with respect to the étale topology.

On the contrary, we recall the ∞-category of derived (algebraic) stacks is defined to be $\mathsf{Sh}\left(\mathsf{dAff}_{\mathbb{C}}, \tau_{\text{ét}}\right)^{\wedge}$, denoted by $\mathsf{dSt}_{\mathbb{C}}$. There is a subcategory of $\mathsf{dSt}_{\mathbb{C}}$ given by the almost finitely presented condition: $\mathsf{dSt}_{\mathbb{C}}^{\mathrm{afp}} := \mathsf{Sh}\left(\mathsf{dAff}_{\mathbb{C}}^{\mathrm{afp}}, \tau_{\text{ét}}\right)^{\wedge}$, where $\mathsf{dAff}_{\mathbb{C}}^{\mathrm{afp}}$ is the category of almost finitely presented derived affine schemes, which can be regarded as a subcategory of ${}^{R}\mathsf{Top}\left(\mathcal{T}_{\text{ét}}\right)$. We have the following adjunction for categories of structured topoi induced by functor between pregeometries: $(-)^{\mathrm{an}} : \mathcal{T}_{\text{ét}} \to \mathcal{T}_{\mathrm{an}}$:

$$\begin{tikzcd} {}^{R}\mathsf{Top}\left(\mathcal{T}_{\mathrm{an}}\right) \arrow[r, bend left, "(-)^{\mathrm{alg}}"] \arrow[r, phantom, "\perp"] & {}^{R}\mathsf{Top}\left(\mathcal{T}_{\text{ét}}\right) \arrow[l, bend left, "(-)^{\mathrm{an}}"] \end{tikzcd}$$

Then it restricts to the functor $(-)^{\mathrm{an}} : \mathsf{dAff}_{\mathbb{C}}^{\mathrm{afp}} \to \mathsf{dAn}_{\mathbb{C}}$. The analytification functor $(-)^{\mathrm{an}} : \mathsf{dSt}_{\mathbb{C}} \to \mathsf{dAnSt}_{\mathbb{C}}$ is then induced from right Kan extension $\mathsf{dAff}_{\mathbb{C}}^{\mathrm{afp}} \hookrightarrow \mathsf{dAff}_{\mathbb{C}}$ applied to the following functor:

$$(-)^{\mathrm{an}} : \mathsf{dSt}_{\mathbb{C}}^{\mathrm{afp}} = \mathsf{Shv}\left(\mathsf{dAff}_{\mathbb{C}}^{\mathrm{afp}}, \tau_{\text{ét}}\right)^{\wedge} \longrightarrow \mathsf{dAnSt}_{\mathbb{C}} = \mathsf{Shv}\left(\mathsf{dAfd}_{\mathbb{C}}, \tau_{\text{ét}}\right)^{\wedge}$$

For our purpose, we need the derived analytic stack of Higgs sheaves, which is expected to be the analytification of the (derived) stack of Higgs sheaves $\mathcal{R}\mathbf{Higgs} := \mathbf{Coh}(X_{\mathsf{Dol}})$. We recall [HP25]:

Example B.6. For X geometric derived stack, consider its tangent bundle defined as

$$TX := \mathrm{Spec}_X\left(\mathrm{Sym}_{\mathcal{O}_X}\left(\mathbb{L}_X\right)\right) \longrightarrow X$$

where $\mathbb{L}_X$ is the cotangent complex of X. Along the zero section of the projection $TX \to X$ we take the formal completion and denote by $\widehat{TX}$, in other words, $\widehat{TX} := (X)_{\mathrm{dR}} \times_{(TX)_{\mathrm{dR}}} TX$. Also, the projection $TX \to X$ gives a relative group structure on X so that we can construct a simplicial space $T^{\bullet}X : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{dSt}_{\mathbb{C}}$ that sends n to $T^nX = TX \times_X TX \times_X \ldots \times_X TX$, similarly we can construct $\widehat{T^{\bullet}X} : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{dSt}_{\mathbb{C}}$. We define the *Dolbeault shape* of X to be $X_{\mathsf{Dol}} := \left|\widehat{T^{\bullet}X}\right| \in (\mathsf{dSt}_{\mathbb{C}})_{/X}$ the geometric realization of $\widehat{T^{\bullet}X}$ and the *nilpotent Dolbeault shape* of X to be $X_{\mathsf{Dol}}^{\mathsf{nil}} := |T^{\bullet}X| \in (\mathsf{dSt}_{\mathbb{C}})_{/X}$, with canonical map $X_{\mathsf{Dol}} \to X_{\mathsf{Dol}}^{\mathsf{nil}}$.

Proposition B.7. *Assume the cotangent complex of X is dualizable, i.e. there exists a tangent complex $\mathbb{T}_X$ dual to $\mathbb{L}_X$ then we have equivalence of category*

$$\mathsf{QCoh}\left(X_{\mathsf{Dol}}\right) \cong \mathsf{Mod}_{\mathrm{Sym}_{\mathcal{O}_X}(\mathbb{T}_X)}\left(\mathsf{QCoh}(X)\right)$$

Also, $\mathsf{Coh}(X_{\mathsf{Dol}})$ (resp. $\mathsf{Coh}(X_{\mathsf{Dol}}^{\mathsf{nil}})$) can be identified with the category of (resp. nilpotent) Higgs sheaves on X.

On the other hand, we can formalize the same procedure to construct Dolbeault shape of an analytic stack. We have the following fact: For X a locally afp geometric stack, $(X_{\mathsf{Dol}})^{\mathrm{an}} \cong (X^{\mathrm{an}})_{\mathsf{Dol}}$.

Example B.8. Another example is the stack of perfect modules/coherent sheaves on a derived stack $\mathcal{X}$. Recall for $A \in \mathsf{dAff}$, then perfect A-modules are compact objects in the category $A\text{-}\mathsf{Mod} =: \mathsf{QCoh}\left(\operatorname{Spec} A\right)$, almost perfect A-modules are those $M \in A\text{-}\mathsf{Mod}$ such that $\pi_i(M) = 0$ for $i \ll 0$ and that the truncation $\tau^{\leqslant n}(M)$ is compact in $A\text{-}\mathsf{mod}^{\leqslant n}$ and coherent sheaves are almost perfect A-modules that with tor-amplitude $\leqslant 0$. For $\mathcal{X} \in \mathsf{dSt}$, the corresponding ∞-categories are given by

$$\mathsf{QCoh}(\mathcal{X}) = \varprojlim_{\operatorname{Spec} A \to \mathcal{X}} \mathsf{QCoh}\left(\operatorname{Spec} A\right); \quad \mathsf{Perf}(\mathcal{X}) = \varprojlim_{\operatorname{Spec} A \to \mathcal{X}} \mathsf{Perf}\left(\operatorname{Spec} A\right);$$

$$\mathsf{APerf}(\mathcal{X}) = \varprojlim_{\operatorname{Spec} A \to \mathcal{X}} \mathsf{APerf}\left(\operatorname{Spec} A\right); \quad \mathsf{Coh}(\mathcal{X}) = \mathsf{APerf}^{\leqslant 0}(X).$$

Correspondingly, we can define the moduli stacks of perfect modules/coherent sheaves as functors:

$$\mathbf{Perf}(\mathcal{X}) : \mathsf{dAff}^{\mathsf{op}} \longrightarrow \mathsf{Spc}; S \mapsto \mathsf{Perf}(\mathcal{X} \times S)^{\simeq} \quad \mathbf{APerf}(\mathcal{X}) : \mathsf{dAff}^{\mathsf{op}} \longrightarrow \mathsf{Spc}; S \mapsto \mathsf{APerf}(\mathcal{X} \times S)^{\simeq}$$

$$\mathbf{Perf}(\mathcal{X}) : \mathsf{dAff}^{\mathsf{op}} \longrightarrow \mathsf{Spc}; S \mapsto \mathsf{Perf}(\mathcal{X} \times S)^{\simeq} \quad \mathbf{APerf}(\mathcal{X}) : \mathsf{dAff}^{\mathsf{op}} \longrightarrow \mathsf{Spc}; S \mapsto \mathsf{APerf}(\mathcal{X} \times S)^{\simeq}$$

$$\mathbf{Coh}(\mathcal{X}) : \mathsf{dAff}^{\mathsf{op}} \longrightarrow \mathsf{Spc}; S \mapsto \mathsf{Coh}_S(\mathcal{X} \times S)^{\simeq} = \mathsf{APerf}_S^{\leqslant 0}\left(\mathcal{X} \times S\right)^{\simeq}$$

where $\mathsf{APerf}_S^{\leqslant 0}$ means the tor-amplitude $\leqslant 0$ relative to S.

Proposition B.9. *Assume $\mathcal{X}$ is a smooth proper scheme over $\mathbb{C}$, let $T^*\mathcal{X} = \operatorname{Spec}\left(\operatorname{Sym}(\mathbb{T}_X)\right)$ be the cotangent space of $\mathcal{X}$, then*

$$\mathsf{Perf}\left(\mathcal{X}_{\mathsf{Dol}}\right) = \mathsf{Perf}_{\mathrm{prop}}\left(T^*\mathcal{X}\right); \quad \mathsf{Perf}\left(\mathcal{X}_{\mathsf{Dol}}^{\mathsf{nil}}\right) = \mathsf{Perf}_{\mathcal{X}}\left(T^*\mathcal{X}\right)$$

where $\mathsf{Perf}_{\mathrm{prop}}(T^*\mathcal{X})$ means perfect complexes on $T^*\mathcal{X}$ with proper support with respect to the projection $T^*\mathcal{X} \to \mathcal{X}$ and $\mathsf{Perf}_{\mathcal{X}}\left(T^*\mathcal{X}\right) \subseteq \mathsf{Perf}_{\mathrm{prop}}(T^*\mathcal{X})$ is subcategory of those support on the zero section.

On the other hand, we have the analytic counterpart of above construction, by replacing dAff with $\mathsf{dAfd}_{\mathbb{C}}$. We denote by $\mathbf{AnPerf}(\mathcal{X})$ and $\mathbf{AnCoh}(\mathcal{X})$ the corresponding stack of analytic perfect modules/coherent sheaves on $\mathcal{X} \in \mathsf{dAnSt}_{\mathbb{C}}$.

Theorem B.10. *[HP25] (1) Assume $\mathcal{X} \in \mathsf{dSt}_{\mathbb{C}}$ is locally almost finitely presented and proper and that $\mathbf{Maps}\left(\mathcal{X}, \mathbf{Perf}_{\mathbb{C}}\right)$ is locally geometric and locally almost finitely presented, then there is an equivalence:*

$$\left(\mathbf{Maps}\left(\mathcal{X}, \mathbf{Perf}\right)\right)^{\mathrm{an}} \xrightarrow{\cong} \mathbf{AnMaps}\left(\mathcal{X}^{\mathrm{an}}, \mathbf{AnPerf}_{\mathbb{C}}\right)$$

(2) Assume X is a proper scheme or one of the Simpson shapes of a proper scheme, then there is an equivalence:

$$\left(\mathbf{Coh}(\mathcal{X})\right)^{\mathrm{an}} \xrightarrow{\cong} \mathbf{AnCoh}\left(\mathcal{X}^{\mathrm{an}}\right)$$

Combining Theorem B.10 with Example B.6, we have the following analytification of derived stack of Higgs sheaves (resp. nilpotent Higgs sheaves) on curve X:

$$\left(\mathcal{R}\mathbf{Higgs}\right)^{\mathrm{an}} = \left(\mathbf{Coh}\left(X_{\mathsf{Dol}}\right)\right)^{\mathrm{an}} \cong \mathbf{AnCoh}\left(\left(X_{\mathsf{Dol}}\right)^{\mathrm{an}}\right) \cong \mathbf{AnCoh}\left(\left(X^{\mathrm{an}}\right)_{\mathsf{Dol}}\right)$$

$$\left(\mathcal{R}\boldsymbol{\Lambda}\right)^{\mathrm{an}} = \left(\mathbf{Coh}\left(X_{\mathsf{Dol}}^{\mathsf{nil}}\right)\right)^{\mathrm{an}} \cong \mathbf{AnCoh}\left(\left(X_{\mathsf{Dol}}^{\mathsf{nil}}\right)^{\mathrm{an}}\right) \cong \mathbf{AnCoh}\left(\left(X^{\mathrm{an}}\right)_{\mathsf{Dol}}^{\mathsf{nil}}\right)$$

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